

Global Benchmark

ÁTILA LOPES COELHO

# COMPUTATIONAL HARMONIC ATLAS

420 SCALES  
× 17 TONIC VARIANTS  

---

7,140 INSTANCES

V.1  
2026

Included: dataset.py



*“Carry a handful of earth every day and you will  
build a mountain.”*

(Confucius)

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## INTRODUCTION

### About This Document

All scales in this document were generated automatically by the **Musical Oracle**.

***Technical Note:** No scale was manually edited or corrected.*

### What is the Musical Oracle?

The Musical Oracle is a deterministic computational system capable of generating harmonic structures (scales and chords) in real time from formal rules, without reliance on precomputed lookup tables.

***Technical Note:** The system does not generate microtonal structures.*

### Why Was This Document Conceived?

To demonstrate, in a structured and verifiable manner, the consistency, reproducibility, and scope of the theorem that underlies the Oracle.

## **Market Impact**

### **Immediate Structural Effects:**

- The end of the data monopoly
- musical scales have effectively become commodities.

### **Competitive advantage now shifts entirely toward:**

- How structures are generated.
- How they are validated.
- How the system evolves.
- ...

### **Within this context:**

- The database levels the playing field;
- The engine defines leadership.

## IMPLEMENTATION & ARCHITECTURE

### Nature of the System

The Oracle can be described as a quasi-DSL (Domain-Specific Language), as it already encapsulates the semantic rules of the musical domain, although it does not yet possess its own formal syntax.

***Technical Note:*** *The Oracle was developed entirely using native language resources. Therefore, it does not depend on external libraries, frameworks, or specialized packages.*

## Core Architecture

**Table 1** – Core Architecture

| COMPONENT                   | SPECIFICATION               |
|-----------------------------|-----------------------------|
| Language                    | Python                      |
| Design                      | Independent Modules         |
| Database (Structural Seeds) | 147KB – 1,768 Lines of Code |
| Interface (Optional)        | 96KB – 2,177 Lines of Code  |
| Kernel                      | 25KB – 430 Lines of Code    |

## Kernel Code Distribution

**Table 2** – Kernel Code Distribution

| COMPONENT                                                      | SPECIFICATION                        |
|----------------------------------------------------------------|--------------------------------------|
| Imports                                                        | 4 Lines of Code                      |
| Constants                                                      | 16 Lines of Code                     |
| Infrastructure<br>(TypedDict, Dataclass, Input Resolver, etc.) | Remaining Lines                      |
| Self-Validation (Optional)                                     | 78 Lines of Code                     |
| Structural Calculation                                         | 35 Lines of Code (Properly Isolated) |

**Technical Note:** The theorem responsible for generating musical structures is materialized in just 35 lines of code. Everything else constitutes standard software infrastructure and contains no relevant musical logic.

## Performance Metrics

**Table 3** – Test Environment

| COMPONENT | SPECIFICATION       |
|-----------|---------------------|
| Hardware  | Lenovo Ideapad S145 |
| Processor | Intel i5-1035G1     |
| Memory    | 4 GB of RAM         |
| System    | Windows 11          |

**Table 4** – Performance Metrics

| METRIC                            | RESULT      |
|-----------------------------------|-------------|
| Generation & Validation Precision | 99.999%     |
| Average Response Time             | < 0.4399 ms |
| Memory Consumption                | < 4.85 KB   |

**Technical Note:** The current codebase has not yet undergone an advanced optimization process. Consequently, there remains significant potential to reduce code size, improve memory efficiency, and bring processing time closer to its theoretical minimum.

## Scalability

**Table 5** – Atlas V.1 Published Capacity

| ENTITY           | VOLUME |
|------------------|--------|
| Scales           | 420    |
| Tonic Variations | x 17   |
| Instances        | 7,140  |

**Table 6** – Current Oracle Capacity

| ENTITY              | VOLUME |
|---------------------|--------|
| Scales              | 420    |
| Tonic Variations    | x 35   |
| Generated Instances | 14,700 |

**Table 7** – Theoretical Oracle Capacity

| ENTITY              | VOLUME    |
|---------------------|-----------|
| Scales              | Unbounded |
| Tonic Variations    | x 35      |
| Generated Instances | Unbounded |

**Technical Note:** Average time required to register a scale and verify the 35 variations: between 1 and 5 minutes.



## Automatic Validation System

1. Generate the scale using one program (e.g., Musical Oracle);
2. Analyze the intervals of the scale using another program (e.g., Bidirectional Interval Analyzer);
3. Compare the intervals (Reference Database vs. Interval Analyzer).

If the scale is incorrect, the intervals will not match. The inconsistency will be detected automatically, and the system will indicate both the error and the correction suggestion.

Example:

- **Position in Structure:** 2
- **Reference Interval (Database/User):** m2
- **Obtained Interval (Analyzer):** M2
- **Error:** Qualitative.

***Technical Note:** This mechanism is only viable because the harmonic generator and the interval analyzer are completely independent; that is, they share neither internal logic nor reciprocal "knowledge." Furthermore, both operate with 99.999% precision. If this precision is reduced, validation becomes an automated opinion. With both operating at 99.999% precision, validation ceases to be interpretative and becomes deterministic—that is, an executable law.*

***Technical Note:** To prevent complacency, a footnote was added: “Automatic validation with 99.999% precision does not replace critical human analysis.”*

***Technical Note:** The harmonic generator, the bidirectional analyzer, and their respective calculation traces can be constructed in just 3 steps each.*

## System Versatility & Data Integrity

The Oracle is capable of generating scales with different cardinalities, levels of complexity, and geographic origins.

**Table 8** – Scale Distribution

| NUMBER OF NOTES | NUMBER OF SCALES |
|-----------------|------------------|
| 3               | 4                |
| 4               | 10               |
| 5               | 88               |
| 6               | 97               |
| 7               | 149              |
| 8               | 49               |
| 9               | 15               |
| 10              | 6                |
| 11              | 0                |
| 12              | 2                |
| <b>TOTAL</b>    | <b>420</b>       |

**Table 9** – Duplicate Structures

| NUMBER OF NOTES | DUPLICATE GROUPS | STRUCTURES INVOLVED |
|-----------------|------------------|---------------------|
| 5               | 9                | 19                  |
| 6               | 2                | 4                   |
| 7               | 17               | 42                  |
| 8               | 6                | 15                  |
| <b>TOTAL</b>    | <b>34</b>        | <b>80</b>           |

**Technical Note:** Scales generally contain between 3 and 12 notes; therefore, this Atlas restricts itself to that spectrum. The Oracle, however, generates structures with 2 to  $X$  notes.

**Technical Note:** Eleven-note scales were omitted due to insufficient reliable references.

**Technical Note:** Structurally equivalent scales (duplicates) were included in the count.

## Tonal Matrix

The Oracle generates 35 tonal variations per scale — that is, the complete tonal matrix. However, this edition of the Computational Harmonic Atlas V.1 deliberately presents only 17 tonal variations per scale, selected according to conventional market standards, in order to enable direct comparison with the solutions currently available.

**Table 10** – Tonal Matrix

| NATURAL | SHARP | FLAT | DOUBLE SHARP | DOUBLE FLAT |
|---------|-------|------|--------------|-------------|
| C       | C#    | Cb   | C##          | Cbb         |
| D       | D#    | Db   | D##          | Dbb         |
| E       | E#    | Eb   | E##          | Ebb         |
| F       | F#    | Fb   | F##          | Fbb         |
| G       | G#    | Gb   | G##          | Gbb         |
| A       | A#    | Ab   | A##          | Abb         |
| B       | B#    | Bb   | B##          | Bbb         |

**Spoiler:** *Atlas V.2 will contain the complete tonal matrix.*

## Interval Analysis

**Table 11** – Interval Volumetric Summary

| METRIC                     | VALUE |
|----------------------------|-------|
| Total structures analyzed  | 420   |
| Total interval occurrences | 3148  |
| Distinct interval types    | 20    |

**Table 12** – Interval Distribution

| INTERVAL | OCCURRENCES |
|----------|-------------|
| P1       | 420         |
| Aug1     | 3           |
| m2       | 156         |
| M2       | 208         |
| Aug2     | 14          |
| m3       | 183         |
| M3       | 206         |
| dim4     | 7           |
| P4       | 245         |
| Aug4     | 56          |
| dim5     | 139         |
| P5       | 299         |
| Aug5     | 16          |
| m6       | 180         |
| M6       | 198         |
| Aug6     | 12          |
| dim7     | 5           |
| m7       | 195         |
| M7       | 186         |
| P8       | 420         |

**Table 13** — Interval Legend

| PERFECT | MAJOR | MINOR | DIMINISHED | AUGMENTED |
|---------|-------|-------|------------|-----------|
| P1      | M2    | m2    | dim4       | Aug1      |
| P4      | M3    | m3    | dim5       | Aug2      |
| P5      | M6    | m6    | dim7       | Aug4      |
| P8      | M7    | m7    |            | Aug5      |
|         |       |       |            | Aug6      |

**Technical Note:** The Oracle was deliberately limited to 10× diminished (10xD) and 10× augmented (10xA).

**Table 14** – Overall Ranking by Frequency

| <b>RANK</b> | <b>INTERVAL</b> | <b>OCCURRENCES</b> |
|-------------|-----------------|--------------------|
| 1           | P1              | 420                |
| 2           | P8              | 420                |
| 3           | P5              | 299                |
| 4           | P4              | 245                |
| 5           | M2              | 208                |
| 6           | M3              | 206                |
| 7           | M6              | 198                |
| 8           | m7              | 195                |
| 9           | M7              | 186                |
| 10          | m3              | 183                |
| 11          | m6              | 180                |
| 12          | m2              | 156                |
| 13          | dim5            | 139                |
| 14          | Aug4            | 56                 |
| 15          | Aug5            | 16                 |
| 16          | Aug2            | 14                 |
| 17          | Aug6            | 12                 |
| 18          | dim4            | 7                  |
| 19          | dim7            | 5                  |
| 20          | Aug1            | 3                  |

**Technical Note:** P1 and P8 occur in all structures; therefore, they should be excluded from analytical ranking.

## DATA GOVERNANCE & TAXONOMY

### Data Organization

The scales were organized by Cardinality (number of notes) and by Alphabetical Order within each group.

### Content Structure

#### 1. STRUCTURAL IDENTIFICATION

- Major

#### 2. STRUCTURAL SEEDS:

- Step Pattern: 2, 2, 1, 2, 2, 2, 1
- Intervals (Ascending): P1, M2, M3, P4, P5, M6, M7, P8

**Technical Note:** Semitone = 1, Tone = 2.

#### 3. GENERATED OUTPUT

- C, D, E, F, G, A, B, C
- C#, D#, E#, F#, G#, A#, B#, C#

**Technical Note:** The notation adopted in this document is strictly theoretical. Objective example: (E#) is not replaced with (F). Although this approach reduces readability, it prevents harmonic paradoxes and interval inconsistencies. Notational simplifications must be handled exclusively at the upper layer (interface).

## GLOBAL CHALLENGE

**Find a single notation error that contradicts the underlying theory. Good luck.**

***Technical Note:** To facilitate verification, the use of music software is recommended, for example, Guitar Pro (GP8), as both systems have the same scales in the same keys. For automated validation, use the `dataset.py` file.*

### Benchmark: Oracle vs. GP8

**Table 15** – Benchmark: Oracle vs. GP8

| METRIC         | RESULT                       |
|----------------|------------------------------|
| Scope          | 7,140 instances              |
| Correspondence | Full                         |
| Exceptions     | GP8 notation simplifications |

### What if I find one wrong note?

Finding an error in the Oracle is equivalent to finding an error in music theory itself. Analogy: if a calculator correctly computes  $1 \times 1$ , it will also correctly compute  $12345 \times 67890$ . Therefore, changing the values does not alter the logic. Any eventual errors are associated exclusively with poor computational implementation—for example, inconsistencies within the database—and never with the theorem itself. However, should the impossible occur—that is, if the error originates from the theorem itself—this would still not compromise its value, because the system would remain, by a wide margin, more precise than the solutions currently available on the market. Furthermore, due to its lean and deterministic logical core, any flaw would be detectable, isolatable, and correctable. Therefore, the theorem is not founded upon an unrealistic promise of absolute perfection, but rather upon consistency and capacity for evolution, remaining technically superior even in the presence of eventual imperfections.

### 3 – NOTE SCALES



**Peruvian Tritonic M6**

Step pattern: 3, 6, 3

Ascending intervals: P1, m3, M6, P8

C, Eb, A, C

C#, E, A#, C#

Db, Fb, Bb, Db

D, F, B, D

D#, F#, B#, D#

Eb, Gb, C, Eb

E, G, C#, E

F, Ab, D, F

F#, A, D#, F#

Gb, Bbb, Eb, Gb

G, Bb, E, G

G#, B, E#, G#

Ab, Cb, F, Ab

A, C, F#, A

A#, C#, F##, A#

Bb, Db, G, Bb

B, D, G#, B

**Sansagari**

Step pattern: 5, 5, 2

Ascending intervals: P1, P4, Aug6, P8

C, F, A#, C

C#, F#, A##, C#

Db, Gb, B, Db

D, G, B#, D

D#, G#, B##, D#

Eb, Ab, C#, Eb

E, A, C##, E

F, Bb, D#, F

F#, B, D##, F#

Gb, Cb, E, Gb

G, C, E#, G

G#, C#, E##, G#

Ab, Db, F#, Ab

A, D, F##, A

A#, D#, F###, A#

Bb, Eb, G#, Bb

B, E, G##, B

**Ute Tritonic**

Step pattern: 3, 7, 2

Ascending intervals: P1, m3, m7, P8

C, Eb, Bb, C

C#, E, B, C#

Db, Fb, Cb, Db

D, F, C, D

D#, F#, C#, D#

Eb, Gb, Db, Eb

E, G, D, E

F, Ab, Eb, F

F#, A, E, F#

Gb, Bbb, Fb, Gb

G, Bb, F, G

G#, B, F#, G#

Ab, Cb, Gb, Ab

A, C, G, A

A#, C#, G#, A#

Bb, Db, Ab, Bb

B, D, A, B

**Warao Tritonic**

Step pattern: 5, 2, 5

Ascending intervals: P1, P4, P5, P8

C, F, G, C

C#, F#, G#, C#

Db, Gb, Ab, Db

D, G, A, D

D#, G#, A#, D#

Eb, Ab, Bb, Eb

E, A, B, E

F, Bb, C, F

F#, B, C#, F#

Gb, Cb, Db, Gb

G, C, D, G

G#, C#, D#, G#

Ab, Db, Eb, Ab

A, D, E, A

A#, D#, E#, A#

Bb, Eb, F, Bb

B, E, F#, B

## 4 – NOTE SCALES

**Genus Primum**

Step pattern: 2, 3, 2, 5

Ascending intervals: P1, M2, P4, P5, P8

C, D, F, G, C

C#, D#, F#, G#, C#

Db, Eb, Gb, Ab, Db

D, E, G, A, D

D#, E#, G#, A#, D#

Eb, F, Ab, Bb, Eb

E, F#, A, B, E

F, G, Bb, C, F

F#, G#, B, C#, F#

Gb, Ab, Cb, Db, Gb

G, A, C, D, G

G#, A#, C#, D#, G#

Ab, Bb, Db, Eb, Ab

A, B, D, E, A

A#, B#, D#, E#, A#

Bb, C, Eb, F, Bb

B, C#, E, F#, B

**Gowleeswari**

Step pattern: 1, 4, 3, 4

Ascending intervals: P1, m2, P4, m6, P8

C, Db, F, Ab, C

C#, D, F#, A, C#

Db, Ebb, Gb, Bbb, Db

D, Eb, G, Bb, D

D#, E, G#, B, D#

Eb, Fb, Ab, Cb, Eb

E, F, A, C, E

F, Gb, Bb, Db, F

F#, G, B, D, F#

Gb, Abb, Cb, Ebb, Gb

G, Ab, C, Eb, G

G#, A, C#, E, G#

Ab, Bbb, Db, Fb, Ab

A, Bb, D, F, A

A#, B, D#, F#, A#

Bb, Cb, Eb, Gb, Bb

B, C, E, G, B

**Lavangi**

Step pattern: 1, 6, 3, 2

Ascending intervals: P1, m2, P5, m7, P8

C, Db, G, Bb, C

C#, D, G#, B, C#

Db, Ebb, Ab, Cb, Db

D, Eb, A, C, D

D#, E, A#, C#, D#

Eb, Fb, Bb, Db, Eb

E, F, B, D, E

F, Gb, C, Eb, F

F#, G, C#, E, F#

Gb, Abb, Db, Fb, Gb

G, Ab, D, F, G

G#, A, D#, F#, G#

Ab, Bbb, Eb, Gb, Ab

A, Bb, E, G, A

A#, B, E#, G#, A#

Bb, Cb, F, Ab, Bb

B, C, F#, A, B



**Messiaen Truncated Mode 5**

Step pattern: 1, 5, 1, 5

Ascending intervals: P1, m2, Aug4, P5, P8

C, Db, F#, G, C

C#, D, F##, G#, C#

Db, Ebb, G, Ab, Db

D, Eb, G#, A, D

D#, E, G##, A#, D#

Eb, Fb, A, Bb, Eb

E, F, A#, B, E

F, Gb, B, C, F

F#, G, B#, C#, F#

Gb, Abb, C, Db, Gb

G, Ab, C#, D, G

G#, A, C##, D#, G#

Ab, Bbb, D, Eb, Ab

A, Bb, D#, E, A

A#, B, D##, E#, A#

Bb, Cb, E, F, Bb

B, C, E#, F#, B

## Messiaen Truncated Mode 5 Inverse

Step pattern: 5, 1, 5, 1

Ascending intervals: P1, P4, Aug4, M7, P8

C, F, F#, B, C

C#, F#, F##, B#, C#

Db, Gb, G, C, Db

D, G, G#, C#, D

D#, G#, G##, C##, D#

Eb, Ab, A, D, Eb

E, A, A#, D#, E

F, Bb, B, E, F

F#, B, B#, E#, F#

Gb, Cb, C, F, Gb

G, C, C#, F#, G

G#, C#, C##, F##, G#

Ab, Db, D, G, Ab

A, D, D#, G#, A

A#, D#, D##, G##, A#

Bb, Eb, E, A, Bb

B, E, E#, A#, B

## Messiaen Truncated Mode 6

Step pattern: 2, 4, 2, 4

Ascending intervals: P1, M2, Aug4, Aug5, P8

C, D, F#, G#, C

C#, D#, F##, G##, C#

Db, Eb, G, A, Db

D, E, G#, A#, D

D#, E#, G##, A##, D#

Eb, F, A, B, Eb

E, F#, A#, B#, E

F, G, B, C#, F

F#, G#, B#, C##, F#

Gb, Ab, C, D, Gb

G, A, C#, D#, G

G#, A#, C##, D##, G#

Ab, Bb, D, E, Ab

A, B, D#, E#, A

A#, B#, D##, E##, A#

Bb, C, E, F#, Bb

B, C#, E#, F##, B

## Messiaen Truncated Mode 6 Inverse

Step pattern: 4, 2, 4, 2

Ascending intervals: P1, M3, Aug4, Aug6, P8

C, E, F#, A#, C

C#, E#, F##, A##, C#

Db, F, G, B, Db

D, F#, G#, B#, D

D#, F##, G##, B##, D#

Eb, G, A, C#, Eb

E, G#, A#, C##, E

F, A, B, D#, F

F#, A#, B#, D##, F#

Gb, Bb, C, E, Gb

G, B, C#, E#, G

G#, B#, C##, E##, G#

Ab, C, D, F#, Ab

A, C#, D#, F##, A

A#, C##, D##, F###, A#

Bb, D, E, G#, Bb

B, D#, E#, G##, B

## Nigamagamini

Step pattern: 4, 2, 5, 1

Ascending intervals: P1, M3, dim5, M7, P8

C, E, Gb, B, C

C#, E#, G, B#, C#

Db, F, Abb, C, Db

D, F#, Ab, C#, D

D#, F##, A, C##, D#

Eb, G, Bbb, D, Eb

E, G#, Bb, D#, E

F, A, Cb, E, F

F#, A#, C, E#, F#

Gb, Bb, Dbb, F, Gb

G, B, Db, F#, G

G#, B#, D, F##, G#

Ab, C, Ebb, G, Ab

A, C#, Eb, G#, A

A#, C##, E, G##, A#

Bb, D, Fb, A, Bb

B, D#, F, A#, B

**Samukam**

Step pattern: 2, 4, 5, 1

Ascending intervals: P1, M2, Aug4, M7, P8

C, D, F#, B, C

C#, D#, F##, B#, C#

Db, Eb, G, C, Db

D, E, G#, C#, D

D#, E#, G##, C##, D#

Eb, F, A, D, Eb

E, F#, A#, D#, E

F, G, B, E, F

F#, G#, B#, E#, F#

Gb, Ab, C, F, Gb

G, A, C#, F#, G

G#, A#, C##, F##, G#

Ab, Bb, D, G, Ab

A, B, D#, G#, A

A#, B#, D##, G##, A#

Bb, C, E, A, Bb

B, C#, E#, A#, B

## Three Semitone Scale

Step pattern: 3, 3, 3, 3

Ascending intervals: P1, Aug2, Aug4, M6, P8

C, D#, F#, A, C

C#, D##, F##, A#, C#

Db, E, G, Bb, Db

D, E#, G#, B, D

D#, E##, G##, B#, D#

Eb, F#, A, C, Eb

E, F##, A#, C#, E

F, G#, B, D, F

F#, G##, B#, D#, F#

Gb, A, C, Eb, Gb

G, A#, C#, E, G

G#, A##, C##, E#, G#

Ab, B, D, F, Ab

A, B#, D#, F#, A

A#, B##, D##, F##, A#

Bb, C#, E, G, Bb

B, C##, E#, G#, B

## 5 – NOTE SCALES



## Abhogi

Step pattern: 2, 1, 2, 4, 3

Ascending intervals: P1, M2, m3, P4, M6, P8

C, D, Eb, F, A, C

C#, D#, E, F#, A#, C#

Db, Eb, Fb, Gb, Bb, Db

D, E, F, G, B, D

D#, E#, F#, G#, B#, D#

Eb, F, Gb, Ab, C, Eb

E, F#, G, A, C#, E

F, G, Ab, Bb, D, F

F#, G#, A, B, D#, F#

Gb, Ab, Bbb, Cb, Eb, Gb

G, A, Bb, C, E, G

G#, A#, B, C#, E#, G#

Ab, Bb, Cb, Db, F, Ab

A, B, C, D, F#, A

A#, B#, C#, D#, F##, A#

Bb, C, Db, Eb, G, Bb

B, C#, D, E, G#, B

**Aeolian Pentatonic (5th mode of major pentatonic)**

Step pattern: 3, 2, 2, 3, 2

Ascending intervals: P1, m3, P4, P5, m7, P8

C, Eb, F, G, Bb, C

C#, E, F#, G#, B, C#

Db, Fb, Gb, Ab, Cb, Db

D, F, G, A, C, D

D#, F#, G#, A#, C#, D#

Eb, Gb, Ab, Bb, Db, Eb

E, G, A, B, D, E

F, Ab, Bb, C, Eb, F

F#, A, B, C#, E, F#

Gb, Bbb, Cb, Db, Fb, Gb

G, Bb, C, D, F, G

G#, B, C#, D#, F#, G#

Ab, Cb, Db, Eb, Gb, Ab

A, C, D, E, G, A

A#, C#, D#, E#, G#, A#

Bb, Db, Eb, F, Ab, Bb

B, D, E, F#, A, B

**Akebono**

Step pattern: 2, 1, 4, 2, 3

Ascending intervals: P1, M2, m3, P5, M6, P8

C, D, Eb, G, A, C

C#, D#, E, G#, A#, C#

Db, Eb, Fb, Ab, Bb, Db

D, E, F, A, B, D

D#, E#, F#, A#, B#, D#

Eb, F, Gb, Bb, C, Eb

E, F#, G, B, C#, E

F, G, Ab, C, D, F

F#, G#, A, C#, D#, F#

Gb, Ab, Bbb, Db, Eb, Gb

G, A, Bb, D, E, G

G#, A#, B, D#, E#, G#

Ab, Bb, Cb, Eb, F, Ab

A, B, C, E, F#, A

A#, B#, C#, E#, F##, A#

Bb, C, Db, F, G, Bb

B, C#, D, F#, G#, B

**Akebono (suspended)**

Step pattern: 1, 4, 2, 1, 4

Ascending intervals: P1, m2, P4, P5, m6, P8

C, Db, F, G, Ab, C

C#, D, F#, G#, A, C#

Db, Ebb, Gb, Ab, Bbb, Db

D, Eb, G, A, Bb, D

D#, E, G#, A#, B, D#

Eb, Fb, Ab, Bb, Cb, Eb

E, F, A, B, C, E

F, Gb, Bb, C, Db, F

F#, G, B, C#, D, F#

Gb, Abb, Cb, Db, Ebb, Gb

G, Ab, C, D, Eb, G

G#, A, C#, D#, E, G#

Ab, Bbb, Db, Eb, Fb, Ab

A, Bb, D, E, F, A

A#, B, D#, E#, F#, A#

Bb, Cb, Eb, F, Gb, Bb

B, C, E, F#, G, B

## Altered Pentatonic

Step pattern: 1, 4, 2, 2, 3

Ascending intervals: P1, m2, P4, P5, M6, P8

C, Db, F, G, A, C

C#, D, F#, G#, A#, C#

Db, Ebb, Gb, Ab, Bb, Db

D, Eb, G, A, B, D

D#, E, G#, A#, B#, D#

Eb, Fb, Ab, Bb, C, Eb

E, F, A, B, C#, E

F, Gb, Bb, C, D, F

F#, G, B, C#, D#, F#

Gb, Abb, Cb, Db, Eb, Gb

G, Ab, C, D, E, G

G#, A, C#, D#, E#, G#

Ab, Bbb, Db, Eb, F, Ab

A, Bb, D, E, F#, A

A#, B, D#, E#, F##, A#

Bb, Cb, Eb, F, G, Bb

B, C, E, F#, G#, B

## Audav Tukhari

Step pattern: 2, 1, 2, 3, 4

Ascending intervals: P1, M2, m3, P4, m6, P8

C, D, Eb, F, Ab, C

C#, D#, E, F#, A, C#

Db, Eb, Fb, Gb, Bbb, Db

D, E, F, G, Bb, D

D#, E#, F#, G#, B, D#

Eb, F, Gb, Ab, Cb, Eb

E, F#, G, A, C, E

F, G, Ab, Bb, Db, F

F#, G#, A, B, D, F#

Gb, Ab, Bbb, Cb, Ebb, Gb

G, A, Bb, C, Eb, G

G#, A#, B, C#, E, G#

Ab, Bb, Cb, Db, Fb, Ab

A, B, C, D, F, A

A#, B#, C#, D#, F#, A#

Bb, C, Db, Eb, Gb, Bb

B, C#, D, E, G, B

**Bacovia (Romania)**

Step pattern: 4, 1, 3, 3, 1

Ascending intervals: P1, M3, P4, m6, M7, P8

C, E, F, Ab, B, C

C#, E#, F#, A, B#, C#

Db, F, Gb, Bbb, C, Db

D, F#, G, Bb, C#, D

D#, F##, G#, B, C##, D#

Eb, G, Ab, Cb, D, Eb

E, G#, A, C, D#, E

F, A, Bb, Db, E, F

F#, A#, B, D, E#, F#

Gb, Bb, Cb, Ebb, F, Gb

G, B, C, Eb, F#, G

G#, B#, C#, E, F##, G#

Ab, C, Db, Fb, G, Ab

A, C#, D, F, G#, A

A#, C##, D#, F#, G##, A#

Bb, D, Eb, Gb, A, Bb

B, D#, E, G, A#, B

## Balinese Pelog

Step pattern: 1, 2, 4, 1, 4

Ascending intervals: P1, m2, m3, P5, m6, P8

C, Db, Eb, G, Ab, C

C#, D, E, G#, A, C#

Db, Ebb, Fb, Ab, Bbb, Db

D, Eb, F, A, Bb, D

D#, E, F#, A#, B, D#

Eb, Fb, Gb, Bb, Cb, Eb

E, F, G, B, C, E

F, Gb, Ab, C, Db, F

F#, G, A, C#, D, F#

Gb, Abb, Bbb, Db, Ebb, Gb

G, Ab, Bb, D, Eb, G

G#, A, B, D#, E, G#

Ab, Bbb, Cb, Eb, Fb, Ab

A, Bb, C, E, F, A

A#, B, C#, E#, F#, A#

Bb, Cb, Db, F, Gb, Bb

B, C, D, F#, G, B



## Blues Pentatonic

Step pattern: 3, 2, 1, 4, 2

Ascending intervals: P1, m3, P4, dim5, m7, P8

C, Eb, F, Gb, Bb, C

C#, E, F#, G, B, C#

Db, Fb, Gb, Abb, Cb, Db

D, F, G, Ab, C, D

D#, F#, G#, A, C#, D#

Eb, Gb, Ab, Bbb, Db, Eb

E, G, A, Bb, D, E

F, Ab, Bb, Cb, Eb, F

F#, A, B, C, E, F#

Gb, Bbb, Cb, Dbb, Fb, Gb

G, Bb, C, Db, F, G

G#, B, C#, D, F#, G#

Ab, Cb, Db, Ebb, Gb, Ab

A, C, D, Eb, G, A

A#, C#, D#, E, G#, A#

Bb, Db, Eb, Fb, Ab, Bb

B, D, E, F, A, B

**Budhamanohari**

Step pattern: 2, 2, 1, 2, 5

Ascending intervals: P1, M2, M3, P4, P5, P8

C, D, E, F, G, C

C#, D#, E#, F#, G#, C#

Db, Eb, F, Gb, Ab, Db

D, E, F#, G, A, D

D#, E#, F##, G#, A#, D#

Eb, F, G, Ab, Bb, Eb

E, F#, G#, A, B, E

F, G, A, Bb, C, F

F#, G#, A#, B, C#, F#

Gb, Ab, Bb, Cb, Db, Gb

G, A, B, C, D, G

G#, A#, B#, C#, D#, G#

Ab, Bb, C, Db, Eb, Ab

A, B, C#, D, E, A

A#, B#, C##, D#, E#, A#

Bb, C, D, Eb, F, Bb

B, C#, D#, E, F#, B

**Chad Gadyo**

Step pattern: 2, 1, 2, 2, 5

Ascending intervals: P1, M2, m3, P4, P5, P8

C, D, Eb, F, G, C

C#, D#, E, F#, G#, C#

Db, Eb, Fb, Gb, Ab, Db

D, E, F, G, A, D

D#, E#, F#, G#, A#, D#

Eb, F, Gb, Ab, Bb, Eb

E, F#, G, A, B, E

F, G, Ab, Bb, C, F

F#, G#, A, B, C#, F#

Gb, Ab, Bbb, Cb, Db, Gb

G, A, Bb, C, D, G

G#, A#, B, C#, D#, G#

Ab, Bb, Cb, Db, Eb, Ab

A, B, C, D, E, A

A#, B#, C#, D#, E#, A#

Bb, C, Db, Eb, F, Bb

B, C#, D, E, F#, B

## Chaio (Pentatonic)

Step pattern: 2, 3, 3, 2, 2

Ascending intervals: P1, M2, P4, m6, m7, P8

C, D, F, Ab, Bb, C

C#, D#, F#, A, B, C#

Db, Eb, Gb, Bbb, Cb, Db

D, E, G, Bb, C, D

D#, E#, G#, B, C#, D#

Eb, F, Ab, Cb, Db, Eb

E, F#, A, C, D, E

F, G, Bb, Db, Eb, F

F#, G#, B, D, E, F#

Gb, Ab, Cb, Ebb, Fb, Gb

G, A, C, Eb, F, G

G#, A#, C#, E, F#, G#

Ab, Bb, Db, Fb, Gb, Ab

A, B, D, F, G, A

A#, B#, D#, F#, G#, A#

Bb, C, Eb, Gb, Ab, Bb

B, C#, E, G, A, B

## Chandrakauns

Step pattern: 3, 2, 3, 3, 1

Ascending intervals: P1, m3, P4, m6, M7, P8

C, Eb, F, Ab, B, C

C#, E, F#, A, B#, C#

Db, Fb, Gb, Bbb, C, Db

D, F, G, Bb, C#, D

D#, F#, G#, B, C##, D#

Eb, Gb, Ab, Cb, D, Eb

E, G, A, C, D#, E

F, Ab, Bb, Db, E, F

F#, A, B, D, E#, F#

Gb, Bbb, Cb, Ebb, F, Gb

G, Bb, C, Eb, F#, G

G#, B, C#, E, F##, G#

Ab, Cb, Db, Fb, G, Ab

A, C, D, F, G#, A

A#, C#, D#, F#, G##, A#

Bb, Db, Eb, Gb, A, Bb

B, D, E, G, A#, B

## Chaya Todi

Step pattern: 1, 2, 3, 2, 4

Ascending intervals: P1, m2, m3, dim5, m6, P8

C, Db, Eb, Gb, Ab, C

C#, D, E, G, A, C#

Db, Ebb, Fb, Abb, Bbb, Db

D, Eb, F, Ab, Bb, D

D#, E, F#, A, B, D#

Eb, Fb, Gb, Bbb, Cb, Eb

E, F, G, Bb, C, E

F, Gb, Ab, Cb, Db, F

F#, G, A, C, D, F#

Gb, Abb, Bbb, Dbb, Ebb, Gb

G, Ab, Bb, Db, Eb, G

G#, A, B, D, E, G#

Ab, Bbb, Cb, Ebb, Fb, Ab

A, Bb, C, Eb, F, A

A#, B, C#, E, F#, A#

Bb, Cb, Db, Fb, Gb, Bb

B, C, D, F, G, B

**Chin**

Step pattern: 3, 3, 2, 2, 2

Ascending intervals: P1, m3, dim5, m6, m7, P8

C, Eb, Gb, Ab, Bb, C

C#, E, G, A, B, C#

Db, Fb, Abb, Bbb, Cb, Db

D, F, Ab, Bb, C, D

D#, F#, A, B, C#, D#

Eb, Gb, Bbb, Cb, Db, Eb

E, G, Bb, C, D, E

F, Ab, Cb, Db, Eb, F

F#, A, C, D, E, F#

Gb, Bbb, Dbb, Ebb, Fb, Gb

G, Bb, Db, Eb, F, G

G#, B, D, E, F#, G#

Ab, Cb, Ebb, Fb, Gb, Ab

A, C, Eb, F, G, A

A#, C#, E, F#, G#, A#

Bb, Db, Fb, Gb, Ab, Bb

B, D, F, G, A, B

**Chitthakarshini**

Step pattern: 1, 2, 2, 3, 4

Ascending intervals: P1, m2, m3, P4, m6, P8

C, Db, Eb, F, Ab, C

C#, D, E, F#, A, C#

Db, Ebb, Fb, Gb, Bbb, Db

D, Eb, F, G, Bb, D

D#, E, F#, G#, B, D#

Eb, Fb, Gb, Ab, Cb, Eb

E, F, G, A, C, E

F, Gb, Ab, Bb, Db, F

F#, G, A, B, D, F#

Gb, Abb, Bbb, Cb, Ebb, Gb

G, Ab, Bb, C, Eb, G

G#, A, B, C#, E, G#

Ab, Bbb, Cb, Db, Fb, Ab

A, Bb, C, D, F, A

A#, B, C#, D#, F#, A#

Bb, Cb, Db, Eb, Gb, Bb

B, C, D, E, G, B



## Deshgaur

Step pattern: 1, 6, 1, 3, 1

Ascending intervals: P1, m2, P5, m6, M7, P8

C, Db, G, Ab, B, C

C#, D, G#, A, B#, C#

Db, Ebb, Ab, Bbb, C, Db

D, Eb, A, Bb, C#, D

D#, E, A#, B, C##, D#

Eb, Fb, Bb, Cb, D, Eb

E, F, B, C, D#, E

F, Gb, C, Db, E, F

F#, G, C#, D, E#, F#

Gb, Abb, Db, Ebb, F, Gb

G, Ab, D, Eb, F#, G

G#, A, D#, E, F##, G#

Ab, Bbb, Eb, Fb, G, Ab

A, Bb, E, F, G#, A

A#, B, E#, F#, G##, A#

Bb, Cb, F, Gb, A, Bb

B, C, F#, G, A#, B

## Devaranjani

Step pattern: 5, 2, 1, 3, 1

Ascending intervals: P1, P4, P5, m6, M7, P8

C, F, G, Ab, B, C

C#, F#, G#, A, B#, C#

Db, Gb, Ab, Bbb, C, Db

D, G, A, Bb, C#, D

D#, G#, A#, B, C##, D#

Eb, Ab, Bb, Cb, D, Eb

E, A, B, C, D#, E

F, Bb, C, Db, E, F

F#, B, C#, D, E#, F#

Gb, Cb, Db, Ebb, F, Gb

G, C, D, Eb, F#, G

G#, C#, D#, E, F##, G#

Ab, Db, Eb, Fb, G, Ab

A, D, E, F, G#, A

A#, D#, E#, F#, G##, A#

Bb, Eb, F, Gb, A, Bb

B, E, F#, G, A#, B

**Dhavalashri**

Step pattern: 4, 2, 1, 2, 3

Ascending intervals: P1, M3, Aug4, P5, M6, P8

C, E, F#, G, A, C

C#, E#, F##, G#, A#, C#

Db, F, G, Ab, Bb, Db

D, F#, G#, A, B, D

D#, F##, G##, A#, B#, D#

Eb, G, A, Bb, C, Eb

E, G#, A#, B, C#, E

F, A, B, C, D, F

F#, A#, B#, C#, D#, F#

Gb, Bb, C, Db, Eb, Gb

G, B, C#, D, E, G

G#, B#, C##, D#, E#, G#

Ab, C, D, Eb, F, Ab

A, C#, D#, E, F#, A

A#, C##, D##, E#, F##, A#

Bb, D, E, F, G, Bb

B, D#, E#, F#, G#, B

**Dorian Pentatonic (2nd mode of major pentatonic)**

Step pattern: 2, 3, 2, 3, 2

Ascending intervals: P1, M2, P4, P5, m7, P8

C, D, F, G, Bb, C

C#, D#, F#, G#, B, C#

Db, Eb, Gb, Ab, Cb, Db

D, E, G, A, C, D

D#, E#, G#, A#, C#, D#

Eb, F, Ab, Bb, Db, Eb

E, F#, A, B, D, E

F, G, Bb, C, Eb, F

F#, G#, B, C#, E, F#

Gb, Ab, Cb, Db, Fb, Gb

G, A, C, D, F, G

G#, A#, C#, D#, F#, G#

Ab, Bb, Db, Eb, Gb, Ab

A, B, D, E, G, A

A#, B#, D#, E#, G#, A#

Bb, C, Eb, F, Ab, Bb

B, C#, E, F#, A, B

## Five-note Prometheus

Step pattern: 2, 2, 2, 4, 2

Ascending intervals: P1, M2, M3, dim5, m7, P8

C, D, E, Gb, Bb, C

C#, D#, E#, G, B, C#

Db, Eb, F, Abb, Cb, Db

D, E, F#, Ab, C, D

D#, E#, F##, A, C#, D#

Eb, F, G, Bbb, Db, Eb

E, F#, G#, Bb, D, E

F, G, A, Cb, Eb, F

F#, G#, A#, C, E, F#

Gb, Ab, Bb, Dbb, Fb, Gb

G, A, B, Db, F, G

G#, A#, B#, D, F#, G#

Ab, Bb, C, Ebb, Gb, Ab

A, B, C#, Eb, G, A

A#, B#, C##, E, G#, A#

Bb, C, D, Fb, Ab, Bb

B, C#, D#, F, A, B

**Gauri**

Step pattern: 1, 4, 2, 4, 1

Ascending intervals: P1, m2, P4, P5, M7, P8

C, Db, F, G, B, C

C#, D, F#, G#, B#, C#

Db, Ebb, Gb, Ab, C, Db

D, Eb, G, A, C#, D

D#, E, G#, A#, C##, D#

Eb, Fb, Ab, Bb, D, Eb

E, F, A, B, D#, E

F, Gb, Bb, C, E, F

F#, G, B, C#, E#, F#

Gb, Abb, Cb, Db, F, Gb

G, Ab, C, D, F#, G

G#, A, C#, D#, F##, G#

Ab, Bbb, Db, Eb, G, Ab

A, Bb, D, E, G#, A

A#, B, D#, E#, G##, A#

Bb, Cb, Eb, F, A, Bb

B, C, E, F#, A#, B

## Guhamanohari

Step pattern: 2, 3, 4, 1, 2

Ascending intervals: P1, M2, P4, M6, m7, P8

C, D, F, A, Bb, C

C#, D#, F#, A#, B, C#

Db, Eb, Gb, Bb, Cb, Db

D, E, G, B, C, D

D#, E#, G#, B#, C#, D#

Eb, F, Ab, C, Db, Eb

E, F#, A, C#, D, E

F, G, Bb, D, Eb, F

F#, G#, B, D#, E, F#

Gb, Ab, Cb, Eb, Fb, Gb

G, A, C, E, F, G

G#, A#, C#, E#, F#, G#

Ab, Bb, Db, F, Gb, Ab

A, B, D, F#, G, A

A#, B#, D#, F##, G#, A#

Bb, C, Eb, G, Ab, Bb

B, C#, E, G#, A, B

## Hamsadhvani

Step pattern: 2, 2, 3, 4, 1

Ascending intervals: P1, M2, M3, P5, M7, P8

C, D, E, G, B, C

C#, D#, E#, G#, B#, C#

Db, Eb, F, Ab, C, Db

D, E, F#, A, C#, D

D#, E#, F##, A#, C##, D#

Eb, F, G, Bb, D, Eb

E, F#, G#, B, D#, E

F, G, A, C, E, F

F#, G#, A#, C#, E#, F#

Gb, Ab, Bb, Db, F, Gb

G, A, B, D, F#, G

G#, A#, B#, D#, F##, G#

Ab, Bb, C, Eb, G, Ab

A, B, C#, E, G#, A

A#, B#, C##, E#, G##, A#

Bb, C, D, F, A, Bb

B, C#, D#, F#, A#, B



## Hamsanada

Step pattern: 2, 4, 1, 4, 1

Ascending intervals: P1, M2, Aug4, P5, M7, P8

C, D, F#, G, B, C

C#, D#, F##, G#, B#, C#

Db, Eb, G, Ab, C, Db

D, E, G#, A, C#, D

D#, E#, G##, A#, C##, D#

Eb, F, A, Bb, D, Eb

E, F#, A#, B, D#, E

F, G, B, C, E, F

F#, G#, B#, C#, E#, F#

Gb, Ab, C, Db, F, Gb

G, A, C#, D, F#, G

G#, A#, C##, D#, F##, G#

Ab, Bb, D, Eb, G, Ab

A, B, D#, E, G#, A

A#, B#, D##, E#, G##, A#

Bb, C, E, F, A, Bb

B, C#, E#, F#, A#, B

**Han-Iwato**

Step pattern: 1, 4, 2, 3, 2

Ascending intervals: P1, m2, P4, P5, m7, P8

C, Db, F, G, Bb, C

C#, D, F#, G#, B, C#

Db, Ebb, Gb, Ab, Cb, Db

D, Eb, G, A, C, D

D#, E, G#, A#, C#, D#

Eb, Fb, Ab, Bb, Db, Eb

E, F, A, B, D, E

F, Gb, Bb, C, Eb, F

F#, G, B, C#, E, F#

Gb, Abb, Cb, Db, Fb, Gb

G, Ab, C, D, F, G

G#, A, C#, D#, F#, G#

Ab, Bbb, Db, Eb, Gb, Ab

A, Bb, D, E, G, A

A#, B, D#, E#, G#, A#

Bb, Cb, Eb, F, Ab, Bb

B, C, E, F#, A, B

**Han-Komoi**

Step pattern: 2, 3, 2, 1, 4

Ascending intervals: P1, M2, P4, P5, m6, P8

C, D, F, G, Ab, C

C#, D#, F#, G#, A, C#

Db, Eb, Gb, Ab, Bbb, Db

D, E, G, A, Bb, D

D#, E#, G#, A#, B, D#

Eb, F, Ab, Bb, Cb, Eb

E, F#, A, B, C, E

F, G, Bb, C, Db, F

F#, G#, B, C#, D, F#

Gb, Ab, Cb, Db, Ebb, Gb

G, A, C, D, Eb, G

G#, A#, C#, D#, E, G#

Ab, Bb, Db, Eb, Fb, Ab

A, B, D, E, F, A

A#, B#, D#, E#, F#, A#

Bb, C, Eb, F, Gb, Bb

B, C#, E, F#, G, B

## Hansadhvani

Step pattern: 2, 2, 3, 4, 1

Ascending intervals: P1, M2, M3, P5, M7, P8

C, D, E, G, B, C

C#, D#, E#, G#, B#, C#

Db, Eb, F, Ab, C, Db

D, E, F#, A, C#, D

D#, E#, F##, A#, C##, D#

Eb, F, G, Bb, D, Eb

E, F#, G#, B, D#, E

F, G, A, C, E, F

F#, G#, A#, C#, E#, F#

Gb, Ab, Bb, Db, F, Gb

G, A, B, D, F#, G

G#, A#, B#, D#, F##, G#

Ab, Bb, C, Eb, G, Ab

A, B, C#, E, G#, A

A#, B#, C##, E#, G##, A#

Bb, C, D, F, A, Bb

B, C#, D#, F#, A#, B

## Japanese Mode

Step pattern: 2, 1, 4, 1, 4

Ascending intervals: P1, M2, m3, P5, m6, P8

C, D, Eb, G, Ab, C

C#, D#, E, G#, A, C#

Db, Eb, Fb, Ab, Bbb, Db

D, E, F, A, Bb, D

D#, E#, F#, A#, B, D#

Eb, F, Gb, Bb, Cb, Eb

E, F#, G, B, C, E

F, G, Ab, C, Db, F

F#, G#, A, C#, D, F#

Gb, Ab, Bbb, Db, Ebb, Gb

G, A, Bb, D, Eb, G

G#, A#, B, D#, E, G#

Ab, Bb, Cb, Eb, Fb, Ab

A, B, C, E, F, A

A#, B#, C#, E#, F#, A#

Bb, C, Db, F, Gb, Bb

B, C#, D, F#, G, B

## Khamaji Durga

Step pattern: 4, 1, 4, 1, 2

Ascending intervals: P1, M3, P4, M6, m7, P8

C, E, F, A, Bb, C

C#, E#, F#, A#, B, C#

Db, F, Gb, Bb, Cb, Db

D, F#, G, B, C, D

D#, F##, G#, B#, C#, D#

Eb, G, Ab, C, Db, Eb

E, G#, A, C#, D, E

F, A, Bb, D, Eb, F

F#, A#, B, D#, E, F#

Gb, Bb, Cb, Eb, Fb, Gb

G, B, C, E, F, G

G#, B#, C#, E#, F#, G#

Ab, C, Db, F, Gb, Ab

A, C#, D, F#, G, A

A#, C##, D#, F##, G#, A#

Bb, D, Eb, G, Ab, Bb

B, D#, E, G#, A, B

## Kokil Pancham

Step pattern: 3, 2, 2, 1, 4

Ascending intervals: P1, m3, P4, P5, m6, P8

C, Eb, F, G, Ab, C

C#, E, F#, G#, A, C#

Db, Fb, Gb, Ab, Bbb, Db

D, F, G, A, Bb, D

D#, F#, G#, A#, B, D#

Eb, Gb, Ab, Bb, Cb, Eb

E, G, A, B, C, E

F, Ab, Bb, C, Db, F

F#, A, B, C#, D, F#

Gb, Bbb, Cb, Db, Ebb, Gb

G, Bb, C, D, Eb, G

G#, B, C#, D#, E, G#

Ab, Cb, Db, Eb, Fb, Ab

A, C, D, E, F, A

A#, C#, D#, E#, F#, A#

Bb, Db, Eb, F, Gb, Bb

B, D, E, F#, G, B

## Kshanika

Step pattern: 1, 4, 3, 3, 1

Ascending intervals: P1, m2, P4, m6, M7, P8

C, Db, F, Ab, B, C

C#, D, F#, A, B#, C#

Db, Ebb, Gb, Bbb, C, Db

D, Eb, G, Bb, C#, D

D#, E, G#, B, C##, D#

Eb, Fb, Ab, Cb, D, Eb

E, F, A, C, D#, E

F, Gb, Bb, Db, E, F

F#, G, B, D, E#, F#

Gb, Abb, Cb, Ebb, F, Gb

G, Ab, C, Eb, F#, G

G#, A, C#, E, F##, G#

Ab, Bbb, Db, Fb, G, Ab

A, Bb, D, F, G#, A

A#, B, D#, F#, G##, A#

Bb, Cb, Eb, Gb, A, Bb

B, C, E, G, A#, B



## Kumarapriya

Step pattern: 1, 1, 6, 3, 1

Ascending intervals: P1, m2, M2, m6, M7, P8

C, Db, D, Ab, B, C

C#, D, D#, A, B#, C#

Db, Ebb, Eb, Bbb, C, Db

D, Eb, E, Bb, C#, D

D#, E, E#, B, C##, D#

Eb, Fb, F, Cb, D, Eb

E, F, F#, C, D#, E

F, Gb, G, Db, E, F

F#, G, G#, D, E#, F#

Gb, Abb, Ab, Ebb, F, Gb

G, Ab, A, Eb, F#, G

G#, A, A#, E, F##, G#

Ab, Bbb, Bb, Fb, G, Ab

A, Bb, B, F, G#, A

A#, B, B#, F#, G##, A#

Bb, Cb, C, Gb, A, Bb

B, C, C#, G, A#, B

**Kumudki**

Step pattern: 2, 2, 2, 5, 1

Ascending intervals: P1, M2, M3, dim5, M7, P8

C, D, E, Gb, B, C

C#, D#, E#, G, B#, C#

Db, Eb, F, Abb, C, Db

D, E, F#, Ab, C#, D

D#, E#, F##, A, C##, D#

Eb, F, G, Bbb, D, Eb

E, F#, G#, Bb, D#, E

F, G, A, Cb, E, F

F#, G#, A#, C, E#, F#

Gb, Ab, Bb, Dbb, F, Gb

G, A, B, Db, F#, G

G#, A#, B#, D, F##, G#

Ab, Bb, C, Ebb, G, Ab

A, B, C#, Eb, G#, A

A#, B#, C##, E, G##, A#

Bb, C, D, Fb, A, Bb

B, C#, D#, F, A#, B

## Kung (pentatonic)

Step pattern: 2, 2, 2, 3, 3

Ascending intervals: P1, M2, M3, Aug4, M6, P8

C, D, E, F#, A, C

C#, D#, E#, F##, A#, C#

Db, Eb, F, G, Bb, Db

D, E, F#, G#, B, D

D#, E#, F##, G##, B#, D#

Eb, F, G, A, C, Eb

E, F#, G#, A#, C#, E

F, G, A, B, D, F

F#, G#, A#, B#, D#, F#

Gb, Ab, Bb, C, Eb, Gb

G, A, B, C#, E, G

G#, A#, B#, C##, E#, G#

Ab, Bb, C, D, F, Ab

A, B, C#, D#, F#, A

A#, B#, C##, D##, F##, A#

Bb, C, D, E, G, Bb

B, C#, D#, E#, G#, B

## Kuntalavarali

Step pattern: 5, 2, 2, 1, 2

Ascending intervals: P1, P4, P5, M6, m7, P8

C, F, G, A, Bb, C

C#, F#, G#, A#, B, C#

Db, Gb, Ab, Bb, Cb, Db

D, G, A, B, C, D

D#, G#, A#, B#, C#, D#

Eb, Ab, Bb, C, Db, Eb

E, A, B, C#, D, E

F, Bb, C, D, Eb, F

F#, B, C#, D#, E, F#

Gb, Cb, Db, Eb, Fb, Gb

G, C, D, E, F, G

G#, C#, D#, E#, F#, G#

Ab, Db, Eb, F, Gb, Ab

A, D, E, F#, G, A

A#, D#, E#, F##, G#, A#

Bb, Eb, F, G, Ab, Bb

B, E, F#, G#, A, B

## Madhukauns (pentatonic)

Step pattern: 3, 3, 1, 3, 2

Ascending intervals: P1, m3, dim5, P5, m7, P8

C, Eb, Gb, G, Bb, C

C#, E, G, G#, B, C#

Db, Fb, Abb, Ab, Cb, Db

D, F, Ab, A, C, D

D#, F#, A, A#, C#, D#

Eb, Gb, Bbb, Bb, Db, Eb

E, G, Bb, B, D, E

F, Ab, Cb, C, Eb, F

F#, A, C, C#, E, F#

Gb, Bbb, Dbb, Db, Fb, Gb

G, Bb, Db, D, F, G

G#, B, D, D#, F#, G#

Ab, Cb, Ebb, Eb, Gb, Ab

A, C, Eb, E, G, A

A#, C#, E, E#, G#, A#

Bb, Db, Fb, F, Ab, Bb

B, D, F, F#, A, B

**Major Pentatonic (1st mode of major pentatonic)**

Step pattern: 2, 2, 3, 2, 3

Ascending intervals: P1, M2, M3, P5, M6, P8

C, D, E, G, A, C

C#, D#, E#, G#, A#, C#

Db, Eb, F, Ab, Bb, Db

D, E, F#, A, B, D

D#, E#, F##, A#, B#, D#

Eb, F, G, Bb, C, Eb

E, F#, G#, B, C#, E

F, G, A, C, D, F

F#, G#, A#, C#, D#, F#

Gb, Ab, Bb, Db, Eb, Gb

G, A, B, D, E, G

G#, A#, B#, D#, E#, G#

Ab, Bb, C, Eb, F, Ab

A, B, C#, E, F#, A

A#, B#, C##, E#, F##, A#

Bb, C, D, F, G, Bb

B, C#, D#, F#, G#, B

**Mamata**

Step pattern: 4, 3, 2, 2, 1

Ascending intervals: P1, M3, P5, M6, M7, P8

C, E, G, A, B, C

C#, E#, G#, A#, B#, C#

Db, F, Ab, Bb, C, Db

D, F#, A, B, C#, D

D#, F##, A#, B#, C##, D#

Eb, G, Bb, C, D, Eb

E, G#, B, C#, D#, E

F, A, C, D, E, F

F#, A#, C#, D#, E#, F#

Gb, Bb, Db, Eb, F, Gb

G, B, D, E, F#, G

G#, B#, D#, E#, F##, G#

Ab, C, Eb, F, G, Ab

A, C#, E, F#, G#, A

A#, C##, E#, F##, G##, A#

Bb, D, F, G, A, Bb

B, D#, F#, G#, A#, B

**Manaranjani (major)**

Step pattern: 1, 3, 3, 3, 2

Ascending intervals: P1, m2, M3, P5, m7, P8

C, Db, E, G, Bb, C

C#, D, E#, G#, B, C#

Db, Ebb, F, Ab, Cb, Db

D, Eb, F#, A, C, D

D#, E, F##, A#, C#, D#

Eb, Fb, G, Bb, Db, Eb

E, F, G#, B, D, E

F, Gb, A, C, Eb, F

F#, G, A#, C#, E, F#

Gb, Abb, Bb, Db, Fb, Gb

G, Ab, B, D, F, G

G#, A, B#, D#, F#, G#

Ab, Bbb, C, Eb, Gb, Ab

A, Bb, C#, E, G, A

A#, B, C##, E#, G#, A#

Bb, Cb, D, F, Ab, Bb

B, C, D#, F#, A, B



**Mand**

Step pattern: 4, 1, 2, 2, 3

Ascending intervals: P1, M3, P4, P5, M6, P8

C, E, F, G, A, C

C#, E#, F#, G#, A#, C#

Db, F, Gb, Ab, Bb, Db

D, F#, G, A, B, D

D#, F##, G#, A#, B#, D#

Eb, G, Ab, Bb, C, Eb

E, G#, A, B, C#, E

F, A, Bb, C, D, F

F#, A#, B, C#, D#, F#

Gb, Bb, Cb, Db, Eb, Gb

G, B, C, D, E, G

G#, B#, C#, D#, E#, G#

Ab, C, Db, Eb, F, Ab

A, C#, D, E, F#, A

A#, C##, D#, E#, F##, A#

Bb, D, Eb, F, G, Bb

B, D#, E, F#, G#, B

## Matha Kokila

Step pattern: 2, 5, 2, 1, 2

Ascending intervals: P1, M2, P5, M6, m7, P8

C, D, G, A, Bb, C

C#, D#, G#, A#, B, C#

Db, Eb, Ab, Bb, Cb, Db

D, E, A, B, C, D

D#, E#, A#, B#, C#, D#

Eb, F, Bb, C, Db, Eb

E, F#, B, C#, D, E

F, G, C, D, Eb, F

F#, G#, C#, D#, E, F#

Gb, Ab, Db, Eb, Fb, Gb

G, A, D, E, F, G

G#, A#, D#, E#, F#, G#

Ab, Bb, Eb, F, Gb, Ab

A, B, E, F#, G, A

A#, B#, E#, F##, G#, A#

Bb, C, F, G, Ab, Bb

B, C#, F#, G#, A, B

## Megharanji

Step pattern: 1, 3, 1, 6, 1

Ascending intervals: P1, m2, M3, P4, M7, P8

C, Db, E, F, B, C

C#, D, E#, F#, B#, C#

Db, Ebb, F, Gb, C, Db

D, Eb, F#, G, C#, D

D#, E, F##, G#, C##, D#

Eb, Fb, G, Ab, D, Eb

E, F, G#, A, D#, E

F, Gb, A, Bb, E, F

F#, G, A#, B, E#, F#

Gb, Abb, Bb, Cb, F, Gb

G, Ab, B, C, F#, G

G#, A, B#, C#, F##, G#

Ab, Bbb, C, Db, G, Ab

A, Bb, C#, D, G#, A

A#, B, C##, D#, G##, A#

Bb, Cb, D, Eb, A, Bb

B, C, D#, E, A#, B

**Minor Pentatonic (5th mode of major pentatonic)**

Step pattern: 3, 2, 2, 3, 2

Ascending intervals: P1, m3, P4, P5, m7, P8

C, Eb, F, G, Bb, C

C#, E, F#, G#, B, C#

Db, Fb, Gb, Ab, Cb, Db

D, F, G, A, C, D

D#, F#, G#, A#, C#, D#

Eb, Gb, Ab, Bb, Db, Eb

E, G, A, B, D, E

F, Ab, Bb, C, Eb, F

F#, A, B, C#, E, F#

Gb, Bbb, Cb, Db, Fb, Gb

G, Bb, C, D, F, G

G#, B, C#, D#, F#, G#

Ab, Cb, Db, Eb, Gb, Ab

A, C, D, E, G, A

A#, C#, D#, E#, G#, A#

Bb, Db, Eb, F, Ab, Bb

B, D, E, F#, A, B

**Mixolydian Pentatonic (4th mode of major pentatonic)**

Step pattern: 2, 3, 2, 2, 3

Ascending intervals: P1, M2, P4, P5, M6, P8

C, D, F, G, A, C

C#, D#, F#, G#, A#, C#

Db, Eb, Gb, Ab, Bb, Db

D, E, G, A, B, D

D#, E#, G#, A#, B#, D#

Eb, F, Ab, Bb, C, Eb

E, F#, A, B, C#, E

F, G, Bb, C, D, F

F#, G#, B, C#, D#, F#

Gb, Ab, Cb, Db, Eb, Gb

G, A, C, D, E, G

G#, A#, C#, D#, E#, G#

Ab, Bb, Db, Eb, F, Ab

A, B, D, E, F#, A

A#, B#, D#, E#, F##, A#

Bb, C, Eb, F, G, Bb

B, C#, E, F#, G#, B

## Mohanangi

Step pattern: 3, 1, 3, 2, 3

Ascending intervals: P1, m3, M3, P5, M6, P8

C, Eb, E, G, A, C

C#, E, E#, G#, A#, C#

Db, Fb, F, Ab, Bb, Db

D, F, F#, A, B, D

D#, F#, F##, A#, B#, D#

Eb, Gb, G, Bb, C, Eb

E, G, G#, B, C#, E

F, Ab, A, C, D, F

F#, A, A#, C#, D#, F#

Gb, Bbb, Bb, Db, Eb, Gb

G, Bb, B, D, E, G

G#, B, B#, D#, E#, G#

Ab, Cb, C, Eb, F, Ab

A, C, C#, E, F#, A

A#, C#, C##, E#, F##, A#

Bb, Db, D, F, G, Bb

B, D, D#, F#, G#, B

## Multani (pentatonic)

Step pattern: 3, 3, 1, 4, 1

Ascending intervals: P1, m3, dim5, P5, M7, P8

C, Eb, Gb, G, B, C

C#, E, G, G#, B#, C#

Db, Fb, Abb, Ab, C, Db

D, F, Ab, A, C#, D

D#, F#, A, A#, C##, D#

Eb, Gb, Bbb, Bb, D, Eb

E, G, Bb, B, D#, E

F, Ab, Cb, C, E, F

F#, A, C, C#, E#, F#

Gb, Bbb, Dbb, Db, F, Gb

G, Bb, Db, D, F#, G

G#, B, D, D#, F##, G#

Ab, Cb, Ebb, Eb, G, Ab

A, C, Eb, E, G#, A

A#, C#, E, E#, G##, A#

Bb, Db, Fb, F, A, Bb

B, D, F, F#, A#, B

## Nabhomani (pentatonic)

Step pattern: 1, 1, 4, 1, 5

Ascending intervals: P1, m2, M2, dim5, P5, P8

C, Db, D, Gb, G, C

C#, D, D#, G, G#, C#

Db, Ebb, Eb, Abb, Ab, Db

D, Eb, E, Ab, A, D

D#, E, E#, A, A#, D#

Eb, Fb, F, Bbb, Bb, Eb

E, F, F#, Bb, B, E

F, Gb, G, Cb, C, F

F#, G, G#, C, C#, F#

Gb, Abb, Ab, Dbb, Db, Gb

G, Ab, A, Db, D, G

G#, A, A#, D, D#, G#

Ab, Bbb, Bb, Ebb, Eb, Ab

A, Bb, B, Eb, E, A

A#, B, B#, E, E#, A#

Bb, Cb, C, Fb, F, Bb

B, C, C#, F, F#, B



**Nata**

Step pattern: 3, 2, 2, 4, 1

Ascending intervals: P1, m3, P4, P5, M7, P8

C, Eb, F, G, B, C

C#, E, F#, G#, B#, C#

Db, Fb, Gb, Ab, C, Db

D, F, G, A, C#, D

D#, F#, G#, A#, C##, D#

Eb, Gb, Ab, Bb, D, Eb

E, G, A, B, D#, E

F, Ab, Bb, C, E, F

F#, A, B, C#, E#, F#

Gb, Bbb, Cb, Db, F, Gb

G, Bb, C, D, F#, G

G#, B, C#, D#, F##, G#

Ab, Cb, Db, Eb, G, Ab

A, C, D, E, G#, A

A#, C#, D#, E#, G##, A#

Bb, Db, Eb, F, A, Bb

B, D, E, F#, A#, B

## Neroshta

Step pattern: 2, 2, 5, 2, 1

Ascending intervals: P1, M2, M3, M6, M7, P8

C, D, E, A, B, C

C#, D#, E#, A#, B#, C#

Db, Eb, F, Bb, C, Db

D, E, F#, B, C#, D

D#, E#, F##, B#, C##, D#

Eb, F, G, C, D, Eb

E, F#, G#, C#, D#, E

F, G, A, D, E, F

F#, G#, A#, D#, E#, F#

Gb, Ab, Bb, Eb, F, Gb

G, A, B, E, F#, G

G#, A#, B#, E#, F##, G#

Ab, Bb, C, F, G, Ab

A, B, C#, F#, G#, A

A#, B#, C##, F##, G##, A#

Bb, C, D, G, A, Bb

B, C#, D#, G#, A#, B

**Pelog**

Step pattern: 1, 2, 4, 3, 2

Ascending intervals: P1, m2, m3, P5, m7, P8

C, Db, Eb, G, Bb, C

C#, D, E, G#, B, C#

Db, Ebb, Fb, Ab, Cb, Db

D, Eb, F, A, C, D

D#, E, F#, A#, C#, D#

Eb, Fb, Gb, Bb, Db, Eb

E, F, G, B, D, E

F, Gb, Ab, C, Eb, F

F#, G, A, C#, E, F#

Gb, Abb, Bbb, Db, Fb, Gb

G, Ab, Bb, D, F, G

G#, A, B, D#, F#, G#

Ab, Bbb, Cb, Eb, Gb, Ab

A, Bb, C, E, G, A

A#, B, C#, E#, G#, A#

Bb, Cb, Db, F, Ab, Bb

B, C, D, F#, A, B

**Pentatonic Major #2 b7**

Step pattern: 3, 1, 3, 3, 2

Ascending intervals: P1, Aug2, M3, P5, m7, P8

C, D#, E, G, Bb, C

C#, D##, E#, G#, B, C#

Db, E, F, Ab, Cb, Db

D, E#, F#, A, C, D

D#, E##, F##, A#, C#, D#

Eb, F#, G, Bb, Db, Eb

E, F##, G#, B, D, E

F, G#, A, C, Eb, F

F#, G##, A#, C#, E, F#

Gb, A, Bb, Db, Fb, Gb

G, A#, B, D, F, G

G#, A##, B#, D#, F#, G#

Ab, B, C, Eb, Gb, Ab

A, B#, C#, E, G, A

A#, B##, C##, E#, G#, A#

Bb, C#, D, F, Ab, Bb

B, C##, D#, F#, A, B

**Pentatonic Major b2**

Step pattern: 1, 3, 3, 2, 3

Ascending intervals: P1, m2, M3, P5, M6, P8

C, Db, E, G, A, C

C#, D, E#, G#, A#, C#

Db, Ebb, F, Ab, Bb, Db

D, Eb, F#, A, B, D

D#, E, F##, A#, B#, D#

Eb, Fb, G, Bb, C, Eb

E, F, G#, B, C#, E

F, Gb, A, C, D, F

F#, G, A#, C#, D#, F#

Gb, Abb, Bb, Db, Eb, Gb

G, Ab, B, D, E, G

G#, A, B#, D#, E#, G#

Ab, Bbb, C, Eb, F, Ab

A, Bb, C#, E, F#, A

A#, B, C##, E#, F##, A#

Bb, Cb, D, F, G, Bb

B, C, D#, F#, G#, B

**Pentatonic Major b2 b5**

Step pattern: 1, 3, 2, 3, 3

Ascending intervals: P1, m2, M3, dim5, M6, P8

C, Db, E, Gb, A, C

C#, D, E#, G, A#, C#

Db, Ebb, F, Abb, Bb, Db

D, Eb, F#, Ab, B, D

D#, E, F##, A, B#, D#

Eb, Fb, G, Bbb, C, Eb

E, F, G#, Bb, C#, E

F, Gb, A, Cb, D, F

F#, G, A#, C, D#, F#

Gb, Abb, Bb, Dbb, Eb, Gb

G, Ab, B, Db, E, G

G#, A, B#, D, E#, G#

Ab, Bbb, C, Ebb, F, Ab

A, Bb, C#, Eb, F#, A

A#, B, C##, E, F##, A#

Bb, Cb, D, Fb, G, Bb

B, C, D#, F, G#, B

## Pentatonic Major b2 b6

Step pattern: 1, 3, 3, 1, 4

Ascending intervals: P1, m2, M3, P5, m6, P8

C, Db, E, G, Ab, C

C#, D, E#, G#, A, C#

Db, Ebb, F, Ab, Bbb, Db

D, Eb, F#, A, Bb, D

D#, E, F##, A#, B, D#

Eb, Fb, G, Bb, Cb, Eb

E, F, G#, B, C, E

F, Gb, A, C, Db, F

F#, G, A#, C#, D, F#

Gb, Abb, Bb, Db, Ebb, Gb

G, Ab, B, D, Eb, G

G#, A, B#, D#, E, G#

Ab, Bbb, C, Eb, Fb, Ab

A, Bb, C#, E, F, A

A#, B, C##, E#, F#, A#

Bb, Cb, D, F, Gb, Bb

B, C, D#, F#, G, B

## Pentatonic Major b5

Step pattern: 2, 2, 2, 3, 3

Ascending intervals: P1, M2, M3, dim5, M6, P8

C, D, E, Gb, A, C

C#, D#, E#, G, A#, C#

Db, Eb, F, Abb, Bb, Db

D, E, F#, Ab, B, D

D#, E#, F##, A, B#, D#

Eb, F, G, Bbb, C, Eb

E, F#, G#, Bb, C#, E

F, G, A, Cb, D, F

F#, G#, A#, C, D#, F#

Gb, Ab, Bb, Dbb, Eb, Gb

G, A, B, Db, E, G

G#, A#, B#, D, E#, G#

Ab, Bb, C, Ebb, F, Ab

A, B, C#, Eb, F#, A

A#, B#, C##, E, F##, A#

Bb, C, D, Fb, G, Bb

B, C#, D#, F, G#, B



**Pentatonic Major b6**

Step pattern: 2, 2, 3, 1, 4

Ascending intervals: P1, M2, M3, P5, m6, P8

C, D, E, G, Ab, C

C#, D#, E#, G#, A, C#

Db, Eb, F, Ab, Bbb, Db

D, E, F#, A, Bb, D

D#, E#, F##, A#, B, D#

Eb, F, G, Bb, Cb, Eb

E, F#, G#, B, C, E

F, G, A, C, Db, F

F#, G#, A#, C#, D, F#

Gb, Ab, Bb, Db, Ebb, Gb

G, A, B, D, Eb, G

G#, A#, B#, D#, E, G#

Ab, Bb, C, Eb, Fb, Ab

A, B, C#, E, F, A

A#, B#, C##, E#, F#, A#

Bb, C, D, F, Gb, Bb

B, C#, D#, F#, G, B

**Pentatonic Minor (major 6)**

Step pattern: 3, 2, 2, 2, 3

Ascending intervals: P1, m3, P4, P5, M6, P8

C, Eb, F, G, A, C

C#, E, F#, G#, A#, C#

Db, Fb, Gb, Ab, Bb, Db

D, F, G, A, B, D

D#, F#, G#, A#, B#, D#

Eb, Gb, Ab, Bb, C, Eb

E, G, A, B, C#, E

F, Ab, Bb, C, D, F

F#, A, B, C#, D#, F#

Gb, Bbb, Cb, Db, Eb, Gb

G, Bb, C, D, E, G

G#, B, C#, D#, E#, G#

Ab, Cb, Db, Eb, F, Ab

A, C, D, E, F#, A

A#, C#, D#, E#, F##, A#

Bb, Db, Eb, F, G, Bb

B, D, E, F#, G#, B

**Pentatonic Minor 7 b5**

Step pattern: 3, 2, 1, 4, 2

Ascending intervals: P1, m3, P4, dim5, m7, P8

C, Eb, F, Gb, Bb, C

C#, E, F#, G, B, C#

Db, Fb, Gb, Abb, Cb, Db

D, F, G, Ab, C, D

D#, F#, G#, A, C#, D#

Eb, Gb, Ab, Bbb, Db, Eb

E, G, A, Bb, D, E

F, Ab, Bb, Cb, Eb, F

F#, A, B, C, E, F#

Gb, Bbb, Cb, Dbb, Fb, Gb

G, Bb, C, Db, F, G

G#, B, C#, D, F#, G#

Ab, Cb, Db, Ebb, Gb, Ab

A, C, D, Eb, G, A

A#, C#, D#, E, G#, A#

Bb, Db, Eb, Fb, Ab, Bb

B, D, E, F, A, B

## Pentatonic Minor m2 dim5 M6

Step pattern: 1, 2, 3, 3, 3

Ascending intervals: P1, m2, m3, dim5, M6, P8

C, Db, Eb, Gb, A, C

C#, D, E, G, A#, C#

Db, Ebb, Fb, Abb, Bb, Db

D, Eb, F, Ab, B, D

D#, E, F#, A, B#, D#

Eb, Fb, Gb, Bbb, C, Eb

E, F, G, Bb, C#, E

F, Gb, Ab, Cb, D, F

F#, G, A, C, D#, F#

Gb, Abb, Bbb, Dbb, Eb, Gb

G, Ab, Bb, Db, E, G

G#, A, B, D, E#, G#

Ab, Bbb, Cb, Ebb, F, Ab

A, Bb, C, Eb, F#, A

A#, B, C#, E, F##, A#

Bb, Cb, Db, Fb, G, Bb

B, C, D, F, G#, B

## Pentatonic whole tone

Step pattern: 4, 2, 2, 2, 2

Ascending intervals: P1, M3, Aug4, Aug5, Aug6, P8

C, E, F#, G#, A#, C

C#, E#, F##, G##, A##, C#

Db, F, G, A, B, Db

D, F#, G#, A#, B#, D

D#, F##, G##, A##, B##, D#

Eb, G, A, B, C#, Eb

E, G#, A#, B#, C##, E

F, A, B, C#, D#, F

F#, A#, B#, C##, D##, F#

Gb, Bb, C, D, E, Gb

G, B, C#, D#, E#, G

G#, B#, C##, D##, E##, G#

Ab, C, D, E, F#, Ab

A, C#, D#, E#, F##, A

A#, C##, D##, E##, F###, A#

Bb, D, E, F#, G#, Bb

B, D#, E#, F##, G##, B

**Phrygian Pentatonic (3rd mode of major pentatonic)**

Step pattern: 3, 2, 3, 2, 2

Ascending intervals: P1, m3, P4, m6, m7, P8

C, Eb, F, Ab, Bb, C

C#, E, F#, A, B, C#

Db, Fb, Gb, Bbb, Cb, Db

D, F, G, Bb, C, D

D#, F#, G#, B, C#, D#

Eb, Gb, Ab, Cb, Db, Eb

E, G, A, C, D, E

F, Ab, Bb, Db, Eb, F

F#, A, B, D, E, F#

Gb, Bbb, Cb, Ebb, Fb, Gb

G, Bb, C, Eb, F, G

G#, B, C#, E, F#, G#

Ab, Cb, Db, Fb, Gb, Ab

A, C, D, F, G, A

A#, C#, D#, F#, G#, A#

Bb, Db, Eb, Gb, Ab, Bb

B, D, E, G, A, B

**Priyadharshini**

Step pattern: 2, 3, 3, 3, 1

Ascending intervals: P1, M2, P4, m6, M7, P8

C, D, F, Ab, B, C

C#, D#, F#, A, B#, C#

Db, Eb, Gb, Bbb, C, Db

D, E, G, Bb, C#, D

D#, E#, G#, B, C##, D#

Eb, F, Ab, Cb, D, Eb

E, F#, A, C, D#, E

F, G, Bb, Db, E, F

F#, G#, B, D, E#, F#

Gb, Ab, Cb, Ebb, F, Gb

G, A, C, Eb, F#, G

G#, A#, C#, E, F##, G#

Ab, Bb, Db, Fb, G, Ab

A, B, D, F, G#, A

A#, B#, D#, F#, G##, A#

Bb, C, Eb, Gb, A, Bb

B, C#, E, G, A#, B

## Puruhutika

Step pattern: 5, 2, 2, 2, 1

Ascending intervals: P1, P4, P5, M6, M7, P8

C, F, G, A, B, C

C#, F#, G#, A#, B#, C#

Db, Gb, Ab, Bb, C, Db

D, G, A, B, C#, D

D#, G#, A#, B#, C##, D#

Eb, Ab, Bb, C, D, Eb

E, A, B, C#, D#, E

F, Bb, C, D, E, F

F#, B, C#, D#, E#, F#

Gb, Cb, Db, Eb, F, Gb

G, C, D, E, F#, G

G#, C#, D#, E#, F##, G#

Ab, Db, Eb, F, G, Ab

A, D, E, F#, G#, A

A#, D#, E#, F##, G##, A#

Bb, Eb, F, G, A, Bb

B, E, F#, G#, A#, B



**Putrika**

Step pattern: 1, 1, 6, 1, 3

Ascending intervals: P1, m2, M2, m6, M6, P8

C, Db, D, Ab, A, C

C#, D, D#, A, A#, C#

Db, Ebb, Eb, Bbb, Bb, Db

D, Eb, E, Bb, B, D

D#, E, E#, B, B#, D#

Eb, Fb, F, Cb, C, Eb

E, F, F#, C, C#, E

F, Gb, G, Db, D, F

F#, G, G#, D, D#, F#

Gb, Abb, Ab, Ebb, Eb, Gb

G, Ab, A, Eb, E, G

G#, A, A#, E, E#, G#

Ab, Bbb, Bb, Fb, F, Ab

A, Bb, B, F, F#, A

A#, B, B#, F#, F##, A#

Bb, Cb, C, Gb, G, Bb

B, C, C#, G, G#, B

**Rajeshwari**

Step pattern: 3, 2, 4, 2, 1

Ascending intervals: P1, m3, P4, M6, M7, P8

C, Eb, F, A, B, C

C#, E, F#, A#, B#, C#

Db, Fb, Gb, Bb, C, Db

D, F, G, B, C#, D

D#, F#, G#, B#, C##, D#

Eb, Gb, Ab, C, D, Eb

E, G, A, C#, D#, E

F, Ab, Bb, D, E, F

F#, A, B, D#, E#, F#

Gb, Bbb, Cb, Eb, F, Gb

G, Bb, C, E, F#, G

G#, B, C#, E#, F##, G#

Ab, Cb, Db, F, G, Ab

A, C, D, F#, G#, A

A#, C#, D#, F##, G##, A#

Bb, Db, Eb, G, A, Bb

B, D, E, G#, A#, B

## Rasranjani

Step pattern: 2, 3, 4, 2, 1

Ascending intervals: P1, M2, P4, M6, M7, P8

C, D, F, A, B, C

C#, D#, F#, A#, B#, C#

Db, Eb, Gb, Bb, C, Db

D, E, G, B, C#, D

D#, E#, G#, B#, C##, D#

Eb, F, Ab, C, D, Eb

E, F#, A, C#, D#, E

F, G, Bb, D, E, F

F#, G#, B, D#, E#, F#

Gb, Ab, Cb, Eb, F, Gb

G, A, C, E, F#, G

G#, A#, C#, E#, F##, G#

Ab, Bb, Db, F, G, Ab

A, B, D, F#, G#, A

A#, B#, D#, F##, G##, A#

Bb, C, Eb, G, A, Bb

B, C#, E, G#, A#, B

## Ryukyu

Step pattern: 4, 1, 2, 4, 1

Ascending intervals: P1, M3, P4, P5, M7, P8

C, E, F, G, B, C

C#, E#, F#, G#, B#, C#

Db, F, Gb, Ab, C, Db

D, F#, G, A, C#, D

D#, F##, G#, A#, C##, D#

Eb, G, Ab, Bb, D, Eb

E, G#, A, B, D#, E

F, A, Bb, C, E, F

F#, A#, B, C#, E#, F#

Gb, Bb, Cb, Db, F, Gb

G, B, C, D, F#, G

G#, B#, C#, D#, F##, G#

Ab, C, Db, Eb, G, Ab

A, C#, D, E, G#, A

A#, C##, D#, E#, G##, A#

Bb, D, Eb, F, A, Bb

B, D#, E, F#, A#, B

## Sanjh ka Hindol

Step pattern: 4, 2, 3, 2, 1

Ascending intervals: P1, M3, Aug4, M6, M7, P8

C, E, F#, A, B, C

C#, E#, F##, A#, B#, C#

Db, F, G, Bb, C, Db

D, F#, G#, B, C#, D

D#, F##, G##, B#, C##, D#

Eb, G, A, C, D, Eb

E, G#, A#, C#, D#, E

F, A, B, D, E, F

F#, A#, B#, D#, E#, F#

Gb, Bb, C, Eb, F, Gb

G, B, C#, E, F#, G

G#, B#, C##, E#, F##, G#

Ab, C, D, F, G, Ab

A, C#, D#, F#, G#, A

A#, C##, D##, F##, G##, A#

Bb, D, E, G, A, Bb

B, D#, E#, G#, A#, B

## Saugandhini

Step pattern: 1, 5, 1, 1, 4

Ascending intervals: P1, m2, dim5, P5, m6, P8

C, Db, Gb, G, Ab, C

C#, D, G, G#, A, C#

Db, Ebb, Abb, Ab, Bbb, Db

D, Eb, Ab, A, Bb, D

D#, E, A, A#, B, D#

Eb, Fb, Bbb, Bb, Cb, Eb

E, F, Bb, B, C, E

F, Gb, Cb, C, Db, F

F#, G, C, C#, D, F#

Gb, Abb, Dbb, Db, Ebb, Gb

G, Ab, Db, D, Eb, G

G#, A, D, D#, E, G#

Ab, Bbb, Ebb, Eb, Fb, Ab

A, Bb, Eb, E, F, A

A#, B, E, E#, F#, A#

Bb, Cb, Fb, F, Gb, Bb

B, C, F, F#, G, B

## Shri Kalyan

Step pattern: 2, 4, 1, 2, 3

Ascending intervals: P1, M2, Aug4, P5, M6, P8

C, D, F#, G, A, C

C#, D#, F##, G#, A#, C#

Db, Eb, G, Ab, Bb, Db

D, E, G#, A, B, D

D#, E#, G##, A#, B#, D#

Eb, F, A, Bb, C, Eb

E, F#, A#, B, C#, E

F, G, B, C, D, F

F#, G#, B#, C#, D#, F#

Gb, Ab, C, Db, Eb, Gb

G, A, C#, D, E, G

G#, A#, C##, D#, E#, G#

Ab, Bb, D, Eb, F, Ab

A, B, D#, E, F#, A

A#, B#, D##, E#, F##, A#

Bb, C, E, F, G, Bb

B, C#, E#, F#, G#, B

## Shubravarni

Step pattern: 2, 4, 3, 1, 2

Ascending intervals: P1, M2, dim5, M6, m7, P8

C, D, Gb, A, Bb, C

C#, D#, G, A#, B, C#

Db, Eb, Abb, Bb, Cb, Db

D, E, Ab, B, C, D

D#, E#, A, B#, C#, D#

Eb, F, Bbb, C, Db, Eb

E, F#, Bb, C#, D, E

F, G, Cb, D, Eb, F

F#, G#, C, D#, E, F#

Gb, Ab, Dbb, Eb, Fb, Gb

G, A, Db, E, F, G

G#, A#, D, E#, F#, G#

Ab, Bb, Ebb, F, Gb, Ab

A, B, Eb, F#, G, A

A#, B#, E, F##, G#, A#

Bb, C, Fb, G, Ab, Bb

B, C#, F, G#, A, B



## Suspended Pentatonic (2nd mode of major pentatonic)

Step pattern: 2, 3, 2, 3, 2

Ascending intervals: P1, M2, P4, P5, m7, P8

C, D, F, G, Bb, C

C#, D#, F#, G#, B, C#

Db, Eb, Gb, Ab, Cb, Db

D, E, G, A, C, D

D#, E#, G#, A#, C#, D#

Eb, F, Ab, Bb, Db, Eb

E, F#, A, B, D, E

F, G, Bb, C, Eb, F

F#, G#, B, C#, E, F#

Gb, Ab, Cb, Db, Fb, Gb

G, A, C, D, F, G

G#, A#, C#, D#, F#, G#

Ab, Bb, Db, Eb, Gb, Ab

A, B, D, E, G, A

A#, B#, D#, E#, G#, A#

Bb, C, Eb, F, Ab, Bb

B, C#, E, F#, A, B

## Syrian Pentatonic

Step pattern: 1, 3, 1, 3, 4

Ascending intervals: P1, m2, M3, P4, m6, P8

C, Db, E, F, Ab, C

C#, D, E#, F#, A, C#

Db, Ebb, F, Gb, Bbb, Db

D, Eb, F#, G, Bb, D

D#, E, F##, G#, B, D#

Eb, Fb, G, Ab, Cb, Eb

E, F, G#, A, C, E

F, Gb, A, Bb, Db, F

F#, G, A#, B, D, F#

Gb, Abb, Bb, Cb, Ebb, Gb

G, Ab, B, C, Eb, G

G#, A, B#, C#, E, G#

Ab, Bbb, C, Db, Fb, Ab

A, Bb, C#, D, F, A

A#, B, C##, D#, F#, A#

Bb, Cb, D, Eb, Gb, Bb

B, C, D#, E, G, B

**Tcherepnin Major Pentatonic Mode I**

Step pattern: 2, 2, 3, 2, 3

Ascending intervals: P1, M2, M3, P5, M6, P8

C, D, E, G, A, C

C#, D#, E#, G#, A#, C#

Db, Eb, F, Ab, Bb, Db

D, E, F#, A, B, D

D#, E#, F##, A#, B#, D#

Eb, F, G, Bb, C, Eb

E, F#, G#, B, C#, E

F, G, A, C, D, F

F#, G#, A#, C#, D#, F#

Gb, Ab, Bb, Db, Eb, Gb

G, A, B, D, E, G

G#, A#, B#, D#, E#, G#

Ab, Bb, C, Eb, F, Ab

A, B, C#, E, F#, A

A#, B#, C##, E#, F##, A#

Bb, C, D, F, G, Bb

B, C#, D#, F#, G#, B

## Tcherepnin Major Pentatonic Mode II

Step pattern: 2, 3, 2, 4, 1

Ascending intervals: P1, M2, P4, P5, M7, P8

C, D, F, G, B, C

C#, D#, F#, G#, B#, C#

Db, Eb, Gb, Ab, C, Db

D, E, G, A, C#, D

D#, E#, G#, A#, C##, D#

Eb, F, Ab, Bb, D, Eb

E, F#, A, B, D#, E

F, G, Bb, C, E, F

F#, G#, B, C#, E#, F#

Gb, Ab, Cb, Db, F, Gb

G, A, C, D, F#, G

G#, A#, C#, D#, F##, G#

Ab, Bb, Db, Eb, G, Ab

A, B, D, E, G#, A

A#, B#, D#, E#, G##, A#

Bb, C, Eb, F, A, Bb

B, C#, E, F#, A#, B

### Tcherepnin Major Pentatonic Mode III

Step pattern: 3, 2, 3, 2, 2

Ascending intervals: P1, m3, P4, m6, m7, P8

C, Eb, F, Ab, Bb, C

C#, E, F#, A, B, C#

Db, Fb, Gb, Bbb, Cb, Db

D, F, G, Bb, C, D

D#, F#, G#, B, C#, D#

Eb, Gb, Ab, Cb, Db, Eb

E, G, A, C, D, E

F, Ab, Bb, Db, Eb, F

F#, A, B, D, E, F#

Gb, Bbb, Cb, Ebb, Fb, Gb

G, Bb, C, Eb, F, G

G#, B, C#, E, F#, G#

Ab, Cb, Db, Fb, Gb, Ab

A, C, D, F, G, A

A#, C#, D#, F#, G#, A#

Bb, Db, Eb, Gb, Ab, Bb

B, D, E, G, A, B

**Tcherepnin Major Pentatonic Mode IV**

Step pattern: 2, 3, 2, 2, 3

Ascending intervals: P1, M2, P4, P5, M6, P8

C, D, F, G, A, C

C#, D#, F#, G#, A#, C#

Db, Eb, Gb, Ab, Bb, Db

D, E, G, A, B, D

D#, E#, G#, A#, B#, D#

Eb, F, Ab, Bb, C, Eb

E, F#, A, B, C#, E

F, G, Bb, C, D, F

F#, G#, B, C#, D#, F#

Gb, Ab, Cb, Db, Eb, Gb

G, A, C, D, E, G

G#, A#, C#, D#, E#, G#

Ab, Bb, Db, Eb, F, Ab

A, B, D, E, F#, A

A#, B#, D#, E#, F##, A#

Bb, C, Eb, F, G, Bb

B, C#, E, F#, G#, B

**Tcherepnin Major Pentatonic Mode V**

Step pattern: 3, 2, 2, 3, 2

Ascending intervals: P1, m3, P4, P5, m7, P8

C, Eb, F, G, Bb, C

C#, E, F#, G#, B, C#

Db, Fb, Gb, Ab, Cb, Db

D, F, G, A, C, D

D#, F#, G#, A#, C#, D#

Eb, Gb, Ab, Bb, Db, Eb

E, G, A, B, D, E

F, Ab, Bb, C, Eb, F

F#, A, B, C#, E, F#

Gb, Bbb, Cb, Db, Fb, Gb

G, Bb, C, D, F, G

G#, B, C#, D#, F#, G#

Ab, Cb, Db, Eb, Gb, Ab

A, C, D, E, G, A

A#, C#, D#, E#, G#, A#

Bb, Db, Eb, F, Ab, Bb

B, D, E, F#, A, B

## Tcherepnin Minor Pentatonic Mode I

Step pattern: 2, 1, 4, 1, 4

Ascending intervals: P1, M2, m3, P5, m6, P8

C, D, Eb, G, Ab, C

C#, D#, E, G#, A, C#

Db, Eb, Fb, Ab, Bbb, Db

D, E, F, A, Bb, D

D#, E#, F#, A#, B, D#

Eb, F, Gb, Bb, Cb, Eb

E, F#, G, B, C, E

F, G, Ab, C, Db, F

F#, G#, A, C#, D, F#

Gb, Ab, Bbb, Db, Ebb, Gb

G, A, Bb, D, Eb, G

G#, A#, B, D#, E, G#

Ab, Bb, Cb, Eb, Fb, Ab

A, B, C, E, F, A

A#, B#, C#, E#, F#, A#

Bb, C, Db, F, Gb, Bb

B, C#, D, F#, G, B



## Tcherepnin Minor Pentatonic Mode II

Step pattern: 1, 4, 1, 4, 2

Ascending intervals: P1, m2, P4, dim5, m7, P8

C, Db, F, Gb, Bb, C

C#, D, F#, G, B, C#

Db, Ebb, Gb, Abb, Cb, Db

D, Eb, G, Ab, C, D

D#, E, G#, A, C#, D#

Eb, Fb, Ab, Bbb, Db, Eb

E, F, A, Bb, D, E

F, Gb, Bb, Cb, Eb, F

F#, G, B, C, E, F#

Gb, Abb, Cb, Dbb, Fb, Gb

G, Ab, C, Db, F, G

G#, A, C#, D, F#, G#

Ab, Bbb, Db, Ebb, Gb, Ab

A, Bb, D, Eb, G, A

A#, B, D#, E, G#, A#

Bb, Cb, Eb, Fb, Ab, Bb

B, C, E, F, A, B

**Tcherepnin Minor Pentatonic Mode III**

Step pattern: 4, 1, 4, 2, 1

Ascending intervals: P1, M3, P4, M6, M7, P8

C, E, F, A, B, C

C#, E#, F#, A#, B#, C#

Db, F, Gb, Bb, C, Db

D, F#, G, B, C#, D

D#, F##, G#, B#, C##, D#

Eb, G, Ab, C, D, Eb

E, G#, A, C#, D#, E

F, A, Bb, D, E, F

F#, A#, B, D#, E#, F#

Gb, Bb, Cb, Eb, F, Gb

G, B, C, E, F#, G

G#, B#, C#, E#, F##, G#

Ab, C, Db, F, G, Ab

A, C#, D, F#, G#, A

A#, C##, D#, F##, G##, A#

Bb, D, Eb, G, A, Bb

B, D#, E, G#, A#, B

## Tcherepnin Minor Pentatonic Mode IV

Step pattern: 1, 4, 2, 1, 4

Ascending intervals: P1, m2, P4, P5, m6, P8

C, Db, F, G, Ab, C

C#, D, F#, G#, A, C#

Db, Ebb, Gb, Ab, Bbb, Db

D, Eb, G, A, Bb, D

D#, E, G#, A#, B, D#

Eb, Fb, Ab, Bb, Cb, Eb

E, F, A, B, C, E

F, Gb, Bb, C, Db, F

F#, G, B, C#, D, F#

Gb, Abb, Cb, Db, Ebb, Gb

G, Ab, C, D, Eb, G

G#, A, C#, D#, E, G#

Ab, Bbb, Db, Eb, Fb, Ab

A, Bb, D, E, F, A

A#, B, D#, E#, F#, A#

Bb, Cb, Eb, F, Gb, Bb

B, C, E, F#, G, B

## Tcherepnin Minor Pentatonic Mode V

Step pattern: 4, 2, 1, 4, 1

Ascending intervals: P1, M3, Aug4, P5, M7, P8

C, E, F#, G, B, C

C#, E#, F##, G#, B#, C#

Db, F, G, Ab, C, Db

D, F#, G#, A, C#, D

D#, F##, G##, A#, C##, D#

Eb, G, A, Bb, D, Eb

E, G#, A#, B, D#, E

F, A, B, C, E, F

F#, A#, B#, C#, E#, F#

Gb, Bb, C, Db, F, Gb

G, B, C#, D, F#, G

G#, B#, C##, D#, F##, G#

Ab, C, D, Eb, G, Ab

A, C#, D#, E, G#, A

A#, C##, D##, E#, G##, A#

Bb, D, E, F, A, Bb

B, D#, E#, F#, A#, B

**Valaji**

Step pattern: 4, 3, 2, 1, 2

Ascending intervals: P1, M3, P5, M6, m7, P8

C, E, G, A, Bb, C

C#, E#, G#, A#, B, C#

Db, F, Ab, Bb, Cb, Db

D, F#, A, B, C, D

D#, F##, A#, B#, C#, D#

Eb, G, Bb, C, Db, Eb

E, G#, B, C#, D, E

F, A, C, D, Eb, F

F#, A#, C#, D#, E, F#

Gb, Bb, Db, Eb, Fb, Gb

G, B, D, E, F, G

G#, B#, D#, E#, F#, G#

Ab, C, Eb, F, Gb, Ab

A, C#, E, F#, G, A

A#, C##, E#, F##, G#, A#

Bb, D, F, G, Ab, Bb

B, D#, F#, G#, A, B

## Varamu

Step pattern: 3, 2, 4, 1, 2

Ascending intervals: P1, m3, P4, M6, m7, P8

C, Eb, F, A, Bb, C

C#, E, F#, A#, B, C#

Db, Fb, Gb, Bb, Cb, Db

D, F, G, B, C, D

D#, F#, G#, B#, C#, D#

Eb, Gb, Ab, C, Db, Eb

E, G, A, C#, D, E

F, Ab, Bb, D, Eb, F

F#, A, B, D#, E, F#

Gb, Bbb, Cb, Eb, Fb, Gb

G, Bb, C, E, F, G

G#, B, C#, E#, F#, G#

Ab, Cb, Db, F, Gb, Ab

A, C, D, F#, G, A

A#, C#, D#, F###, G#, A#

Bb, Db, Eb, G, Ab, Bb

B, D, E, G#, A, B

## Varini

Step pattern: 3, 4, 1, 2, 2

Ascending intervals: P1, m3, P5, m6, m7, P8

C, Eb, G, Ab, Bb, C

C#, E, G#, A, B, C#

Db, Fb, Ab, Bbb, Cb, Db

D, F, A, Bb, C, D

D#, F#, A#, B, C#, D#

Eb, Gb, Bb, Cb, Db, Eb

E, G, B, C, D, E

F, Ab, C, Db, Eb, F

F#, A, C#, D, E, F#

Gb, Bbb, Db, Ebb, Fb, Gb

G, Bb, D, Eb, F, G

G#, B, D#, E, F#, G#

Ab, Cb, Eb, Fb, Gb, Ab

A, C, E, F, G, A

A#, C#, E#, F#, G#, A#

Bb, Db, F, Gb, Ab, Bb

B, D, F#, G, A, B

**Zilaf**

Step pattern: 4, 1, 2, 1, 4

Ascending intervals: P1, M3, P4, P5, m6, P8

C, E, F, G, Ab, C

C#, E#, F#, G#, A, C#

Db, F, Gb, Ab, Bbb, Db

D, F#, G, A, Bb, D

D#, F##, G#, A#, B, D#

Eb, G, Ab, Bb, Cb, Eb

E, G#, A, B, C, E

F, A, Bb, C, Db, F

F#, A#, B, C#, D, F#

Gb, Bb, Cb, Db, Ebb, Gb

G, B, C, D, Eb, G

G#, B#, C#, D#, E, G#

Ab, C, Db, Eb, Fb, Ab

A, C#, D, E, F, A

A#, C##, D#, E#, F#, A#

Bb, D, Eb, F, Gb, Bb

B, D#, E, F#, G, B



## 6 – NOTE SCALES

**Amarasenapriya**

Step pattern: 2, 1, 3, 1, 4, 1

Ascending intervals: P1, M2, m3, dim5, P5, M7, P8

C, D, Eb, Gb, G, B, C

C#, D#, E, G, G#, B#, C#

Db, Eb, Fb, Abb, Ab, C, Db

D, E, F, Ab, A, C#, D

D#, E#, F#, A, A#, C##, D#

Eb, F, Gb, Bbb, Bb, D, Eb

E, F#, G, Bb, B, D#, E

F, G, Ab, Cb, C, E, F

F#, G#, A, C, C#, E#, F#

Gb, Ab, Bbb, Dbb, Db, F, Gb

G, A, Bb, Db, D, F#, G

G#, A#, B, D, D#, F##, G#

Ab, Bb, Cb, Ebb, Eb, G, Ab

A, B, C, Eb, E, G#, A

A#, B#, C#, E, E#, G##, A#

Bb, C, Db, Fb, F, A, Bb

B, C#, D, F, F#, A#, B

## Ancient Chinese

Step pattern: 2, 2, 2, 1, 2, 3

Ascending intervals: P1, M2, M3, Aug4, P5, M6, P8

C, D, E, F#, G, A, C

C#, D#, E#, F##, G#, A#, C#

Db, Eb, F, G, Ab, Bb, Db

D, E, F#, G#, A, B, D

D#, E#, F##, G##, A#, B#, D#

Eb, F, G, A, Bb, C, Eb

E, F#, G#, A#, B, C#, E

F, G, A, B, C, D, F

F#, G#, A#, B#, C#, D#, F#

Gb, Ab, Bb, C, Db, Eb, Gb

G, A, B, C#, D, E, G

G#, A#, B#, C##, D#, E#, G#

Ab, Bb, C, D, Eb, F, Ab

A, B, C#, D#, E, F#, A

A#, B#, C##, D##, E#, F##, A#

Bb, C, D, E, F, G, Bb

B, C#, D#, E#, F#, G#, B

## Augmented

Step pattern: 3, 1, 3, 1, 3, 1

Ascending intervals: P1, Aug2, M3, P5, m6, M7, P8

C, D#, E, G, Ab, B, C

C#, D##, E#, G#, A, B#, C#

Db, E, F, Ab, Bbb, C, Db

D, E#, F#, A, Bb, C#, D

D#, E##, F##, A#, B, C##, D#

Eb, F#, G, Bb, Cb, D, Eb

E, F##, G#, B, C, D#, E

F, G#, A, C, Db, E, F

F#, G##, A#, C#, D, E#, F#

Gb, A, Bb, Db, Ebb, F, Gb

G, A#, B, D, Eb, F#, G

G#, A##, B#, D#, E, F##, G#

Ab, B, C, Eb, Fb, G, Ab

A, B#, C#, E, F, G#, A

A#, B##, C##, E#, F#, G##, A#

Bb, C#, D, F, Gb, A, Bb

B, C##, D#, F#, G, A#, B

**Bauli**

Step pattern: 1, 3, 3, 1, 3, 1

Ascending intervals: P1, m2, M3, P5, m6, M7, P8

C, Db, E, G, Ab, B, C

C#, D, E#, G#, A, B#, C#

Db, Ebb, F, Ab, Bbb, C, Db

D, Eb, F#, A, Bb, C#, D

D#, E, F##, A#, B, C##, D#

Eb, Fb, G, Bb, Cb, D, Eb

E, F, G#, B, C, D#, E

F, Gb, A, C, Db, E, F

F#, G, A#, C#, D, E#, F#

Gb, Abb, Bb, Db, Ebb, F, Gb

G, Ab, B, D, Eb, F#, G

G#, A, B#, D#, E, F##, G#

Ab, Bbb, C, Eb, Fb, G, Ab

A, Bb, C#, E, F, G#, A

A#, B, C##, E#, F#, G##, A#

Bb, Cb, D, F, Gb, A, Bb

B, C, D#, F#, G, A#, B

## Bhinna Pancama

Step pattern: 2, 3, 2, 1, 3, 1

Ascending intervals: P1, M2, P4, P5, m6, M7, P8

C, D, F, G, Ab, B, C

C#, D#, F#, G#, A, B#, C#

Db, Eb, Gb, Ab, Bbb, C, Db

D, E, G, A, Bb, C#, D

D#, E#, G#, A#, B, C##, D#

Eb, F, Ab, Bb, Cb, D, Eb

E, F#, A, B, C, D#, E

F, G, Bb, C, Db, E, F

F#, G#, B, C#, D, E#, F#

Gb, Ab, Cb, Db, Ebb, F, Gb

G, A, C, D, Eb, F#, G

G#, A#, C#, D#, E, F##, G#

Ab, Bb, Db, Eb, Fb, G, Ab

A, B, D, E, F, G#, A

A#, B#, D#, E#, F#, G##, A#

Bb, C, Eb, F, Gb, A, Bb

B, C#, E, F#, G, A#, B

## Blues Major Hexatonic

Step pattern: 2, 1, 1, 3, 2, 3

Ascending intervals: P1, M2, Aug2, M3, P5, M6, P8

C, D, D#, E, G, A, C

C#, D#, D##, E#, G#, A#, C#

Db, Eb, E, F, Ab, Bb, Db

D, E, E#, F#, A, B, D

D#, E#, E##, F##, A#, B#, D#

Eb, F, F#, G, Bb, C, Eb

E, F#, F##, G#, B, C#, E

F, G, G#, A, C, D, F

F#, G#, G##, A#, C#, D#, F#

Gb, Ab, A, Bb, Db, Eb, Gb

G, A, A#, B, D, E, G

G#, A#, A##, B#, D#, E#, G#

Ab, Bb, B, C, Eb, F, Ab

A, B, B#, C#, E, F#, A

A#, B#, B##, C##, E#, F##, A#

Bb, C, C#, D, F, G, Bb

B, C#, C##, D#, F#, G#, B

## Blues Minor Hexatonic

Step pattern: 3, 2, 1, 1, 3, 2

Ascending intervals: P1, m3, P4, dim5, P5, m7, P8

C, Eb, F, Gb, G, Bb, C

C#, E, F#, G, G#, B, C#

Db, Fb, Gb, Abb, Ab, Cb, Db

D, F, G, Ab, A, C, D

D#, F#, G#, A, A#, C#, D#

Eb, Gb, Ab, Bbb, Bb, Db, Eb

E, G, A, Bb, B, D, E

F, Ab, Bb, Cb, C, Eb, F

F#, A, B, C, C#, E, F#

Gb, Bbb, Cb, Dbb, Db, Fb, Gb

G, Bb, C, Db, D, F, G

G#, B, C#, D, D#, F#, G#

Ab, Cb, Db, Ebb, Eb, Gb, Ab

A, C, D, Eb, E, G, A

A#, C#, D#, E, E#, G#, A#

Bb, Db, Eb, Fb, F, Ab, Bb

B, D, E, F, F#, A, B



## Brindabani Sarang

Step pattern: 2, 3, 2, 3, 1, 1

Ascending intervals: P1, M2, P4, P5, m7, M7, P8

C, D, F, G, Bb, B, C

C#, D#, F#, G#, B, B#, C#

Db, Eb, Gb, Ab, Cb, C, Db

D, E, G, A, C, C#, D

D#, E#, G#, A#, C#, C##, D#

Eb, F, Ab, Bb, Db, D, Eb

E, F#, A, B, D, D#, E

F, G, Bb, C, Eb, E, F

F#, G#, B, C#, E, E#, F#

Gb, Ab, Cb, Db, Fb, F, Gb

G, A, C, D, F, F#, G

G#, A#, C#, D#, F#, F##, G#

Ab, Bb, Db, Eb, Gb, G, Ab

A, B, D, E, G, G#, A

A#, B#, D#, E#, G#, G##, A#

Bb, C, Eb, F, Ab, A, Bb

B, C#, E, F#, A, A#, B

## Chandrajyoti

Step pattern: 1, 1, 4, 1, 2, 3

Ascending intervals: P1, m2, M2, dim5, P5, M6, P8

C, Db, D, Gb, G, A, C

C#, D, D#, G, G#, A#, C#

Db, Ebb, Eb, Abb, Ab, Bb, Db

D, Eb, E, Ab, A, B, D

D#, E, E#, A, A#, B#, D#

Eb, Fb, F, Bbb, Bb, C, Eb

E, F, F#, Bb, B, C#, E

F, Gb, G, Cb, C, D, F

F#, G, G#, C, C#, D#, F#

Gb, Abb, Ab, Dbb, Db, Eb, Gb

G, Ab, A, Db, D, E, G

G#, A, A#, D, D#, E#, G#

Ab, Bbb, Bb, Ebb, Eb, F, Ab

A, Bb, B, Eb, E, F#, A

A#, B, B#, E, E#, F##, A#

Bb, Cb, C, Fb, F, G, Bb

B, C, C#, F, F#, G#, B

## Chaya Vati

Step pattern: 1, 3, 1, 4, 2, 1

Ascending intervals: P1, m2, M3, P4, M6, M7, P8

C, Db, E, F, A, B, C

C#, D, E#, F#, A#, B#, C#

Db, Ebb, F, Gb, Bb, C, Db

D, Eb, F#, G, B, C#, D

D#, E, F##, G#, B#, C##, D#

Eb, Fb, G, Ab, C, D, Eb

E, F, G#, A, C#, D#, E

F, Gb, A, Bb, D, E, F

F#, G, A#, B, D#, E#, F#

Gb, Abb, Bb, Cb, Eb, F, Gb

G, Ab, B, C, E, F#, G

G#, A, B#, C#, E#, F##, G#

Ab, Bbb, C, Db, F, G, Ab

A, Bb, C#, D, F#, G#, A

A#, B, C##, D#, F##, G##, A#

Bb, Cb, D, Eb, G, A, Bb

B, C, D#, E, G#, A#, B

**Dipak**

Step pattern: 2, 2, 1, 1, 1, 5

Ascending intervals: P1, M2, M3, P4, dim5, P5, P8

C, D, E, F, Gb, G, C

C#, D#, E#, F#, G, G#, C#

Db, Eb, F, Gb, Abb, Ab, Db

D, E, F#, G, Ab, A, D

D#, E#, F##, G#, A, A#, D#

Eb, F, G, Ab, Bbb, Bb, Eb

E, F#, G#, A, Bb, B, E

F, G, A, Bb, Cb, C, F

F#, G#, A#, B, C, C#, F#

Gb, Ab, Bb, Cb, Dbb, Db, Gb

G, A, B, C, Db, D, G

G#, A#, B#, C#, D, D#, G#

Ab, Bb, C, Db, Ebb, Eb, Ab

A, B, C#, D, Eb, E, A

A#, B#, C##, D#, E, E#, A#

Bb, C, D, Eb, Fb, F, Bb

B, C#, D#, E, F, F#, B

## Double-Phrygian Hexatonic

Step pattern: 1, 2, 2, 1, 3, 3

Ascending intervals: P1, m2, m3, P4, dim5, M6, P8

C, Db, Eb, F, Gb, A, C

C#, D, E, F#, G, A#, C#

Db, Ebb, Fb, Gb, Abb, Bb, Db

D, Eb, F, G, Ab, B, D

D#, E, F#, G#, A, B#, D#

Eb, Fb, Gb, Ab, Bbb, C, Eb

E, F, G, A, Bb, C#, E

F, Gb, Ab, Bb, Cb, D, F

F#, G, A, B, C, D#, F#

Gb, Abb, Bbb, Cb, Dbb, Eb, Gb

G, Ab, Bb, C, Db, E, G

G#, A, B, C#, D, E#, G#

Ab, Bbb, Cb, Db, Ebb, F, Ab

A, Bb, C, D, Eb, F#, A

A#, B, C#, D#, E, F##, A#

Bb, Cb, Db, Eb, Fb, G, Bb

B, C, D, E, F, G#, B

## Eskimo Hexatonic

Step pattern: 2, 2, 2, 2, 3, 1

Ascending intervals: P1, M2, M3, dim5, m6, M7, P8

C, D, E, Gb, Ab, B, C

C#, D#, E#, G, A, B#, C#

Db, Eb, F, Abb, Bbb, C, Db

D, E, F#, Ab, Bb, C#, D

D#, E#, F##, A, B, C##, D#

Eb, F, G, Bbb, Cb, D, Eb

E, F#, G#, Bb, C, D#, E

F, G, A, Cb, Db, E, F

F#, G#, A#, C, D, E#, F#

Gb, Ab, Bb, Dbb, Ebb, F, Gb

G, A, B, Db, Eb, F#, G

G#, A#, B#, D, E, F##, G#

Ab, Bb, C, Ebb, Fb, G, Ab

A, B, C#, Eb, F, G#, A

A#, B#, C##, E, F#, G##, A#

Bb, C, D, Fb, Gb, A, Bb

B, C#, D#, F, G, A#, B

## Gandharavam

Step pattern: 1, 2, 2, 2, 3, 2

Ascending intervals: P1, m2, m3, P4, P5, m7, P8

C, Db, Eb, F, G, Bb, C

C#, D, E, F#, G#, B, C#

Db, Ebb, Fb, Gb, Ab, Cb, Db

D, Eb, F, G, A, C, D

D#, E, F#, G#, A#, C#, D#

Eb, Fb, Gb, Ab, Bb, Db, Eb

E, F, G, A, B, D, E

F, Gb, Ab, Bb, C, Eb, F

F#, G, A, B, C#, E, F#

Gb, Abb, Bbb, Cb, Db, Fb, Gb

G, Ab, Bb, C, D, F, G

G#, A, B, C#, D#, F#, G#

Ab, Bbb, Cb, Db, Eb, Gb, Ab

A, Bb, C, D, E, G, A

A#, B, C#, D#, E#, G#, A#

Bb, Cb, Db, Eb, F, Ab, Bb

B, C, D, E, F#, A, B

## Gangatarangini

Step pattern: 4, 1, 1, 2, 3, 1

Ascending intervals: P1, M3, P4, dim5, m6, M7, P8

C, E, F, Gb, Ab, B, C

C#, E#, F#, G, A, B#, C#

Db, F, Gb, Abb, Bbb, C, Db

D, F#, G, Ab, Bb, C#, D

D#, F##, G#, A, B, C##, D#

Eb, G, Ab, Bbb, Cb, D, Eb

E, G#, A, Bb, C, D#, E

F, A, Bb, Cb, Db, E, F

F#, A#, B, C, D, E#, F#

Gb, Bb, Cb, Dbb, Ebb, F, Gb

G, B, C, Db, Eb, F#, G

G#, B#, C#, D, E, F##, G#

Ab, C, Db, Ebb, Fb, G, Ab

A, C#, D, Eb, F, G#, A

A#, C##, D#, E, F#, G##, A#

Bb, D, Eb, Fb, Gb, A, Bb

B, D#, E, F, G, A#, B



## Gaula

Step pattern: 1, 3, 1, 2, 3, 2

Ascending intervals: P1, m2, M3, P4, P5, m7, P8

C, Db, E, F, G, Bb, C

C#, D, E#, F#, G#, B, C#

Db, Ebb, F, Gb, Ab, Cb, Db

D, Eb, F#, G, A, C, D

D#, E, F##, G#, A#, C#, D#

Eb, Fb, G, Ab, Bb, Db, Eb

E, F, G#, A, B, D, E

F, Gb, A, Bb, C, Eb, F

F#, G, A#, B, C#, E, F#

Gb, Abb, Bb, Cb, Db, Fb, Gb

G, Ab, B, C, D, F, G

G#, A, B#, C#, D#, F#, G#

Ab, Bbb, C, Db, Eb, Gb, Ab

A, Bb, C#, D, E, G, A

A#, B, C##, D#, E#, G#, A#

Bb, Cb, D, Eb, F, Ab, Bb

B, C, D#, E, F#, A, B

## Genus Secundum

Step pattern: 4, 1, 2, 2, 2, 1

Ascending intervals: P1, M3, P4, P5, M6, M7, P8

C, E, F, G, A, B, C

C#, E#, F#, G#, A#, B#, C#

Db, F, Gb, Ab, Bb, C, Db

D, F#, G, A, B, C#, D

D#, F##, G#, A#, B#, C##, D#

Eb, G, Ab, Bb, C, D, Eb

E, G#, A, B, C#, D#, E

F, A, Bb, C, D, E, F

F#, A#, B, C#, D#, E#, F#

Gb, Bb, Cb, Db, Eb, F, Gb

G, B, C, D, E, F#, G

G#, B#, C#, D#, E#, F##, G#

Ab, C, Db, Eb, F, G, Ab

A, C#, D, E, F#, G#, A

A#, C##, D#, E#, F##, G##, A#

Bb, D, Eb, F, G, A, Bb

B, D#, E, F#, G#, A#, B

## Genus Tertium

Step pattern: 3, 1, 3, 1, 3, 1

Ascending intervals: P1, m3, M3, P5, m6, M7, P8

C, Eb, E, G, Ab, B, C

C#, E, E#, G#, A, B#, C#

Db, Fb, F, Ab, Bbb, C, Db

D, F, F#, A, Bb, C#, D

D#, F#, F##, A#, B, C##, D#

Eb, Gb, G, Bb, Cb, D, Eb

E, G, G#, B, C, D#, E

F, Ab, A, C, Db, E, F

F#, A, A#, C#, D, E#, F#

Gb, Bbb, Bb, Db, Ebb, F, Gb

G, Bb, B, D, Eb, F#, G

G#, B, B#, D#, E, F##, G#

Ab, Cb, C, Eb, Fb, G, Ab

A, C, C#, E, F, G#, A

A#, C#, C##, E#, F#, G##, A#

Bb, Db, D, F, Gb, A, Bb

B, D, D#, F#, G, A#, B

## Ghantana

Step pattern: 2, 1, 2, 3, 3, 1

Ascending intervals: P1, M2, m3, P4, m6, M7, P8

C, D, Eb, F, Ab, B, C

C#, D#, E, F#, A, B#, C#

Db, Eb, Fb, Gb, Bbb, C, Db

D, E, F, G, Bb, C#, D

D#, E#, F#, G#, B, C##, D#

Eb, F, Gb, Ab, Cb, D, Eb

E, F#, G, A, C, D#, E

F, G, Ab, Bb, Db, E, F

F#, G#, A, B, D, E#, F#

Gb, Ab, Bbb, Cb, Ebb, F, Gb

G, A, Bb, C, Eb, F#, G

G#, A#, B, C#, E, F##, G#

Ab, Bb, Cb, Db, Fb, G, Ab

A, B, C, D, F, G#, A

A#, B#, C#, D#, F#, G##, A#

Bb, C, Db, Eb, Gb, A, Bb

B, C#, D, E, G, A#, B

## Gopikatilaka

Step pattern: 2, 1, 3, 1, 3, 2

Ascending intervals: P1, M2, m3, dim5, P5, m7, P8

C, D, Eb, Gb, G, Bb, C

C#, D#, E, G, G#, B, C#

Db, Eb, Fb, Abb, Ab, Cb, Db

D, E, F, Ab, A, C, D

D#, E#, F#, A, A#, C#, D#

Eb, F, Gb, Bbb, Bb, Db, Eb

E, F#, G, Bb, B, D, E

F, G, Ab, Cb, C, Eb, F

F#, G#, A, C, C#, E, F#

Gb, Ab, Bbb, Dbb, Db, Fb, Gb

G, A, Bb, Db, D, F, G

G#, A#, B, D, D#, F#, G#

Ab, Bb, Cb, Ebb, Eb, Gb, Ab

A, B, C, Eb, E, G, A

A#, B#, C#, E, E#, G#, A#

Bb, C, Db, Fb, F, Ab, Bb

B, C#, D, F, F#, A, B

## Gurjari Todi

Step pattern: 1, 2, 3, 2, 2, 2

Ascending intervals: P1, m2, m3, dim5, m6, m7, P8

C, Db, Eb, Gb, Ab, Bb, C

C#, D, E, G, A, B, C#

Db, Ebb, Fb, Abb, Bbb, Cb, Db

D, Eb, F, Ab, Bb, C, D

D#, E, F#, A, B, C#, D#

Eb, Fb, Gb, Bbb, Cb, Db, Eb

E, F, G, Bb, C, D, E

F, Gb, Ab, Cb, Db, Eb, F

F#, G, A, C, D, E, F#

Gb, Abb, Bbb, Dbb, Ebb, Fb, Gb

G, Ab, Bb, Db, Eb, F, G

G#, A, B, D, E, F#, G#

Ab, Bbb, Cb, Ebb, Fb, Gb, Ab

A, Bb, C, Eb, F, G, A

A#, B, C#, E, F#, G#, A#

Bb, Cb, Db, Fb, Gb, Ab, Bb

B, C, D, F, G, A, B

## Hamsa Vinodini

Step pattern: 2, 2, 1, 4, 2, 1

Ascending intervals: P1, M2, M3, P4, M6, M7, P8

C, D, E, F, A, B, C

C#, D#, E#, F#, A#, B#, C#

Db, Eb, F, Gb, Bb, C, Db

D, E, F#, G, B, C#, D

D#, E#, F##, G#, B#, C##, D#

Eb, F, G, Ab, C, D, Eb

E, F#, G#, A, C#, D#, E

F, G, A, Bb, D, E, F

F#, G#, A#, B, D#, E#, F#

Gb, Ab, Bb, Cb, Eb, F, Gb

G, A, B, C, E, F#, G

G#, A#, B#, C#, E#, F##, G#

Ab, Bb, C, Db, F, G, Ab

A, B, C#, D, F#, G#, A

A#, B#, C##, D#, F##, G##, A#

Bb, C, D, Eb, G, A, Bb

B, C#, D#, E, G#, A#, B

## Hawaiian (hexatonic)

Step pattern: 2, 1, 4, 2, 2, 1

Ascending intervals: P1, M2, m3, P5, M6, M7, P8

C, D, Eb, G, A, B, C

C#, D#, E, G#, A#, B#, C#

Db, Eb, Fb, Ab, Bb, C, Db

D, E, F, A, B, C#, D

D#, E#, F#, A#, B#, C##, D#

Eb, F, Gb, Bb, C, D, Eb

E, F#, G, B, C#, D#, E

F, G, Ab, C, D, E, F

F#, G#, A, C#, D#, E#, F#

Gb, Ab, Bbb, Db, Eb, F, Gb

G, A, Bb, D, E, F#, G

G#, A#, B, D#, E#, F##, G#

Ab, Bb, Cb, Eb, F, G, Ab

A, B, C, E, F#, G#, A

A#, B#, C#, E#, F##, G##, A#

Bb, C, Db, F, G, A, Bb

B, C#, D, F#, G#, A#, B



## Hejjajji

Step pattern: 1, 3, 2, 2, 1, 3

Ascending intervals: P1, m2, M3, dim5, m6, M6, P8

C, Db, E, Gb, Ab, A, C

C#, D, E#, G, A, A#, C#

Db, Ebb, F, Abb, Bbb, Bb, Db

D, Eb, F#, Ab, Bb, B, D

D#, E, F##, A, B, B#, D#

Eb, Fb, G, Bbb, Cb, C, Eb

E, F, G#, Bb, C, C#, E

F, Gb, A, Cb, Db, D, F

F#, G, A#, C, D, D#, F#

Gb, Abb, Bb, Dbb, Ebb, Eb, Gb

G, Ab, B, Db, Eb, E, G

G#, A, B#, D, E, E#, G#

Ab, Bbb, C, Ebb, Fb, F, Ab

A, Bb, C#, Eb, F, F#, A

A#, B, C##, E, F#, F##, A#

Bb, Cb, D, Fb, Gb, G, Bb

B, C, D#, F, G, G#, B

## Honchoshi Plagal From

Step pattern: 1, 2, 2, 1, 4, 2

Ascending intervals: P1, m2, m3, P4, dim5, m7, P8

C, Db, Eb, F, Gb, Bb, C

C#, D, E, F#, G, B, C#

Db, Ebb, Fb, Gb, Abb, Cb, Db

D, Eb, F, G, Ab, C, D

D#, E, F#, G#, A, C#, D#

Eb, Fb, Gb, Ab, Bbb, Db, Eb

E, F, G, A, Bb, D, E

F, Gb, Ab, Bb, Cb, Eb, F

F#, G, A, B, C, E, F#

Gb, Abb, Bbb, Cb, Dbb, Fb, Gb

G, Ab, Bb, C, Db, F, G

G#, A, B, C#, D, F#, G#

Ab, Bbb, Cb, Db, Ebb, Gb, Ab

A, Bb, C, D, Eb, G, A

A#, B, C#, D#, E, G#, A#

Bb, Cb, Db, Eb, Fb, Ab, Bb

B, C, D, E, F, A, B

**Insen**

Step pattern: 1, 4, 2, 1, 2, 2

Ascending intervals: P1, m2, P4, P5, m6, m7, P8

C, Db, F, G, Ab, Bb, C

C#, D, F#, G#, A, B, C#

Db, Ebb, Gb, Ab, Bbb, Cb, Db

D, Eb, G, A, Bb, C, D

D#, E, G#, A#, B, C#, D#

Eb, Fb, Ab, Bb, Cb, Db, Eb

E, F, A, B, C, D, E

F, Gb, Bb, C, Db, Eb, F

F#, G, B, C#, D, E, F#

Gb, Abb, Cb, Db, Ebb, Fb, Gb

G, Ab, C, D, Eb, F, G

G#, A, C#, D#, E, F#, G#

Ab, Bbb, Db, Eb, Fb, Gb, Ab

A, Bb, D, E, F, G, A

A#, B, D#, E#, F#, G#, A#

Bb, Cb, Eb, F, Gb, Ab, Bb

B, C, E, F#, G, A, B

## Inverted Augmented

Step pattern: 1, 3, 1, 3, 1, 3

Ascending intervals: P1, m2, M3, P4, m6, M6, P8

C, Db, E, F, Ab, A, C

C#, D, E#, F#, A, A#, C#

Db, Ebb, F, Gb, Bbb, Bb, Db

D, Eb, F#, G, Bb, B, D

D#, E, F##, G#, B, B#, D#

Eb, Fb, G, Ab, Cb, C, Eb

E, F, G#, A, C, C#, E

F, Gb, A, Bb, Db, D, F

F#, G, A#, B, D, D#, F#

Gb, Abb, Bb, Cb, Ebb, Eb, Gb

G, Ab, B, C, Eb, E, G

G#, A, B#, C#, E, E#, G#

Ab, Bbb, C, Db, Fb, F, Ab

A, Bb, C#, D, F, F#, A

A#, B, C##, D#, F#, F##, A#

Bb, Cb, D, Eb, Gb, G, Bb

B, C, D#, E, G, G#, B

## Jaganmohanam

Step pattern: 2, 4, 1, 1, 2, 2

Ascending intervals: P1, M2, dim5, P5, m6, m7, P8

C, D, Gb, G, Ab, Bb, C

C#, D#, G, G#, A, B, C#

Db, Eb, Abb, Ab, Bbb, Cb, Db

D, E, Ab, A, Bb, C, D

D#, E#, A, A#, B, C#, D#

Eb, F, Bbb, Bb, Cb, Db, Eb

E, F#, Bb, B, C, D, E

F, G, Cb, C, Db, Eb, F

F#, G#, C, C#, D, E, F#

Gb, Ab, Dbb, Db, Ebb, Fb, Gb

G, A, Db, D, Eb, F, G

G#, A#, D, D#, E, F#, G#

Ab, Bb, Ebb, Eb, Fb, Gb, Ab

A, B, Eb, E, F, G, A

A#, B#, E, E#, F#, G#, A#

Bb, C, Fb, F, Gb, Ab, Bb

B, C#, F, F#, G, A, B

**Jivantika**

Step pattern: 1, 4, 2, 2, 2, 1

Ascending intervals: P1, m2, P4, P5, M6, M7, P8

C, Db, F, G, A, B, C

C#, D, F#, G#, A#, B#, C#

Db, Ebb, Gb, Ab, Bb, C, Db

D, Eb, G, A, B, C#, D

D#, E, G#, A#, B#, C##, D#

Eb, Fb, Ab, Bb, C, D, Eb

E, F, A, B, C#, D#, E

F, Gb, Bb, C, D, E, F

F#, G, B, C#, D#, E#, F#

Gb, Abb, Cb, Db, Eb, F, Gb

G, Ab, C, D, E, F#, G

G#, A, C#, D#, E#, F##, G#

Ab, Bbb, Db, Eb, F, G, Ab

A, Bb, D, E, F#, G#, A

A#, B, D#, E#, F##, G##, A#

Bb, Cb, Eb, F, G, A, Bb

B, C, E, F#, G#, A#, B

**Jivantini**

Step pattern: 3, 3, 1, 3, 1, 1

Ascending intervals: P1, m3, dim5, P5, m7, M7, P8

C, Eb, Gb, G, Bb, B, C

C#, E, G, G#, B, B#, C#

Db, Fb, Abb, Ab, Cb, C, Db

D, F, Ab, A, C, C#, D

D#, F#, A, A#, C#, C##, D#

Eb, Gb, Bbb, Bb, Db, D, Eb

E, G, Bb, B, D, D#, E

F, Ab, Cb, C, Eb, E, F

F#, A, C, C#, E, E#, F#

Gb, Bbb, Dbb, Db, Fb, F, Gb

G, Bb, Db, D, F, F#, G

G#, B, D, D#, F#, F##, G#

Ab, Cb, Ebb, Eb, Gb, G, Ab

A, C, Eb, E, G, G#, A

A#, C#, E, E#, G#, G##, A#

Bb, Db, Fb, F, Ab, A, Bb

B, D, F, F#, A, A#, B

**Jog**

Step pattern: 3, 1, 1, 2, 3, 2

Ascending intervals: P1, m3, M3, P4, P5, m7, P8

C, Eb, E, F, G, Bb, C

C#, E, E#, F#, G#, B, C#

Db, Fb, F, Gb, Ab, Cb, Db

D, F, F#, G, A, C, D

D#, F#, F##, G#, A#, C#, D#

Eb, Gb, G, Ab, Bb, Db, Eb

E, G, G#, A, B, D, E

F, Ab, A, Bb, C, Eb, F

F#, A, A#, B, C#, E, F#

Gb, Bbb, Bb, Cb, Db, Fb, Gb

G, Bb, B, C, D, F, G

G#, B, B#, C#, D#, F#, G#

Ab, Cb, C, Db, Eb, Gb, Ab

A, C, C#, D, E, G, A

A#, C#, C##, D#, E#, G#, A#

Bb, Db, D, Eb, F, Ab, Bb

B, D, D#, E, F#, A, B



**Jyoti**

Step pattern: 4, 2, 1, 1, 2, 2

Ascending intervals: P1, M3, dim5, P5, m6, m7, P8

C, E, Gb, G, Ab, Bb, C

C#, E#, G, G#, A, B, C#

Db, F, Abb, Ab, Bbb, Cb, Db

D, F#, Ab, A, Bb, C, D

D#, F##, A, A#, B, C#, D#

Eb, G, Bbb, Bb, Cb, Db, Eb

E, G#, Bb, B, C, D, E

F, A, Cb, C, Db, Eb, F

F#, A#, C, C#, D, E, F#

Gb, Bb, Dbb, Db, Ebb, Fb, Gb

G, B, Db, D, Eb, F, G

G#, B#, D, D#, E, F#, G#

Ab, C, Ebb, Eb, Fb, Gb, Ab

A, C#, Eb, E, F, G, A

A#, C##, E, E#, F#, G#, A#

Bb, D, Fb, F, Gb, Ab, Bb

B, D#, F, F#, G, A, B

## Kalagada

Step pattern: 1, 3, 3, 1, 1, 3

Ascending intervals: P1, m2, M3, P5, m6, M6, P8

C, Db, E, G, Ab, A, C

C#, D, E#, G#, A, A#, C#

Db, Ebb, F, Ab, Bbb, Bb, Db

D, Eb, F#, A, Bb, B, D

D#, E, F##, A#, B, B#, D#

Eb, Fb, G, Bb, Cb, C, Eb

E, F, G#, B, C, C#, E

F, Gb, A, C, Db, D, F

F#, G, A#, C#, D, D#, F#

Gb, Abb, Bb, Db, Ebb, Eb, Gb

G, Ab, B, D, Eb, E, G

G#, A, B#, D#, E, E#, G#

Ab, Bbb, C, Eb, Fb, F, Ab

A, Bb, C#, E, F, F#, A

A#, B, C##, E#, F#, F##, A#

Bb, Cb, D, F, Gb, G, Bb

B, C, D#, F#, G, G#, B

**Kalakanthi (hexatonic)**

Step pattern: 1, 4, 2, 1, 1, 3

Ascending intervals: P1, m2, P4, P5, m6, M6, P8

C, Db, F, G, Ab, A, C

C#, D, F#, G#, A, A#, C#

Db, Ebb, Gb, Ab, Bbb, Bb, Db

D, Eb, G, A, Bb, B, D

D#, E, G#, A#, B, B#, D#

Eb, Fb, Ab, Bb, Cb, C, Eb

E, F, A, B, C, C#, E

F, Gb, Bb, C, Db, D, F

F#, G, B, C#, D, D#, F#

Gb, Abb, Cb, Db, Ebb, Eb, Gb

G, Ab, C, D, Eb, E, G

G#, A, C#, D#, E, E#, G#

Ab, Bbb, Db, Eb, Fb, F, Ab

A, Bb, D, E, F, F#, A

A#, B, D#, E#, F#, F##, A#

Bb, Cb, Eb, F, Gb, G, Bb

B, C, E, F#, G, G#, B

## Kalavati

Step pattern: 1, 3, 1, 2, 2, 3

Ascending intervals: P1, m2, M3, P4, P5, M6, P8

C, Db, E, F, G, A, C

C#, D, E#, F#, G#, A#, C#

Db, Ebb, F, Gb, Ab, Bb, Db

D, Eb, F#, G, A, B, D

D#, E, F##, G#, A#, B#, D#

Eb, Fb, G, Ab, Bb, C, Eb

E, F, G#, A, B, C#, E

F, Gb, A, Bb, C, D, F

F#, G, A#, B, C#, D#, F#

Gb, Abb, Bb, Cb, Db, Eb, Gb

G, Ab, B, C, D, E, G

G#, A, B#, C#, D#, E#, G#

Ab, Bbb, C, Db, Eb, F, Ab

A, Bb, C#, D, E, F#, A

A#, B, C##, D#, E#, F##, A#

Bb, Cb, D, Eb, F, G, Bb

B, C, D#, E, F#, G#, B

**Kamalamanohari (with m7)**

Step pattern: 4, 1, 2, 1, 2, 2

Ascending intervals: P1, M3, P4, P5, m6, m7, P8

C, E, F, G, Ab, Bb, C

C#, E#, F#, G#, A, B, C#

Db, F, Gb, Ab, Bbb, Cb, Db

D, F#, G, A, Bb, C, D

D#, F##, G#, A#, B, C#, D#

Eb, G, Ab, Bb, Cb, Db, Eb

E, G#, A, B, C, D, E

F, A, Bb, C, Db, Eb, F

F#, A#, B, C#, D, E, F#

Gb, Bb, Cb, Db, Ebb, Fb, Gb

G, B, C, D, Eb, F, G

G#, B#, C#, D#, E, F#, G#

Ab, C, Db, Eb, Fb, Gb, Ab

A, C#, D, E, F, G, A

A#, C##, D#, E#, F#, G#, A#

Bb, D, Eb, F, Gb, Ab, Bb

B, D#, E, F#, G, A, B

## Kashyapi

Step pattern: 1, 2, 4, 1, 2, 2

Ascending intervals: P1, m2, m3, P5, m6, m7, P8

C, Db, Eb, G, Ab, Bb, C

C#, D, E, G#, A, B, C#

Db, Ebb, Fb, Ab, Bbb, Cb, Db

D, Eb, F, A, Bb, C, D

D#, E, F#, A#, B, C#, D#

Eb, Fb, Gb, Bb, Cb, Db, Eb

E, F, G, B, C, D, E

F, Gb, Ab, C, Db, Eb, F

F#, G, A, C#, D, E, F#

Gb, Abb, Bbb, Db, Ebb, Fb, Gb

G, Ab, Bb, D, Eb, F, G

G#, A, B, D#, E, F#, G#

Ab, Bbb, Cb, Eb, Fb, Gb, Ab

A, Bb, C, E, F, G, A

A#, B, C#, E#, F#, G#, A#

Bb, Cb, Db, F, Gb, Ab, Bb

B, C, D, F#, G, A, B

## Khamas

Step pattern: 4, 1, 2, 2, 1, 2

Ascending intervals: P1, M3, P4, P5, M6, m7, P8

C, E, F, G, A, Bb, C

C#, E#, F#, G#, A#, B, C#

Db, F, Gb, Ab, Bb, Cb, Db

D, F#, G, A, B, C, D

D#, F##, G#, A#, B#, C#, D#

Eb, G, Ab, Bb, C, Db, Eb

E, G#, A, B, C#, D, E

F, A, Bb, C, D, Eb, F

F#, A#, B, C#, D#, E, F#

Gb, Bb, Cb, Db, Eb, Fb, Gb

G, B, C, D, E, F, G

G#, B#, C#, D#, E#, F#, G#

Ab, C, Db, Eb, F, Gb, Ab

A, C#, D, E, F#, G, A

A#, C##, D#, E#, F##, G#, A#

Bb, D, Eb, F, G, Ab, Bb

B, D#, E, F#, G#, A, B

**Kumud**

Step pattern: 2, 2, 3, 2, 2, 1

Ascending intervals: P1, M2, M3, P5, M6, M7, P8

C, D, E, G, A, B, C

C#, D#, E#, G#, A#, B#, C#

Db, Eb, F, Ab, Bb, C, Db

D, E, F#, A, B, C#, D

D#, E#, F##, A#, B#, C##, D#

Eb, F, G, Bb, C, D, Eb

E, F#, G#, B, C#, D#, E

F, G, A, C, D, E, F

F#, G#, A#, C#, D#, E#, F#

Gb, Ab, Bb, Db, Eb, F, Gb

G, A, B, D, E, F#, G

G#, A#, B#, D#, E#, F##, G#

Ab, Bb, C, Eb, F, G, Ab

A, B, C#, E, F#, G#, A

A#, B#, C##, E#, F##, G##, A#

Bb, C, D, F, G, A, Bb

B, C#, D#, F#, G#, A#, B



**Lalit Bhairav**

Step pattern: 1, 3, 1, 3, 2, 2

Ascending intervals: P1, m2, M3, P4, m6, m7, P8

C, Db, E, F, Ab, Bb, C

C#, D, E#, F#, A, B, C#

Db, Ebb, F, Gb, Bbb, Cb, Db

D, Eb, F#, G, Bb, C, D

D#, E, F##, G#, B, C#, D#

Eb, Fb, G, Ab, Cb, Db, Eb

E, F, G#, A, C, D, E

F, Gb, A, Bb, Db, Eb, F

F#, G, A#, B, D, E, F#

Gb, Abb, Bb, Cb, Ebb, Fb, Gb

G, Ab, B, C, Eb, F, G

G#, A, B#, C#, E, F#, G#

Ab, Bbb, C, Db, Fb, Gb, Ab

A, Bb, C#, D, F, G, A

A#, B, C##, D#, F#, G#, A#

Bb, Cb, D, Eb, Gb, Ab, Bb

B, C, D#, E, G, A, B

**Latika**

Step pattern: 2, 2, 3, 1, 3, 1

Ascending intervals: P1, M2, M3, P5, m6, M7, P8

C, D, E, G, Ab, B, C

C#, D#, E#, G#, A, B#, C#

Db, Eb, F, Ab, Bbb, C, Db

D, E, F#, A, Bb, C#, D

D#, E#, F##, A#, B, C##, D#

Eb, F, G, Bb, Cb, D, Eb

E, F#, G#, B, C, D#, E

F, G, A, C, Db, E, F

F#, G#, A#, C#, D, E#, F#

Gb, Ab, Bb, Db, Ebb, F, Gb

G, A, B, D, Eb, F#, G

G#, A#, B#, D#, E, F##, G#

Ab, Bb, C, Eb, Fb, G, Ab

A, B, C#, E, F, G#, A

A#, B#, C##, E#, F#, G##, A#

Bb, C, D, F, Gb, A, Bb

B, C#, D#, F#, G, A#, B

## Lydian Hexatonic

Step pattern: 2, 3, 2, 2, 2, 1

Ascending intervals: P1, M2, P4, P5, M6, M7, P8

C, D, F, G, A, B, C

C#, D#, F#, G#, A#, B#, C#

Db, Eb, Gb, Ab, Bb, C, Db

D, E, G, A, B, C#, D

D#, E#, G#, A#, B#, C##, D#

Eb, F, Ab, Bb, C, D, Eb

E, F#, A, B, C#, D#, E

F, G, Bb, C, D, E, F

F#, G#, B, C#, D#, E#, F#

Gb, Ab, Cb, Db, Eb, F, Gb

G, A, C, D, E, F#, G

G#, A#, C#, D#, E#, F##, G#

Ab, Bb, Db, Eb, F, G, Ab

A, B, D, E, F#, G#, A

A#, B#, D#, E#, F##, G##, A#

Bb, C, Eb, F, G, A, Bb

B, C#, E, F#, G#, A#, B

**Madhukauns (hexatonic)**

Step pattern: 3, 3, 1, 2, 1, 2

Ascending intervals: P1, m3, dim5, P5, M6, m7, P8

C, Eb, Gb, G, A, Bb, C

C#, E, G, G#, A#, B, C#

Db, Fb, Abb, Ab, Bb, Cb, Db

D, F, Ab, A, B, C, D

D#, F#, A, A#, B#, C#, D#

Eb, Gb, Bbb, Bb, C, Db, Eb

E, G, Bb, B, C#, D, E

F, Ab, Cb, C, D, Eb, F

F#, A, C, C#, D#, E, F#

Gb, Bbb, Dbb, Db, Eb, Fb, Gb

G, Bb, Db, D, E, F, G

G#, B, D, D#, E#, F#, G#

Ab, Cb, Ebb, Eb, F, Gb, Ab

A, C, Eb, E, F#, G, A

A#, C#, E, E#, F##, G#, A#

Bb, Db, Fb, F, G, Ab, Bb

B, D, F, F#, G#, A, B

## Major Diatonic Hexachord

Step pattern: 2, 2, 1, 2, 2, 3

Ascending intervals: P1, M2, M3, P4, P5, M6, P8

C, D, E, F, G, A, C

C#, D#, E#, F#, G#, A#, C#

Db, Eb, F, Gb, Ab, Bb, Db

D, E, F#, G, A, B, D

D#, E#, F##, G#, A#, B#, D#

Eb, F, G, Ab, Bb, C, Eb

E, F#, G#, A, B, C#, E

F, G, A, Bb, C, D, F

F#, G#, A#, B, C#, D#, F#

Gb, Ab, Bb, Cb, Db, Eb, Gb

G, A, B, C, D, E, G

G#, A#, B#, C#, D#, E#, G#

Ab, Bb, C, Db, Eb, F, Ab

A, B, C#, D, E, F#, A

A#, B#, C##, D#, E#, F##, A#

Bb, C, D, Eb, F, G, Bb

B, C#, D#, E, F#, G#, B

## Major Pentablues

Step pattern: 2, 1, 1, 3, 2, 3

Ascending intervals: P1, M2, m3, M3, P5, M6, P8

C, D, Eb, E, G, A, C

C#, D#, E, E#, G#, A#, C#

Db, Eb, Fb, F, Ab, Bb, Db

D, E, F, F#, A, B, D

D#, E#, F#, F##, A#, B#, D#

Eb, F, Gb, G, Bb, C, Eb

E, F#, G, G#, B, C#, E

F, G, Ab, A, C, D, F

F#, G#, A, A#, C#, D#, F#

Gb, Ab, Bbb, Bb, Db, Eb, Gb

G, A, Bb, B, D, E, G

G#, A#, B, B#, D#, E#, G#

Ab, Bb, Cb, C, Eb, F, Ab

A, B, C, C#, E, F#, A

A#, B#, C#, C##, E#, F##, A#

Bb, C, Db, D, F, G, Bb

B, C#, D, D#, F#, G#, B

**Malahari**

Step pattern: 1, 3, 1, 2, 1, 4

Ascending intervals: P1, m2, M3, P4, P5, m6, P8

C, Db, E, F, G, Ab, C

C#, D, E#, F#, G#, A, C#

Db, Ebb, F, Gb, Ab, Bbb, Db

D, Eb, F#, G, A, Bb, D

D#, E, F##, G#, A#, B, D#

Eb, Fb, G, Ab, Bb, Cb, Eb

E, F, G#, A, B, C, E

F, Gb, A, Bb, C, Db, F

F#, G, A#, B, C#, D, F#

Gb, Abb, Bb, Cb, Db, Ebb, Gb

G, Ab, B, C, D, Eb, G

G#, A, B#, C#, D#, E, G#

Ab, Bbb, C, Db, Eb, Fb, Ab

A, Bb, C#, D, E, F, A

A#, B, C##, D#, E#, F#, A#

Bb, Cb, D, Eb, F, Gb, Bb

B, C, D#, E, F#, G, B

**Malarani**

Step pattern: 2, 4, 1, 3, 1, 1

Ascending intervals: P1, M2, dim5, P5, m7, M7, P8

C, D, Gb, G, Bb, B, C

C#, D#, G, G#, B, B#, C#

Db, Eb, Abb, Ab, Cb, C, Db

D, E, Ab, A, C, C#, D

D#, E#, A, A#, C#, C##, D#

Eb, F, Bbb, Bb, Db, D, Eb

E, F#, Bb, B, D, D#, E

F, G, Cb, C, Eb, E, F

F#, G#, C, C#, E, E#, F#

Gb, Ab, Dbb, Db, Fb, F, Gb

G, A, Db, D, F, F#, G

G#, A#, D, D#, F#, F##, G#

Ab, Bb, Ebb, Eb, Gb, G, Ab

A, B, Eb, E, G, G#, A

A#, B#, E, E#, G#, G##, A#

Bb, C, Fb, F, Ab, A, Bb

B, C#, F, F#, A, A#, B



## Malayamarutam

Step pattern: 1, 3, 3, 2, 1, 2

Ascending intervals: P1, m2, M3, P5, M6, m7, P8

C, Db, E, G, A, Bb, C

C#, D, E#, G#, A#, B, C#

Db, Ebb, F, Ab, Bb, Cb, Db

D, Eb, F#, A, B, C, D

D#, E, F##, A#, B#, C#, D#

Eb, Fb, G, Bb, C, Db, Eb

E, F, G#, B, C#, D, E

F, Gb, A, C, D, Eb, F

F#, G, A#, C#, D#, E, F#

Gb, Abb, Bb, Db, Eb, Fb, Gb

G, Ab, B, D, E, F, G

G#, A, B#, D#, E#, F#, G#

Ab, Bbb, C, Eb, F, Gb, Ab

A, Bb, C#, E, F#, G, A

A#, B, C##, E#, F##, G#, A#

Bb, Cb, D, F, G, Ab, Bb

B, C, D#, F#, G#, A, B

**Manavi**

Step pattern: 2, 1, 4, 2, 1, 2

Ascending intervals: P1, M2, m3, P5, M6, m7, P8

C, D, Eb, G, A, Bb, C

C#, D#, E, G#, A#, B, C#

Db, Eb, Fb, Ab, Bb, Cb, Db

D, E, F, A, B, C, D

D#, E#, F#, A#, B#, C#, D#

Eb, F, Gb, Bb, C, Db, Eb

E, F#, G, B, C#, D, E

F, G, Ab, C, D, Eb, F

F#, G#, A, C#, D#, E, F#

Gb, Ab, Bbb, Db, Eb, Fb, Gb

G, A, Bb, D, E, F, G

G#, A#, B, D#, E#, F#, G#

Ab, Bb, Cb, Eb, F, Gb, Ab

A, B, C, E, F#, G, A

A#, B#, C#, E#, F##, G#, A#

Bb, C, Db, F, G, Ab, Bb

B, C#, D, F#, G#, A, B

## Mandari

Step pattern: 1, 3, 2, 1, 4, 1

Ascending intervals: P1, m2, M3, dim5, P5, M7, P8

C, Db, E, Gb, G, B, C

C#, D, E#, G, G#, B#, C#

Db, Ebb, F, Abb, Ab, C, Db

D, Eb, F#, Ab, A, C#, D

D#, E, F##, A, A#, C##, D#

Eb, Fb, G, Bbb, Bb, D, Eb

E, F, G#, Bb, B, D#, E

F, Gb, A, Cb, C, E, F

F#, G, A#, C, C#, E#, F#

Gb, Abb, Bb, Dbb, Db, F, Gb

G, Ab, B, Db, D, F#, G

G#, A, B#, D, D#, F##, G#

Ab, Bbb, C, Ebb, Eb, G, Ab

A, Bb, C#, Eb, E, G#, A

A#, B, C##, E, E#, G##, A#

Bb, Cb, D, Fb, F, A, Bb

B, C, D#, F, F#, A#, B

**Marva**

Step pattern: 1, 3, 2, 3, 2, 1

Ascending intervals: P1, m2, M3, dim5, M6, M7, P8

C, Db, E, Gb, A, B, C

C#, D, E#, G, A#, B#, C#

Db, Ebb, F, Abb, Bb, C, Db

D, Eb, F#, Ab, B, C#, D

D#, E, F##, A, B#, C##, D#

Eb, Fb, G, Bbb, C, D, Eb

E, F, G#, Bb, C#, D#, E

F, Gb, A, Cb, D, E, F

F#, G, A#, C, D#, E#, F#

Gb, Abb, Bb, Dbb, Eb, F, Gb

G, Ab, B, Db, E, F#, G

G#, A, B#, D, E#, F##, G#

Ab, Bbb, C, Ebb, F, G, Ab

A, Bb, C#, Eb, F#, G#, A

A#, B, C##, E, F##, G##, A#

Bb, Cb, D, Fb, G, A, Bb

B, C, D#, F, G#, A#, B

### Messiaen Mode 1 (6 groups of 2 notes each)

Step pattern: 2, 2, 2, 2, 2, 2

Ascending intervals: P1, M2, M3, Aug4, Aug5, Aug6, P8

C, D, E, F#, G#, A#, C

C#, D#, E#, F##, G###, A###, C#

Db, Eb, F, G, A, B, Db

D, E, F#, G#, A#, B#, D

D#, E#, F##, G###, A###, B##, D#

Eb, F, G, A, B, C#, Eb

E, F#, G#, A#, B#, C##, E

F, G, A, B, C#, D#, F

F#, G#, A#, B#, C##, D##, F#

Gb, Ab, Bb, C, D, E, Gb

G, A, B, C#, D#, E#, G

G#, A#, B#, C##, D##, E##, G#

Ab, Bb, C, D, E, F#, Ab

A, B, C#, D#, E#, F##, A

A#, B#, C##, D##, E##, F####, A#

Bb, C, D, E, F#, G#, Bb

B, C#, D#, E#, F##, G###, B

**Minor Hexatonic**

Step pattern: 2, 1, 2, 2, 3, 2

Ascending intervals: P1, M2, m3, P4, P5, m7, P8

C, D, Eb, F, G, Bb, C

C#, D#, E, F#, G#, B, C#

Db, Eb, Fb, Gb, Ab, Cb, Db

D, E, F, G, A, C, D

D#, E#, F#, G#, A#, C#, D#

Eb, F, Gb, Ab, Bb, Db, Eb

E, F#, G, A, B, D, E

F, G, Ab, Bb, C, Eb, F

F#, G#, A, B, C#, E, F#

Gb, Ab, Bbb, Cb, Db, Fb, Gb

G, A, Bb, C, D, F, G

G#, A#, B, C#, D#, F#, G#

Ab, Bb, Cb, Db, Eb, Gb, Ab

A, B, C, D, E, G, A

A#, B#, C#, D#, E#, G#, A#

Bb, C, Db, Eb, F, Ab, Bb

B, C#, D, E, F#, A, B

### Mixolydian Hexatonic (suspended)

Step pattern: 2, 3, 2, 2, 1, 2

Ascending intervals: P1, M2, P4, P5, M6, m7, P8

C, D, F, G, A, Bb, C

C#, D#, F#, G#, A#, B, C#

Db, Eb, Gb, Ab, Bb, Cb, Db

D, E, G, A, B, C, D

D#, E#, G#, A#, B#, C#, D#

Eb, F, Ab, Bb, C, Db, Eb

E, F#, A, B, C#, D, E

F, G, Bb, C, D, Eb, F

F#, G#, B, C#, D#, E, F#

Gb, Ab, Cb, Db, Eb, Fb, Gb

G, A, C, D, E, F, G

G#, A#, C#, D#, E#, F#, G#

Ab, Bb, Db, Eb, F, Gb, Ab

A, B, D, E, F#, G, A

A#, B#, D#, E#, F##, G#, A#

Bb, C, Eb, F, G, Ab, Bb

B, C#, E, F#, G#, A, B

## Mruganandana

Step pattern: 2, 2, 2, 3, 2, 1

Ascending intervals: P1, M2, M3, Aug4, M6, M7, P8

C, D, E, F#, A, B, C

C#, D#, E#, F##, A#, B#, C#

Db, Eb, F, G, Bb, C, Db

D, E, F#, G#, B, C#, D

D#, E#, F##, G##, B#, C##, D#

Eb, F, G, A, C, D, Eb

E, F#, G#, A#, C#, D#, E

F, G, A, B, D, E, F

F#, G#, A#, B#, D#, E#, F#

Gb, Ab, Bb, C, Eb, F, Gb

G, A, B, C#, E, F#, G

G#, A#, B#, C##, E#, F##, G#

Ab, Bb, C, D, F, G, Ab

A, B, C#, D#, F#, G#, A

A#, B#, C##, D##, F##, G##, A#

Bb, C, D, E, G, A, Bb

B, C#, D#, E#, G#, A#, B



## Nattaikurinji

Step pattern: 2, 2, 1, 4, 1, 2

Ascending intervals: P1, M2, M3, P4, M6, m7, P8

C, D, E, F, A, Bb, C

C#, D#, E#, F#, A#, B, C#

Db, Eb, F, Gb, Bb, Cb, Db

D, E, F#, G, B, C, D

D#, E#, F##, G#, B#, C#, D#

Eb, F, G, Ab, C, Db, Eb

E, F#, G#, A, C#, D, E

F, G, A, Bb, D, Eb, F

F#, G#, A#, B, D#, E, F#

Gb, Ab, Bb, Cb, Eb, Fb, Gb

G, A, B, C, E, F, G

G#, A#, B#, C#, E#, F#, G#

Ab, Bb, C, Db, F, Gb, Ab

A, B, C#, D, F#, G, A

A#, B#, C##, D#, F##, G#, A#

Bb, C, D, Eb, G, Ab, Bb

B, C#, D#, E, G#, A, B

## Navamanohari

Step pattern: 2, 3, 2, 1, 2, 2

Ascending intervals: P1, M2, P4, P5, m6, m7, P8

C, D, F, G, Ab, Bb, C

C#, D#, F#, G#, A, B, C#

Db, Eb, Gb, Ab, Bbb, Cb, Db

D, E, G, A, Bb, C, D

D#, E#, G#, A#, B, C#, D#

Eb, F, Ab, Bb, Cb, Db, Eb

E, F#, A, B, C, D, E

F, G, Bb, C, Db, Eb, F

F#, G#, B, C#, D, E, F#

Gb, Ab, Cb, Db, Ebb, Fb, Gb

G, A, C, D, Eb, F, G

G#, A#, C#, D#, E, F#, G#

Ab, Bb, Db, Eb, Fb, Gb, Ab

A, B, D, E, F, G, A

A#, B#, D#, E#, F#, G#, A#

Bb, C, Eb, F, Gb, Ab, Bb

B, C#, E, F#, G, A, B

**Neelangi**

Step pattern: 2, 1, 3, 2, 1, 3

Ascending intervals: P1, M2, m3, dim5, m6, M6, P8

C, D, Eb, Gb, Ab, A, C

C#, D#, E, G, A, A#, C#

Db, Eb, Fb, Abb, Bbb, Bb, Db

D, E, F, Ab, Bb, B, D

D#, E#, F#, A, B, B#, D#

Eb, F, Gb, Bbb, Cb, C, Eb

E, F#, G, Bb, C, C#, E

F, G, Ab, Cb, Db, D, F

F#, G#, A, C, D, D#, F#

Gb, Ab, Bbb, Dbb, Ebb, Eb, Gb

G, A, Bb, Db, Eb, E, G

G#, A#, B, D, E, E#, G#

Ab, Bb, Cb, Ebb, Fb, F, Ab

A, B, C, Eb, F, F#, A

A#, B#, C#, E, F#, F##, A#

Bb, C, Db, Fb, Gb, G, Bb

B, C#, D, F, G, G#, B

## Nishadi

Step pattern: 2, 4, 1, 2, 2, 1

Ascending intervals: P1, M2, Aug4, P5, M6, M7, P8

C, D, F#, G, A, B, C

C#, D#, F##, G#, A#, B#, C#

Db, Eb, G, Ab, Bb, C, Db

D, E, G#, A, B, C#, D

D#, E#, G##, A#, B#, C##, D#

Eb, F, A, Bb, C, D, Eb

E, F#, A#, B, C#, D#, E

F, G, B, C, D, E, F

F#, G#, B#, C#, D#, E#, F#

Gb, Ab, C, Db, Eb, F, Gb

G, A, C#, D, E, F#, G

G#, A#, C##, D#, E#, F##, G#

Ab, Bb, D, Eb, F, G, Ab

A, B, D#, E, F#, G#, A

A#, B#, D##, E#, F##, G##, A#

Bb, C, E, F, G, A, Bb

B, C#, E#, F#, G#, A#, B

**Padi**

Step pattern: 1, 4, 2, 1, 3, 1

Ascending intervals: P1, m2, P4, P5, m6, M7, P8

C, Db, F, G, Ab, B, C

C#, D, F#, G#, A, B#, C#

Db, Ebb, Gb, Ab, Bbb, C, Db

D, Eb, G, A, Bb, C#, D

D#, E, G#, A#, B, C##, D#

Eb, Fb, Ab, Bb, Cb, D, Eb

E, F, A, B, C, D#, E

F, Gb, Bb, C, Db, E, F

F#, G, B, C#, D, E#, F#

Gb, Abb, Cb, Db, Ebb, F, Gb

G, Ab, C, D, Eb, F#, G

G#, A, C#, D#, E, F##, G#

Ab, Bbb, Db, Eb, Fb, G, Ab

A, Bb, D, E, F, G#, A

A#, B, D#, E#, F#, G##, A#

Bb, Cb, Eb, F, Gb, A, Bb

B, C, E, F#, G, A#, B

## Paraju

Step pattern: 4, 1, 2, 1, 3, 1

Ascending intervals: P1, M3, P4, P5, m6, M7, P8

C, E, F, G, Ab, B, C

C#, E#, F#, G#, A, B#, C#

Db, F, Gb, Ab, Bbb, C, Db

D, F#, G, A, Bb, C#, D

D#, F##, G#, A#, B, C##, D#

Eb, G, Ab, Bb, Cb, D, Eb

E, G#, A, B, C, D#, E

F, A, Bb, C, Db, E, F

F#, A#, B, C#, D, E#, F#

Gb, Bb, Cb, Db, Ebb, F, Gb

G, B, C, D, Eb, F#, G

G#, B#, C#, D#, E, F##, G#

Ab, C, Db, Eb, Fb, G, Ab

A, C#, D, E, F, G#, A

A#, C##, D#, E#, F#, G##, A#

Bb, D, Eb, F, Gb, A, Bb

B, D#, E, F#, G, A#, B

## Pentatonic Minor added 6

Step pattern: 3, 2, 2, 2, 1, 2

Ascending intervals: P1, m3, P4, P5, M6, m7, P8

C, Eb, F, G, A, Bb, C

C#, E, F#, G#, A#, B, C#

Db, Fb, Gb, Ab, Bb, Cb, Db

D, F, G, A, B, C, D

D#, F#, G#, A#, B#, C#, D#

Eb, Gb, Ab, Bb, C, Db, Eb

E, G, A, B, C#, D, E

F, Ab, Bb, C, D, Eb, F

F#, A, B, C#, D#, E, F#

Gb, Bbb, Cb, Db, Eb, Fb, Gb

G, Bb, C, D, E, F, G

G#, B, C#, D#, E#, F#, G#

Ab, Cb, Db, Eb, F, Gb, Ab

A, C, D, E, F#, G, A

A#, C#, D#, E#, F##, G#, A#

Bb, Db, Eb, F, G, Ab, Bb

B, D, E, F#, G#, A, B

## Phrygian Hexatonic

Step pattern: 3, 2, 2, 1, 2, 2

Ascending intervals: P1, m3, P4, P5, m6, m7, P8

C, Eb, F, G, Ab, Bb, C

C#, E, F#, G#, A, B, C#

Db, Fb, Gb, Ab, Bbb, Cb, Db

D, F, G, A, Bb, C, D

D#, F#, G#, A#, B, C#, D#

Eb, Gb, Ab, Bb, Cb, Db, Eb

E, G, A, B, C, D, E

F, Ab, Bb, C, Db, Eb, F

F#, A, B, C#, D, E, F#

Gb, Bbb, Cb, Db, Ebb, Fb, Gb

G, Bb, C, D, Eb, F, G

G#, B, C#, D#, E, F#, G#

Ab, Cb, Db, Eb, Fb, Gb, Ab

A, C, D, E, F, G, A

A#, C#, D#, E#, F#, G#, A#

Bb, Db, Eb, F, Gb, Ab, Bb

B, D, E, F#, G, A, B



## Prometheus

Step pattern: 2, 2, 2, 3, 1, 2

Ascending intervals: P1, M2, M3, dim5, M6, m7, P8

C, D, E, Gb, A, Bb, C

C#, D#, E#, G, A#, B, C#

Db, Eb, F, Abb, Bb, Cb, Db

D, E, F#, Ab, B, C, D

D#, E#, F##, A, B#, C#, D#

Eb, F, G, Bbb, C, Db, Eb

E, F#, G#, Bb, C#, D, E

F, G, A, Cb, D, Eb, F

F#, G#, A#, C, D#, E, F#

Gb, Ab, Bb, Dbb, Eb, Fb, Gb

G, A, B, Db, E, F, G

G#, A#, B#, D, E#, F#, G#

Ab, Bb, C, Ebb, F, Gb, Ab

A, B, C#, Eb, F#, G, A

A#, B#, C##, E, F##, G#, A#

Bb, C, D, Fb, G, Ab, Bb

B, C#, D#, F, G#, A, B

## Prometheus Neapolitan

Step pattern: 1, 3, 2, 3, 1, 2

Ascending intervals: P1, m2, M3, dim5, M6, m7, P8

C, Db, E, Gb, A, Bb, C

C#, D, E#, G, A#, B, C#

Db, Ebb, F, Abb, Bb, Cb, Db

D, Eb, F#, Ab, B, C, D

D#, E, F##, A, B#, C#, D#

Eb, Fb, G, Bbb, C, Db, Eb

E, F, G#, Bb, C#, D, E

F, Gb, A, Cb, D, Eb, F

F#, G, A#, C, D#, E, F#

Gb, Abb, Bb, Dbb, Eb, Fb, Gb

G, Ab, B, Db, E, F, G

G#, A, B#, D, E#, F#, G#

Ab, Bbb, C, Ebb, F, Gb, Ab

A, Bb, C#, Eb, F#, G, A

A#, B, C##, E, F##, G#, A#

Bb, Cb, D, Fb, G, Ab, Bb

B, C, D#, F, G#, A, B

## Pyramid Hexatonic

Step pattern: 2, 1, 2, 1, 3, 3

Ascending intervals: P1, M2, m3, P4, dim5, M6, P8

C, D, Eb, F, Gb, A, C

C#, D#, E, F#, G, A#, C#

Db, Eb, Fb, Gb, Abb, Bb, Db

D, E, F, G, Ab, B, D

D#, E#, F#, G#, A, B#, D#

Eb, F, Gb, Ab, Bbb, C, Eb

E, F#, G, A, Bb, C#, E

F, G, Ab, Bb, Cb, D, F

F#, G#, A, B, C, D#, F#

Gb, Ab, Bbb, Cb, Dbb, Eb, Gb

G, A, Bb, C, Db, E, G

G#, A#, B, C#, D, E#, G#

Ab, Bb, Cb, Db, Ebb, F, Ab

A, B, C, D, Eb, F#, A

A#, B#, C#, D#, E, F##, A#

Bb, C, Db, Eb, Fb, G, Bb

B, C#, D, E, F, G#, B

## Ranjani

Step pattern: 2, 1, 3, 3, 2, 1

Ascending intervals: P1, M2, m3, dim5, M6, M7, P8

C, D, Eb, Gb, A, B, C

C#, D#, E, G, A#, B#, C#

Db, Eb, Fb, Abb, Bb, C, Db

D, E, F, Ab, B, C#, D

D#, E#, F#, A, B#, C##, D#

Eb, F, Gb, Bbb, C, D, Eb

E, F#, G, Bb, C#, D#, E

F, G, Ab, Cb, D, E, F

F#, G#, A, C, D#, E#, F#

Gb, Ab, Bbb, Dbb, Eb, F, Gb

G, A, Bb, Db, E, F#, G

G#, A#, B, D, E#, F##, G#

Ab, Bb, Cb, Ebb, F, G, Ab

A, B, C, Eb, F#, G#, A

A#, B#, C#, E, F##, G##, A#

Bb, C, Db, Fb, G, A, Bb

B, C#, D, F, G#, A#, B

## Rasamanjari

Step pattern: 3, 1, 2, 1, 4, 1

Ascending intervals: P1, m3, M3, dim5, P5, M7, P8

C, Eb, E, Gb, G, B, C

C#, E, E#, G, G#, B#, C#

Db, Fb, F, Abb, Ab, C, Db

D, F, F#, Ab, A, C#, D

D#, F#, F##, A, A#, C##, D#

Eb, Gb, G, Bbb, Bb, D, Eb

E, G, G#, Bb, B, D#, E

F, Ab, A, Cb, C, E, F

F#, A, A#, C, C#, E#, F#

Gb, Bbb, Bb, Dbb, Db, F, Gb

G, Bb, B, Db, D, F#, G

G#, B, B#, D, D#, F##, G#

Ab, Cb, C, Ebb, Eb, G, Ab

A, C, C#, Eb, E, G#, A

A#, C#, C##, E, E#, G##, A#

Bb, Db, D, Fb, F, A, Bb

B, D, D#, F, F#, A#, B

## Rasavali

Step pattern: 1, 4, 2, 2, 1, 2

Ascending intervals: P1, m2, P4, P5, M6, m7, P8

C, Db, F, G, A, Bb, C

C#, D, F#, G#, A#, B, C#

Db, Ebb, Gb, Ab, Bb, Cb, Db

D, Eb, G, A, B, C, D

D#, E, G#, A#, B#, C#, D#

Eb, Fb, Ab, Bb, C, Db, Eb

E, F, A, B, C#, D, E

F, Gb, Bb, C, D, Eb, F

F#, G, B, C#, D#, E, F#

Gb, Abb, Cb, Db, Eb, Fb, Gb

G, Ab, C, D, E, F, G

G#, A, C#, D#, E#, F#, G#

Ab, Bbb, Db, Eb, F, Gb, Ab

A, Bb, D, E, F#, G, A

A#, B, D#, E#, F##, G#, A#

Bb, Cb, Eb, F, G, Ab, Bb

B, C, E, F#, G#, A, B

**Ratnakanthi**

Step pattern: 2, 2, 2, 1, 4, 1

Ascending intervals: P1, M2, M3, Aug4, P5, M7, P8

C, D, E, F#, G, B, C

C#, D#, E#, F##, G#, B#, C#

Db, Eb, F, G, Ab, C, Db

D, E, F#, G#, A, C#, D

D#, E#, F##, G##, A#, C##, D#

Eb, F, G, A, Bb, D, Eb

E, F#, G#, A#, B, D#, E

F, G, A, B, C, E, F

F#, G#, A#, B#, C#, E#, F#

Gb, Ab, Bb, C, Db, F, Gb

G, A, B, C#, D, F#, G

G#, A#, B#, C##, D#, F##, G#

Ab, Bb, C, D, Eb, G, Ab

A, B, C#, D#, E, G#, A

A#, B#, C##, D##, E#, G##, A#

Bb, C, D, E, F, A, Bb

B, C#, D#, E#, F#, A#, B

**Ritsu**

Step pattern: 1, 2, 2, 3, 2, 2

Ascending intervals: P1, m2, m3, P4, m6, m7, P8

C, Db, Eb, F, Ab, Bb, C

C#, D, E, F#, A, B, C#

Db, Ebb, Fb, Gb, Bbb, Cb, Db

D, Eb, F, G, Bb, C, D

D#, E, F#, G#, B, C#, D#

Eb, Fb, Gb, Ab, Cb, Db, Eb

E, F, G, A, C, D, E

F, Gb, Ab, Bb, Db, Eb, F

F#, G, A, B, D, E, F#

Gb, Abb, Bbb, Cb, Ebb, Fb, Gb

G, Ab, Bb, C, Eb, F, G

G#, A, B, C#, E, F#, G#

Ab, Bbb, Cb, Db, Fb, Gb, Ab

A, Bb, C, D, F, G, A

A#, B, C#, D#, F#, G#, A#

Bb, Cb, Db, Eb, Gb, Ab, Bb

B, C, D, E, G, A, B



## Rundra Pancama

Step pattern: 1, 3, 1, 4, 1, 2

Ascending intervals: P1, m2, M3, P4, M6, m7, P8

C, Db, E, F, A, Bb, C

C#, D, E#, F#, A#, B, C#

Db, Ebb, F, Gb, Bb, Cb, Db

D, Eb, F#, G, B, C, D

D#, E, F##, G#, B#, C#, D#

Eb, Fb, G, Ab, C, Db, Eb

E, F, G#, A, C#, D, E

F, Gb, A, Bb, D, Eb, F

F#, G, A#, B, D#, E, F#

Gb, Abb, Bb, Cb, Eb, Fb, Gb

G, Ab, B, C, E, F, G

G#, A, B#, C#, E#, F#, G#

Ab, Bbb, C, Db, F, Gb, Ab

A, Bb, C#, D, F#, G, A

A#, B, C##, D#, F##, G#, A#

Bb, Cb, D, Eb, G, Ab, Bb

B, C, D#, E, G#, A, B

## Salagavarali

Step pattern: 1, 2, 4, 2, 1, 2

Ascending intervals: P1, m2, m3, P5, M6, m7, P8

C, Db, Eb, G, A, Bb, C

C#, D, E, G#, A#, B, C#

Db, Ebb, Fb, Ab, Bb, Cb, Db

D, Eb, F, A, B, C, D

D#, E, F#, A#, B#, C#, D#

Eb, Fb, Gb, Bb, C, Db, Eb

E, F, G, B, C#, D, E

F, Gb, Ab, C, D, Eb, F

F#, G, A, C#, D#, E, F#

Gb, Abb, Bbb, Db, Eb, Fb, Gb

G, Ab, Bb, D, E, F, G

G#, A, B, D#, E#, F#, G#

Ab, Bbb, Cb, Eb, F, Gb, Ab

A, Bb, C, E, F#, G, A

A#, B, C#, E#, F##, G#, A#

Bb, Cb, Db, F, G, Ab, Bb

B, C, D, F#, G#, A, B

## Sarasanana

Step pattern: 2, 2, 1, 3, 3, 1

Ascending intervals: P1, M2, M3, P4, m6, M7, P8

C, D, E, F, Ab, B, C

C#, D#, E#, F#, A, B#, C#

Db, Eb, F, Gb, Bbb, C, Db

D, E, F#, G, Bb, C#, D

D#, E#, F##, G#, B, C##, D#

Eb, F, G, Ab, Cb, D, Eb

E, F#, G#, A, C, D#, E

F, G, A, Bb, Db, E, F

F#, G#, A#, B, D, E#, F#

Gb, Ab, Bb, Cb, Ebb, F, Gb

G, A, B, C, Eb, F#, G

G#, A#, B#, C#, E, F##, G#

Ab, Bb, C, Db, Fb, G, Ab

A, B, C#, D, F, G#, A

A#, B#, C##, D#, F#, G##, A#

Bb, C, D, Eb, Gb, A, Bb

B, C#, D#, E, G, A#, B

## Sarasvati

Step pattern: 2, 4, 1, 2, 1, 2

Ascending intervals: P1, M2, Aug4, P5, M6, m7, P8

C, D, F#, G, A, Bb, C

C#, D#, F##, G#, A#, B, C#

Db, Eb, G, Ab, Bb, Cb, Db

D, E, G#, A, B, C, D

D#, E#, G##, A#, B#, C#, D#

Eb, F, A, Bb, C, Db, Eb

E, F#, A#, B, C#, D, E

F, G, B, C, D, Eb, F

F#, G#, B#, C#, D#, E, F#

Gb, Ab, C, Db, Eb, Fb, Gb

G, A, C#, D, E, F, G

G#, A#, C##, D#, E#, F#, G#

Ab, Bb, D, Eb, F, Gb, Ab

A, B, D#, E, F#, G, A

A#, B#, D##, E#, F##, G#, A#

Bb, C, E, F, G, Ab, Bb

B, C#, E#, F#, G#, A, B

## Saravati

Step pattern: 4, 1, 2, 1, 1, 3

Ascending intervals: P1, M3, P4, P5, m6, M6, P8

C, E, F, G, Ab, A, C

C#, E#, F#, G#, A, A#, C#

Db, F, Gb, Ab, Bbb, Bb, Db

D, F#, G, A, Bb, B, D

D#, F##, G#, A#, B, B#, D#

Eb, G, Ab, Bb, Cb, C, Eb

E, G#, A, B, C, C#, E

F, A, Bb, C, Db, D, F

F#, A#, B, C#, D, D#, F#

Gb, Bb, Cb, Db, Ebb, Eb, Gb

G, B, C, D, Eb, E, G

G#, B#, C#, D#, E, E#, G#

Ab, C, Db, Eb, Fb, F, Ab

A, C#, D, E, F, F#, A

A#, C##, D#, E#, F#, F##, A#

Bb, D, Eb, F, Gb, G, Bb

B, D#, E, F#, G, G#, B

## Sindhura Kafi

Step pattern: 2, 1, 2, 2, 4, 1

Ascending intervals: P1, M2, m3, P4, P5, M7, P8

C, D, Eb, F, G, B, C

C#, D#, E, F#, G#, B#, C#

Db, Eb, Fb, Gb, Ab, C, Db

D, E, F, G, A, C#, D

D#, E#, F#, G#, A#, C##, D#

Eb, F, Gb, Ab, Bb, D, Eb

E, F#, G, A, B, D#, E

F, G, Ab, Bb, C, E, F

F#, G#, A, B, C#, E#, F#

Gb, Ab, Bbb, Cb, Db, F, Gb

G, A, Bb, C, D, F#, G

G#, A#, B, C#, D#, F##, G#

Ab, Bb, Cb, Db, Eb, G, Ab

A, B, C, D, E, G#, A

A#, B#, C#, D#, E#, G##, A#

Bb, C, Db, Eb, F, A, Bb

B, C#, D, E, F#, A#, B

**Sohini**

Step pattern: 1, 3, 1, 3, 3, 1

Ascending intervals: P1, m2, M3, P4, m6, M7, P8

C, Db, E, F, Ab, B, C

C#, D, E#, F#, A, B#, C#

Db, Ebb, F, Gb, Bbb, C, Db

D, Eb, F#, G, Bb, C#, D

D#, E, F##, G#, B, C##, D#

Eb, Fb, G, Ab, Cb, D, Eb

E, F, G#, A, C, D#, E

F, Gb, A, Bb, Db, E, F

F#, G, A#, B, D, E#, F#

Gb, Abb, Bb, Cb, Ebb, F, Gb

G, Ab, B, C, Eb, F#, G

G#, A, B#, C#, E, F##, G#

Ab, Bbb, C, Db, Fb, G, Ab

A, Bb, C#, D, F, G#, A

A#, B, C##, D#, F#, G##, A#

Bb, Cb, D, Eb, Gb, A, Bb

B, C, D#, E, G, A#, B

**Sriranjani**

Step pattern: 2, 1, 2, 4, 1, 2

Ascending intervals: P1, M2, m3, P4, M6, m7, P8

C, D, Eb, F, A, Bb, C

C#, D#, E, F#, A#, B, C#

Db, Eb, Fb, Gb, Bb, Cb, Db

D, E, F, G, B, C, D

D#, E#, F#, G#, B#, C#, D#

Eb, F, Gb, Ab, C, Db, Eb

E, F#, G, A, C#, D, E

F, G, Ab, Bb, D, Eb, F

F#, G#, A, B, D#, E, F#

Gb, Ab, Bbb, Cb, Eb, Fb, Gb

G, A, Bb, C, E, F, G

G#, A#, B, C#, E#, F#, G#

Ab, Bb, Cb, Db, F, Gb, Ab

A, B, C, D, F#, G, A

A#, B#, C#, D#, F##, G#, A#

Bb, C, Db, Eb, G, Ab, Bb

B, C#, D, E, G#, A, B



## Suddha Bangala

Step pattern: 2, 1, 2, 2, 2, 3

Ascending intervals: P1, M2, m3, P4, P5, M6, P8

C, D, Eb, F, G, A, C

C#, D#, E, F#, G#, A#, C#

Db, Eb, Fb, Gb, Ab, Bb, Db

D, E, F, G, A, B, D

D#, E#, F#, G#, A#, B#, D#

Eb, F, Gb, Ab, Bb, C, Eb

E, F#, G, A, B, C#, E

F, G, Ab, Bb, C, D, F

F#, G#, A, B, C#, D#, F#

Gb, Ab, Bbb, Cb, Db, Eb, Gb

G, A, Bb, C, D, E, G

G#, A#, B, C#, D#, E#, G#

Ab, Bb, Cb, Db, Eb, F, Ab

A, B, C, D, E, F#, A

A#, B#, C#, D#, E#, F##, A#

Bb, C, Db, Eb, F, G, Bb

B, C#, D, E, F#, G#, B

## Suddha Mukhari

Step pattern: 1, 1, 3, 3, 1, 3

Ascending intervals: P1, m2, M2, P4, m6, M6, P8

C, Db, D, F, Ab, A, C

C#, D, D#, F#, A, A#, C#

Db, Ebb, Eb, Gb, Bbb, Bb, Db

D, Eb, E, G, Bb, B, D

D#, E, E#, G#, B, B#, D#

Eb, Fb, F, Ab, Cb, C, Eb

E, F, F#, A, C, C#, E

F, Gb, G, Bb, Db, D, F

F#, G, G#, B, D, D#, F#

Gb, Abb, Ab, Cb, Ebb, Eb, Gb

G, Ab, A, C, Eb, E, G

G#, A, A#, C#, E, E#, G#

Ab, Bbb, Bb, Db, Fb, F, Ab

A, Bb, B, D, F, F#, A

A#, B, B#, D#, F#, F##, A#

Bb, Cb, C, Eb, Gb, G, Bb

B, C, C#, E, G, G#, B

## Suddha Simantini

Step pattern: 1, 2, 2, 2, 1, 4

Ascending intervals: P1, m2, m3, P4, P5, m6, P8

C, Db, Eb, F, G, Ab, C

C#, D, E, F#, G#, A, C#

Db, Ebb, Fb, Gb, Ab, Bbb, Db

D, Eb, F, G, A, Bb, D

D#, E, F#, G#, A#, B, D#

Eb, Fb, Gb, Ab, Bb, Cb, Eb

E, F, G, A, B, C, E

F, Gb, Ab, Bb, C, Db, F

F#, G, A, B, C#, D, F#

Gb, Abb, Bbb, Cb, Db, Ebb, Gb

G, Ab, Bb, C, D, Eb, G

G#, A, B, C#, D#, E, G#

Ab, Bbb, Cb, Db, Eb, Fb, Ab

A, Bb, C, D, E, F, A

A#, B, C#, D#, E#, F#, A#

Bb, Cb, Db, Eb, F, Gb, Bb

B, C, D, E, F#, G, B

## Syamalam

Step pattern: 2, 1, 3, 1, 1, 4

Ascending intervals: P1, M2, m3, dim5, P5, m6, P8

C, D, Eb, Gb, G, Ab, C

C#, D#, E, G, G#, A, C#

Db, Eb, Fb, Abb, Ab, Bbb, Db

D, E, F, Ab, A, Bb, D

D#, E#, F#, A, A#, B, D#

Eb, F, Gb, Bbb, Bb, Cb, Eb

E, F#, G, Bb, B, C, E

F, G, Ab, Cb, C, Db, F

F#, G#, A, C, C#, D, F#

Gb, Ab, Bbb, Dbb, Db, Ebb, Gb

G, A, Bb, Db, D, Eb, G

G#, A#, B, D, D#, E, G#

Ab, Bb, Cb, Ebb, Eb, Fb, Ab

A, B, C, Eb, E, F, A

A#, B#, C#, E, E#, F#, A#

Bb, C, Db, Fb, F, Gb, Bb

B, C#, D, F, F#, G, B

### Takemitsu Tree Line Mode (m7)

Step pattern: 2, 1, 3, 2, 2, 2

Ascending intervals: P1, M2, m3, dim5, m6, m7, P8

C, D, Eb, Gb, Ab, Bb, C

C#, D#, E, G, A, B, C#

Db, Eb, Fb, Abb, Bbb, Cb, Db

D, E, F, Ab, Bb, C, D

D#, E#, F#, A, B, C#, D#

Eb, F, Gb, Bbb, Cb, Db, Eb

E, F#, G, Bb, C, D, E

F, G, Ab, Cb, Db, Eb, F

F#, G#, A, C, D, E, F#

Gb, Ab, Bbb, Dbb, Ebb, Fb, Gb

G, A, Bb, Db, Eb, F, G

G#, A#, B, D, E, F#, G#

Ab, Bb, Cb, Ebb, Fb, Gb, Ab

A, B, C, Eb, F, G, A

A#, B#, C#, E, F#, G#, A#

Bb, C, Db, Fb, Gb, Ab, Bb

B, C#, D, F, G, A, B

### Takemitsu Tree Line Mode (M7)

Step pattern: 2, 1, 3, 2, 3, 1

Ascending intervals: P1, M2, m3, dim5, m6, M7, P8

C, D, Eb, Gb, Ab, B, C

C#, D#, E, G, A, B#, C#

Db, Eb, Fb, Abb, Bbb, C, Db

D, E, F, Ab, Bb, C#, D

D#, E#, F#, A, B, C##, D#

Eb, F, Gb, Bbb, Cb, D, Eb

E, F#, G, Bb, C, D#, E

F, G, Ab, Cb, Db, E, F

F#, G#, A, C, D, E#, F#

Gb, Ab, Bbb, Dbb, Ebb, F, Gb

G, A, Bb, Db, Eb, F#, G

G#, A#, B, D, E, F##, G#

Ab, Bb, Cb, Ebb, Fb, G, Ab

A, B, C, Eb, F, G#, A

A#, B#, C#, E, F#, G##, A#

Bb, C, Db, Fb, Gb, A, Bb

B, C#, D, F, G, A#, B

**Takka**

Step pattern: 3, 2, 2, 1, 3, 1

Ascending intervals: P1, m3, P4, P5, m6, M7, P8

C, Eb, F, G, Ab, B, C

C#, E, F#, G#, A, B#, C#

Db, Fb, Gb, Ab, Bbb, C, Db

D, F, G, A, Bb, C#, D

D#, F#, G#, A#, B, C##, D#

Eb, Gb, Ab, Bb, Cb, D, Eb

E, G, A, B, C, D#, E

F, Ab, Bb, C, Db, E, F

F#, A, B, C#, D, E#, F#

Gb, Bbb, Cb, Db, Ebb, F, Gb

G, Bb, C, D, Eb, F#, G

G#, B, C#, D#, E, F##, G#

Ab, Cb, Db, Eb, Fb, G, Ab

A, C, D, E, F, G#, A

A#, C#, D#, E#, F#, G##, A#

Bb, Db, Eb, F, Gb, A, Bb

B, D, E, F#, G, A#, B

## Tilang

Step pattern: 4, 1, 2, 3, 1, 1

Ascending intervals: P1, M3, P4, P5, m7, M7, P8

C, E, F, G, Bb, B, C

C#, E#, F#, G#, B, B#, C#

Db, F, Gb, Ab, Cb, C, Db

D, F#, G, A, C, C#, D

D#, F##, G#, A#, C#, C##, D#

Eb, G, Ab, Bb, Db, D, Eb

E, G#, A, B, D, D#, E

F, A, Bb, C, Eb, E, F

F#, A#, B, C#, E, E#, F#

Gb, Bb, Cb, Db, Fb, F, Gb

G, B, C, D, F, F#, G

G#, B#, C#, D#, F#, F##, G#

Ab, C, Db, Eb, Gb, G, Ab

A, C#, D, E, G, G#, A

A#, C##, D#, E#, G#, G##, A#

Bb, D, Eb, F, Ab, A, Bb

B, D#, E, F#, A, A#, B



**Trimurti**

Step pattern: 2, 1, 4, 1, 2, 2

Ascending intervals: P1, M2, m3, P5, m6, m7, P8

C, D, Eb, G, Ab, Bb, C

C#, D#, E, G#, A, B, C#

Db, Eb, Fb, Ab, Bbb, Cb, Db

D, E, F, A, Bb, C, D

D#, E#, F#, A#, B, C#, D#

Eb, F, Gb, Bb, Cb, Db, Eb

E, F#, G, B, C, D, E

F, G, Ab, C, Db, Eb, F

F#, G#, A, C#, D, E, F#

Gb, Ab, Bbb, Db, Ebb, Fb, Gb

G, A, Bb, D, Eb, F, G

G#, A#, B, D#, E, F#, G#

Ab, Bb, Cb, Eb, Fb, Gb, Ab

A, B, C, E, F, G, A

A#, B#, C#, E#, F#, G#, A#

Bb, C, Db, F, Gb, Ab, Bb

B, C#, D, F#, G, A, B

## Vijayanagari

Step pattern: 2, 1, 3, 1, 2, 3

Ascending intervals: P1, M2, m3, dim5, P5, M6, P8

C, D, Eb, Gb, G, A, C

C#, D#, E, G, G#, A#, C#

Db, Eb, Fb, Abb, Ab, Bb, Db

D, E, F, Ab, A, B, D

D#, E#, F#, A, A#, B#, D#

Eb, F, Gb, Bbb, Bb, C, Eb

E, F#, G, Bb, B, C#, E

F, G, Ab, Cb, C, D, F

F#, G#, A, C, C#, D#, F#

Gb, Ab, Bbb, Dbb, Db, Eb, Gb

G, A, Bb, Db, D, E, G

G#, A#, B, D, D#, E#, G#

Ab, Bb, Cb, Ebb, Eb, F, Ab

A, B, C, Eb, E, F#, A

A#, B#, C#, E, E#, F##, A#

Bb, C, Db, Fb, F, G, Bb

B, C#, D, F, F#, G#, B

## Vijayasri

Step pattern: 1, 1, 4, 1, 4, 1

Ascending intervals: P1, m2, M2, dim5, P5, M7, P8

C, Db, D, Gb, G, B, C

C#, D, D#, G, G#, B#, C#

Db, Ebb, Eb, Abb, Ab, C, Db

D, Eb, E, Ab, A, C#, D

D#, E, E#, A, A#, C##, D#

Eb, Fb, F, Bbb, Bb, D, Eb

E, F, F#, Bb, B, D#, E

F, Gb, G, Cb, C, E, F

F#, G, G#, C, C#, E#, F#

Gb, Abb, Ab, Dbb, Db, F, Gb

G, Ab, A, Db, D, F#, G

G#, A, A#, D, D#, F##, G#

Ab, Bbb, Bb, Ebb, Eb, G, Ab

A, Bb, B, Eb, E, G#, A

A#, B, B#, E, E#, G##, A#

Bb, Cb, C, Fb, F, A, Bb

B, C, C#, F, F#, A#, B

## Vijayavasanta

Step pattern: 4, 2, 1, 3, 1, 1

Ascending intervals: P1, M3, Aug4, P5, m7, M7, P8

C, E, F#, G, Bb, B, C

C#, E#, F##, G#, B, B#, C#

Db, F, G, Ab, Cb, C, Db

D, F#, G#, A, C, C#, D

D#, F##, G##, A#, C#, C##, D#

Eb, G, A, Bb, Db, D, Eb

E, G#, A#, B, D, D#, E

F, A, B, C, Eb, E, F

F#, A#, B#, C#, E, E#, F#

Gb, Bb, C, Db, Fb, F, Gb

G, B, C#, D, F, F#, G

G#, B#, C##, D#, F#, F##, G#

Ab, C, D, Eb, Gb, G, Ab

A, C#, D#, E, G, G#, A

A#, C##, D##, E#, G#, G##, A#

Bb, D, E, F, Ab, A, Bb

B, D#, E#, F#, A, A#, B

## Vilasini

Step pattern: 2, 2, 1, 2, 4, 1

Ascending intervals: P1, M2, M3, P4, P5, M7, P8

C, D, E, F, G, B, C

C#, D#, E#, F#, G#, B#, C#

Db, Eb, F, Gb, Ab, C, Db

D, E, F#, G, A, C#, D

D#, E#, F##, G#, A#, C##, D#

Eb, F, G, Ab, Bb, D, Eb

E, F#, G#, A, B, D#, E

F, G, A, Bb, C, E, F

F#, G#, A#, B, C#, E#, F#

Gb, Ab, Bb, Cb, Db, F, Gb

G, A, B, C, D, F#, G

G#, A#, B#, C#, D#, F##, G#

Ab, Bb, C, Db, Eb, G, Ab

A, B, C#, D, E, G#, A

A#, B#, C##, D#, E#, G##, A#

Bb, C, D, Eb, F, A, Bb

B, C#, D#, E, F#, A#, B

## Vivardhini

Step pattern: 2, 2, 1, 2, 3, 2

Ascending intervals: P1, M2, M3, P4, P5, m7, P8

C, D, E, F, G, Bb, C

C#, D#, E#, F#, G#, B, C#

Db, Eb, F, Gb, Ab, Cb, Db

D, E, F#, G, A, C, D

D#, E#, F##, G#, A#, C#, D#

Eb, F, G, Ab, Bb, Db, Eb

E, F#, G#, A, B, D, E

F, G, A, Bb, C, Eb, F

F#, G#, A#, B, C#, E, F#

Gb, Ab, Bb, Cb, Db, Fb, Gb

G, A, B, C, D, F, G

G#, A#, B#, C#, D#, F#, G#

Ab, Bb, C, Db, Eb, Gb, Ab

A, B, C#, D, E, G, A

A#, B#, C##, D#, E#, G#, A#

Bb, C, D, Eb, F, Ab, Bb

B, C#, D#, E, F#, A, B

## Viyogavarali

Step pattern: 1, 2, 2, 3, 3, 1

Ascending intervals: P1, m2, m3, P4, m6, M7, P8

C, Db, Eb, F, Ab, B, C

C#, D, E, F#, A, B#, C#

Db, Ebb, Fb, Gb, Bbb, C, Db

D, Eb, F, G, Bb, C#, D

D#, E, F#, G#, B, C##, D#

Eb, Fb, Gb, Ab, Cb, D, Eb

E, F, G, A, C, D#, E

F, Gb, Ab, Bb, Db, E, F

F#, G, A, B, D, E#, F#

Gb, Abb, Bbb, Cb, Ebb, F, Gb

G, Ab, Bb, C, Eb, F#, G

G#, A, B, C#, E, F##, G#

Ab, Bbb, Cb, Db, Fb, G, Ab

A, Bb, C, D, F, G#, A

A#, B, C#, D#, F#, G##, A#

Bb, Cb, Db, Eb, Gb, A, Bb

B, C, D, E, G, A#, B

## Vutari

Step pattern: 4, 2, 1, 2, 1, 2

Ascending intervals: P1, M3, Aug4, P5, M6, m7, P8

C, E, F#, G, A, Bb, C

C#, E#, F##, G#, A#, B, C#

Db, F, G, Ab, Bb, Cb, Db

D, F#, G#, A, B, C, D

D#, F##, G##, A#, B#, C#, D#

Eb, G, A, Bb, C, Db, Eb

E, G#, A#, B, C#, D, E

F, A, B, C, D, Eb, F

F#, A#, B#, C#, D#, E, F#

Gb, Bb, C, Db, Eb, Fb, Gb

G, B, C#, D, E, F, G

G#, B#, C##, D#, E#, F#, G#

Ab, C, D, Eb, F, Gb, Ab

A, C#, D#, E, F#, G, A

A#, C##, D##, E#, F##, G#, A#

Bb, D, E, F, G, Ab, Bb

B, D#, E#, F#, G#, A, B



## Whole-tone

Step pattern: 2, 2, 2, 2, 2, 2

Ascending intervals: P1, M2, M3, Aug4, Aug5, Aug6, P8

C, D, E, F#, G#, A#, C

C#, D#, E#, F##, G##, A##, C#

Db, Eb, F, G, A, B, Db

D, E, F#, G#, A#, B#, D

D#, E#, F##, G##, A##, B##, D#

Eb, F, G, A, B, C#, Eb

E, F#, G#, A#, B#, C##, E

F, G, A, B, C#, D#, F

F#, G#, A#, B#, C##, D##, F#

Gb, Ab, Bb, C, D, E, Gb

G, A, B, C#, D#, E#, G

G#, A#, B#, C##, D##, E##, G#

Ab, Bb, C, D, E, F#, Ab

A, B, C#, D#, E#, F##, A

A#, B#, C##, D##, E##, F###, A#

Bb, C, D, E, F#, G#, Bb

B, C#, D#, E#, F##, G##, B

**Yosen**

Step pattern: 2, 3, 2, 2, 1, 2

Ascending intervals: P1, M2, P4, P5, M6, m7, P8

C, D, F, G, A, Bb, C

C#, D#, F#, G#, A#, B, C#

Db, Eb, Gb, Ab, Bb, Cb, Db

D, E, G, A, B, C, D

D#, E#, G#, A#, B#, C#, D#

Eb, F, Ab, Bb, C, Db, Eb

E, F#, A, B, C#, D, E

F, G, Bb, C, D, Eb, F

F#, G#, B, C#, D#, E, F#

Gb, Ab, Cb, Db, Eb, Fb, Gb

G, A, C, D, E, F, G

G#, A#, C#, D#, E#, F#, G#

Ab, Bb, Db, Eb, F, Gb, Ab

A, B, D, E, F#, G, A

A#, B#, D#, E#, F##, G#, A#

Bb, C, Eb, F, G, Ab, Bb

B, C#, E, F#, G#, A, B

## 7 – NOTE SCALES

## Acoustic Scale

Step pattern: 2, 2, 2, 1, 2, 1, 2

Ascending intervals: P1, M2, M3, Aug4, P5, M6, m7, P8

C, D, E, F#, G, A, Bb, C

C#, D#, E#, F##, G#, A#, B, C#

Db, Eb, F, G, Ab, Bb, Cb, Db

D, E, F#, G#, A, B, C, D

D#, E#, F##, G##, A#, B#, C#, D#

Eb, F, G, A, Bb, C, Db, Eb

E, F#, G#, A#, B, C#, D, E

F, G, A, B, C, D, Eb, F

F#, G#, A#, B#, C#, D#, E, F#

Gb, Ab, Bb, C, Db, Eb, Fb, Gb

G, A, B, C#, D, E, F, G

G#, A#, B#, C##, D#, E#, F#, G#

Ab, Bb, C, D, Eb, F, Gb, Ab

A, B, C#, D#, E, F#, G, A

A#, B#, C##, D##, E#, F##, G#, A#

Bb, C, D, E, F, G, Ab, Bb

B, C#, D#, E#, F#, G#, A, B

**Aeolian (6th mode of major)**

Step pattern: 2, 1, 2, 2, 1, 2, 2

Ascending intervals: P1, M2, m3, P4, P5, m6, m7, P8

C, D, Eb, F, G, Ab, Bb, C

C#, D#, E, F#, G#, A, B, C#

Db, Eb, Fb, Gb, Ab, Bbb, Cb, Db

D, E, F, G, A, Bb, C, D

D#, E#, F#, G#, A#, B, C#, D#

Eb, F, Gb, Ab, Bb, Cb, Db, Eb

E, F#, G, A, B, C, D, E

F, G, Ab, Bb, C, Db, Eb, F

F#, G#, A, B, C#, D, E, F#

Gb, Ab, Bbb, Cb, Db, Ebb, Fb, Gb

G, A, Bb, C, D, Eb, F, G

G#, A#, B, C#, D#, E, F#, G#

Ab, Bb, Cb, Db, Eb, Fb, Gb, Ab

A, B, C, D, E, F, G, A

A#, B#, C#, D#, E#, F#, G#, A#

Bb, C, Db, Eb, F, Gb, Ab, Bb

B, C#, D, E, F#, G, A, B

### **Aeolian Harmonic (6th mode of harmonic minor)**

Step pattern: 3, 1, 2, 1, 2, 2, 1

Ascending intervals: P1, Aug2, M3, Aug4, P5, M6, M7, P8

C, D#, E, F#, G, A, B, C

C#, D##, E#, F##, G#, A#, B#, C#

Db, E, F, G, Ab, Bb, C, Db

D, E#, F#, G#, A, B, C#, D

D#, E##, F##, G##, A#, B#, C##, D#

Eb, F#, G, A, Bb, C, D, Eb

E, F##, G#, A#, B, C#, D#, E

F, G#, A, B, C, D, E, F

F#, G##, A#, B#, C#, D#, E#, F#

Gb, A, Bb, C, Db, Eb, F, Gb

G, A#, B, C#, D, E, F#, G

G#, A##, B#, C##, D#, E#, F##, G#

Ab, B, C, D, Eb, F, G, Ab

A, B#, C#, D#, E, F#, G#, A

A#, B##, C##, D##, E#, F##, G##, A#

Bb, C#, D, E, F, G, A, Bb

B, C##, D#, E#, F#, G#, A#, B

**Aeolian Major (5th mode of melodic minor)**

Step pattern: 2, 2, 1, 2, 1, 2, 2

Ascending intervals: P1, M2, M3, P4, P5, m6, m7, P8

C, D, E, F, G, Ab, Bb, C

C#, D#, E#, F#, G#, A, B, C#

Db, Eb, F, Gb, Ab, Bbb, Cb, Db

D, E, F#, G, A, Bb, C, D

D#, E#, F##, G#, A#, B, C#, D#

Eb, F, G, Ab, Bb, Cb, Db, Eb

E, F#, G#, A, B, C, D, E

F, G, A, Bb, C, Db, Eb, F

F#, G#, A#, B, C#, D, E, F#

Gb, Ab, Bb, Cb, Db, Ebb, Fb, Gb

G, A, B, C, D, Eb, F, G

G#, A#, B#, C#, D#, E, F#, G#

Ab, Bb, C, Db, Eb, Fb, Gb, Ab

A, B, C#, D, E, F, G, A

A#, B#, C##, D#, E#, F#, G#, A#

Bb, C, D, Eb, F, Gb, Ab, Bb

B, C#, D#, E, F#, G, A, B

### Altered Diminished (7th mode of harmonic minor)

Step pattern: 1, 2, 1, 2, 2, 1, 3

Ascending intervals: P1, m2, m3, dim4, dim5, m6, dim7, P8

C, Db, Eb, Fb, Gb, Ab, Bbb, C

C#, D, E, F, G, A, Bb, C#

Db, Ebb, Fb, Gbb, Abb, Bbb, Cbb, Db

D, Eb, F, Gb, Ab, Bb, Cb, D

D#, E, F#, G, A, B, C, D#

Eb, Fb, Gb, Abb, Bbb, Cb, Dbb, Eb

E, F, G, Ab, Bb, C, Db, E

F, Gb, Ab, Bbb, Cb, Db, Ebb, F

F#, G, A, Bb, C, D, Eb, F#

Gb, Abb, Bbb, Cbb, Dbb, Ebb, Fbb, Gb

G, Ab, Bb, Cb, Db, Eb, Fb, G

G#, A, B, C, D, E, F, G#

Ab, Bbb, Cb, Dbb, Ebb, Fb, Gbb, Ab

A, Bb, C, Db, Eb, F, Gb, A

A#, B, C#, D, E, F#, G, A#

Bb, Cb, Db, Ebb, Fb, Gb, Abb, Bb

B, C, D, Eb, F, G, Ab, B



**Altered Dominant (7th mode of melodic minor)**

Step pattern: 1, 2, 1, 2, 2, 2, 2

Ascending intervals: P1, m2, m3, dim4, dim5, m6, m7, P8

C, Db, Eb, Fb, Gb, Ab, Bb, C

C#, D, E, F, G, A, B, C#

Db, Ebb, Fb, Gbb, Abb, Bbb, Cb, Db

D, Eb, F, Gb, Ab, Bb, C, D

D#, E, F#, G, A, B, C#, D#

Eb, Fb, Gb, Abb, Bbb, Cb, Db, Eb

E, F, G, Ab, Bb, C, D, E

F, Gb, Ab, Bbb, Cb, Db, Eb, F

F#, G, A, Bb, C, D, E, F#

Gb, Abb, Bbb, Cbb, Dbb, Ebb, Fb, Gb

G, Ab, Bb, Cb, Db, Eb, F, G

G#, A, B, C, D, E, F#, G#

Ab, Bbb, Cb, Dbb, Ebb, Fb, Gb, Ab

A, Bb, C, Db, Eb, F, G, A

A#, B, C#, D, E, F#, G#, A#

Bb, Cb, Db, Ebb, Fb, Gb, Ab, Bb

B, C, D, Eb, F, G, A, B

### Altered Scale (7th mode of melodic minor)

Step pattern: 1, 2, 1, 2, 2, 2, 2

Ascending intervals: P1, m2, m3, dim4, dim5, m6, m7, P8

C, Db, Eb, Fb, Gb, Ab, Bb, C

C#, D, E, F, G, A, B, C#

Db, Ebb, Fb, Gbb, Abb, Bbb, Cb, Db

D, Eb, F, Gb, Ab, Bb, C, D

D#, E, F#, G, A, B, C#, D#

Eb, Fb, Gb, Abb, Bbb, Cb, Db, Eb

E, F, G, Ab, Bb, C, D, E

F, Gb, Ab, Bbb, Cb, Db, Eb, F

F#, G, A, Bb, C, D, E, F#

Gb, Abb, Bbb, Cbb, Dbb, Ebb, Fb, Gb

G, Ab, Bb, Cb, Db, Eb, F, G

G#, A, B, C, D, E, F#, G#

Ab, Bbb, Cb, Dbb, Ebb, Fb, Gb, Ab

A, Bb, C, Db, Eb, F, G, A

A#, B, C#, D, E, F#, G#, A#

Bb, Cb, Db, Ebb, Fb, Gb, Ab, Bb

B, C, D, Eb, F, G, A, B

## Ambika

Step pattern: 2, 1, 3, 1, 2, 2, 1

Ascending intervals: P1, M2, m3, dim5, P5, M6, M7, P8

C, D, Eb, Gb, G, A, B, C

C#, D#, E, G, G#, A#, B#, C#

Db, Eb, Fb, Abb, Ab, Bb, C, Db

D, E, F, Ab, A, B, C#, D

D#, E#, F#, A, A#, B#, C##, D#

Eb, F, Gb, Bbb, Bb, C, D, Eb

E, F#, G, Bb, B, C#, D#, E

F, G, Ab, Cb, C, D, E, F

F#, G#, A, C, C#, D#, E#, F#

Gb, Ab, Bbb, Dbb, Db, Eb, F, Gb

G, A, Bb, Db, D, E, F#, G

G#, A#, B, D, D#, E#, F##, G#

Ab, Bb, Cb, Ebb, Eb, F, G, Ab

A, B, C, Eb, E, F#, G#, A

A#, B#, C#, E, E#, F##, G##, A#

Bb, C, Db, Fb, F, G, A, Bb

B, C#, D, F, F#, G#, A#, B

**Arabic (5th mode harmonic minor)**

Step pattern: 1, 3, 1, 2, 1, 2, 2

Ascending intervals: P1, m2, M3, P4, P5, m6, m7, P8

C, Db, E, F, G, Ab, Bb, C

C#, D, E#, F#, G#, A, B, C#

Db, Ebb, F, Gb, Ab, Bbb, Cb, Db

D, Eb, F#, G, A, Bb, C, D

D#, E, F##, G#, A#, B, C#, D#

Eb, Fb, G, Ab, Bb, Cb, Db, Eb

E, F, G#, A, B, C, D, E

F, Gb, A, Bb, C, Db, Eb, F

F#, G, A#, B, C#, D, E, F#

Gb, Abb, Bb, Cb, Db, Ebb, Fb, Gb

G, Ab, B, C, D, Eb, F, G

G#, A, B#, C#, D#, E, F#, G#

Ab, Bbb, C, Db, Eb, Fb, Gb, Ab

A, Bb, C#, D, E, F, G, A

A#, B, C##, D#, E#, F#, G#, A#

Bb, Cb, D, Eb, F, Gb, Ab, Bb

B, C, D#, E, F#, G, A, B

**Banshikicho**

Step pattern: 2, 1, 1, 3, 2, 1, 2

Ascending intervals: P1, M2, m3, dim4, P5, M6, m7, P8

C, D, Eb, Fb, G, A, Bb, C

C#, D#, E, F, G#, A#, B, C#

Db, Eb, Fb, Gbb, Ab, Bb, Cb, Db

D, E, F, Gb, A, B, C, D

D#, E#, F#, G, A#, B#, C#, D#

Eb, F, Gb, Abb, Bb, C, Db, Eb

E, F#, G, Ab, B, C#, D, E

F, G, Ab, Bbb, C, D, Eb, F

F#, G#, A, Bb, C#, D#, E, F#

Gb, Ab, Bbb, Cbb, Db, Eb, Fb, Gb

G, A, Bb, Cb, D, E, F, G

G#, A#, B, C, D#, E#, F#, G#

Ab, Bb, Cb, Dbb, Eb, F, Gb, Ab

A, B, C, Db, E, F#, G, A

A#, B#, C#, D, E#, F##, G#, A#

Bb, C, Db, Ebb, F, G, Ab, Bb

B, C#, D, Eb, F#, G#, A, B

**Bartok**

Step pattern: 2, 2, 2, 1, 2, 1, 2

Ascending intervals: P1, M2, M3, Aug4, P5, M6, m7, P8

C, D, E, F#, G, A, Bb, C

C#, D#, E#, F##, G#, A#, B, C#

Db, Eb, F, G, Ab, Bb, Cb, Db

D, E, F#, G#, A, B, C, D

D#, E#, F##, G##, A#, B#, C#, D#

Eb, F, G, A, Bb, C, Db, Eb

E, F#, G#, A#, B, C#, D, E

F, G, A, B, C, D, Eb, F

F#, G#, A#, B#, C#, D#, E, F#

Gb, Ab, Bb, C, Db, Eb, Fb, Gb

G, A, B, C#, D, E, F, G

G#, A#, B#, C##, D#, E#, F#, G#

Ab, Bb, C, D, Eb, F, Gb, Ab

A, B, C#, D#, E, F#, G, A

A#, B#, C##, D##, E#, F##, G#, A#

Bb, C, D, E, F, G, Ab, Bb

B, C#, D#, E#, F#, G#, A, B

## Bhairubahar Thaata

Step pattern: 1, 3, 1, 2, 2, 2, 1

Ascending intervals: P1, m2, M3, P4, P5, M6, M7, P8

C, Db, E, F, G, A, B, C

C#, D, E#, F#, G#, A#, B#, C#

Db, Ebb, F, Gb, Ab, Bb, C, Db

D, Eb, F#, G, A, B, C#, D

D#, E, F##, G#, A#, B#, C##, D#

Eb, Fb, G, Ab, Bb, C, D, Eb

E, F, G#, A, B, C#, D#, E

F, Gb, A, Bb, C, D, E, F

F#, G, A#, B, C#, D#, E#, F#

Gb, Abb, Bb, Cb, Db, Eb, F, Gb

G, Ab, B, C, D, E, F#, G

G#, A, B#, C#, D#, E#, F##, G#

Ab, Bbb, C, Db, Eb, F, G, Ab

A, Bb, C#, D, E, F#, G#, A

A#, B, C##, D#, E#, F##, G##, A#

Bb, Cb, D, Eb, F, G, A, Bb

B, C, D#, E, F#, G#, A#, B

**Bhankar**

Step pattern: 1, 3, 1, 1, 2, 1, 3

Ascending intervals: P1, m2, M3, P4, dim5, m6, M6, P8

C, Db, E, F, Gb, Ab, A, C

C#, D, E#, F#, G, A, A#, C#

Db, Ebb, F, Gb, Abb, Bbb, Bb, Db

D, Eb, F#, G, Ab, Bb, B, D

D#, E, F##, G#, A, B, B#, D#

Eb, Fb, G, Ab, Bbb, Cb, C, Eb

E, F, G#, A, Bb, C, C#, E

F, Gb, A, Bb, Cb, Db, D, F

F#, G, A#, B, C, D, D#, F#

Gb, Abb, Bb, Cb, Dbb, Ebb, Eb, Gb

G, Ab, B, C, Db, Eb, E, G

G#, A, B#, C#, D, E, E#, G#

Ab, Bbb, C, Db, Ebb, Fb, F, Ab

A, Bb, C#, D, Eb, F, F#, A

A#, B, C##, D#, E, F#, F##, A#

Bb, Cb, D, Eb, Fb, Gb, G, Bb

B, C, D#, E, F, G, G#, B



**Bhanumati**

Step pattern: 1, 1, 3, 2, 2, 1, 2

Ascending intervals: P1, m2, M2, P4, P5, M6, m7, P8

C, Db, D, F, G, A, Bb, C

C#, D, D#, F#, G#, A#, B, C#

Db, Ebb, Eb, Gb, Ab, Bb, Cb, Db

D, Eb, E, G, A, B, C, D

D#, E, E#, G#, A#, B#, C#, D#

Eb, Fb, F, Ab, Bb, C, Db, Eb

E, F, F#, A, B, C#, D, E

F, Gb, G, Bb, C, D, Eb, F

F#, G, G#, B, C#, D#, E, F#

Gb, Abb, Ab, Cb, Db, Eb, Fb, Gb

G, Ab, A, C, D, E, F, G

G#, A, A#, C#, D#, E#, F#, G#

Ab, Bbb, Bb, Db, Eb, F, Gb, Ab

A, Bb, B, D, E, F#, G, A

A#, B, B#, D#, E#, F##, G#, A#

Bb, Cb, C, Eb, F, G, Ab, Bb

B, C, C#, E, F#, G#, A, B

**Bhavani (heptatonic)**

Step pattern: 1, 2, 3, 1, 1, 2, 2

Ascending intervals: P1, m2, m3, dim5, P5, m6, m7, P8

C, Db, Eb, Gb, G, Ab, Bb, C

C#, D, E, G, G#, A, B, C#

Db, Ebb, Fb, Abb, Ab, Bbb, Cb, Db

D, Eb, F, Ab, A, Bb, C, D

D#, E, F#, A, A#, B, C#, D#

Eb, Fb, Gb, Bbb, Bb, Cb, Db, Eb

E, F, G, Bb, B, C, D, E

F, Gb, Ab, Cb, C, Db, Eb, F

F#, G, A, C, C#, D, E, F#

Gb, Abb, Bbb, Dbb, Db, Ebb, Fb, Gb

G, Ab, Bb, Db, D, Eb, F, G

G#, A, B, D, D#, E, F#, G#

Ab, Bbb, Cb, Ebb, Eb, Fb, Gb, Ab

A, Bb, C, Eb, E, F, G, A

A#, B, C#, E, E#, F#, G#, A#

Bb, Cb, Db, Fb, F, Gb, Ab, Bb

B, C, D, F, F#, G, A, B

**Bhusavati**

Step pattern: 2, 2, 2, 1, 2, 1, 2

Ascending intervals: P1, M2, M3, dim5, P5, M6, m7, P8

C, D, E, Gb, G, A, Bb, C

C#, D#, E#, G, G#, A#, B, C#

Db, Eb, F, Abb, Ab, Bb, Cb, Db

D, E, F#, Ab, A, B, C, D

D#, E#, F##, A, A#, B#, C#, D#

Eb, F, G, Bbb, Bb, C, Db, Eb

E, F#, G#, Bb, B, C#, D, E

F, G, A, Cb, C, D, Eb, F

F#, G#, A#, C, C#, D#, E, F#

Gb, Ab, Bb, Dbb, Db, Eb, Fb, Gb

G, A, B, Db, D, E, F, G

G#, A#, B#, D, D#, E#, F#, G#

Ab, Bb, C, Ebb, Eb, F, Gb, Ab

A, B, C#, Eb, E, F#, G, A

A#, B#, C##, E, E#, F##, G#, A#

Bb, C, D, Fb, F, G, Ab, Bb

B, C#, D#, F, F#, G#, A, B

**Blues (add M7)**

Step pattern: 3, 2, 1, 1, 3, 1, 1

Ascending intervals: P1, m3, P4, dim5, P5, m7, M7, P8

C, Eb, F, Gb, G, Bb, B, C

C#, E, F#, G, G#, B, B#, C#

Db, Fb, Gb, Abb, Ab, Cb, C, Db

D, F, G, Ab, A, C, C#, D

D#, F#, G#, A, A#, C#, C##, D#

Eb, Gb, Ab, Bbb, Bb, Db, D, Eb

E, G, A, Bb, B, D, D#, E

F, Ab, Bb, Cb, C, Eb, E, F

F#, A, B, C, C#, E, E#, F#

Gb, Bbb, Cb, Dbb, Db, Fb, F, Gb

G, Bb, C, Db, D, F, F#, G

G#, B, C#, D, D#, F#, F##, G#

Ab, Cb, Db, Ebb, Eb, Gb, G, Ab

A, C, D, Eb, E, G, G#, A

A#, C#, D#, E, E#, G#, G##, A#

Bb, Db, Eb, Fb, F, Ab, A, Bb

B, D, E, F, F#, A, A#, B

## Blues Heptatonic

Step pattern: 3, 1, 1, 1, 1, 3, 2

Ascending intervals: P1, m3, M3, P4, dim5, P5, m7, P8

C, Eb, E, F, Gb, G, Bb, C

C#, E, E#, F#, G, G#, B, C#

Db, Fb, F, Gb, Abb, Ab, Cb, Db

D, F, F#, G, Ab, A, C, D

D#, F#, F##, G#, A, A#, C#, D#

Eb, Gb, G, Ab, Bbb, Bb, Db, Eb

E, G, G#, A, Bb, B, D, E

F, Ab, A, Bb, Cb, C, Eb, F

F#, A, A#, B, C, C#, E, F#

Gb, Bbb, Bb, Cb, Dbb, Db, Fb, Gb

G, Bb, B, C, Db, D, F, G

G#, B, B#, C#, D, D#, F#, G#

Ab, Cb, C, Db, Ebb, Eb, Gb, Ab

A, C, C#, D, Eb, E, G, A

A#, C#, C##, D#, E, E#, G#, A#

Bb, Db, D, Eb, Fb, F, Ab, Bb

B, D, D#, E, F, F#, A, B

## Blues Minor Heptatonic (add M2)

Step pattern: 2, 1, 2, 1, 1, 3, 2

Ascending intervals: P1, M2, m3, P4, dim5, P5, m7, P8

C, D, Eb, F, Gb, G, Bb, C

C#, D#, E, F#, G, G#, B, C#

Db, Eb, Fb, Gb, Abb, Ab, Cb, Db

D, E, F, G, Ab, A, C, D

D#, E#, F#, G#, A, A#, C#, D#

Eb, F, Gb, Ab, Bbb, Bb, Db, Eb

E, F#, G, A, Bb, B, D, E

F, G, Ab, Bb, Cb, C, Eb, F

F#, G#, A, B, C, C#, E, F#

Gb, Ab, Bbb, Cb, Dbb, Db, Fb, Gb

G, A, Bb, C, Db, D, F, G

G#, A#, B, C#, D, D#, F#, G#

Ab, Bb, Cb, Db, Ebb, Eb, Gb, Ab

A, B, C, D, Eb, E, G, A

A#, B#, C#, D#, E, E#, G#, A#

Bb, C, Db, Eb, Fb, F, Ab, Bb

B, C#, D, E, F, F#, A, B

## Bluesy

Step pattern: 3, 1, 1, 2, 2, 1, 2

Ascending intervals: P1, m3, M3, P4, P5, M6, m7, P8

C, Eb, E, F, G, A, Bb, C

C#, E, E#, F#, G#, A#, B, C#

Db, Fb, F, Gb, Ab, Bb, Cb, Db

D, F, F#, G, A, B, C, D

D#, F#, F##, G#, A#, B#, C#, D#

Eb, Gb, G, Ab, Bb, C, Db, Eb

E, G, G#, A, B, C#, D, E

F, Ab, A, Bb, C, D, Eb, F

F#, A, A#, B, C#, D#, E, F#

Gb, Bbb, Bb, Cb, Db, Eb, Fb, Gb

G, Bb, B, C, D, E, F, G

G#, B, B#, C#, D#, E#, F#, G#

Ab, Cb, C, Db, Eb, F, Gb, Ab

A, C, C#, D, E, F#, G, A

A#, C#, C##, D#, E#, F##, G#, A#

Bb, Db, D, Eb, F, G, Ab, Bb

B, D, D#, E, F#, G#, A, B

## Byzantine Scale

Step pattern: 1, 3, 1, 2, 1, 3, 1

Ascending intervals: P1, m2, M3, P4, P5, m6, M7, P8

C, Db, E, F, G, Ab, B, C

C#, D, E#, F#, G#, A, B#, C#

Db, Ebb, F, Gb, Ab, Bbb, C, Db

D, Eb, F#, G, A, Bb, C#, D

D#, E, F##, G#, A#, B, C##, D#

Eb, Fb, G, Ab, Bb, Cb, D, Eb

E, F, G#, A, B, C, D#, E

F, Gb, A, Bb, C, Db, E, F

F#, G, A#, B, C#, D, E#, F#

Gb, Abb, Bb, Cb, Db, Ebb, F, Gb

G, Ab, B, C, D, Eb, F#, G

G#, A, B#, C#, D#, E, F##, G#

Ab, Bbb, C, Db, Eb, Fb, G, Ab

A, Bb, C#, D, E, F, G#, A

A#, B, C##, D#, E#, F#, G##, A#

Bb, Cb, D, Eb, F, Gb, A, Bb

B, C, D#, E, F#, G, A#, B



**Camara**

Step pattern: 2, 1, 3, 1, 1, 2, 2

Ascending intervals: P1, M2, m3, dim5, P5, m6, m7, P8

C, D, Eb, Gb, G, Ab, Bb, C

C#, D#, E, G, G#, A, B, C#

Db, Eb, Fb, Abb, Ab, Bbb, Cb, Db

D, E, F, Ab, A, Bb, C, D

D#, E#, F#, A, A#, B, C#, D#

Eb, F, Gb, Bbb, Bb, Cb, Db, Eb

E, F#, G, Bb, B, C, D, E

F, G, Ab, Cb, C, Db, Eb, F

F#, G#, A, C, C#, D, E, F#

Gb, Ab, Bbb, Dbb, Db, Ebb, Fb, Gb

G, A, Bb, Db, D, Eb, F, G

G#, A#, B, D, D#, E, F#, G#

Ab, Bb, Cb, Ebb, Eb, Fb, Gb, Ab

A, B, C, Eb, E, F, G, A

A#, B#, C#, E, E#, F#, G#, A#

Bb, C, Db, Fb, F, Gb, Ab, Bb

B, C#, D, F, F#, G, A, B

## Chaturangini

Step pattern: 2, 2, 2, 1, 3, 1, 1

Ascending intervals: P1, M2, M3, Aug4, P5, m7, M7, P8

C, D, E, F#, G, Bb, B, C

C#, D#, E#, F##, G#, B, B#, C#

Db, Eb, F, G, Ab, Cb, C, Db

D, E, F#, G#, A, C, C#, D

D#, E#, F##, G##, A#, C#, C##, D#

Eb, F, G, A, Bb, Db, D, Eb

E, F#, G#, A#, B, D, D#, E

F, G, A, B, C, Eb, E, F

F#, G#, A#, B#, C#, E, E#, F#

Gb, Ab, Bb, C, Db, Fb, F, Gb

G, A, B, C#, D, F, F#, G

G#, A#, B#, C##, D#, F#, F##, G#

Ab, Bb, C, D, Eb, Gb, G, Ab

A, B, C#, D#, E, G, G#, A

A#, B#, C##, D##, E#, G#, G##, A#

Bb, C, D, E, F, Ab, A, Bb

B, C#, D#, E#, F#, A, A#, B

## Chromantic Hypodorian

Step pattern: 2, 1, 1, 3, 1, 1, 3

Ascending intervals: P1, M2, m3, M3, P5, m6, M6, P8

C, D, Eb, E, G, Ab, A, C

C#, D#, E, E#, G#, A, A#, C#

Db, Eb, Fb, F, Ab, Bbb, Bb, Db

D, E, F, F#, A, Bb, B, D

D#, E#, F#, F##, A#, B, B#, D#

Eb, F, Gb, G, Bb, Cb, C, Eb

E, F#, G, G#, B, C, C#, E

F, G, Ab, A, C, Db, D, F

F#, G#, A, A#, C#, D, D#, F#

Gb, Ab, Bbb, Bb, Db, Ebb, Eb, Gb

G, A, Bb, B, D, Eb, E, G

G#, A#, B, B#, D#, E, E#, G#

Ab, Bb, Cb, C, Eb, Fb, F, Ab

A, B, C, C#, E, F, F#, A

A#, B#, C#, C##, E#, F#, F##, A#

Bb, C, Db, D, F, Gb, G, Bb

B, C#, D, D#, F#, G, G#, B

## Chromantic Hypolydian

Step pattern: 1, 3, 2, 1, 1, 3, 1

Ascending intervals: P1, m2, M3, dim5, P5, m6, M7, P8

C, Db, E, Gb, G, Ab, B, C

C#, D, E#, G, G#, A, B#, C#

Db, Ebb, F, Abb, Ab, Bbb, C, Db

D, Eb, F#, Ab, A, Bb, C#, D

D#, E, F##, A, A#, B, C##, D#

Eb, Fb, G, Bbb, Bb, Cb, D, Eb

E, F, G#, Bb, B, C, D#, E

F, Gb, A, Cb, C, Db, E, F

F#, G, A#, C, C#, D, E#, F#

Gb, Abb, Bb, Dbb, Db, Ebb, F, Gb

G, Ab, B, Db, D, Eb, F#, G

G#, A, B#, D, D#, E, F##, G#

Ab, Bbb, C, Ebb, Eb, Fb, G, Ab

A, Bb, C#, Eb, E, F, G#, A

A#, B, C##, E, E#, F#, G##, A#

Bb, Cb, D, Fb, F, Gb, A, Bb

B, C, D#, F, F#, G, A#, B

## Chromantic Hypolydian Inverse

Step pattern: 1, 3, 1, 1, 2, 3, 1

Ascending intervals: P1, m2, M3, P4, dim5, m6, M7, P8

C, Db, E, F, Gb, Ab, B, C

C#, D, E#, F#, G, A, B#, C#

Db, Ebb, F, Gb, Abb, Bbb, C, Db

D, Eb, F#, G, Ab, Bb, C#, D

D#, E, F##, G#, A, B, C##, D#

Eb, Fb, G, Ab, Bbb, Cb, D, Eb

E, F, G#, A, Bb, C, D#, E

F, Gb, A, Bb, Cb, Db, E, F

F#, G, A#, B, C, D, E#, F#

Gb, Abb, Bb, Cb, Dbb, Ebb, F, Gb

G, Ab, B, C, Db, Eb, F#, G

G#, A, B#, C#, D, E, F##, G#

Ab, Bbb, C, Db, Ebb, Fb, G, Ab

A, Bb, C#, D, Eb, F, G#, A

A#, B, C##, D#, E, F#, G##, A#

Bb, Cb, D, Eb, Fb, Gb, A, Bb

B, C, D#, E, F, G, A#, B

## Chromantic Hypophrygian Inverse

Step pattern: 1, 1, 3, 1, 1, 2, 3

Ascending intervals: P1, m2, M2, P4, dim5, P5, M6, P8

C, Db, D, F, Gb, G, A, C

C#, D, D#, F#, G, G#, A#, C#

Db, Ebb, Eb, Gb, Abb, Ab, Bb, Db

D, Eb, E, G, Ab, A, B, D

D#, E, E#, G#, A, A#, B#, D#

Eb, Fb, F, Ab, Bbb, Bb, C, Eb

E, F, F#, A, Bb, B, C#, E

F, Gb, G, Bb, Cb, C, D, F

F#, G, G#, B, C, C#, D#, F#

Gb, Abb, Ab, Cb, Dbb, Db, Eb, Gb

G, Ab, A, C, Db, D, E, G

G#, A, A#, C#, D, D#, E#, G#

Ab, Bbb, Bb, Db, Ebb, Eb, F, Ab

A, Bb, B, D, Eb, E, F#, A

A#, B, B#, D#, E, E#, F##, A#

Bb, Cb, C, Eb, Fb, F, G, Bb

B, C, C#, E, F, F#, G#, B

## Chromatic Phrygian

Step pattern: 3, 1, 1, 3, 2, 1, 1

Ascending intervals: P1, m3, M3, P4, m6, m7, M7, P8

C, Eb, E, F, Ab, Bb, B, C

C#, E, E#, F#, A, B, B#, C#

Db, Fb, F, Gb, Bbb, Cb, C, Db

D, F, F#, G, Bb, C, C#, D

D#, F#, F##, G#, B, C#, C##, D#

Eb, Gb, G, Ab, Cb, Db, D, Eb

E, G, G#, A, C, D, D#, E

F, Ab, A, Bb, Db, Eb, E, F

F#, A, A#, B, D, E, E#, F#

Gb, Bbb, Bb, Cb, Ebb, Fb, F, Gb

G, Bb, B, C, Eb, F, F#, G

G#, B, B#, C#, E, F#, F##, G#

Ab, Cb, C, Db, Fb, Gb, G, Ab

A, C, C#, D, F, G, G#, A

A#, C#, C##, D#, F#, G#, G##, A#

Bb, Db, D, Eb, Gb, Ab, A, Bb

B, D, D#, E, G, A, A#, B

## Chromantic Phrygian Inverse

Step pattern: 1, 1, 2, 3, 1, 1, 3

Ascending intervals: P1, m2, M2, M3, P5, m6, M6, P8

C, Db, D, E, G, Ab, A, C

C#, D, D#, E#, G#, A, A#, C#

Db, Ebb, Eb, F, Ab, Bbb, Bb, Db

D, Eb, E, F#, A, Bb, B, D

D#, E, E#, F##, A#, B, B#, D#

Eb, Fb, F, G, Bb, Cb, C, Eb

E, F, F#, G#, B, C, C#, E

F, Gb, G, A, C, Db, D, F

F#, G, G#, A#, C#, D, D#, F#

Gb, Abb, Ab, Bb, Db, Ebb, Eb, Gb

G, Ab, A, B, D, Eb, E, G

G#, A, A#, B#, D#, E, E#, G#

Ab, Bbb, Bb, C, Eb, Fb, F, Ab

A, Bb, B, C#, E, F, F#, A

A#, B, B#, C##, E#, F#, F##, A#

Bb, Cb, C, D, F, Gb, G, Bb

B, C, C#, D#, F#, G, G#, B



## Chromatic Dorian

Step pattern: 1, 1, 3, 2, 1, 1, 3

Ascending intervals: P1, m2, M2, P4, P5, m6, M6, P8

C, Db, D, F, G, Ab, A, C

C#, D, D#, F#, G#, A, A#, C#

Db, Ebb, Eb, Gb, Ab, Bbb, Bb, Db

D, Eb, E, G, A, Bb, B, D

D#, E, E#, G#, A#, B, B#, D#

Eb, Fb, F, Ab, Bb, Cb, C, Eb

E, F, F#, A, B, C, C#, E

F, Gb, G, Bb, C, Db, D, F

F#, G, G#, B, C#, D, D#, F#

Gb, Abb, Ab, Cb, Db, Ebb, Eb, Gb

G, Ab, A, C, D, Eb, E, G

G#, A, A#, C#, D#, E, E#, G#

Ab, Bbb, Bb, Db, Eb, Fb, F, Ab

A, Bb, B, D, E, F, F#, A

A#, B, B#, D#, E#, F#, F##, A#

Bb, Cb, C, Eb, F, Gb, G, Bb

B, C, C#, E, F#, G, G#, B

## Chromatic Hypodorian Inverse

Step pattern: 3, 1, 1, 3, 1, 1, 2

Ascending intervals: P1, m3, M3, P4, m6, M6, m7, P8

C, Eb, E, F, Ab, A, Bb, C

C#, E, E#, F#, A, A#, B, C#

Db, Fb, F, Gb, Bbb, Bb, Cb, Db

D, F, F#, G, Bb, B, C, D

D#, F#, F##, G#, B, B#, C#, D#

Eb, Gb, G, Ab, Cb, C, Db, Eb

E, G, G#, A, C, C#, D, E

F, Ab, A, Bb, Db, D, Eb, F

F#, A, A#, B, D, D#, E, F#

Gb, Bbb, Bb, Cb, Ebb, Eb, Fb, Gb

G, Bb, B, C, Eb, E, F, G

G#, B, B#, C#, E, E#, F#, G#

Ab, Cb, C, Db, Fb, F, Gb, Ab

A, C, C#, D, F, F#, G, A

A#, C#, C##, D#, F#, F##, G#, A#

Bb, Db, D, Eb, Gb, G, Ab, Bb

B, D, D#, E, G, G#, A, B

## Chromatic Lydian

Step pattern: 1, 3, 1, 1, 3, 2, 1

Ascending intervals: P1, m2, M3, P4, dim5, M6, M7, P8

C, Db, E, F, Gb, A, B, C

C#, D, E#, F#, G, A#, B#, C#

Db, Ebb, F, Gb, Abb, Bb, C, Db

D, Eb, F#, G, Ab, B, C#, D

D#, E, F##, G#, A, B#, C##, D#

Eb, Fb, G, Ab, Bbb, C, D, Eb

E, F, G#, A, Bb, C#, D#, E

F, Gb, A, Bb, Cb, D, E, F

F#, G, A#, B, C, D#, E#, F#

Gb, Abb, Bb, Cb, Dbb, Eb, F, Gb

G, Ab, B, C, Db, E, F#, G

G#, A, B#, C#, D, E#, F##, G#

Ab, Bbb, C, Db, Ebb, F, G, Ab

A, Bb, C#, D, Eb, F#, G#, A

A#, B, C##, D#, E, F##, G##, A#

Bb, Cb, D, Eb, Fb, G, A, Bb

B, C, D#, E, F, G#, A#, B

## Chromatic Lydian Inverse

Step pattern: 1, 2, 3, 1, 1, 3, 1

Ascending intervals: P1, m2, m3, dim5, P5, m6, M7, P8

C, Db, Eb, Gb, G, Ab, B, C

C#, D, E, G, G#, A, B#, C#

Db, Ebb, Fb, Abb, Ab, Bbb, C, Db

D, Eb, F, Ab, A, Bb, C#, D

D#, E, F#, A, A#, B, C##, D#

Eb, Fb, Gb, Bbb, Bb, Cb, D, Eb

E, F, G, Bb, B, C, D#, E

F, Gb, Ab, Cb, C, Db, E, F

F#, G, A, C, C#, D, E#, F#

Gb, Abb, Bbb, Dbb, Db, Ebb, F, Gb

G, Ab, Bb, Db, D, Eb, F#, G

G#, A, B, D, D#, E, F##, G#

Ab, Bbb, Cb, Ebb, Eb, Fb, G, Ab

A, Bb, C, Eb, E, F, G#, A

A#, B, C#, E, E#, F#, G##, A#

Bb, Cb, Db, Fb, F, Gb, A, Bb

B, C, D, F, F#, G, A#, B

## Chromatic Mixolydian

Step pattern: 1, 1, 3, 1, 1, 3, 2

Ascending intervals: P1, m2, M2, P4, dim5, P5, m7, P8

C, Db, D, F, Gb, G, Bb, C

C#, D, D#, F#, G, G#, B, C#

Db, Ebb, Eb, Gb, Abb, Ab, Cb, Db

D, Eb, E, G, Ab, A, C, D

D#, E, E#, G#, A, A#, C#, D#

Eb, Fb, F, Ab, Bbb, Bb, Db, Eb

E, F, F#, A, Bb, B, D, E

F, Gb, G, Bb, Cb, C, Eb, F

F#, G, G#, B, C, C#, E, F#

Gb, Abb, Ab, Cb, Dbb, Db, Fb, Gb

G, Ab, A, C, Db, D, F, G

G#, A, A#, C#, D, D#, F#, G#

Ab, Bbb, Bb, Db, Ebb, Eb, Gb, Ab

A, Bb, B, D, Eb, E, G, A

A#, B, B#, D#, E, E#, G#, A#

Bb, Cb, C, Eb, Fb, F, Ab, Bb

B, C, C#, E, F, F#, A, B

## Chromatic Mixolydian Inverse

Step pattern: 2, 3, 1, 1, 3, 1, 1

Ascending intervals: P1, M2, P4, dim5, P5, m7, M7, P8

C, D, F, Gb, G, Bb, B, C

C#, D#, F#, G, G#, B, B#, C#

Db, Eb, Gb, Abb, Ab, Cb, C, Db

D, E, G, Ab, A, C, C#, D

D#, E#, G#, A, A#, C#, C##, D#

Eb, F, Ab, Bbb, Bb, Db, D, Eb

E, F#, A, Bb, B, D, D#, E

F, G, Bb, Cb, C, Eb, E, F

F#, G#, B, C, C#, E, E#, F#

Gb, Ab, Cb, Dbb, Db, Fb, F, Gb

G, A, C, Db, D, F, F#, G

G#, A#, C#, D, D#, F#, F##, G#

Ab, Bb, Db, Ebb, Eb, Gb, G, Ab

A, B, D, Eb, E, G, G#, A

A#, B#, D#, E, E#, G#, G##, A#

Bb, C, Eb, Fb, F, Ab, A, Bb

B, C#, E, F, F#, A, A#, B

## Citrambari

Step pattern: 2, 2, 2, 1, 3, 1, 1

Ascending intervals: P1, M2, M3, dim5, P5, m7, M7, P8

C, D, E, Gb, G, Bb, B, C

C#, D#, E#, G, G#, B, B#, C#

Db, Eb, F, Abb, Ab, Cb, C, Db

D, E, F#, Ab, A, C, C#, D

D#, E#, F##, A, A#, C#, C##, D#

Eb, F, G, Bbb, Bb, Db, D, Eb

E, F#, G#, Bb, B, D, D#, E

F, G, A, Cb, C, Eb, E, F

F#, G#, A#, C, C#, E, E#, F#

Gb, Ab, Bb, Dbb, Db, Fb, F, Gb

G, A, B, Db, D, F, F#, G

G#, A#, B#, D, D#, F#, F##, G#

Ab, Bb, C, Ebb, Eb, Gb, G, Ab

A, B, C#, Eb, E, G, G#, A

A#, B#, C##, E, E#, G#, G##, A#

Bb, C, D, Fb, F, Ab, A, Bb

B, C#, D#, F, F#, A, A#, B

## Cudamani

Step pattern: 3, 1, 1, 2, 1, 2, 2

Ascending intervals: P1, m3, M3, P4, P5, m6, m7, P8

C, Eb, E, F, G, Ab, Bb, C

C#, E, E#, F#, G#, A, B, C#

Db, Fb, F, Gb, Ab, Bbb, Cb, Db

D, F, F#, G, A, Bb, C, D

D#, F#, F##, G#, A#, B, C#, D#

Eb, Gb, G, Ab, Bb, Cb, Db, Eb

E, G, G#, A, B, C, D, E

F, Ab, A, Bb, C, Db, Eb, F

F#, A, A#, B, C#, D, E, F#

Gb, Bbb, Bb, Cb, Db, Ebb, Fb, Gb

G, Bb, B, C, D, Eb, F, G

G#, B, B#, C#, D#, E, F#, G#

Ab, Cb, C, Db, Eb, Fb, Gb, Ab

A, C, C#, D, E, F, G, A

A#, C#, C##, D#, E#, F#, G#, A#

Bb, Db, D, Eb, F, Gb, Ab, Bb

B, D, D#, E, F#, G, A, B



## Devarashtra

Step pattern: 3, 1, 2, 1, 1, 3, 1

Ascending intervals: P1, m3, M3, dim5, P5, m6, M7, P8

C, Eb, E, Gb, G, Ab, B, C

C#, E, E#, G, G#, A, B#, C#

Db, Fb, F, Abb, Ab, Bbb, C, Db

D, F, F#, Ab, A, Bb, C#, D

D#, F#, F##, A, A#, B, C##, D#

Eb, Gb, G, Bbb, Bb, Cb, D, Eb

E, G, G#, Bb, B, C, D#, E

F, Ab, A, Cb, C, Db, E, F

F#, A, A#, C, C#, D, E#, F#

Gb, Bbb, Bb, Dbb, Db, Ebb, F, Gb

G, Bb, B, Db, D, Eb, F#, G

G#, B, B#, D, D#, E, F##, G#

Ab, Cb, C, Ebb, Eb, Fb, G, Ab

A, C, C#, Eb, E, F, G#, A

A#, C#, C##, E, E#, F#, G##, A#

Bb, Db, D, Fb, F, Gb, A, Bb

B, D, D#, F, F#, G, A#, B

### Diminished b9 (7th mode of harmonic major)

Step pattern: 1, 2, 2, 1, 2, 1, 3

Ascending intervals: P1, m2, m3, P4, dim5, m6, dim7, P8

C, Db, Eb, F, Gb, Ab, Bbb, C

C#, D, E, F#, G, A, Bb, C#

Db, Ebb, Fb, Gb, Abb, Bbb, Cbb, Db

D, Eb, F, G, Ab, Bb, Cb, D

D#, E, F#, G#, A, B, C, D#

Eb, Fb, Gb, Ab, Bbb, Cb, Dbb, Eb

E, F, G, A, Bb, C, Db, E

F, Gb, Ab, Bb, Cb, Db, Ebb, F

F#, G, A, B, C, D, Eb, F#

Gb, Abb, Bbb, Cb, Dbb, Ebb, Fbb, Gb

G, Ab, Bb, C, Db, Eb, Fb, G

G#, A, B, C#, D, E, F, G#

Ab, Bbb, Cb, Db, Ebb, Fb, Gbb, Ab

A, Bb, C, D, Eb, F, Gb, A

A#, B, C#, D#, E, F#, G, A#

Bb, Cb, Db, Eb, Fb, Gb, Abb, Bb

B, C, D, E, F, G, Ab, B

### Diminished whole-tone (7th mode of melodic minor)

Step pattern: 1, 2, 1, 2, 2, 2, 2

Ascending intervals: P1, m2, Aug2, M3, Aug4, Aug5, m7, P8

C, Db, D#, E, F#, G#, Bb, C

C#, D, D##, E#, F##, G##, B, C#

Db, Ebb, E, F, G, A, Cb, Db

D, Eb, E#, F#, G#, A#, C, D

D#, E, E##, F##, G##, A##, C#, D#

Eb, Fb, F#, G, A, B, Db, Eb

E, F, F##, G#, A#, B#, D, E

F, Gb, G#, A, B, C#, Eb, F

F#, G, G##, A#, B#, C##, E, F#

Gb, Abb, A, Bb, C, D, Fb, Gb

G, Ab, A#, B, C#, D#, F, G

G#, A, A##, B#, C##, D##, F#, G#

Ab, Bbb, B, C, D, E, Gb, Ab

A, Bb, B#, C#, D#, E#, G, A

A#, B, B##, C##, D##, E##, G#, A#

Bb, Cb, C#, D, E, F#, Ab, Bb

B, C, C##, D#, E#, F##, A, B

**Dorian #4 (4th mode of harmonic minor)**

Step pattern: 2, 1, 3, 1, 2, 1, 2

Ascending intervals: P1, M2, m3, Aug4, P5, M6, m7, P8

C, D, Eb, F#, G, A, Bb, C

C#, D#, E, F##, G#, A#, B, C#

Db, Eb, Fb, G, Ab, Bb, Cb, Db

D, E, F, G#, A, B, C, D

D#, E#, F#, G##, A#, B#, C#, D#

Eb, F, Gb, A, Bb, C, Db, Eb

E, F#, G, A#, B, C#, D, E

F, G, Ab, B, C, D, Eb, F

F#, G#, A, B#, C#, D#, E, F#

Gb, Ab, Bbb, C, Db, Eb, Fb, Gb

G, A, Bb, C#, D, E, F, G

G#, A#, B, C##, D#, E#, F#, G#

Ab, Bb, Cb, D, Eb, F, Gb, Ab

A, B, C, D#, E, F#, G, A

A#, B#, C#, D##, E#, F##, G#, A#

Bb, C, Db, E, F, G, Ab, Bb

B, C#, D, E#, F#, G#, A, B

**Dorian (2nd mode major)**

Step pattern: 2, 1, 2, 2, 2, 1, 2

Ascending intervals: P1, M2, m3, P4, P5, M6, m7, P8

C, D, Eb, F, G, A, Bb, C

C#, D#, E, F#, G#, A#, B, C#

Db, Eb, Fb, Gb, Ab, Bb, Cb, Db

D, E, F, G, A, B, C, D

D#, E#, F#, G#, A#, B#, C#, D#

Eb, F, Gb, Ab, Bb, C, Db, Eb

E, F#, G, A, B, C#, D, E

F, G, Ab, Bb, C, D, Eb, F

F#, G#, A, B, C#, D#, E, F#

Gb, Ab, Bbb, Cb, Db, Eb, Fb, Gb

G, A, Bb, C, D, E, F, G

G#, A#, B, C#, D#, E#, F#, G#

Ab, Bb, Cb, Db, Eb, F, Gb, Ab

A, B, C, D, E, F#, G, A

A#, B#, C#, D#, E#, F##, G#, A#

Bb, C, Db, Eb, F, G, Ab, Bb

B, C#, D, E, F#, G#, A, B

**Dorian b2 (2nd mode of melodic minor)**

Step pattern: 1, 2, 2, 2, 2, 1, 2

Ascending intervals: P1, m2, m3, P4, P5, M6, m7, P8

C, Db, Eb, F, G, A, Bb, C

C#, D, E, F#, G#, A#, B, C#

Db, Ebb, Fb, Gb, Ab, Bb, Cb, Db

D, Eb, F, G, A, B, C, D

D#, E, F#, G#, A#, B#, C#, D#

Eb, Fb, Gb, Ab, Bb, C, Db, Eb

E, F, G, A, B, C#, D, E

F, Gb, Ab, Bb, C, D, Eb, F

F#, G, A, B, C#, D#, E, F#

Gb, Abb, Bbb, Cb, Db, Eb, Fb, Gb

G, Ab, Bb, C, D, E, F, G

G#, A, B, C#, D#, E#, F#, G#

Ab, Bbb, Cb, Db, Eb, F, Gb, Ab

A, Bb, C, D, E, F#, G, A

A#, B, C#, D#, E#, F##, G#, A#

Bb, Cb, Db, Eb, F, G, Ab, Bb

B, C, D, E, F#, G#, A, B

**Dorian b2 b5 (2nd mode of harmonic minor)**

Step pattern: 1, 2, 2, 1, 3, 1, 2

Ascending intervals: P1, m2, m3, P4, dim5, M6, m7, P8

C, Db, Eb, F, Gb, A, Bb, C

C#, D, E, F#, G, A#, B, C#

Db, Ebb, Fb, Gb, Abb, Bb, Cb, Db

D, Eb, F, G, Ab, B, C, D

D#, E, F#, G#, A, B#, C#, D#

Eb, Fb, Gb, Ab, Bbb, C, Db, Eb

E, F, G, A, Bb, C#, D, E

F, Gb, Ab, Bb, Cb, D, Eb, F

F#, G, A, B, C, D#, E, F#

Gb, Abb, Bbb, Cb, Dbb, Eb, Fb, Gb

G, Ab, Bb, C, Db, E, F, G

G#, A, B, C#, D, E#, F#, G#

Ab, Bbb, Cb, Db, Ebb, F, Gb, Ab

A, Bb, C, D, Eb, F#, G, A

A#, B, C#, D#, E, F##, G#, A#

Bb, Cb, Db, Eb, Fb, G, Ab, Bb

B, C, D, E, F, G#, A, B

**Dorian b5 (2nd mode of harmonic major)**

Step pattern: 2, 1, 2, 1, 3, 1, 2

Ascending intervals: P1, M2, m3, P4, dim5, M6, m7, P8

C, D, Eb, F, Gb, A, Bb, C

C#, D#, E, F#, G, A#, B, C#

Db, Eb, Fb, Gb, Abb, Bb, Cb, Db

D, E, F, G, Ab, B, C, D

D#, E#, F#, G#, A, B#, C#, D#

Eb, F, Gb, Ab, Bbb, C, Db, Eb

E, F#, G, A, Bb, C#, D, E

F, G, Ab, Bb, Cb, D, Eb, F

F#, G#, A, B, C, D#, E, F#

Gb, Ab, Bbb, Cb, Dbb, Eb, Fb, Gb

G, A, Bb, C, Db, E, F, G

G#, A#, B, C#, D, E#, F#, G#

Ab, Bb, Cb, Db, Ebb, F, Gb, Ab

A, B, C, D, Eb, F#, G, A

A#, B#, C#, D#, E, F##, G#, A#

Bb, C, Db, Eb, Fb, G, Ab, Bb

B, C#, D, E, F, G#, A, B



**Dorian b9 (2nd mode of melodic minor)**

Step pattern: 1, 2, 2, 2, 2, 1, 2

Ascending intervals: P1, m2, m3, P4, P5, M6, m7, P8

C, Db, Eb, F, G, A, Bb, C

C#, D, E, F#, G#, A#, B, C#

Db, Ebb, Fb, Gb, Ab, Bb, Cb, Db

D, Eb, F, G, A, B, C, D

D#, E, F#, G#, A#, B#, C#, D#

Eb, Fb, Gb, Ab, Bb, C, Db, Eb

E, F, G, A, B, C#, D, E

F, Gb, Ab, Bb, C, D, Eb, F

F#, G, A, B, C#, D#, E, F#

Gb, Abb, Bbb, Cb, Db, Eb, Fb, Gb

G, Ab, Bb, C, D, E, F, G

G#, A, B, C#, D#, E#, F#, G#

Ab, Bbb, Cb, Db, Eb, F, Gb, Ab

A, Bb, C, D, E, F#, G, A

A#, B, C#, D#, E#, F##, G#, A#

Bb, Cb, Db, Eb, F, G, Ab, Bb

B, C, D, E, F#, G#, A, B

### Double Harmonic Major (5th mode harmonic minor)

Step pattern: 1, 3, 1, 2, 1, 3, 1

Ascending intervals: P1, m2, M3, P4, P5, m6, M7, P8

C, Db, E, F, G, Ab, B, C

C#, D, E#, F#, G#, A, B#, C#

Db, Ebb, F, Gb, Ab, Bbb, C, Db

D, Eb, F#, G, A, Bb, C#, D

D#, E, F##, G#, A#, B, C##, D#

Eb, Fb, G, Ab, Bb, Cb, D, Eb

E, F, G#, A, B, C, D#, E

F, Gb, A, Bb, C, Db, E, F

F#, G, A#, B, C#, D, E#, F#

Gb, Abb, Bb, Cb, Db, Ebb, F, Gb

G, Ab, B, C, D, Eb, F#, G

G#, A, B#, C#, D#, E, F##, G#

Ab, Bbb, C, Db, Eb, Fb, G, Ab

A, Bb, C#, D, E, F, G#, A

A#, B, C##, D#, E#, F#, G##, A#

Bb, Cb, D, Eb, F, Gb, A, Bb

B, C, D#, E, F#, G, A#, B

## Double Harmonic Minor

Step pattern: 2, 1, 3, 1, 1, 3, 1

Ascending intervals: P1, M2, m3, Aug4, P5, m6, M7, P8

C, D, Eb, F#, G, Ab, B, C

C#, D#, E, F##, G#, A, B#, C#

Db, Eb, Fb, G, Ab, Bbb, C, Db

D, E, F, G#, A, Bb, C#, D

D#, E#, F#, G##, A#, B, C##, D#

Eb, F, Gb, A, Bb, Cb, D, Eb

E, F#, G, A#, B, C, D#, E

F, G, Ab, B, C, Db, E, F

F#, G#, A, B#, C#, D, E#, F#

Gb, Ab, Bbb, C, Db, Ebb, F, Gb

G, A, Bb, C#, D, Eb, F#, G

G#, A#, B, C##, D#, E, F##, G#

Ab, Bb, Cb, D, Eb, Fb, G, Ab

A, B, C, D#, E, F, G#, A

A#, B#, C#, D##, E#, F#, G##, A#

Bb, C, Db, E, F, Gb, A, Bb

B, C#, D, E#, F#, G, A#, B

## Enigmatic Scale

Step pattern: 1, 3, 2, 2, 2, 1, 1

Ascending intervals: P1, m2, M3, Aug4, Aug5, Aug6, M7, P8

C, Db, E, F#, G#, A#, B, C

C#, D, E#, F##, G##, A##, B#, C#

Db, Ebb, F, G, A, B, C, Db

D, Eb, F#, G#, A#, B#, C#, D

D#, E, F##, G##, A##, B##, C##, D#

Eb, Fb, G, A, B, C#, D, Eb

E, F, G#, A#, B#, C##, D#, E

F, Gb, A, B, C#, D#, E, F

F#, G, A#, B#, C##, D##, E#, F#

Gb, Abb, Bb, C, D, E, F, Gb

G, Ab, B, C#, D#, E#, F#, G

G#, A, B#, C##, D##, E##, F##, G#

Ab, Bbb, C, D, E, F#, G, Ab

A, Bb, C#, D#, E#, F##, G#, A

A#, B, C##, D##, E##, F####, G##, A#

Bb, Cb, D, E, F#, G#, A, Bb

B, C, D#, E#, F##, G##, A#, B

## Ethiopian

Step pattern: 2, 2, 1, 2, 1, 3, 1

Ascending intervals: P1, M2, M3, P4, P5, m6, M7, P8

C, D, E, F, G, Ab, B, C

C#, D#, E#, F#, G#, A, B#, C#

Db, Eb, F, Gb, Ab, Bbb, C, Db

D, E, F#, G, A, Bb, C#, D

D#, E#, F##, G#, A#, B, C##, D#

Eb, F, G, Ab, Bb, Cb, D, Eb

E, F#, G#, A, B, C, D#, E

F, G, A, Bb, C, Db, E, F

F#, G#, A#, B, C#, D, E#, F#

Gb, Ab, Bb, Cb, Db, Ebb, F, Gb

G, A, B, C, D, Eb, F#, G

G#, A#, B#, C#, D#, E, F##, G#

Ab, Bb, C, Db, Eb, Fb, G, Ab

A, B, C#, D, E, F, G#, A

A#, B#, C##, D#, E#, F#, G##, A#

Bb, C, D, Eb, F, Gb, A, Bb

B, C#, D#, E, F#, G, A#, B

## Ganasamavarali

Step pattern: 1, 1, 3, 2, 1, 3, 1

Ascending intervals: P1, m2, M2, P4, P5, m6, M7, P8

C, Db, D, F, G, Ab, B, C

C#, D, D#, F#, G#, A, B#, C#

Db, Ebb, Eb, Gb, Ab, Bbb, C, Db

D, Eb, E, G, A, Bb, C#, D

D#, E, E#, G#, A#, B, C##, D#

Eb, Fb, F, Ab, Bb, Cb, D, Eb

E, F, F#, A, B, C, D#, E

F, Gb, G, Bb, C, Db, E, F

F#, G, G#, B, C#, D, E#, F#

Gb, Abb, Ab, Cb, Db, Ebb, F, Gb

G, Ab, A, C, D, Eb, F#, G

G#, A, A#, C#, D#, E, F##, G#

Ab, Bbb, Bb, Db, Eb, Fb, G, Ab

A, Bb, B, D, E, F, G#, A

A#, B, B#, D#, E#, F#, G##, A#

Bb, Cb, C, Eb, F, Gb, A, Bb

B, C, C#, E, F#, G, A#, B

**Girvani**

Step pattern: 1, 2, 3, 1, 1, 1, 3

Ascending intervals: P1, m2, m3, dim5, P5, m6, M6, P8

C, Db, Eb, Gb, G, Ab, A, C

C#, D, E, G, G#, A, A#, C#

Db, Ebb, Fb, Abb, Ab, Bbb, Bb, Db

D, Eb, F, Ab, A, Bb, B, D

D#, E, F#, A, A#, B, B#, D#

Eb, Fb, Gb, Bbb, Bb, Cb, C, Eb

E, F, G, Bb, B, C, C#, E

F, Gb, Ab, Cb, C, Db, D, F

F#, G, A, C, C#, D, D#, F#

Gb, Abb, Bbb, Dbb, Db, Ebb, Eb, Gb

G, Ab, Bb, Db, D, Eb, E, G

G#, A, B, D, D#, E, E#, G#

Ab, Bbb, Cb, Ebb, Eb, Fb, F, Ab

A, Bb, C, Eb, E, F, F#, A

A#, B, C#, E, E#, F#, F##, A#

Bb, Cb, Db, Fb, F, Gb, G, Bb

B, C, D, F, F#, G, G#, B

## Gypsy Heptatonic

Step pattern: 1, 3, 1, 2, 1, 1, 3

Ascending intervals: P1, m2, M3, P4, P5, m6, M6, P8

C, Db, E, F, G, Ab, A, C

C#, D, E#, F#, G#, A, A#, C#

Db, Ebb, F, Gb, Ab, Bbb, Bb, Db

D, Eb, F#, G, A, Bb, B, D

D#, E, F##, G#, A#, B, B#, D#

Eb, Fb, G, Ab, Bb, Cb, C, Eb

E, F, G#, A, B, C, C#, E

F, Gb, A, Bb, C, Db, D, F

F#, G, A#, B, C#, D, D#, F#

Gb, Abb, Bb, Cb, Db, Ebb, Eb, Gb

G, Ab, B, C, D, Eb, E, G

G#, A, B#, C#, D#, E, E#, G#

Ab, Bbb, C, Db, Eb, Fb, F, Ab

A, Bb, C#, D, E, F, F#, A

A#, B, C##, D#, E#, F#, F##, A#

Bb, Cb, D, Eb, F, Gb, G, Bb

B, C, D#, E, F#, G, G#, B



**Half-diminished (6th mode of melodic minor)**

Step pattern: 2, 1, 2, 1, 2, 2, 2

Ascending intervals: P1, M2, m3, P4, dim5, m6, m7, P8

C, D, Eb, F, Gb, Ab, Bb, C

C#, D#, E, F#, G, A, B, C#

Db, Eb, Fb, Gb, Abb, Bbb, Cb, Db

D, E, F, G, Ab, Bb, C, D

D#, E#, F#, G#, A, B, C#, D#

Eb, F, Gb, Ab, Bbb, Cb, Db, Eb

E, F#, G, A, Bb, C, D, E

F, G, Ab, Bb, Cb, Db, Eb, F

F#, G#, A, B, C, D, E, F#

Gb, Ab, Bbb, Cb, Dbb, Ebb, Fb, Gb

G, A, Bb, C, Db, Eb, F, G

G#, A#, B, C#, D, E, F#, G#

Ab, Bb, Cb, Db, Ebb, Fb, Gb, Ab

A, B, C, D, Eb, F, G, A

A#, B#, C#, D#, E, F#, G#, A#

Bb, C, Db, Eb, Fb, Gb, Ab, Bb

B, C#, D, E, F, G, A, B

## Hamsagiri

Step pattern: 3, 1, 2, 1, 3, 1, 1

Ascending intervals: P1, m3, M3, dim5, P5, m7, M7, P8

C, Eb, E, Gb, G, Bb, B, C

C#, E, E#, G, G#, B, B#, C#

Db, Fb, F, Abb, Ab, Cb, C, Db

D, F, F#, Ab, A, C, C#, D

D#, F#, F##, A, A#, C#, C##, D#

Eb, Gb, G, Bbb, Bb, Db, D, Eb

E, G, G#, Bb, B, D, D#, E

F, Ab, A, Cb, C, Eb, E, F

F#, A, A#, C, C#, E, E#, F#

Gb, Bbb, Bb, Dbb, Db, Fb, F, Gb

G, Bb, B, Db, D, F, F#, G

G#, B, B#, D, D#, F#, F##, G#

Ab, Cb, C, Ebb, Eb, Gb, G, Ab

A, C, C#, Eb, E, G, G#, A

A#, C#, C##, E, E#, G#, G##, A#

Bb, Db, D, Fb, F, Ab, A, Bb

B, D, D#, F, F#, A, A#, B

## Hamsalata

Step pattern: 2, 2, 2, 1, 1, 3, 1

Ascending intervals: P1, M2, M3, dim5, P5, m6, M7, P8

C, D, E, Gb, G, Ab, B, C

C#, D#, E#, G, G#, A, B#, C#

Db, Eb, F, Abb, Ab, Bbb, C, Db

D, E, F#, Ab, A, Bb, C#, D

D#, E#, F##, A, A#, B, C##, D#

Eb, F, G, Bbb, Bb, Cb, D, Eb

E, F#, G#, Bb, B, C, D#, E

F, G, A, Cb, C, Db, E, F

F#, G#, A#, C, C#, D, E#, F#

Gb, Ab, Bb, Dbb, Db, Ebb, F, Gb

G, A, B, Db, D, Eb, F#, G

G#, A#, B#, D, D#, E, F##, G#

Ab, Bb, C, Ebb, Eb, Fb, G, Ab

A, B, C#, Eb, E, F, G#, A

A#, B#, C##, E, E#, F#, G##, A#

Bb, C, D, Fb, F, Gb, A, Bb

B, C#, D#, F, F#, G, A#, B

**Harmonic Major (1st mode harmonic major)**

Step pattern: 2, 2, 1, 2, 1, 3, 1

Ascending intervals: P1, M2, M3, P4, P5, m6, M7, P8

C, D, E, F, G, Ab, B, C

C#, D#, E#, F#, G#, A, B#, C#

Db, Eb, F, Gb, Ab, Bbb, C, Db

D, E, F#, G, A, Bb, C#, D

D#, E#, F##, G#, A#, B, C##, D#

Eb, F, G, Ab, Bb, Cb, D, Eb

E, F#, G#, A, B, C, D#, E

F, G, A, Bb, C, Db, E, F

F#, G#, A#, B, C#, D, E#, F#

Gb, Ab, Bb, Cb, Db, Ebb, F, Gb

G, A, B, C, D, Eb, F#, G

G#, A#, B#, C#, D#, E, F##, G#

Ab, Bb, C, Db, Eb, Fb, G, Ab

A, B, C#, D, E, F, G#, A

A#, B#, C##, D#, E#, F#, G##, A#

Bb, C, D, Eb, F, Gb, A, Bb

B, C#, D#, E, F#, G, A#, B

### Harmonic Major (3rd mode harmonic minor)

Step pattern: 2, 2, 1, 2, 1, 3, 1

Ascending intervals: P1, M2, M3, P4, P5, m6, M7, P8

C, D, E, F, G, Ab, B, C

C#, D#, E#, F#, G#, A, B#, C#

Db, Eb, F, Gb, Ab, Bbb, C, Db

D, E, F#, G, A, Bb, C#, D

D#, E#, F##, G#, A#, B, C##, D#

Eb, F, G, Ab, Bb, Cb, D, Eb

E, F#, G#, A, B, C, D#, E

F, G, A, Bb, C, Db, E, F

F#, G#, A#, B, C#, D, E#, F#

Gb, Ab, Bb, Cb, Db, Ebb, F, Gb

G, A, B, C, D, Eb, F#, G

G#, A#, B#, C#, D#, E, F##, G#

Ab, Bb, C, Db, Eb, Fb, G, Ab

A, B, C#, D, E, F, G#, A

A#, B#, C##, D#, E#, F#, G##, A#

Bb, C, D, Eb, F, Gb, A, Bb

B, C#, D#, E, F#, G, A#, B

### Harmonic Minor (1st mode of harmonic minor)

Step pattern: 2, 1, 2, 2, 1, 3, 1

Ascending intervals: P1, M2, m3, P4, P5, m6, M7, P8

C, D, Eb, F, G, Ab, B, C

C#, D#, E, F#, G#, A, B#, C#

Db, Eb, Fb, Gb, Ab, Bbb, C, Db

D, E, F, G, A, Bb, C#, D

D#, E#, F#, G#, A#, B, C##, D#

Eb, F, Gb, Ab, Bb, Cb, D, Eb

E, F#, G, A, B, C, D#, E

F, G, Ab, Bb, C, Db, E, F

F#, G#, A, B, C#, D, E#, F#

Gb, Ab, Bbb, Cb, Db, Ebb, F, Gb

G, A, Bb, C, D, Eb, F#, G

G#, A#, B, C#, D#, E, F##, G#

Ab, Bb, Cb, Db, Eb, Fb, G, Ab

A, B, C, D, E, F, G#, A

A#, B#, C#, D#, E#, F#, G##, A#

Bb, C, Db, Eb, F, Gb, A, Bb

B, C#, D, E, F#, G, A#, B

## Hijaz

Step pattern: 1, 4, 1, 2, 1, 1, 2

Ascending intervals: P1, m2, P4, dim5, m6, M6, m7, P8

C, Db, F, Gb, Ab, A, Bb, C

C#, D, F#, G, A, A#, B, C#

Db, Ebb, Gb, Abb, Bbb, Bb, Cb, Db

D, Eb, G, Ab, Bb, B, C, D

D#, E, G#, A, B, B#, C#, D#

Eb, Fb, Ab, Bbb, Cb, C, Db, Eb

E, F, A, Bb, C, C#, D, E

F, Gb, Bb, Cb, Db, D, Eb, F

F#, G, B, C, D, D#, E, F#

Gb, Abb, Cb, Dbb, Ebb, Eb, Fb, Gb

G, Ab, C, Db, Eb, E, F, G

G#, A, C#, D, E, E#, F#, G#

Ab, Bbb, Db, Ebb, Fb, F, Gb, Ab

A, Bb, D, Eb, F, F#, G, A

A#, B, D#, E, F#, F##, G#, A#

Bb, Cb, Eb, Fb, Gb, G, Ab, Bb

B, C, E, F, G, G#, A, B

### Hindu scale (5th mode of melodic minor)

Step pattern: 2, 2, 1, 2, 1, 2, 2

Ascending intervals: P1, M2, M3, P4, P5, m6, m7, P8

C, D, E, F, G, Ab, Bb, C

C#, D#, E#, F#, G#, A, B, C#

Db, Eb, F, Gb, Ab, Bbb, Cb, Db

D, E, F#, G, A, Bb, C, D

D#, E#, F##, G#, A#, B, C#, D#

Eb, F, G, Ab, Bb, Cb, Db, Eb

E, F#, G#, A, B, C, D, E

F, G, A, Bb, C, Db, Eb, F

F#, G#, A#, B, C#, D, E, F#

Gb, Ab, Bb, Cb, Db, Ebb, Fb, Gb

G, A, B, C, D, Eb, F, G

G#, A#, B#, C#, D#, E, F#, G#

Ab, Bb, C, Db, Eb, Fb, Gb, Ab

A, B, C#, D, E, F, G, A

A#, B#, C##, D#, E#, F#, G#, A#

Bb, C, D, Eb, F, Gb, Ab, Bb

B, C#, D#, E, F#, G, A, B



## Houzam

Step pattern: 3, 1, 1, 2, 2, 2, 1

Ascending intervals: P1, Aug2, M3, P4, P5, M6, M7, P8

C, D#, E, F, G, A, B, C

C#, D##, E#, F#, G#, A#, B#, C#

Db, E, F, Gb, Ab, Bb, C, Db

D, E#, F#, G, A, B, C#, D

D#, E##, F##, G#, A#, B#, C##, D#

Eb, F#, G, Ab, Bb, C, D, Eb

E, F##, G#, A, B, C#, D#, E

F, G#, A, Bb, C, D, E, F

F#, G##, A#, B, C#, D#, E#, F#

Gb, A, Bb, Cb, Db, Eb, F, Gb

G, A#, B, C, D, E, F#, G

G#, A##, B#, C#, D#, E#, F##, G#

Ab, B, C, Db, Eb, F, G, Ab

A, B#, C#, D, E, F#, G#, A

A#, B##, C##, D#, E#, F##, G##, A#

Bb, C#, D, Eb, F, G, A, Bb

B, C##, D#, E, F#, G#, A#, B

## Hungarian Major

Step pattern: 3, 1, 2, 1, 2, 1, 2

Ascending intervals: P1, Aug2, M3, Aug4, P5, M6, m7, P8

C, D#, E, F#, G, A, Bb, C

C#, D##, E#, F##, G#, A#, B, C#

Db, E, F, G, Ab, Bb, Cb, Db

D, E#, F#, G#, A, B, C, D

D#, E##, F##, G##, A#, B#, C#, D#

Eb, F#, G, A, Bb, C, Db, Eb

E, F##, G#, A#, B, C#, D, E

F, G#, A, B, C, D, Eb, F

F#, G##, A#, B#, C#, D#, E, F#

Gb, A, Bb, C, Db, Eb, Fb, Gb

G, A#, B, C#, D, E, F, G

G#, A##, B#, C##, D#, E#, F#, G#

Ab, B, C, D, Eb, F, Gb, Ab

A, B#, C#, D#, E, F#, G, A

A#, B##, C##, D##, E#, F##, G#, A#

Bb, C#, D, E, F, G, Ab, Bb

B, C##, D#, E#, F#, G#, A, B

## Hungarian Minor

Step pattern: 2, 1, 3, 1, 1, 3, 1

Ascending intervals: P1, M2, m3, Aug4, P5, m6, M7, P8

C, D, Eb, F#, G, Ab, B, C

C#, D#, E, F##, G#, A, B#, C#

Db, Eb, Fb, G, Ab, Bbb, C, Db

D, E, F, G#, A, Bb, C#, D

D#, E#, F#, G##, A#, B, C##, D#

Eb, F, Gb, A, Bb, Cb, D, Eb

E, F#, G, A#, B, C, D#, E

F, G, Ab, B, C, Db, E, F

F#, G#, A, B#, C#, D, E#, F#

Gb, Ab, Bbb, C, Db, Ebb, F, Gb

G, A, Bb, C#, D, Eb, F#, G

G#, A#, B, C##, D#, E, F##, G#

Ab, Bb, Cb, D, Eb, Fb, G, Ab

A, B, C, D#, E, F, G#, A

A#, B#, C#, D##, E#, F#, G##, A#

Bb, C, Db, E, F, Gb, A, Bb

B, C#, D, E#, F#, G, A#, B

**Ionian #5**

Step pattern: 2, 2, 1, 3, 1, 2, 1

Ascending intervals: P1, M2, M3, P4, Aug5, M6, M7, P8

C, D, E, F, G#, A, B, C

C#, D#, E#, F#, G##, A#, B#, C#

Db, Eb, F, Gb, A, Bb, C, Db

D, E, F#, G, A#, B, C#, D

D#, E#, F##, G#, A##, B#, C##, D#

Eb, F, G, Ab, B, C, D, Eb

E, F#, G#, A, B#, C#, D#, E

F, G, A, Bb, C#, D, E, F

F#, G#, A#, B, C##, D#, E#, F#

Gb, Ab, Bb, Cb, D, Eb, F, Gb

G, A, B, C, D#, E, F#, G

G#, A#, B#, C#, D##, E#, F##, G#

Ab, Bb, C, Db, E, F, G, Ab

A, B, C#, D, E#, F#, G#, A

A#, B#, C##, D#, E##, F##, G##, A#

Bb, C, D, Eb, F#, G, A, Bb

B, C#, D#, E, F##, G#, A#, B

**Ionian (1st mode major)**

Step pattern: 2, 2, 1, 2, 2, 2, 1

Ascending intervals: P1, M2, M3, P4, P5, M6, M7, P8

C, D, E, F, G, A, B, C

C#, D#, E#, F#, G#, A#, B#, C#

Db, Eb, F, Gb, Ab, Bb, C, Db

D, E, F#, G, A, B, C#, D

D#, E#, F##, G#, A#, B#, C##, D#

Eb, F, G, Ab, Bb, C, D, Eb

E, F#, G#, A, B, C#, D#, E

F, G, A, Bb, C, D, E, F

F#, G#, A#, B, C#, D#, E#, F#

Gb, Ab, Bb, Cb, Db, Eb, F, Gb

G, A, B, C, D, E, F#, G

G#, A#, B#, C#, D#, E#, F##, G#

Ab, Bb, C, Db, Eb, F, G, Ab

A, B, C#, D, E, F#, G#, A

A#, B#, C##, D#, E#, F##, G##, A#

Bb, C, D, Eb, F, G, A, Bb

B, C#, D#, E, F#, G#, A#, B

### **Ionian Augmented (3rd mode of harmonic minor)**

Step pattern: 2, 2, 1, 3, 1, 2, 1

Ascending intervals: P1, M2, M3, P4, Aug5, M6, M7, P8

C, D, E, F, G#, A, B, C

C#, D#, E#, F#, G##, A#, B#, C#

Db, Eb, F, Gb, A, Bb, C, Db

D, E, F#, G, A#, B, C#, D

D#, E#, F##, G#, A##, B#, C##, D#

Eb, F, G, Ab, B, C, D, Eb

E, F#, G#, A, B#, C#, D#, E

F, G, A, Bb, C#, D, E, F

F#, G#, A#, B, C##, D#, E#, F#

Gb, Ab, Bb, Cb, D, Eb, F, Gb

G, A, B, C, D#, E, F#, G

G#, A#, B#, C#, D##, E#, F##, G#

Ab, Bb, C, Db, E, F, G, Ab

A, B, C#, D, E#, F#, G#, A

A#, B#, C##, D#, E##, F##, G##, A#

Bb, C, D, Eb, F#, G, A, Bb

B, C#, D#, E, F##, G#, A#, B

## Jalarnava

Step pattern: 1, 1, 4, 1, 1, 2, 2

Ascending intervals: P1, m2, M2, dim5, P5, m6, m7, P8

C, Db, D, Gb, G, Ab, Bb, C

C#, D, D#, G, G#, A, B, C#

Db, Ebb, Eb, Abb, Ab, Bbb, Cb, Db

D, Eb, E, Ab, A, Bb, C, D

D#, E, E#, A, A#, B, C#, D#

Eb, Fb, F, Bbb, Bb, Cb, Db, Eb

E, F, F#, Bb, B, C, D, E

F, Gb, G, Cb, C, Db, Eb, F

F#, G, G#, C, C#, D, E, F#

Gb, Abb, Ab, Dbb, Db, Ebb, Fb, Gb

G, Ab, A, Db, D, Eb, F, G

G#, A, A#, D, D#, E, F#, G#

Ab, Bbb, Bb, Ebb, Eb, Fb, Gb, Ab

A, Bb, B, Eb, E, F, G, A

A#, B, B#, E, E#, F#, G#, A#

Bb, Cb, C, Fb, F, Gb, Ab, Bb

B, C, C#, F, F#, G, A, B

**Jeths**

Step pattern: 2, 1, 2, 1, 3, 2, 1

Ascending intervals: P1, M2, m3, P4, dim5, M6, M7, P8

C, D, Eb, F, Gb, A, B, C

C#, D#, E, F#, G, A#, B#, C#

Db, Eb, Fb, Gb, Abb, Bb, C, Db

D, E, F, G, Ab, B, C#, D

D#, E#, F#, G#, A, B#, C##, D#

Eb, F, Gb, Ab, Bbb, C, D, Eb

E, F#, G, A, Bb, C#, D#, E

F, G, Ab, Bb, Cb, D, E, F

F#, G#, A, B, C, D#, E#, F#

Gb, Ab, Bbb, Cb, Dbb, Eb, F, Gb

G, A, Bb, C, Db, E, F#, G

G#, A#, B, C#, D, E#, F##, G#

Ab, Bb, Cb, Db, Ebb, F, G, Ab

A, B, C, D, Eb, F#, G#, A

A#, B#, C#, D#, E, F##, G##, A#

Bb, C, Db, Eb, Fb, G, A, Bb

B, C#, D, E, F, G#, A#, B



**Jeyasuddhamalavi**

Step pattern: 1, 3, 1, 2, 3, 1, 1

Ascending intervals: P1, m2, M3, P4, P5, m7, M7, P8

C, Db, E, F, G, Bb, B, C

C#, D, E#, F#, G#, B, B#, C#

Db, Ebb, F, Gb, Ab, Cb, C, Db

D, Eb, F#, G, A, C, C#, D

D#, E, F##, G#, A#, C#, C##, D#

Eb, Fb, G, Ab, Bb, Db, D, Eb

E, F, G#, A, B, D, D#, E

F, Gb, A, Bb, C, Eb, E, F

F#, G, A#, B, C#, E, E#, F#

Gb, Abb, Bb, Cb, Db, Fb, F, Gb

G, Ab, B, C, D, F, F#, G

G#, A, B#, C#, D#, F#, F##, G#

Ab, Bbb, C, Db, Eb, Gb, G, Ab

A, Bb, C#, D, E, G, G#, A

A#, B, C##, D#, E#, G#, G##, A#

Bb, Cb, D, Eb, F, Ab, A, Bb

B, C, D#, E, F#, A, A#, B

## Jhankara Bhramavi

Step pattern: 2, 1, 2, 2, 1, 1, 3

Ascending intervals: P1, M2, m3, P4, P5, m6, M6, P8

C, D, Eb, F, G, Ab, A, C

C#, D#, E, F#, G#, A, A#, C#

Db, Eb, Fb, Gb, Ab, Bbb, Bb, Db

D, E, F, G, A, Bb, B, D

D#, E#, F#, G#, A#, B, B#, D#

Eb, F, Gb, Ab, Bb, Cb, C, Eb

E, F#, G, A, B, C, C#, E

F, G, Ab, Bb, C, Db, D, F

F#, G#, A, B, C#, D, D#, F#

Gb, Ab, Bbb, Cb, Db, Ebb, Eb, Gb

G, A, Bb, C, D, Eb, E, G

G#, A#, B, C#, D#, E, E#, G#

Ab, Bb, Cb, Db, Eb, Fb, F, Ab

A, B, C, D, E, F, F#, A

A#, B#, C#, D#, E#, F#, F##, A#

Bb, C, Db, Eb, F, Gb, G, Bb

B, C#, D, E, F#, G, G#, B

**Jotismatti**

Step pattern: 3, 1, 2, 1, 1, 2, 2

Ascending intervals: P1, m3, M3, dim5, P5, m6, m7, P8

C, Eb, E, Gb, G, Ab, Bb, C

C#, E, E#, G, G#, A, B, C#

Db, Fb, F, Abb, Ab, Bbb, Cb, Db

D, F, F#, Ab, A, Bb, C, D

D#, F#, F##, A, A#, B, C#, D#

Eb, Gb, G, Bbb, Bb, Cb, Db, Eb

E, G, G#, Bb, B, C, D, E

F, Ab, A, Cb, C, Db, Eb, F

F#, A, A#, C, C#, D, E, F#

Gb, Bbb, Bb, Dbb, Db, Ebb, Fb, Gb

G, Bb, B, Db, D, Eb, F, G

G#, B, B#, D, D#, E, F#, G#

Ab, Cb, C, Ebb, Eb, Fb, Gb, Ab

A, C, C#, Eb, E, F, G, A

A#, C#, C##, E, E#, F#, G#, A#

Bb, Db, D, Fb, F, Gb, Ab, Bb

B, D, D#, F, F#, G, A, B

## Kalahamsa

Step pattern: 3, 1, 1, 2, 1, 1, 3

Ascending intervals: P1, m3, M3, P4, P5, m6, M6, P8

C, Eb, E, F, G, Ab, A, C

C#, E, E#, F#, G#, A, A#, C#

Db, Fb, F, Gb, Ab, Bbb, Bb, Db

D, F, F#, G, A, Bb, B, D

D#, F#, F##, G#, A#, B, B#, D#

Eb, Gb, G, Ab, Bb, Cb, C, Eb

E, G, G#, A, B, C, C#, E

F, Ab, A, Bb, C, Db, D, F

F#, A, A#, B, C#, D, D#, F#

Gb, Bbb, Bb, Cb, Db, Ebb, Eb, Gb

G, Bb, B, C, D, Eb, E, G

G#, B, B#, C#, D#, E, E#, G#

Ab, Cb, C, Db, Eb, Fb, F, Ab

A, C, C#, D, E, F, F#, A

A#, C#, C##, D#, E#, F#, F##, A#

Bb, Db, D, Eb, F, Gb, G, Bb

B, D, D#, E, F#, G, G#, B

**Keseri**

Step pattern: 2, 2, 1, 2, 1, 1, 3

Ascending intervals: P1, M2, M3, P4, P5, m6, M6, P8

C, D, E, F, G, Ab, A, C

C#, D#, E#, F#, G#, A, A#, C#

Db, Eb, F, Gb, Ab, Bbb, Bb, Db

D, E, F#, G, A, Bb, B, D

D#, E#, F##, G#, A#, B, B#, D#

Eb, F, G, Ab, Bb, Cb, C, Eb

E, F#, G#, A, B, C, C#, E

F, G, A, Bb, C, Db, D, F

F#, G#, A#, B, C#, D, D#, F#

Gb, Ab, Bb, Cb, Db, Ebb, Eb, Gb

G, A, B, C, D, Eb, E, G

G#, A#, B#, C#, D#, E, E#, G#

Ab, Bb, C, Db, Eb, Fb, F, Ab

A, B, C#, D, E, F, F#, A

A#, B#, C##, D#, E#, F#, F##, A#

Bb, C, D, Eb, F, Gb, G, Bb

B, C#, D#, E, F#, G, G#, B

## Kumbhini

Step pattern: 1, 1, 4, 1, 2, 2, 1

Ascending intervals: P1, m2, M2, dim5, P5, M6, M7, P8

C, Db, D, Gb, G, A, B, C

C#, D, D#, G, G#, A#, B#, C#

Db, Ebb, Eb, Abb, Ab, Bb, C, Db

D, Eb, E, Ab, A, B, C#, D

D#, E, E#, A, A#, B#, C##, D#

Eb, Fb, F, Bbb, Bb, C, D, Eb

E, F, F#, Bb, B, C#, D#, E

F, Gb, G, Cb, C, D, E, F

F#, G, G#, C, C#, D#, E#, F#

Gb, Abb, Ab, Dbb, Db, Eb, F, Gb

G, Ab, A, Db, D, E, F#, G

G#, A, A#, D, D#, E#, F##, G#

Ab, Bbb, Bb, Ebb, Eb, F, G, Ab

A, Bb, B, Eb, E, F#, G#, A

A#, B, B#, E, E#, F##, G##, A#

Bb, Cb, C, Fb, F, G, A, Bb

B, C, C#, F, F#, G#, A#, B

**Kuntala**

Step pattern: 2, 2, 2, 1, 1, 1, 3

Ascending intervals: P1, M2, M3, dim5, P5, m6, M6, P8

C, D, E, Gb, G, Ab, A, C

C#, D#, E#, G, G#, A, A#, C#

Db, Eb, F, Abb, Ab, Bbb, Bb, Db

D, E, F#, Ab, A, Bb, B, D

D#, E#, F##, A, A#, B, B#, D#

Eb, F, G, Bbb, Bb, Cb, C, Eb

E, F#, G#, Bb, B, C, C#, E

F, G, A, Cb, C, Db, D, F

F#, G#, A#, C, C#, D, D#, F#

Gb, Ab, Bb, Dbb, Db, Ebb, Eb, Gb

G, A, B, Db, D, Eb, E, G

G#, A#, B#, D, D#, E, E#, G#

Ab, Bb, C, Ebb, Eb, Fb, F, Ab

A, B, C#, Eb, E, F, F#, A

A#, B#, C##, E, E#, F#, F##, A#

Bb, C, D, Fb, F, Gb, G, Bb

B, C#, D#, F, F#, G, G#, B

**Kusumakaram**

Step pattern: 3, 1, 2, 1, 2, 2, 1

Ascending intervals: P1, m3, M3, dim5, P5, M6, M7, P8

C, Eb, E, Gb, G, A, B, C

C#, E, E#, G, G#, A#, B#, C#

Db, Fb, F, Abb, Ab, Bb, C, Db

D, F, F#, Ab, A, B, C#, D

D#, F#, F##, A, A#, B#, C##, D#

Eb, Gb, G, Bbb, Bb, C, D, Eb

E, G, G#, Bb, B, C#, D#, E

F, Ab, A, Cb, C, D, E, F

F#, A, A#, C, C#, D#, E#, F#

Gb, Bbb, Bb, Dbb, Db, Eb, F, Gb

G, Bb, B, Db, D, E, F#, G

G#, B, B#, D, D#, E#, F##, G#

Ab, Cb, C, Ebb, Eb, F, G, Ab

A, C, C#, Eb, E, F#, G#, A

A#, C#, C##, E, E#, F##, G##, A#

Bb, Db, D, Fb, F, G, A, Bb

B, D, D#, F, F#, G#, A#, B



**Locrian (7th mode of major)**

Step pattern: 1, 2, 2, 1, 2, 2, 2

Ascending intervals: P1, m2, m3, P4, dim5, m6, m7, P8

C, Db, Eb, F, Gb, Ab, Bb, C

C#, D, E, F#, G, A, B, C#

Db, Ebb, Fb, Gb, Abb, Bbb, Cb, Db

D, Eb, F, G, Ab, Bb, C, D

D#, E, F#, G#, A, B, C#, D#

Eb, Fb, Gb, Ab, Bbb, Cb, Db, Eb

E, F, G, A, Bb, C, D, E

F, Gb, Ab, Bb, Cb, Db, Eb, F

F#, G, A, B, C, D, E, F#

Gb, Abb, Bbb, Cb, Dbb, Ebb, Fb, Gb

G, Ab, Bb, C, Db, Eb, F, G

G#, A, B, C#, D, E, F#, G#

Ab, Bbb, Cb, Db, Ebb, Fb, Gb, Ab

A, Bb, C, D, Eb, F, G, A

A#, B, C#, D#, E, F#, G#, A#

Bb, Cb, Db, Eb, Fb, Gb, Ab, Bb

B, C, D, E, F, G, A, B

**Locrian bb7**

Step pattern: 1, 2, 2, 1, 2, 1, 3

Ascending intervals: P1, m2, m3, P4, dim5, m6, dim7, P8

C, Db, Eb, F, Gb, Ab, Bbb, C

C#, D, E, F#, G, A, Bb, C#

Db, Ebb, Fb, Gb, Abb, Bbb, Cbb, Db

D, Eb, F, G, Ab, Bb, Cb, D

D#, E, F#, G#, A, B, C, D#

Eb, Fb, Gb, Ab, Bbb, Cb, Dbb, Eb

E, F, G, A, Bb, C, Db, E

F, Gb, Ab, Bb, Cb, Db, Ebb, F

F#, G, A, B, C, D, Eb, F#

Gb, Abb, Bbb, Cb, Dbb, Ebb, Fbb, Gb

G, Ab, Bb, C, Db, Eb, Fb, G

G#, A, B, C#, D, E, F, G#

Ab, Bbb, Cb, Db, Ebb, Fb, Gbb, Ab

A, Bb, C, D, Eb, F, Gb, A

A#, B, C#, D#, E, F#, G, A#

Bb, Cb, Db, Eb, Fb, Gb, Abb, Bb

B, C, D, E, F, G, Ab, B

# **Locrian Diminished (7th mode of harmonic minor)**

Step pattern: 1, 2, 1, 2, 2, 1, 3

Ascending intervals: P1, m2, m3, dim4, dim5, m6, dim7, P8

C, Db, Eb, Fb, Gb, Ab, Bbb, C

C#, D, E, F, G, A, Bb, C#

Db, Ebb, Fb, Gbb, Abb, Bbb, Cbb, Db

D, Eb, F, Gb, Ab, Bb, Cb, D

D#, E, F#, G, A, B, C, D#

Eb, Fb, Gb, Abb, Bbb, Cb, Dbb, Eb

E, F, G, Ab, Bb, C, Db, E

F, Gb, Ab, Bbb, Cb, Db, Ebb, F

F#, G, A, Bb, C, D, Eb, F#

Gb, Abb, Bbb, Cbb, Dbb, Ebb, Fbb, Gb

G, Ab, Bb, Cb, Db, Eb, Fb, G

G#, A, B, C, D, E, F, G#

Ab, Bbb, Cb, Dbb, Ebb, Fb, Gbb, Ab

A, Bb, C, Db, Eb, F, Gb, A

A#, B, C#, D, E, F#, G, A#

Bb, Cb, Db, Ebb, Fb, Gb, Abb, Bb

B, C, D, Eb, F, G, Ab, B

**Locrian nat.2 (6th mode of melodic minor)**

Step pattern: 2, 1, 2, 1, 2, 2, 2

Ascending intervals: P1, M2, m3, P4, dim5, m6, m7, P8

C, D, Eb, F, Gb, Ab, Bb, C

C#, D#, E, F#, G, A, B, C#

Db, Eb, Fb, Gb, Abb, Bbb, Cb, Db

D, E, F, G, Ab, Bb, C, D

D#, E#, F#, G#, A, B, C#, D#

Eb, F, Gb, Ab, Bbb, Cb, Db, Eb

E, F#, G, A, Bb, C, D, E

F, G, Ab, Bb, Cb, Db, Eb, F

F#, G#, A, B, C, D, E, F#

Gb, Ab, Bbb, Cb, Dbb, Ebb, Fb, Gb

G, A, Bb, C, Db, Eb, F, G

G#, A#, B, C#, D, E, F#, G#

Ab, Bb, Cb, Db, Ebb, Fb, Gb, Ab

A, B, C, D, Eb, F, G, A

A#, B#, C#, D#, E, F#, G#, A#

Bb, C, Db, Eb, Fb, Gb, Ab, Bb

B, C#, D, E, F, G, A, B

**Locrian nat.2 b7**

Step pattern: 2, 1, 2, 1, 2, 1, 3

Ascending intervals: P1, M2, m3, P4, dim5, m6, dim7, P8

C, D, Eb, F, Gb, Ab, Bbb, C

C#, D#, E, F#, G, A, Bb, C#

Db, Eb, Fb, Gb, Abb, Bbb, Cbb, Db

D, E, F, G, Ab, Bb, Cb, D

D#, E#, F#, G#, A, B, C, D#

Eb, F, Gb, Ab, Bbb, Cb, Dbb, Eb

E, F#, G, A, Bb, C, Db, E

F, G, Ab, Bb, Cb, Db, Ebb, F

F#, G#, A, B, C, D, Eb, F#

Gb, Ab, Bbb, Cb, Dbb, Ebb, Fbb, Gb

G, A, Bb, C, Db, Eb, Fb, G

G#, A#, B, C#, D, E, F, G#

Ab, Bb, Cb, Db, Ebb, Fb, Gbb, Ab

A, B, C, D, Eb, F, Gb, A

A#, B#, C#, D#, E, F#, G, A#

Bb, C, Db, Eb, Fb, Gb, Abb, Bb

B, C#, D, E, F, G, Ab, B

**Locrian nat.6 (2nd mode of harmonic minor)**

Step pattern: 1, 2, 2, 1, 3, 1, 2

Ascending intervals: P1, m2, m3, P4, dim5, M6, m7, P8

C, Db, Eb, F, Gb, A, Bb, C

C#, D, E, F#, G, A#, B, C#

Db, Ebb, Fb, Gb, Abb, Bb, Cb, Db

D, Eb, F, G, Ab, B, C, D

D#, E, F#, G#, A, B#, C#, D#

Eb, Fb, Gb, Ab, Bbb, C, Db, Eb

E, F, G, A, Bb, C#, D, E

F, Gb, Ab, Bb, Cb, D, Eb, F

F#, G, A, B, C, D#, E, F#

Gb, Abb, Bbb, Cb, Dbb, Eb, Fb, Gb

G, Ab, Bb, C, Db, E, F, G

G#, A, B, C#, D, E#, F#, G#

Ab, Bbb, Cb, Db, Ebb, F, Gb, Ab

A, Bb, C, D, Eb, F#, G, A

A#, B, C#, D#, E, F##, G#, A#

Bb, Cb, Db, Eb, Fb, G, Ab, Bb

B, C, D, E, F, G#, A, B

**Lydian #2 (6th mode of harmonic minor)**

Step pattern: 3, 1, 2, 1, 2, 2, 1

Ascending intervals: P1, Aug2, M3, Aug4, P5, M6, M7, P8

C, D#, E, F#, G, A, B, C

C#, D##, E#, F##, G#, A#, B#, C#

Db, E, F, G, Ab, Bb, C, Db

D, E#, F#, G#, A, B, C#, D

D#, E##, F##, G##, A#, B#, C##, D#

Eb, F#, G, A, Bb, C, D, Eb

E, F##, G#, A#, B, C#, D#, E

F, G#, A, B, C, D, E, F

F#, G##, A#, B#, C#, D#, E#, F#

Gb, A, Bb, C, Db, Eb, F, Gb

G, A#, B, C#, D, E, F#, G

G#, A##, B#, C##, D#, E#, F##, G#

Ab, B, C, D, Eb, F, G, Ab

A, B#, C#, D#, E, F#, G#, A

A#, B##, C##, D##, E#, F##, G##, A#

Bb, C#, D, E, F, G, A, Bb

B, C##, D#, E#, F#, G#, A#, B

## Lydian (4th mode of major)

Step pattern: 2, 2, 2, 1, 2, 2, 1

Ascending intervals: P1, M2, M3, Aug4, P5, M6, M7, P8

C, D, E, F#, G, A, B, C

C#, D#, E#, F##, G#, A#, B#, C#

Db, Eb, F, G, Ab, Bb, C, Db

D, E, F#, G#, A, B, C#, D

D#, E#, F##, G##, A#, B#, C##, D#

Eb, F, G, A, Bb, C, D, Eb

E, F#, G#, A#, B, C#, D#, E

F, G, A, B, C, D, E, F

F#, G#, A#, B#, C#, D#, E#, F#

Gb, Ab, Bb, C, Db, Eb, F, Gb

G, A, B, C#, D, E, F#, G

G#, A#, B#, C##, D#, E#, F##, G#

Ab, Bb, C, D, Eb, F, G, Ab

A, B, C#, D#, E, F#, G#, A

A#, B#, C##, D##, E#, F##, G##, A#

Bb, C, D, E, F, G, A, Bb

B, C#, D#, E#, F#, G#, A#, B



## Lydian Augmented #2 (6th mode of harmonic major)

Step pattern: 3, 1, 2, 2, 1, 2, 1

Ascending intervals: P1, Aug2, M3, Aug4, Aug5, M6, M7, P8

C, D#, E, F#, G#, A, B, C

C#, D##, E#, F##, G##, A#, B#, C#

Db, E, F, G, A, Bb, C, Db

D, E#, F#, G#, A#, B, C#, D

D#, E##, F##, G##, A##, B#, C##, D#

Eb, F#, G, A, B, C, D, Eb

E, F##, G#, A#, B#, C#, D#, E

F, G#, A, B, C#, D, E, F

F#, G##, A#, B#, C##, D#, E#, F#

Gb, A, Bb, C, D, Eb, F, Gb

G, A#, B, C#, D#, E, F#, G

G#, A##, B#, C##, D##, E#, F##, G#

Ab, B, C, D, E, F, G, Ab

A, B#, C#, D#, E#, F#, G#, A

A#, B##, C##, D##, E##, F##, G##, A#

Bb, C#, D, E, F#, G, A, Bb

B, C##, D#, E#, F##, G#, A#, B

### Lydian Augmented (3rd mode of melodic minor)

Step pattern: 2, 2, 2, 2, 1, 2, 1

Ascending intervals: P1, M2, M3, Aug4, Aug5, M6, M7, P8

C, D, E, F#, G#, A, B, C

C#, D#, E#, F##, G##, A#, B#, C#

Db, Eb, F, G, A, Bb, C, Db

D, E, F#, G#, A#, B, C#, D

D#, E#, F##, G##, A##, B#, C##, D#

Eb, F, G, A, B, C, D, Eb

E, F#, G#, A#, B#, C#, D#, E

F, G, A, B, C#, D, E, F

F#, G#, A#, B#, C##, D#, E#, F#

Gb, Ab, Bb, C, D, Eb, F, Gb

G, A, B, C#, D#, E, F#, G

G#, A#, B#, C##, D##, E#, F##, G#

Ab, Bb, C, D, E, F, G, Ab

A, B, C#, D#, E#, F#, G#, A

A#, B#, C##, D##, E##, F##, G##, A#

Bb, C, D, E, F#, G, A, Bb

B, C#, D#, E#, F##, G#, A#, B

### Lydian Dominant (4th mode of melodic minor)

Step pattern: 2, 2, 2, 1, 2, 1, 2

Ascending intervals: P1, M2, M3, Aug4, P5, M6, m7, P8

C, D, E, F#, G, A, Bb, C

C#, D#, E#, F##, G#, A#, B, C#

Db, Eb, F, G, Ab, Bb, Cb, Db

D, E, F#, G#, A, B, C, D

D#, E#, F##, G##, A#, B#, C#, D#

Eb, F, G, A, Bb, C, Db, Eb

E, F#, G#, A#, B, C#, D, E

F, G, A, B, C, D, Eb, F

F#, G#, A#, B#, C#, D#, E, F#

Gb, Ab, Bb, C, Db, Eb, Fb, Gb

G, A, B, C#, D, E, F, G

G#, A#, B#, C##, D#, E#, F#, G#

Ab, Bb, C, D, Eb, F, Gb, Ab

A, B, C#, D#, E, F#, G, A

A#, B#, C##, D##, E#, F##, G#, A#

Bb, C, D, E, F, G, Ab, Bb

B, C#, D#, E#, F#, G#, A, B

## Lydian Dominant Augmented

Step pattern: 2, 2, 2, 2, 1, 1, 2

Ascending intervals: P1, M2, M3, Aug4, Aug5, M6, m7, P8

C, D, E, F#, G#, A, Bb, C

C#, D#, E#, F##, G##, A#, B, C#

Db, Eb, F, G, A, Bb, Cb, Db

D, E, F#, G#, A#, B, C, D

D#, E#, F##, G##, A##, B#, C#, D#

Eb, F, G, A, B, C, Db, Eb

E, F#, G#, A#, B#, C#, D, E

F, G, A, B, C#, D, Eb, F

F#, G#, A#, B#, C##, D#, E, F#

Gb, Ab, Bb, C, D, Eb, Fb, Gb

G, A, B, C#, D#, E, F, G

G#, A#, B#, C##, D##, E#, F#, G#

Ab, Bb, C, D, E, F, Gb, Ab

A, B, C#, D#, E#, F#, G, A

A#, B#, C##, D##, E##, F##, G#, A#

Bb, C, D, E, F#, G, Ab, Bb

B, C#, D#, E#, F##, G#, A, B

**Madhuri**

Step pattern: 4, 1, 2, 2, 1, 1, 1

Ascending intervals: P1, M3, P4, P5, M6, m7, M7, P8

C, E, F, G, A, Bb, B, C

C#, E#, F#, G#, A#, B, B#, C#

Db, F, Gb, Ab, Bb, Cb, C, Db

D, F#, G, A, B, C, C#, D

D#, F##, G#, A#, B#, C#, C##, D#

Eb, G, Ab, Bb, C, Db, D, Eb

E, G#, A, B, C#, D, D#, E

F, A, Bb, C, D, Eb, E, F

F#, A#, B, C#, D#, E, E#, F#

Gb, Bb, Cb, Db, Eb, Fb, F, Gb

G, B, C, D, E, F, F#, G

G#, B#, C#, D#, E#, F#, F##, G#

Ab, C, Db, Eb, F, Gb, G, Ab

A, C#, D, E, F#, G, G#, A

A#, C##, D#, E#, F##, G#, G##, A#

Bb, D, Eb, F, G, Ab, A, Bb

B, D#, E, F#, G#, A, A#, B

## Major

Step pattern: 2, 2, 1, 2, 2, 2, 1

Ascending intervals: P1, M2, M3, P4, P5, M6, M7, P8

C, D, E, F, G, A, B, C

C#, D#, E#, F#, G#, A#, B#, C#

Db, Eb, F, Gb, Ab, Bb, C, Db

D, E, F#, G, A, B, C#, D

D#, E#, F##, G#, A#, B#, C##, D#

Eb, F, G, Ab, Bb, C, D, Eb

E, F#, G#, A, B, C#, D#, E

F, G, A, Bb, C, D, E, F

F#, G#, A#, B, C#, D#, E#, F#

Gb, Ab, Bb, Cb, Db, Eb, F, Gb

G, A, B, C, D, E, F#, G

G#, A#, B#, C#, D#, E#, F##, G#

Ab, Bb, C, Db, Eb, F, G, Ab

A, B, C#, D, E, F#, G#, A

A#, B#, C##, D#, E#, F##, G##, A#

Bb, C, D, Eb, F, G, A, Bb

B, C#, D#, E, F#, G#, A#, B

## Major Gipsy

Step pattern: 1, 3, 1, 2, 1, 3, 1

Ascending intervals: P1, m2, M3, P4, P5, m6, M7, P8

C, Db, E, F, G, Ab, B, C

C#, D, E#, F#, G#, A, B#, C#

Db, Ebb, F, Gb, Ab, Bbb, C, Db

D, Eb, F#, G, A, Bb, C#, D

D#, E, F##, G#, A#, B, C##, D#

Eb, Fb, G, Ab, Bb, Cb, D, Eb

E, F, G#, A, B, C, D#, E

F, Gb, A, Bb, C, Db, E, F

F#, G, A#, B, C#, D, E#, F#

Gb, Abb, Bb, Cb, Db, Ebb, F, Gb

G, Ab, B, C, D, Eb, F#, G

G#, A, B#, C#, D#, E, F##, G#

Ab, Bbb, C, Db, Eb, Fb, G, Ab

A, Bb, C#, D, E, F, G#, A

A#, B, C##, D#, E#, F#, G##, A#

Bb, Cb, D, Eb, F, Gb, A, Bb

B, C, D#, E, F#, G, A#, B

## Major Locrian

Step pattern: 2, 2, 1, 1, 2, 2, 2

Ascending intervals: P1, M2, M3, P4, dim5, m6, m7, P8

C, D, E, F, Gb, Ab, Bb, C

C#, D#, E#, F#, G, A, B, C#

Db, Eb, F, Gb, Abb, Bbb, Cb, Db

D, E, F#, G, Ab, Bb, C, D

D#, E#, F##, G#, A, B, C#, D#

Eb, F, G, Ab, Bbb, Cb, Db, Eb

E, F#, G#, A, Bb, C, D, E

F, G, A, Bb, Cb, Db, Eb, F

F#, G#, A#, B, C, D, E, F#

Gb, Ab, Bb, Cb, Dbb, Ebb, Fb, Gb

G, A, B, C, Db, Eb, F, G

G#, A#, B#, C#, D, E, F#, G#

Ab, Bb, C, Db, Ebb, Fb, Gb, Ab

A, B, C#, D, Eb, F, G, A

A#, B#, C##, D#, E, F#, G#, A#

Bb, C, D, Eb, Fb, Gb, Ab, Bb

B, C#, D#, E, F, G, A, B



**Malini**

Step pattern: 1, 2, 2, 2, 1, 1, 3

Ascending intervals: P1, m2, m3, P4, P5, m6, M6, P8

C, Db, Eb, F, G, Ab, A, C

C#, D, E, F#, G#, A, A#, C#

Db, Ebb, Fb, Gb, Ab, Bbb, Bb, Db

D, Eb, F, G, A, Bb, B, D

D#, E, F#, G#, A#, B, B#, D#

Eb, Fb, Gb, Ab, Bb, Cb, C, Eb

E, F, G, A, B, C, C#, E

F, Gb, Ab, Bb, C, Db, D, F

F#, G, A, B, C#, D, D#, F#

Gb, Abb, Bbb, Cb, Db, Ebb, Eb, Gb

G, Ab, Bb, C, D, Eb, E, G

G#, A, B, C#, D#, E, E#, G#

Ab, Bbb, Cb, Db, Eb, Fb, F, Ab

A, Bb, C, D, E, F, F#, A

A#, B, C#, D#, E#, F#, F##, A#

Bb, Cb, Db, Eb, F, Gb, G, Bb

B, C, D, E, F#, G, G#, B

**Manoranjani**

Step pattern: 1, 1, 3, 2, 2, 2, 1

Ascending intervals: P1, m2, M2, P4, P5, M6, M7, P8

C, Db, D, F, G, A, B, C

C#, D, D#, F#, G#, A#, B#, C#

Db, Ebb, Eb, Gb, Ab, Bb, C, Db

D, Eb, E, G, A, B, C#, D

D#, E, E#, G#, A#, B#, C##, D#

Eb, Fb, F, Ab, Bb, C, D, Eb

E, F, F#, A, B, C#, D#, E

F, Gb, G, Bb, C, D, E, F

F#, G, G#, B, C#, D#, E#, F#

Gb, Abb, Ab, Cb, Db, Eb, F, Gb

G, Ab, A, C, D, E, F#, G

G#, A, A#, C#, D#, E#, F##, G#

Ab, Bbb, Bb, Db, Eb, F, G, Ab

A, Bb, B, D, E, F#, G#, A

A#, B, B#, D#, E#, F##, G##, A#

Bb, Cb, C, Eb, F, G, A, Bb

B, C, C#, E, F#, G#, A#, B

## Mantra of Will Scale

Step pattern: 1, 3, 2, 1, 1, 1, 3

Ascending intervals: P1, m2, M3, dim5, P5, m6, M6, P8

C, Db, E, Gb, G, Ab, A, C

C#, D, E#, G, G#, A, A#, C#

Db, Ebb, F, Abb, Ab, Bbb, Bb, Db

D, Eb, F#, Ab, A, Bb, B, D

D#, E, F##, A, A#, B, B#, D#

Eb, Fb, G, Bbb, Bb, Cb, C, Eb

E, F, G#, Bb, B, C, C#, E

F, Gb, A, Cb, C, Db, D, F

F#, G, A#, C, C#, D, D#, F#

Gb, Abb, Bb, Dbb, Db, Ebb, Eb, Gb

G, Ab, B, Db, D, Eb, E, G

G#, A, B#, D, D#, E, E#, G#

Ab, Bbb, C, Ebb, Eb, Fb, F, Ab

A, Bb, C#, Eb, E, F, F#, A

A#, B, C##, E, E#, F#, F##, A#

Bb, Cb, D, Fb, F, Gb, G, Bb

B, C, D#, F, F#, G, G#, B

## Maqam Nahawand

Step pattern: 2, 1, 4, 1, 2, 1, 1

Ascending intervals: P1, M2, m3, P5, m6, m7, M7, P8

C, D, Eb, G, Ab, Bb, B, C

C#, D#, E, G#, A, B, B#, C#

Db, Eb, Fb, Ab, Bbb, Cb, C, Db

D, E, F, A, Bb, C, C#, D

D#, E#, F#, A#, B, C#, C##, D#

Eb, F, Gb, Bb, Cb, Db, D, Eb

E, F#, G, B, C, D, D#, E

F, G, Ab, C, Db, Eb, E, F

F#, G#, A, C#, D, E, E#, F#

Gb, Ab, Bbb, Db, Ebb, Fb, F, Gb

G, A, Bb, D, Eb, F, F#, G

G#, A#, B, D#, E, F#, F##, G#

Ab, Bb, Cb, Eb, Fb, Gb, G, Ab

A, B, C, E, F, G, G#, A

A#, B#, C#, E#, F#, G#, G##, A#

Bb, C, Db, F, Gb, Ab, A, Bb

B, C#, D, F#, G, A, A#, B

## Mecakalyani

Step pattern: 2, 2, 2, 1, 2, 2, 1

Ascending intervals: P1, M2, M3, dim5, P5, M6, M7, P8

C, D, E, Gb, G, A, B, C

C#, D#, E#, G, G#, A#, B#, C#

Db, Eb, F, Abb, Ab, Bb, C, Db

D, E, F#, Ab, A, B, C#, D

D#, E#, F##, A, A#, B#, C##, D#

Eb, F, G, Bbb, Bb, C, D, Eb

E, F#, G#, Bb, B, C#, D#, E

F, G, A, Cb, C, D, E, F

F#, G#, A#, C, C#, D#, E#, F#

Gb, Ab, Bb, Dbb, Db, Eb, F, Gb

G, A, B, Db, D, E, F#, G

G#, A#, B#, D, D#, E#, F##, G#

Ab, Bb, C, Ebb, Eb, F, G, Ab

A, B, C#, Eb, E, F#, G#, A

A#, B#, C##, E, E#, F##, G##, A#

Bb, C, D, Fb, F, G, A, Bb

B, C#, D#, F, F#, G#, A#, B

## Melodic Major

Step pattern: 2, 2, 1, 2, 1, 2, 2

Ascending intervals: P1, M2, M3, P4, P5, m6, m7, P8

C, D, E, F, G, Ab, Bb, C

C#, D#, E#, F#, G#, A, B, C#

Db, Eb, F, Gb, Ab, Bbb, Cb, Db

D, E, F#, G, A, Bb, C, D

D#, E#, F##, G#, A#, B, C#, D#

Eb, F, G, Ab, Bb, Cb, Db, Eb

E, F#, G#, A, B, C, D, E

F, G, A, Bb, C, Db, Eb, F

F#, G#, A#, B, C#, D, E, F#

Gb, Ab, Bb, Cb, Db, Ebb, Fb, Gb

G, A, B, C, D, Eb, F, G

G#, A#, B#, C#, D#, E, F#, G#

Ab, Bb, C, Db, Eb, Fb, Gb, Ab

A, B, C#, D, E, F, G, A

A#, B#, C##, D#, E#, F#, G#, A#

Bb, C, D, Eb, F, Gb, Ab, Bb

B, C#, D#, E, F#, G, A, B

**Melodic Minor #11**

Step pattern: 2, 1, 3, 1, 2, 2, 1

Ascending intervals: P1, M2, m3, Aug4, P5, M6, M7, P8

C, D, Eb, F#, G, A, B, C

C#, D#, E, F##, G#, A#, B#, C#

Db, Eb, Fb, G, Ab, Bb, C, Db

D, E, F, G#, A, B, C#, D

D#, E#, F#, G##, A#, B#, C##, D#

Eb, F, Gb, A, Bb, C, D, Eb

E, F#, G, A#, B, C#, D#, E

F, G, Ab, B, C, D, E, F

F#, G#, A, B#, C#, D#, E#, F#

Gb, Ab, Bbb, C, Db, Eb, F, Gb

G, A, Bb, C#, D, E, F#, G

G#, A#, B, C##, D#, E#, F##, G#

Ab, Bb, Cb, D, Eb, F, G, Ab

A, B, C, D#, E, F#, G#, A

A#, B#, C#, D##, E#, F##, G##, A#

Bb, C, Db, E, F, G, A, Bb

B, C#, D, E#, F#, G#, A#, B

**Melodic Minor (1st mode of melodic minor)**

Step pattern: 2, 1, 2, 2, 2, 2, 1

Ascending intervals: P1, M2, m3, P4, P5, M6, M7, P8

C, D, Eb, F, G, A, B, C

C#, D#, E, F#, G#, A#, B#, C#

Db, Eb, Fb, Gb, Ab, Bb, C, Db

D, E, F, G, A, B, C#, D

D#, E#, F#, G#, A#, B#, C##, D#

Eb, F, Gb, Ab, Bb, C, D, Eb

E, F#, G, A, B, C#, D#, E

F, G, Ab, Bb, C, D, E, F

F#, G#, A, B, C#, D#, E#, F#

Gb, Ab, Bbb, Cb, Db, Eb, F, Gb

G, A, Bb, C, D, E, F#, G

G#, A#, B, C#, D#, E#, F##, G#

Ab, Bb, Cb, Db, Eb, F, G, Ab

A, B, C, D, E, F#, G#, A

A#, B#, C#, D#, E#, F##, G##, A#

Bb, C, Db, Eb, F, G, A, Bb

B, C#, D, E, F#, G#, A#, B



**Minor**

Step pattern: 2, 1, 2, 2, 1, 2, 2

Ascending intervals: P1, M2, m3, P4, P5, m6, m7, P8

C, D, Eb, F, G, Ab, Bb, C

C#, D#, E, F#, G#, A, B, C#

Db, Eb, Fb, Gb, Ab, Bbb, Cb, Db

D, E, F, G, A, Bb, C, D

D#, E#, F#, G#, A#, B, C#, D#

Eb, F, Gb, Ab, Bb, Cb, Db, Eb

E, F#, G, A, B, C, D, E

F, G, Ab, Bb, C, Db, Eb, F

F#, G#, A, B, C#, D, E, F#

Gb, Ab, Bbb, Cb, Db, Ebb, Fb, Gb

G, A, Bb, C, D, Eb, F, G

G#, A#, B, C#, D#, E, F#, G#

Ab, Bb, Cb, Db, Eb, Fb, Gb, Ab

A, B, C, D, E, F, G, A

A#, B#, C#, D#, E#, F#, G#, A#

Bb, C, Db, Eb, F, Gb, Ab, Bb

B, C#, D, E, F#, G, A, B

## Minor Gipsy

Step pattern: 2, 1, 3, 1, 1, 3, 1

Ascending intervals: P1, M2, m3, Aug4, P5, m6, M7, P8

C, D, Eb, F#, G, Ab, B, C

C#, D#, E, F##, G#, A, B#, C#

Db, Eb, Fb, G, Ab, Bbb, C, Db

D, E, F, G#, A, Bb, C#, D

D#, E#, F#, G##, A#, B, C##, D#

Eb, F, Gb, A, Bb, Cb, D, Eb

E, F#, G, A#, B, C, D#, E

F, G, Ab, B, C, Db, E, F

F#, G#, A, B#, C#, D, E#, F#

Gb, Ab, Bbb, C, Db, Ebb, F, Gb

G, A, Bb, C#, D, Eb, F#, G

G#, A#, B, C##, D#, E, F##, G#

Ab, Bb, Cb, D, Eb, Fb, G, Ab

A, B, C, D#, E, F, G#, A

A#, B#, C#, D##, E#, F#, G##, A#

Bb, C, Db, E, F, Gb, A, Bb

B, C#, D, E#, F#, G, A#, B

**Mixolydian (5th mode of major)**

Step pattern: 2, 2, 1, 2, 2, 1, 2

Ascending intervals: P1, M2, M3, P4, P5, M6, m7, P8

C, D, E, F, G, A, B $\flat$ , C

C $\sharp$ , D $\sharp$ , E $\sharp$ , F $\sharp$ , G $\sharp$ , A $\sharp$ , B, C $\sharp$

D $\flat$ , E $\flat$ , F, G $\flat$ , A $\flat$ , B $\flat$ , C $\flat$ , D $\flat$

D, E, F $\sharp$ , G, A, B, C, D

D $\sharp$ , E $\sharp$ , F $\sharp\sharp$ , G $\sharp$ , A $\sharp$ , B $\sharp$ , C $\sharp$ , D $\sharp$

E $\flat$ , F, G, A $\flat$ , B $\flat$ , C, D $\flat$ , E $\flat$

E, F $\sharp$ , G $\sharp$ , A, B, C $\sharp$ , D, E

F, G, A, B $\flat$ , C, D, E $\flat$ , F

F $\sharp$ , G $\sharp$ , A $\sharp$ , B, C $\sharp$ , D $\sharp$ , E, F $\sharp$

G $\flat$ , A $\flat$ , B $\flat$ , C $\flat$ , D $\flat$ , E $\flat$ , F $\flat$ , G $\flat$

G, A, B, C, D, E, F, G

G $\sharp$ , A $\sharp$ , B $\sharp$ , C $\sharp$ , D $\sharp$ , E $\sharp$ , F $\sharp$ , G $\sharp$

A $\flat$ , B $\flat$ , C, D $\flat$ , E $\flat$ , F, G $\flat$ , A $\flat$

A, B, C $\sharp$ , D, E, F $\sharp$ , G, A

A $\sharp$ , B $\sharp$ , C $\sharp\sharp$ , D $\sharp$ , E $\sharp$ , F $\sharp\sharp$ , G $\sharp$ , A $\sharp$

B $\flat$ , C, D, E $\flat$ , F, G, A $\flat$ , B $\flat$

B, C $\sharp$ , D $\sharp$ , E, F $\sharp$ , G $\sharp$ , A, B

### Mixolydian b2 (5th mode of harmonic major)

Step pattern: 1, 3, 1, 2, 2, 1, 2

Ascending intervals: P1, m2, M3, P4, P5, M6, m7, P8

C, Db, E, F, G, A, Bb, C

C#, D, E#, F#, G#, A#, B, C#

Db, Ebb, F, Gb, Ab, Bb, Cb, Db

D, Eb, F#, G, A, B, C, D

D#, E, F##, G#, A#, B#, C#, D#

Eb, Fb, G, Ab, Bb, C, Db, Eb

E, F, G#, A, B, C#, D, E

F, Gb, A, Bb, C, D, Eb, F

F#, G, A#, B, C#, D#, E, F#

Gb, Abb, Bb, Cb, Db, Eb, Fb, Gb

G, Ab, B, C, D, E, F, G

G#, A, B#, C#, D#, E#, F#, G#

Ab, Bbb, C, Db, Eb, F, Gb, Ab

A, Bb, C#, D, E, F#, G, A

A#, B, C##, D#, E#, F##, G#, A#

Bb, Cb, D, Eb, F, G, Ab, Bb

B, C, D#, E, F#, G#, A, B

**Mixolydian b6 (5th mode of melodic minor)**

Step pattern: 2, 2, 1, 2, 1, 2, 2

Ascending intervals: P1, M2, M3, P4, P5, m6, m7, P8

C, D, E, F, G, Ab, Bb, C

C#, D#, E#, F#, G#, A, B, C#

Db, Eb, F, Gb, Ab, Bbb, Cb, Db

D, E, F#, G, A, Bb, C, D

D#, E#, F##, G#, A#, B, C#, D#

Eb, F, G, Ab, Bb, Cb, Db, Eb

E, F#, G#, A, B, C, D, E

F, G, A, Bb, C, Db, Eb, F

F#, G#, A#, B, C#, D, E, F#

Gb, Ab, Bb, Cb, Db, Ebb, Fb, Gb

G, A, B, C, D, Eb, F, G

G#, A#, B#, C#, D#, E, F#, G#

Ab, Bb, C, Db, Eb, Fb, Gb, Ab

A, B, C#, D, E, F, G, A

A#, B#, C##, D#, E#, F#, G#, A#

Bb, C, D, Eb, F, Gb, Ab, Bb

B, C#, D#, E, F#, G, A, B

### Nabhomani (heptatonic)

Step pattern: 1, 1, 4, 1, 2, 1, 2

Ascending intervals: P1, m2, M2, dim5, P5, M6, m7, P8

C, Db, D, Gb, G, A, Bb, C

C#, D, D#, G, G#, A#, B, C#

Db, Ebb, Eb, Abb, Ab, Bb, Cb, Db

D, Eb, E, Ab, A, B, C, D

D#, E, E#, A, A#, B#, C#, D#

Eb, Fb, F, Bbb, Bb, C, Db, Eb

E, F, F#, Bb, B, C#, D, E

F, Gb, G, Cb, C, D, Eb, F

F#, G, G#, C, C#, D#, E, F#

Gb, Abb, Ab, Dbb, Db, Eb, Fb, Gb

G, Ab, A, Db, D, E, F, G

G#, A, A#, D, D#, E#, F#, G#

Ab, Bbb, Bb, Ebb, Eb, F, Gb, Ab

A, Bb, B, Eb, E, F#, G, A

A#, B, B#, E, E#, F##, G#, A#

Bb, Cb, C, Fb, F, G, Ab, Bb

B, C, C#, F, F#, G#, A, B

## Naganandini

Step pattern: 2, 2, 1, 2, 3, 1, 1

Ascending intervals: P1, M2, M3, P4, P5, Aug6, M7, P8

C, D, E, F, G, A#, B, C

C#, D#, E#, F#, G#, A###, B#, C#

Db, Eb, F, Gb, Ab, B, C, Db

D, E, F#, G, A, B#, C#, D

D#, E#, F###, G#, A#, B###, C###, D#

Eb, F, G, Ab, Bb, C#, D, Eb

E, F#, G#, A, B, C###, D#, E

F, G, A, Bb, C, D#, E, F

F#, G#, A#, B, C#, D###, E#, F#

Gb, Ab, Bb, Cb, Db, E, F, Gb

G, A, B, C, D, E#, F#, G

G#, A#, B#, C#, D#, E###, F###, G#

Ab, Bb, C, Db, Eb, F#, G, Ab

A, B, C#, D, E, F###, G#, A

A#, B#, C###, D#, E#, F####, G###, A#

Bb, C, D, Eb, F, G#, A, Bb

B, C#, D#, E, F#, G###, A#, B

**Nasamani**

Step pattern: 3, 1, 2, 1, 2, 1, 2

Ascending intervals: P1, m3, M3, dim5, P5, M6, m7, P8

C, Eb, E, Gb, G, A, Bb, C

C#, E, E#, G, G#, A#, B, C#

Db, Fb, F, Abb, Ab, Bb, Cb, Db

D, F, F#, Ab, A, B, C, D

D#, F#, F##, A, A#, B#, C#, D#

Eb, Gb, G, Bbb, Bb, C, Db, Eb

E, G, G#, Bb, B, C#, D, E

F, Ab, A, Cb, C, D, Eb, F

F#, A, A#, C, C#, D#, E, F#

Gb, Bbb, Bb, Dbb, Db, Eb, Fb, Gb

G, Bb, B, Db, D, E, F, G

G#, B, B#, D, D#, E#, F#, G#

Ab, Cb, C, Ebb, Eb, F, Gb, Ab

A, C, C#, Eb, E, F#, G, A

A#, C#, C##, E, E#, F##, G#, A#

Bb, Db, D, Fb, F, G, Ab, Bb

B, D, D#, F, F#, G#, A, B



## Neapolitan m6

Step pattern: 1, 2, 2, 2, 1, 3, 1

Ascending intervals: P1, m2, m3, P4, P5, m6, M7, P8

C, Db, Eb, F, G, Ab, B, C

C#, D, E, F#, G#, A, B#, C#

Db, Ebb, Fb, Gb, Ab, Bbb, C, Db

D, Eb, F, G, A, Bb, C#, D

D#, E, F#, G#, A#, B, C##, D#

Eb, Fb, Gb, Ab, Bb, Cb, D, Eb

E, F, G, A, B, C, D#, E

F, Gb, Ab, Bb, C, Db, E, F

F#, G, A, B, C#, D, E#, F#

Gb, Abb, Bbb, Cb, Db, Ebb, F, Gb

G, Ab, Bb, C, D, Eb, F#, G

G#, A, B, C#, D#, E, F##, G#

Ab, Bbb, Cb, Db, Eb, Fb, G, Ab

A, Bb, C, D, E, F, G#, A

A#, B, C#, D#, E#, F#, G##, A#

Bb, Cb, Db, Eb, F, Gb, A, Bb

B, C, D, E, F#, G, A#, B

## Neapolitan M6

Step pattern: 1, 2, 2, 2, 2, 2, 1

Ascending intervals: P1, m2, m3, P4, P5, M6, M7, P8

C, Db, Eb, F, G, A, B, C

C#, D, E, F#, G#, A#, B#, C#

Db, Ebb, Fb, Gb, Ab, Bb, C, Db

D, Eb, F, G, A, B, C#, D

D#, E, F#, G#, A#, B#, C##, D#

Eb, Fb, Gb, Ab, Bb, C, D, Eb

E, F, G, A, B, C#, D#, E

F, Gb, Ab, Bb, C, D, E, F

F#, G, A, B, C#, D#, E#, F#

Gb, Abb, Bbb, Cb, Db, Eb, F, Gb

G, Ab, Bb, C, D, E, F#, G

G#, A, B, C#, D#, E#, F##, G#

Ab, Bbb, Cb, Db, Eb, F, G, Ab

A, Bb, C, D, E, F#, G#, A

A#, B, C#, D#, E#, F##, G##, A#

Bb, Cb, Db, Eb, F, G, A, Bb

B, C, D, E, F#, G#, A#, B

**Niaventi**

Step pattern: 2, 1, 3, 1, 1, 3, 1

Ascending intervals: P1, M2, m3, dim5, P5, m6, M7, P8

C, D, Eb, Gb, G, Ab, B, C

C#, D#, E, G, G#, A, B#, C#

Db, Eb, Fb, Abb, Ab, Bbb, C, Db

D, E, F, Ab, A, Bb, C#, D

D#, E#, F#, A, A#, B, C##, D#

Eb, F, Gb, Bbb, Bb, Cb, D, Eb

E, F#, G, Bb, B, C, D#, E

F, G, Ab, Cb, C, Db, E, F

F#, G#, A, C, C#, D, E#, F#

Gb, Ab, Bbb, Dbb, Db, Ebb, F, Gb

G, A, Bb, Db, D, Eb, F#, G

G#, A#, B, D, D#, E, F##, G#

Ab, Bb, Cb, Ebb, Eb, Fb, G, Ab

A, B, C, Eb, E, F, G#, A

A#, B#, C#, E, E#, F#, G##, A#

Bb, C, Db, Fb, F, Gb, A, Bb

B, C#, D, F, F#, G, A#, B

## Nisada

Step pattern: 2, 1, 3, 1, 3, 1, 1

Ascending intervals: P1, M2, m3, dim5, P5, m7, M7, P8

C, D, Eb, Gb, G, Bb, B, C

C#, D#, E, G, G#, B, B#, C#

Db, Eb, Fb, Abb, Ab, Cb, C, Db

D, E, F, Ab, A, C, C#, D

D#, E#, F#, A, A#, C#, C##, D#

Eb, F, Gb, Bbb, Bb, Db, D, Eb

E, F#, G, Bb, B, D, D#, E

F, G, Ab, Cb, C, Eb, E, F

F#, G#, A, C, C#, E, E#, F#

Gb, Ab, Bbb, Dbb, Db, Fb, F, Gb

G, A, Bb, Db, D, F, F#, G

G#, A#, B, D, D#, F#, F##, G#

Ab, Bb, Cb, Ebb, Eb, Gb, G, Ab

A, B, C, Eb, E, G, G#, A

A#, B#, C#, E, E#, G#, G##, A#

Bb, C, Db, Fb, F, Ab, A, Bb

B, C#, D, F, F#, A, A#, B

**Nohkan**

Step pattern: 2, 3, 1, 2, 1, 2, 1

Ascending intervals: P1, M2, P4, dim5, m6, M6, M7, P8

C, D, F, Gb, Ab, A, B, C

C#, D#, F#, G, A, A#, B#, C#

Db, Eb, Gb, Abb, Bbb, Bb, C, Db

D, E, G, Ab, Bb, B, C#, D

D#, E#, G#, A, B, B#, C##, D#

Eb, F, Ab, Bbb, Cb, C, D, Eb

E, F#, A, Bb, C, C#, D#, E

F, G, Bb, Cb, Db, D, E, F

F#, G#, B, C, D, D#, E#, F#

Gb, Ab, Cb, Dbb, Ebb, Eb, F, Gb

G, A, C, Db, Eb, E, F#, G

G#, A#, C#, D, E, E#, F##, G#

Ab, Bb, Db, Ebb, Fb, F, G, Ab

A, B, D, Eb, F, F#, G#, A

A#, B#, D#, E, F#, F##, G##, A#

Bb, C, Eb, Fb, Gb, G, A, Bb

B, C#, E, F, G, G#, A#, B

**None**

Step pattern: 3, 1, 1, 2, 3, 1, 1

Ascending intervals: P1, m3, M3, P4, P5, m7, M7, P8

C, Eb, E, F, G, Bb, B, C

C#, E, E#, F#, G#, B, B#, C#

Db, Fb, F, Gb, Ab, Cb, C, Db

D, F, F#, G, A, C, C#, D

D#, F#, F##, G#, A#, C#, C##, D#

Eb, Gb, G, Ab, Bb, Db, D, Eb

E, G, G#, A, B, D, D#, E

F, Ab, A, Bb, C, Eb, E, F

F#, A, A#, B, C#, E, E#, F#

Gb, Bbb, Bb, Cb, Db, Fb, F, Gb

G, Bb, B, C, D, F, F#, G

G#, B, B#, C#, D#, F#, F##, G#

Ab, Cb, C, Db, Eb, Gb, G, Ab

A, C, C#, D, E, G, G#, A

A#, C#, C##, D#, E#, G#, G##, A#

Bb, Db, D, Eb, F, Ab, A, Bb

B, D, D#, E, F#, A, A#, B

# **Oriental (2nd mode of double harmonic minor)**

Step pattern: 1, 3, 1, 1, 3, 1, 2

Ascending intervals: P1, m2, M3, P4, dim5, M6, m7, P8

C, Db, E, F, Gb, A, Bb, C

C#, D, E#, F#, G, A#, B, C#

Db, Ebb, F, Gb, Abb, Bb, Cb, Db

D, Eb, F#, G, Ab, B, C, D

D#, E, F##, G#, A, B#, C#, D#

Eb, Fb, G, Ab, Bbb, C, Db, Eb

E, F, G#, A, Bb, C#, D, E

F, Gb, A, Bb, Cb, D, Eb, F

F#, G, A#, B, C, D#, E, F#

Gb, Abb, Bb, Cb, Dbb, Eb, Fb, Gb

G, Ab, B, C, Db, E, F, G

G#, A, B#, C#, D, E#, F#, G#

Ab, Bbb, C, Db, Ebb, F, Gb, Ab

A, Bb, C#, D, Eb, F#, G, A

A#, B, C##, D#, E, F##, G#, A#

Bb, Cb, D, Eb, Fb, G, Ab, Bb

B, C, D#, E, F, G#, A, B

**Peiraiotikos**

Step pattern: 1, 3, 2, 1, 2, 2, 1

Ascending intervals: P1, m2, M3, dim5, P5, M6, M7, P8

C, Db, E, Gb, G, A, B, C

C#, D, E#, G, G#, A#, B#, C#

Db, Ebb, F, Abb, Ab, Bb, C, Db

D, Eb, F#, Ab, A, B, C#, D

D#, E, F##, A, A#, B#, C##, D#

Eb, Fb, G, Bbb, Bb, C, D, Eb

E, F, G#, Bb, B, C#, D#, E

F, Gb, A, Cb, C, D, E, F

F#, G, A#, C, C#, D#, E#, F#

Gb, Abb, Bb, Dbb, Db, Eb, F, Gb

G, Ab, B, Db, D, E, F#, G

G#, A, B#, D, D#, E#, F##, G#

Ab, Bbb, C, Ebb, Eb, F, G, Ab

A, Bb, C#, Eb, E, F#, G#, A

A#, B, C##, E, E#, F##, G##, A#

Bb, Cb, D, Fb, F, G, A, Bb

B, C, D#, F, F#, G#, A#, B



## Petrushka Chord

Step pattern: 1, 3, 2, 1, 2, 1, 2

Ascending intervals: P1, Aug1, M3, Aug4, P5, M6, Aug6, P8

C, C#, E, F#, G, A, A#, C

C#, C##, E#, F##, G#, A#, A##, C#

Db, D, F, G, Ab, Bb, B, Db

D, D#, F#, G#, A, B, B#, D

D#, D##, F##, G##, A#, B#, B##, D#

Eb, E, G, A, Bb, C, C#, Eb

E, E#, G#, A#, B, C#, C##, E

F, F#, A, B, C, D, D#, F

F#, F##, A#, B#, C#, D#, D##, F#

Gb, G, Bb, C, Db, Eb, E, Gb

G, G#, B, C#, D, E, E#, G

G#, G##, B#, C##, D#, E#, E##, G#

Ab, A, C, D, Eb, F, F#, Ab

A, A#, C#, D#, E, F#, F##, A

A#, A##, C##, D##, E#, F##, F###, A#

Bb, B, D, E, F, G, G#, Bb

B, B#, D#, E#, F#, G#, G##, B

## Phenadyuti

Step pattern: 1, 1, 3, 2, 1, 2, 2

Ascending intervals: P1, m2, M2, P4, P5, m6, m7, P8

C, Db, D, F, G, Ab, Bb, C

C#, D, D#, F#, G#, A, B, C#

Db, Ebb, Eb, Gb, Ab, Bbb, Cb, Db

D, Eb, E, G, A, Bb, C, D

D#, E, E#, G#, A#, B, C#, D#

Eb, Fb, F, Ab, Bb, Cb, Db, Eb

E, F, F#, A, B, C, D, E

F, Gb, G, Bb, C, Db, Eb, F

F#, G, G#, B, C#, D, E, F#

Gb, Abb, Ab, Cb, Db, Ebb, Fb, Gb

G, Ab, A, C, D, Eb, F, G

G#, A, A#, C#, D#, E, F#, G#

Ab, Bbb, Bb, Db, Eb, Fb, Gb, Ab

A, Bb, B, D, E, F, G, A

A#, B, B#, D#, E#, F#, G#, A#

Bb, Cb, C, Eb, F, Gb, Ab, Bb

B, C, C#, E, F#, G, A, B

### Phrygian (3rd mode of major)

Step pattern: 1, 2, 2, 2, 1, 2, 2

Ascending intervals: P1, m2, m3, P4, P5, m6, m7, P8

C, Db, Eb, F, G, Ab, Bb, C

C#, D, E, F#, G#, A, B, C#

Db, Ebb, Fb, Gb, Ab, Bbb, Cb, Db

D, Eb, F, G, A, Bb, C, D

D#, E, F#, G#, A#, B, C#, D#

Eb, Fb, Gb, Ab, Bb, Cb, Db, Eb

E, F, G, A, B, C, D, E

F, Gb, Ab, Bb, C, Db, Eb, F

F#, G, A, B, C#, D, E, F#

Gb, Abb, Bbb, Cb, Db, Ebb, Fb, Gb

G, Ab, Bb, C, D, Eb, F, G

G#, A, B, C#, D#, E, F#, G#

Ab, Bbb, Cb, Db, Eb, Fb, Gb, Ab

A, Bb, C, D, E, F, G, A

A#, B, C#, D#, E#, F#, G#, A#

Bb, Cb, Db, Eb, F, Gb, Ab, Bb

B, C, D, E, F#, G, A, B

### Phrygian b4 (3rd mode of harmonic major)

Step pattern: 1, 2, 1, 3, 1, 2, 2

Ascending intervals: P1, m2, m3, dim4, P5, m6, m7, P8

C, Db, Eb, Fb, G, Ab, Bb, C

C#, D, E, F, G#, A, B, C#

Db, Ebb, Fb, Gbb, Ab, Bbb, Cb, Db

D, Eb, F, Gb, A, Bb, C, D

D#, E, F#, G, A#, B, C#, D#

Eb, Fb, Gb, Abb, Bb, Cb, Db, Eb

E, F, G, Ab, B, C, D, E

F, Gb, Ab, Bbb, C, Db, Eb, F

F#, G, A, Bb, C#, D, E, F#

Gb, Abb, Bbb, Cbb, Db, Ebb, Fb, Gb

G, Ab, Bb, Cb, D, Eb, F, G

G#, A, B, C, D#, E, F#, G#

Ab, Bbb, Cb, Dbb, Eb, Fb, Gb, Ab

A, Bb, C, Db, E, F, G, A

A#, B, C#, D, E#, F#, G#, A#

Bb, Cb, Db, Ebb, F, Gb, Ab, Bb

B, C, D, Eb, F#, G, A, B

### Phrygian Dominant (5th mode of harmonic minor)

Step pattern: 1, 3, 1, 2, 1, 2, 2

Ascending intervals: P1, m2, M3, P4, P5, m6, m7, P8

C, Db, E, F, G, Ab, Bb, C

C#, D, E#, F#, G#, A, B, C#

Db, Ebb, F, Gb, Ab, Bbb, Cb, Db

D, Eb, F#, G, A, Bb, C, D

D#, E, F##, G#, A#, B, C#, D#

Eb, Fb, G, Ab, Bb, Cb, Db, Eb

E, F, G#, A, B, C, D, E

F, Gb, A, Bb, C, Db, Eb, F

F#, G, A#, B, C#, D, E, F#

Gb, Abb, Bb, Cb, Db, Ebb, Fb, Gb

G, Ab, B, C, D, Eb, F, G

G#, A, B#, C#, D#, E, F#, G#

Ab, Bbb, C, Db, Eb, Fb, Gb, Ab

A, Bb, C#, D, E, F, G, A

A#, B, C##, D#, E#, F#, G#, A#

Bb, Cb, D, Eb, F, Gb, Ab, Bb

B, C, D#, E, F#, G, A, B

**Phrygian nat. 6 (2nd mode of melodic minor)**

Step pattern: 1, 2, 2, 2, 2, 1, 2

Ascending intervals: P1, m2, m3, P4, P5, M6, m7, P8

C, Db, Eb, F, G, A, Bb, C

C#, D, E, F#, G#, A#, B, C#

Db, Ebb, Fb, Gb, Ab, Bb, Cb, Db

D, Eb, F, G, A, B, C, D

D#, E, F#, G#, A#, B#, C#, D#

Eb, Fb, Gb, Ab, Bb, C, Db, Eb

E, F, G, A, B, C#, D, E

F, Gb, Ab, Bb, C, D, Eb, F

F#, G, A, B, C#, D#, E, F#

Gb, Abb, Bbb, Cb, Db, Eb, Fb, Gb

G, Ab, Bb, C, D, E, F, G

G#, A, B, C#, D#, E#, F#, G#

Ab, Bbb, Cb, Db, Eb, F, Gb, Ab

A, Bb, C, D, E, F#, G, A

A#, B, C#, D#, E#, F##, G#, A#

Bb, Cb, Db, Eb, F, G, Ab, Bb

B, C, D, E, F#, G#, A, B

## Pratapa

Step pattern: 1, 3, 2, 1, 1, 2, 2

Ascending intervals: P1, m2, M3, dim5, P5, m6, m7, P8

C, Db, E, Gb, G, Ab, Bb, C

C#, D, E#, G, G#, A, B, C#

Db, Ebb, F, Abb, Ab, Bbb, Cb, Db

D, Eb, F#, Ab, A, Bb, C, D

D#, E, F##, A, A#, B, C#, D#

Eb, Fb, G, Bbb, Bb, Cb, Db, Eb

E, F, G#, Bb, B, C, D, E

F, Gb, A, Cb, C, Db, Eb, F

F#, G, A#, C, C#, D, E, F#

Gb, Abb, Bb, Dbb, Db, Ebb, Fb, Gb

G, Ab, B, Db, D, Eb, F, G

G#, A, B#, D, D#, E, F#, G#

Ab, Bbb, C, Ebb, Eb, Fb, Gb, Ab

A, Bb, C#, Eb, E, F, G, A

A#, B, C##, E, E#, F#, G#, A#

Bb, Cb, D, Fb, F, Gb, Ab, Bb

B, C, D#, F, F#, G, A, B

## Rageshri

Step pattern: 2, 2, 1, 4, 1, 1, 1

Ascending intervals: P1, M2, M3, P4, M6, m7, M7, P8

C, D, E, F, A, Bb, B, C

C#, D#, E#, F#, A#, B, B#, C#

Db, Eb, F, Gb, Bb, Cb, C, Db

D, E, F#, G, B, C, C#, D

D#, E#, F##, G#, B#, C#, C##, D#

Eb, F, G, Ab, C, Db, D, Eb

E, F#, G#, A, C#, D, D#, E

F, G, A, Bb, D, Eb, E, F

F#, G#, A#, B, D#, E, E#, F#

Gb, Ab, Bb, Cb, Eb, Fb, F, Gb

G, A, B, C, E, F, F#, G

G#, A#, B#, C#, E#, F#, F##, G#

Ab, Bb, C, Db, F, Gb, G, Ab

A, B, C#, D, F#, G, G#, A

A#, B#, C##, D#, F##, G#, G##, A#

Bb, C, D, Eb, G, Ab, A, Bb

B, C#, D#, E, G#, A, A#, B



## Ramamanohari

Step pattern: 1, 3, 2, 1, 2, 1, 2

Ascending intervals: P1, m2, M3, dim5, P5, M6, m7, P8

C, Db, E, Gb, G, A, Bb, C

C#, D, E#, G, G#, A#, B, C#

Db, Ebb, F, Abb, Ab, Bb, Cb, Db

D, Eb, F#, Ab, A, B, C, D

D#, E, F##, A, A#, B#, C#, D#

Eb, Fb, G, Bbb, Bb, C, Db, Eb

E, F, G#, Bb, B, C#, D, E

F, Gb, A, Cb, C, D, Eb, F

F#, G, A#, C, C#, D#, E, F#

Gb, Abb, Bb, Dbb, Db, Eb, Fb, Gb

G, Ab, B, Db, D, E, F, G

G#, A, B#, D, D#, E#, F#, G#

Ab, Bbb, C, Ebb, Eb, F, Gb, Ab

A, Bb, C#, Eb, E, F#, G, A

A#, B, C##, E, E#, F##, G#, A#

Bb, Cb, D, Fb, F, G, Ab, Bb

B, C, D#, F, F#, G#, A, B

## Ratipriya

Step pattern: 2, 2, 2, 1, 1, 2, 2

Ascending intervals: P1, M2, M3, dim5, P5, m6, m7, P8

C, D, E, Gb, G, Ab, Bb, C

C#, D#, E#, G, G#, A, B, C#

Db, Eb, F, Abb, Ab, Bbb, Cb, Db

D, E, F#, Ab, A, Bb, C, D

D#, E#, F##, A, A#, B, C#, D#

Eb, F, G, Bbb, Bb, Cb, Db, Eb

E, F#, G#, Bb, B, C, D, E

F, G, A, Cb, C, Db, Eb, F

F#, G#, A#, C, C#, D, E, F#

Gb, Ab, Bb, Dbb, Db, Ebb, Fb, Gb

G, A, B, Db, D, Eb, F, G

G#, A#, B#, D, D#, E, F#, G#

Ab, Bb, C, Ebb, Eb, Fb, Gb, Ab

A, B, C#, Eb, E, F, G, A

A#, B#, C##, E, E#, F#, G#, A#

Bb, C, D, Fb, F, Gb, Ab, Bb

B, C#, D#, F, F#, G, A, B

## Ravikriya

Step pattern: 1, 1, 4, 1, 3, 1, 1

Ascending intervals: P1, m2, M2, dim5, P5, m7, M7, P8

C, Db, D, Gb, G, Bb, B, C

C#, D, D#, G, G#, B, B#, C#

Db, Ebb, Eb, Abb, Ab, Cb, C, Db

D, Eb, E, Ab, A, C, C#, D

D#, E, E#, A, A#, C#, C##, D#

Eb, Fb, F, Bbb, Bb, Db, D, Eb

E, F, F#, Bb, B, D, D#, E

F, Gb, G, Cb, C, Eb, E, F

F#, G, G#, C, C#, E, E#, F#

Gb, Abb, Ab, Dbb, Db, Fb, F, Gb

G, Ab, A, Db, D, F, F#, G

G#, A, A#, D, D#, F#, F##, G#

Ab, Bbb, Bb, Ebb, Eb, Gb, G, Ab

A, Bb, B, Eb, E, G, G#, A

A#, B, B#, E, E#, G#, G##, A#

Bb, Cb, C, Fb, F, Ab, A, Bb

B, C, C#, F, F#, A, A#, B

## Romanian Major

Step pattern: 1, 3, 2, 1, 2, 1, 2

Ascending intervals: P1, m2, M3, Aug4, P5, M6, m7, P8

C, Db, E, F#, G, A, Bb, C

C#, D, E#, F##, G#, A#, B, C#

Db, Ebb, F, G, Ab, Bb, Cb, Db

D, Eb, F#, G#, A, B, C, D

D#, E, F##, G##, A#, B#, C#, D#

Eb, Fb, G, A, Bb, C, Db, Eb

E, F, G#, A#, B, C#, D, E

F, Gb, A, B, C, D, Eb, F

F#, G, A#, B#, C#, D#, E, F#

Gb, Abb, Bb, C, Db, Eb, Fb, Gb

G, Ab, B, C#, D, E, F, G

G#, A, B#, C##, D#, E#, F#, G#

Ab, Bbb, C, D, Eb, F, Gb, Ab

A, Bb, C#, D#, E, F#, G, A

A#, B, C##, D##, E#, F##, G#, A#

Bb, Cb, D, E, F, G, Ab, Bb

B, C, D#, E#, F#, G#, A, B

## Rupavati

Step pattern: 1, 2, 2, 2, 3, 1, 1

Ascending intervals: P1, m2, m3, P4, P5, m7, M7, P8

C, Db, Eb, F, G, Bb, B, C

C#, D, E, F#, G#, B, B#, C#

Db, Ebb, Fb, Gb, Ab, Cb, C, Db

D, Eb, F, G, A, C, C#, D

D#, E, F#, G#, A#, C#, C##, D#

Eb, Fb, Gb, Ab, Bb, Db, D, Eb

E, F, G, A, B, D, D#, E

F, Gb, Ab, Bb, C, Eb, E, F

F#, G, A, B, C#, E, E#, F#

Gb, Abb, Bbb, Cb, Db, Fb, F, Gb

G, Ab, Bb, C, D, F, F#, G

G#, A, B, C#, D#, F#, F##, G#

Ab, Bbb, Cb, Db, Eb, Gb, G, Ab

A, Bb, C, D, E, G, G#, A

A#, B, C#, D#, E#, G#, G##, A#

Bb, Cb, Db, Eb, F, Ab, A, Bb

B, C, D, E, F#, A, A#, B

## Sabach

Step pattern: 2, 1, 1, 3, 1, 2, 2

Ascending intervals: P1, M2, m3, M3, P5, m6, m7, P8

C, D, Eb, E, G, Ab, Bb, C

C#, D#, E, E#, G#, A, B, C#

Db, Eb, Fb, F, Ab, Bbb, Cb, Db

D, E, F, F#, A, Bb, C, D

D#, E#, F#, F##, A#, B, C#, D#

Eb, F, Gb, G, Bb, Cb, Db, Eb

E, F#, G, G#, B, C, D, E

F, G, Ab, A, C, Db, Eb, F

F#, G#, A, A#, C#, D, E, F#

Gb, Ab, Bbb, Bb, Db, Ebb, Fb, Gb

G, A, Bb, B, D, Eb, F, G

G#, A#, B, B#, D#, E, F#, G#

Ab, Bb, Cb, C, Eb, Fb, Gb, Ab

A, B, C, C#, E, F, G, A

A#, B#, C#, C##, E#, F#, G#, A#

Bb, C, Db, D, F, Gb, Ab, Bb

B, C#, D, D#, F#, G, A, B

**Salaga**

Step pattern: 1, 1, 4, 1, 1, 1, 3

Ascending intervals: P1, m2, M2, dim5, P5, m6, M6, P8

C, Db, D, Gb, G, Ab, A, C

C#, D, D#, G, G#, A, A#, C#

Db, Ebb, Eb, Abb, Ab, Bbb, Bb, Db

D, Eb, E, Ab, A, Bb, B, D

D#, E, E#, A, A#, B, B#, D#

Eb, Fb, F, Bbb, Bb, Cb, C, Eb

E, F, F#, Bb, B, C, C#, E

F, Gb, G, Cb, C, Db, D, F

F#, G, G#, C, C#, D, D#, F#

Gb, Abb, Ab, Dbb, Db, Ebb, Eb, Gb

G, Ab, A, Db, D, Eb, E, G

G#, A, A#, D, D#, E, E#, G#

Ab, Bbb, Bb, Ebb, Eb, Fb, F, Ab

A, Bb, B, Eb, E, F, F#, A

A#, B, B#, E, E#, F#, F##, A#

Bb, Cb, C, Fb, F, Gb, G, Bb

B, C, C#, F, F#, G, G#, B

**Samanta**

Step pattern: 2, 2, 1, 2, 3, 1, 1

Ascending intervals: P1, M2, M3, P4, P5, m7, M7, P8

C, D, E, F, G, Bb, B, C

C#, D#, E#, F#, G#, B, B#, C#

Db, Eb, F, Gb, Ab, Cb, C, Db

D, E, F#, G, A, C, C#, D

D#, E#, F##, G#, A#, C#, C##, D#

Eb, F, G, Ab, Bb, Db, D, Eb

E, F#, G#, A, B, D, D#, E

F, G, A, Bb, C, Eb, E, F

F#, G#, A#, B, C#, E, E#, F#

Gb, Ab, Bb, Cb, Db, Fb, F, Gb

G, A, B, C, D, F, F#, G

G#, A#, B#, C#, D#, F#, F##, G#

Ab, Bb, C, Db, Eb, Gb, G, Ab

A, B, C#, D, E, G, G#, A

A#, B#, C##, D#, E#, G#, G##, A#

Bb, C, D, Eb, F, Ab, A, Bb

B, C#, D#, E, F#, A, A#, B



## Santanamanjari

Step pattern: 3, 1, 2, 1, 1, 1, 3

Ascending intervals: P1, m3, M3, dim5, P5, m6, M6, P8

C, Eb, E, Gb, G, Ab, A, C

C#, E, E#, G, G#, A, A#, C#

Db, Fb, F, Abb, Ab, Bbb, Bb, Db

D, F, F#, Ab, A, Bb, B, D

D#, F#, F##, A, A#, B, B#, D#

Eb, Gb, G, Bbb, Bb, Cb, C, Eb

E, G, G#, Bb, B, C, C#, E

F, Ab, A, Cb, C, Db, D, F

F#, A, A#, C, C#, D, D#, F#

Gb, Bbb, Bb, Dbb, Db, Ebb, Eb, Gb

G, Bb, B, Db, D, Eb, E, G

G#, B, B#, D, D#, E, E#, G#

Ab, Cb, C, Ebb, Eb, Fb, F, Ab

A, C, C#, Eb, E, F, F#, A

A#, C#, C##, E, E#, F#, F##, A#

Bb, Db, D, Fb, F, Gb, G, Bb

B, D, D#, F, F#, G, G#, B

## Sauviram

Step pattern: 1, 2, 3, 1, 2, 2, 1

Ascending intervals: P1, m2, m3, dim5, P5, M6, M7, P8

C, Db, Eb, Gb, G, A, B, C

C#, D, E, G, G#, A#, B#, C#

Db, Ebb, Fb, Abb, Ab, Bb, C, Db

D, Eb, F, Ab, A, B, C#, D

D#, E, F#, A, A#, B#, C##, D#

Eb, Fb, Gb, Bbb, Bb, C, D, Eb

E, F, G, Bb, B, C#, D#, E

F, Gb, Ab, Cb, C, D, E, F

F#, G, A, C, C#, D#, E#, F#

Gb, Abb, Bbb, Dbb, Db, Eb, F, Gb

G, Ab, Bb, Db, D, E, F#, G

G#, A, B, D, D#, E#, F##, G#

Ab, Bbb, Cb, Ebb, Eb, F, G, Ab

A, Bb, C, Eb, E, F#, G#, A

A#, B, C#, E, E#, F##, G##, A#

Bb, Cb, Db, Fb, F, G, A, Bb

B, C, D, F, F#, G#, A#, B

## Sengah

Step pattern: 3, 1, 1, 2, 1, 3, 1

Ascending intervals: P1, m3, M3, P4, P5, m6, M7, P8

C, Eb, E, F, G, Ab, B, C

C#, E, E#, F#, G#, A, B#, C#

Db, Fb, F, Gb, Ab, Bbb, C, Db

D, F, F#, G, A, Bb, C#, D

D#, F#, F##, G#, A#, B, C##, D#

Eb, Gb, G, Ab, Bb, Cb, D, Eb

E, G, G#, A, B, C, D#, E

F, Ab, A, Bb, C, Db, E, F

F#, A, A#, B, C#, D, E#, F#

Gb, Bbb, Bb, Cb, Db, Ebb, F, Gb

G, Bb, B, C, D, Eb, F#, G

G#, B, B#, C#, D#, E, F##, G#

Ab, Cb, C, Db, Eb, Fb, G, Ab

A, C, C#, D, E, F, G#, A

A#, C#, C##, D#, E#, F#, G##, A#

Bb, Db, D, Eb, F, Gb, A, Bb

B, D, D#, E, F#, G, A#, B

## Sorati

Step pattern: 2, 3, 2, 2, 1, 1, 1

Ascending intervals: P1, M2, P4, P5, M6, m7, M7, P8

C, D, F, G, A, Bb, B, C

C#, D#, F#, G#, A#, B, B#, C#

Db, Eb, Gb, Ab, Bb, Cb, C, Db

D, E, G, A, B, C, C#, D

D#, E#, G#, A#, B#, C#, C##, D#

Eb, F, Ab, Bb, C, Db, D, Eb

E, F#, A, B, C#, D, D#, E

F, G, Bb, C, D, Eb, E, F

F#, G#, B, C#, D#, E, E#, F#

Gb, Ab, Cb, Db, Eb, Fb, F, Gb

G, A, C, D, E, F, F#, G

G#, A#, C#, D#, E#, F#, F##, G#

Ab, Bb, Db, Eb, F, Gb, G, Ab

A, B, D, E, F#, G, G#, A

A#, B#, D#, E#, F##, G#, G##, A#

Bb, C, Eb, F, G, Ab, A, Bb

B, C#, E, F#, G#, A, A#, B

## Spanish Mode

Step pattern: 1, 2, 2, 2, 1, 2, 2

Ascending intervals: P1, m2, m3, P4, P5, m6, m7, P8

C, Db, Eb, F, G, Ab, Bb, C

C#, D, E, F#, G#, A, B, C#

Db, Ebb, Fb, Gb, Ab, Bbb, Cb, Db

D, Eb, F, G, A, Bb, C, D

D#, E, F#, G#, A#, B, C#, D#

Eb, Fb, Gb, Ab, Bb, Cb, Db, Eb

E, F, G, A, B, C, D, E

F, Gb, Ab, Bb, C, Db, Eb, F

F#, G, A, B, C#, D, E, F#

Gb, Abb, Bbb, Cb, Db, Ebb, Fb, Gb

G, Ab, Bb, C, D, Eb, F, G

G#, A, B, C#, D#, E, F#, G#

Ab, Bbb, Cb, Db, Eb, Fb, Gb, Ab

A, Bb, C, D, E, F, G, A

A#, B, C#, D#, E#, F#, G#, A#

Bb, Cb, Db, Eb, F, Gb, Ab, Bb

B, C, D, E, F#, G, A, B

### Super Locrian (7th mode of melodic minor)

Step pattern: 1, 2, 1, 2, 2, 2, 2

Ascending intervals: P1, m2, m3, dim4, dim5, m6, m7, P8

C, Db, Eb, Fb, Gb, Ab, Bb, C

C#, D, E, F, G, A, B, C#

Db, Ebb, Fb, Gbb, Abb, Bbb, Cb, Db

D, Eb, F, Gb, Ab, Bb, C, D

D#, E, F#, G, A, B, C#, D#

Eb, Fb, Gb, Abb, Bbb, Cb, Db, Eb

E, F, G, Ab, Bb, C, D, E

F, Gb, Ab, Bbb, Cb, Db, Eb, F

F#, G, A, Bb, C, D, E, F#

Gb, Abb, Bbb, Cbb, Dbb, Ebb, Fb, Gb

G, Ab, Bb, Cb, Db, Eb, F, G

G#, A, B, C, D, E, F#, G#

Ab, Bbb, Cb, Dbb, Ebb, Fb, Gb, Ab

A, Bb, C, Db, Eb, F, G, A

A#, B, C#, D, E, F#, G#, A#

Bb, Cb, Db, Ebb, Fb, Gb, Ab, Bb

B, C, D, Eb, F, G, A, B

## Tanukirti

Step pattern: 1, 1, 3, 2, 3, 1, 1

Ascending intervals: P1, m2, M2, P4, P5, m7, M7, P8

C, Db, D, F, G, Bb, B, C

C#, D, D#, F#, G#, B, B#, C#

Db, Ebb, Eb, Gb, Ab, Cb, C, Db

D, Eb, E, G, A, C, C#, D

D#, E, E#, G#, A#, C#, C##, D#

Eb, Fb, F, Ab, Bb, Db, D, Eb

E, F, F#, A, B, D, D#, E

F, Gb, G, Bb, C, Eb, E, F

F#, G, G#, B, C#, E, E#, F#

Gb, Abb, Ab, Cb, Db, Fb, F, Gb

G, Ab, A, C, D, F, F#, G

G#, A, A#, C#, D#, F#, F##, G#

Ab, Bbb, Bb, Db, Eb, Gb, G, Ab

A, Bb, B, D, E, G, G#, A

A#, B, B#, D#, E#, G#, G##, A#

Bb, Cb, C, Eb, F, Ab, A, Bb

B, C, C#, E, F#, A, A#, B

## Tivravahini

Step pattern: 1, 2, 3, 1, 2, 1, 2

Ascending intervals: P1, m2, m3, dim5, P5, M6, m7, P8

C, Db, Eb, Gb, G, A, Bb, C

C#, D, E, G, G#, A#, B, C#

Db, Ebb, Fb, Abb, Ab, Bb, Cb, Db

D, Eb, F, Ab, A, B, C, D

D#, E, F#, A, A#, B#, C#, D#

Eb, Fb, Gb, Bbb, Bb, C, Db, Eb

E, F, G, Bb, B, C#, D, E

F, Gb, Ab, Cb, C, D, Eb, F

F#, G, A, C, C#, D#, E, F#

Gb, Abb, Bbb, Dbb, Db, Eb, Fb, Gb

G, Ab, Bb, Db, D, E, F, G

G#, A, B, D, D#, E#, F#, G#

Ab, Bbb, Cb, Ebb, Eb, F, Gb, Ab

A, Bb, C, Eb, E, F#, G, A

A#, B, C#, E, E#, F##, G#, A#

Bb, Cb, Db, Fb, F, G, Ab, Bb

B, C, D, F, F#, G#, A, B



## Trishuli

Step pattern: 3, 1, 1, 2, 2, 2, 1

Ascending intervals: P1, m3, M3, P4, P5, M6, M7, P8

C, Eb, E, F, G, A, B, C

C#, E, E#, F#, G#, A#, B#, C#

Db, Fb, F, Gb, Ab, Bb, C, Db

D, F, F#, G, A, B, C#, D

D#, F#, F##, G#, A#, B#, C##, D#

Eb, Gb, G, Ab, Bb, C, D, Eb

E, G, G#, A, B, C#, D#, E

F, Ab, A, Bb, C, D, E, F

F#, A, A#, B, C#, D#, E#, F#

Gb, Bbb, Bb, Cb, Db, Eb, F, Gb

G, Bb, B, C, D, E, F#, G

G#, B, B#, C#, D#, E#, F##, G#

Ab, Cb, C, Db, Eb, F, G, Ab

A, C, C#, D, E, F#, G#, A

A#, C#, C##, D#, E#, F##, G##, A#

Bb, Db, D, Eb, F, G, A, Bb

B, D, D#, E, F#, G#, A#, B

## Tunisian

Step pattern: 2, 1, 3, 1, 2, 1, 2

Ascending intervals: P1, M2, m3, dim5, P5, M6, m7, P8

C, D, Eb, Gb, G, A, Bb, C

C#, D#, E, G, G#, A#, B, C#

Db, Eb, Fb, Abb, Ab, Bb, Cb, Db

D, E, F, Ab, A, B, C, D

D#, E#, F#, A, A#, B#, C#, D#

Eb, F, Gb, Bbb, Bb, C, Db, Eb

E, F#, G, Bb, B, C#, D, E

F, G, Ab, Cb, C, D, Eb, F

F#, G#, A, C, C#, D#, E, F#

Gb, Ab, Bbb, Dbb, Db, Eb, Fb, Gb

G, A, Bb, Db, D, E, F, G

G#, A#, B, D, D#, E#, F#, G#

Ab, Bb, Cb, Ebb, Eb, F, Gb, Ab

A, B, C, Eb, E, F#, G, A

A#, B#, C#, E, E#, F##, G#, A#

Bb, C, Db, Fb, F, G, Ab, Bb

B, C#, D, F, F#, G#, A, B

**Vamsavathi**

Step pattern: 1, 2, 3, 1, 3, 1, 1

Ascending intervals: P1, m2, m3, dim5, P5, m7, M7, P8

C, Db, Eb, Gb, G, Bb, B, C

C#, D, E, G, G#, B, B#, C#

Db, Ebb, Fb, Abb, Ab, Cb, C, Db

D, Eb, F, Ab, A, C, C#, D

D#, E, F#, A, A#, C#, C##, D#

Eb, Fb, Gb, Bbb, Bb, Db, D, Eb

E, F, G, Bb, B, D, D#, E

F, Gb, Ab, Cb, C, Eb, E, F

F#, G, A, C, C#, E, E#, F#

Gb, Abb, Bbb, Dbb, Db, Fb, F, Gb

G, Ab, Bb, Db, D, F, F#, G

G#, A, B, D, D#, F#, F##, G#

Ab, Bbb, Cb, Ebb, Eb, Gb, G, Ab

A, Bb, C, Eb, E, G, G#, A

A#, B, C#, E, E#, G#, G##, A#

Bb, Cb, Db, Fb, F, Ab, A, Bb

B, C, D, F, F#, A, A#, B

## Varali

Step pattern: 1, 1, 4, 1, 1, 3, 1

Ascending intervals: P1, m2, M2, dim5, P5, m6, M7, P8

C, Db, D, Gb, G, Ab, B, C

C#, D, D#, G, G#, A, B#, C#

Db, Ebb, Eb, Abb, Ab, Bbb, C, Db

D, Eb, E, Ab, A, Bb, C#, D

D#, E, E#, A, A#, B, C##, D#

Eb, Fb, F, Bbb, Bb, Cb, D, Eb

E, F, F#, Bb, B, C, D#, E

F, Gb, G, Cb, C, Db, E, F

F#, G, G#, C, C#, D, E#, F#

Gb, Abb, Ab, Dbb, Db, Ebb, F, Gb

G, Ab, A, Db, D, Eb, F#, G

G#, A, A#, D, D#, E, F##, G#

Ab, Bbb, Bb, Ebb, Eb, Fb, G, Ab

A, Bb, B, Eb, E, F, G#, A

A#, B, B#, E, E#, F#, G##, A#

Bb, Cb, C, Fb, F, Gb, A, Bb

B, C, C#, F, F#, G, A#, B

## Verdis Enigmatic

Step pattern: 1, 3, 2, 2, 2, 1, 1

Ascending intervals: P1, m2, M3, dim5, m6, m7, M7, P8

C, Db, E, Gb, Ab, Bb, B, C

C#, D, E#, G, A, B, B#, C#

Db, Ebb, F, Abb, Bbb, Cb, C, Db

D, Eb, F#, Ab, Bb, C, C#, D

D#, E, F##, A, B, C#, C##, D#

Eb, Fb, G, Bbb, Cb, Db, D, Eb

E, F, G#, Bb, C, D, D#, E

F, Gb, A, Cb, Db, Eb, E, F

F#, G, A#, C, D, E, E#, F#

Gb, Abb, Bb, Dbb, Ebb, Fb, F, Gb

G, Ab, B, Db, Eb, F, F#, G

G#, A, B#, D, E, F#, F##, G#

Ab, Bbb, C, Ebb, Fb, Gb, G, Ab

A, Bb, C#, Eb, F, G, G#, A

A#, B, C##, E, F#, G#, G##, A#

Bb, Cb, D, Fb, Gb, Ab, A, Bb

B, C, D#, F, G, A, A#, B

## Viravasantham

Step pattern: 2, 1, 2, 2, 3, 1, 1

Ascending intervals: P1, M2, m3, P4, P5, m7, M7, P8

C, D, Eb, F, G, Bb, B, C

C#, D#, E, F#, G#, B, B#, C#

Db, Eb, Fb, Gb, Ab, Cb, C, Db

D, E, F, G, A, C, C#, D

D#, E#, F#, G#, A#, C#, C##, D#

Eb, F, Gb, Ab, Bb, Db, D, Eb

E, F#, G, A, B, D, D#, E

F, G, Ab, Bb, C, Eb, E, F

F#, G#, A, B, C#, E, E#, F#

Gb, Ab, Bbb, Cb, Db, Fb, F, Gb

G, A, Bb, C, D, F, F#, G

G#, A#, B, C#, D#, F#, F##, G#

Ab, Bb, Cb, Db, Eb, Gb, G, Ab

A, B, C, D, E, G, G#, A

A#, B#, C#, D#, E#, G#, G##, A#

Bb, C, Db, Eb, F, Ab, A, Bb

B, C#, D, E, F#, A, A#, B

## Visvambhari

Step pattern: 1, 3, 2, 1, 3, 1, 1

Ascending intervals: P1, m2, M3, dim5, P5, m7, M7, P8

C, Db, E, Gb, G, Bb, B, C

C#, D, E#, G, G#, B, B#, C#

Db, Ebb, F, Abb, Ab, Cb, C, Db

D, Eb, F#, Ab, A, C, C#, D

D#, E, F##, A, A#, C#, C##, D#

Eb, Fb, G, Bbb, Bb, Db, D, Eb

E, F, G#, Bb, B, D, D#, E

F, Gb, A, Cb, C, Eb, E, F

F#, G, A#, C, C#, E, E#, F#

Gb, Abb, Bb, Dbb, Db, Fb, F, Gb

G, Ab, B, Db, D, F, F#, G

G#, A, B#, D, D#, F#, F##, G#

Ab, Bbb, C, Ebb, Eb, Gb, G, Ab

A, Bb, C#, Eb, E, G, G#, A

A#, B, C##, E, E#, G#, G##, A#

Bb, Cb, D, Fb, F, Ab, A, Bb

B, C, D#, F, F#, A, A#, B

**Whole-tone add M7**

Step pattern: 2, 2, 2, 2, 2, 1, 1

Ascending intervals: P1, M2, M3, Aug4, Aug5, Aug6, M7, P8

C, D, E, F#, G#, A#, B, C

C#, D#, E#, F##, G##, A##, B#, C#

Db, Eb, F, G, A, B, C, Db

D, E, F#, G#, A#, B#, C#, D

D#, E#, F##, G##, A##, B##, C##, D#

Eb, F, G, A, B, C#, D, Eb

E, F#, G#, A#, B#, C##, D#, E

F, G, A, B, C#, D#, E, F

F#, G#, A#, B#, C##, D##, E#, F#

Gb, Ab, Bb, C, D, E, F, Gb

G, A, B, C#, D#, E#, F#, G

G#, A#, B#, C##, D##, E##, F##, G#

Ab, Bb, C, D, E, F#, G, Ab

A, B, C#, D#, E#, F##, G#, A

A#, B#, C##, D##, E##, F###, G##, A#

Bb, C, D, E, F#, G#, A, Bb

B, C#, D#, E#, F##, G##, A#, B



## 8 – NOTE SCALES

## Adonai Malakh

Step pattern: 1, 1, 1, 2, 2, 2, 1, 2

Ascending intervals: P1, m2, M2, m3, P4, P5, M6, m7, P8

C, Db, D, Eb, F, G, A, Bb, C

C#, D, D#, E, F#, G#, A#, B, C#

Db, Ebb, Eb, Fb, Gb, Ab, Bb, Cb, Db

D, Eb, E, F, G, A, B, C, D

D#, E, E#, F#, G#, A#, B#, C#, D#

Eb, Fb, F, Gb, Ab, Bb, C, Db, Eb

E, F, F#, G, A, B, C#, D, E

F, Gb, G, Ab, Bb, C, D, Eb, F

F#, G, G#, A, B, C#, D#, E, F#

Gb, Abb, Ab, Bbb, Cb, Db, Eb, Fb, Gb

G, Ab, A, Bb, C, D, E, F, G

G#, A, A#, B, C#, D#, E#, F#, G#

Ab, Bbb, Bb, Cb, Db, Eb, F, Gb, Ab

A, Bb, B, C, D, E, F#, G, A

A#, B, B#, C#, D#, E#, F##, G#, A#

Bb, Cb, C, Db, Eb, F, G, Ab, Bb

B, C, C#, D, E, F#, G#, A, B

## Ahava Rabba

Step pattern: 1, 2, 1, 1, 1, 2, 2, 2

Ascending intervals: P1, m2, Aug2, M3, P4, dim5, m6, m7, P8

C, Db, D#, E, F, Gb, Ab, Bb, C

C#, D, D##, E#, F#, G, A, B, C#

Db, Ebb, E, F, Gb, Abb, Bbb, Cb, Db

D, Eb, E#, F#, G, Ab, Bb, C, D

D#, E, E##, F##, G#, A, B, C#, D#

Eb, Fb, F#, G, Ab, Bbb, Cb, Db, Eb

E, F, F##, G#, A, Bb, C, D, E

F, Gb, G#, A, Bb, Cb, Db, Eb, F

F#, G, G##, A#, B, C, D, E, F#

Gb, Abb, A, Bb, Cb, Dbb, Ebb, Fb, Gb

G, Ab, A#, B, C, Db, Eb, F, G

G#, A, A##, B#, C#, D, E, F#, G#

Ab, Bbb, B, C, Db, Ebb, Fb, Gb, Ab

A, Bb, B#, C#, D, Eb, F, G, A

A#, B, B##, C##, D#, E, F#, G#, A#

Bb, Cb, C#, D, Eb, Fb, Gb, Ab, Bb

B, C, C##, D#, E, F, G, A, B

## Algerian

Step pattern: 2, 1, 2, 1, 1, 1, 3, 1

Ascending intervals: P1, M2, m3, P4, Aug4, P5, m6, M7, P8

C, D, Eb, F, F#, G, Ab, B, C

C#, D#, E, F#, F##, G#, A, B#, C#

Db, Eb, Fb, Gb, G, Ab, Bbb, C, Db

D, E, F, G, G#, A, Bb, C#, D

D#, E#, F#, G#, G##, A#, B, C##, D#

Eb, F, Gb, Ab, A, Bb, Cb, D, Eb

E, F#, G, A, A#, B, C, D#, E

F, G, Ab, Bb, B, C, Db, E, F

F#, G#, A, B, B#, C#, D, E#, F#

Gb, Ab, Bbb, Cb, C, Db, Ebb, F, Gb

G, A, Bb, C, C#, D, Eb, F#, G

G#, A#, B, C#, C##, D#, E, F##, G#

Ab, Bb, Cb, Db, D, Eb, Fb, G, Ab

A, B, C, D, D#, E, F, G#, A

A#, B#, C#, D#, D##, E#, F#, G##, A#

Bb, C, Db, Eb, E, F, Gb, A, Bb

B, C#, D, E, E#, F#, G, A#, B

## Altered Mixolydian

Step pattern: 2, 2, 1, 2, 1, 1, 2, 1

Ascending intervals: P1, M2, M3, P4, P5, Aug5, M6, M7, P8

C, D, E, F, G, G#, A, B, C

C#, D#, E#, F#, G#, G##, A#, B#, C#

Db, Eb, F, Gb, Ab, A, Bb, C, Db

D, E, F#, G, A, A#, B, C#, D

D#, E#, F##, G#, A#, A##, B#, C##, D#

Eb, F, G, Ab, Bb, B, C, D, Eb

E, F#, G#, A, B, B#, C#, D#, E

F, G, A, Bb, C, C#, D, E, F

F#, G#, A#, B, C#, C##, D#, E#, F#

Gb, Ab, Bb, Cb, Db, D, Eb, F, Gb

G, A, B, C, D, D#, E, F#, G

G#, A#, B#, C#, D#, D##, E#, F##, G#

Ab, Bb, C, Db, Eb, E, F, G, Ab

A, B, C#, D, E, E#, F#, G#, A

A#, B#, C##, D#, E#, E##, F##, G##, A#

Bb, C, D, Eb, F, F#, G, A, Bb

B, C#, D#, E, F#, F##, G#, A#, B

**Altered Mixolydian M7 (add m6)**

Step pattern: 2, 2, 1, 2, 1, 1, 2, 1

Ascending intervals: P1, M2, M3, P4, P5, m6, M6, M7, P8

C, D, E, F, G, Ab, A, B, C

C#, D#, E#, F#, G#, A, A#, B#, C#

Db, Eb, F, Gb, Ab, Bbb, Bb, C, Db

D, E, F#, G, A, Bb, B, C#, D

D#, E#, F##, G#, A#, B, B#, C##, D#

Eb, F, G, Ab, Bb, Cb, C, D, Eb

E, F#, G#, A, B, C, C#, D#, E

F, G, A, Bb, C, Db, D, E, F

F#, G#, A#, B, C#, D, D#, E#, F#

Gb, Ab, Bb, Cb, Db, Ebb, Eb, F, Gb

G, A, B, C, D, Eb, E, F#, G

G#, A#, B#, C#, D#, E, E#, F##, G#

Ab, Bb, C, Db, Eb, Fb, F, G, Ab

A, B, C#, D, E, F, F#, G#, A

A#, B#, C##, D#, E#, F#, F##, G##, A#

Bb, C, D, Eb, F, Gb, G, A, Bb

B, C#, D#, E, F#, G, G#, A#, B

**Bahar**

Step pattern: 2, 1, 2, 2, 2, 1, 1, 1

Ascending intervals: P1, M2, m3, P4, P5, M6, m7, M7, P8

C, D, Eb, F, G, A, Bb, B, C

C#, D#, E, F#, G#, A#, B, B#, C#

Db, Eb, Fb, Gb, Ab, Bb, Cb, C, Db

D, E, F, G, A, B, C, C#, D

D#, E#, F#, G#, A#, B#, C#, C##, D#

Eb, F, Gb, Ab, Bb, C, Db, D, Eb

E, F#, G, A, B, C#, D, D#, E

F, G, Ab, Bb, C, D, Eb, E, F

F#, G#, A, B, C#, D#, E, E#, F#

Gb, Ab, Bbb, Cb, Db, Eb, Fb, F, Gb

G, A, Bb, C, D, E, F, F#, G

G#, A#, B, C#, D#, E#, F#, F##, G#

Ab, Bb, Cb, Db, Eb, F, Gb, G, Ab

A, B, C, D, E, F#, G, G#, A

A#, B#, C#, D#, E#, F##, G#, G##, A#

Bb, C, Db, Eb, F, G, Ab, A, Bb

B, C#, D, E, F#, G#, A, A#, B

**Bhatiyar**

Step pattern: 1, 3, 1, 1, 1, 2, 2, 1

Ascending intervals: P1, m2, M3, P4, dim5, P5, M6, M7, P8

C, Db, E, F, Gb, G, A, B, C

C#, D, E#, F#, G, G#, A#, B#, C#

Db, Ebb, F, Gb, Abb, Ab, Bb, C, Db

D, Eb, F#, G, Ab, A, B, C#, D

D#, E, F##, G#, A, A#, B#, C##, D#

Eb, Fb, G, Ab, Bbb, Bb, C, D, Eb

E, F, G#, A, Bb, B, C#, D#, E

F, Gb, A, Bb, Cb, C, D, E, F

F#, G, A#, B, C, C#, D#, E#, F#

Gb, Abb, Bb, Cb, Dbb, Db, Eb, F, Gb

G, Ab, B, C, Db, D, E, F#, G

G#, A, B#, C#, D, D#, E#, F##, G#

Ab, Bbb, C, Db, Ebb, Eb, F, G, Ab

A, Bb, C#, D, Eb, E, F#, G#, A

A#, B, C##, D#, E, E#, F##, G##, A#

Bb, Cb, D, Eb, Fb, F, G, A, Bb

B, C, D#, E, F, F#, G#, A#, B



## Blues Octatonic

Step pattern: 2, 1, 2, 1, 1, 2, 1, 2

Ascending intervals: P1, M2, m3, P4, dim5, P5, M6, m7, P8

C, D, Eb, F, Gb, G, A, Bb, C

C#, D#, E, F#, G, G#, A#, B, C#

Db, Eb, Fb, Gb, Abb, Ab, Bb, Cb, Db

D, E, F, G, Ab, A, B, C, D

D#, E#, F#, G#, A, A#, B#, C#, D#

Eb, F, Gb, Ab, Bbb, Bb, C, Db, Eb

E, F#, G, A, Bb, B, C#, D, E

F, G, Ab, Bb, Cb, C, D, Eb, F

F#, G#, A, B, C, C#, D#, E, F#

Gb, Ab, Bbb, Cb, Dbb, Db, Eb, Fb, Gb

G, A, Bb, C, Db, D, E, F, G

G#, A#, B, C#, D, D#, E#, F#, G#

Ab, Bb, Cb, Db, Ebb, Eb, F, Gb, Ab

A, B, C, D, Eb, E, F#, G, A

A#, B#, C#, D#, E, E#, F##, G#, A#

Bb, C, Db, Eb, Fb, F, G, Ab, Bb

B, C#, D, E, F, F#, G#, A, B

### Blues Octatonic (with M3)

Step pattern: 3, 1, 1, 1, 1, 3, 1, 1

Ascending intervals: P1, m3, M3, P4, dim5, P5, m7, M7, P8

C, Eb, E, F, Gb, G, Bb, B, C

C#, E, E#, F#, G, G#, B, B#, C#

Db, Fb, F, Gb, Abb, Ab, Cb, C, Db

D, F, F#, G, Ab, A, C, C#, D

D#, F#, F##, G#, A, A#, C#, C##, D#

Eb, Gb, G, Ab, Bbb, Bb, Db, D, Eb

E, G, G#, A, Bb, B, D, D#, E

F, Ab, A, Bb, Cb, C, Eb, E, F

F#, A, A#, B, C, C#, E, E#, F#

Gb, Bbb, Bb, Cb, Dbb, Db, Fb, F, Gb

G, Bb, B, C, Db, D, F, F#, G

G#, B, B#, C#, D, D#, F#, F##, G#

Ab, Cb, C, Db, Ebb, Eb, Gb, G, Ab

A, C, C#, D, Eb, E, G, G#, A

A#, C#, C##, D#, E, E#, G#, G##, A#

Bb, Db, D, Eb, Fb, F, Ab, A, Bb

B, D, D#, E, F, F#, A, A#, B

## Chinthamani

Step pattern: 2, 1, 3, 1, 1, 1, 1, 2

Ascending intervals: P1, M2, m3, dim5, P5, m6, M6, m7, P8

C, D, Eb, Gb, G, Ab, A, Bb, C

C#, D#, E, G, G#, A, A#, B, C#

Db, Eb, Fb, Abb, Ab, Bbb, Bb, Cb, Db

D, E, F, Ab, A, Bb, B, C, D

D#, E#, F#, A, A#, B, B#, C#, D#

Eb, F, Gb, Bbb, Bb, Cb, C, Db, Eb

E, F#, G, Bb, B, C, C#, D, E

F, G, Ab, Cb, C, Db, D, Eb, F

F#, G#, A, C, C#, D, D#, E, F#

Gb, Ab, Bbb, Dbb, Db, Ebb, Eb, Fb, Gb

G, A, Bb, Db, D, Eb, E, F, G

G#, A#, B, D, D#, E, E#, F#, G#

Ab, Bb, Cb, Ebb, Eb, Fb, F, Gb, Ab

A, B, C, Eb, E, F, F#, G, A

A#, B#, C#, E, E#, F#, F##, G#, A#

Bb, C, Db, Fb, F, Gb, G, Ab, Bb

B, C#, D, F, F#, G, G#, A, B

## Diminished

Step pattern: 2, 1, 2, 1, 2, 1, 2, 1

Ascending intervals: P1, M2, m3, P4, dim5, m6, M6, M7, P8

C, D, Eb, F, Gb, Ab, A, B, C

C#, D#, E, F#, G, A, A#, B#, C#

Db, Eb, Fb, Gb, Abb, Bbb, Bb, C, Db

D, E, F, G, Ab, Bb, B, C#, D

D#, E#, F#, G#, A, B, B#, C##, D#

Eb, F, Gb, Ab, Bbb, Cb, C, D, Eb

E, F#, G, A, Bb, C, C#, D#, E

F, G, Ab, Bb, Cb, Db, D, E, F

F#, G#, A, B, C, D, D#, E#, F#

Gb, Ab, Bbb, Cb, Dbb, Ebb, Eb, F, Gb

G, A, Bb, C, Db, Eb, E, F#, G

G#, A#, B, C#, D, E, E#, F##, G#

Ab, Bb, Cb, Db, Ebb, Fb, F, G, Ab

A, B, C, D, Eb, F, F#, G#, A

A#, B#, C#, D#, E, F#, F##, G##, A#

Bb, C, Db, Eb, Fb, Gb, G, A, Bb

B, C#, D, E, F, G, G#, A#, B

## Dominant Bebop

Step pattern: 2, 2, 1, 2, 2, 1, 1, 1

Ascending intervals: P1, M2, M3, P4, P5, M6, m7, M7, P8

C, D, E, F, G, A, B $\flat$ , B, C

C $\sharp$ , D $\sharp$ , E $\sharp$ , F $\sharp$ , G $\sharp$ , A $\sharp$ , B, B $\sharp$ , C $\sharp$

D $\flat$ , E $\flat$ , F, G $\flat$ , A $\flat$ , B $\flat$ , C $\flat$ , C, D $\flat$

D, E, F $\sharp$ , G, A, B, C, C $\sharp$ , D

D $\sharp$ , E $\sharp$ , F $\sharp\sharp$ , G $\sharp$ , A $\sharp$ , B $\sharp$ , C $\sharp$ , C $\sharp\sharp$ , D $\sharp$

E $\flat$ , F, G, A $\flat$ , B $\flat$ , C, D $\flat$ , D, E $\flat$

E, F $\sharp$ , G $\sharp$ , A, B, C $\sharp$ , D, D $\sharp$ , E

F, G, A, B $\flat$ , C, D, E $\flat$ , E, F

F $\sharp$ , G $\sharp$ , A $\sharp$ , B, C $\sharp$ , D $\sharp$ , E, E $\sharp$ , F $\sharp$

G $\flat$ , A $\flat$ , B $\flat$ , C $\flat$ , D $\flat$ , E $\flat$ , F $\flat$ , F, G $\flat$

G, A, B, C, D, E, F, F $\sharp$ , G

G $\sharp$ , A $\sharp$ , B $\sharp$ , C $\sharp$ , D $\sharp$ , E $\sharp$ , F $\sharp$ , F $\sharp\sharp$ , G $\sharp$

A $\flat$ , B $\flat$ , C, D $\flat$ , E $\flat$ , F, G $\flat$ , G, A $\flat$

A, B, C $\sharp$ , D, E, F $\sharp$ , G, G $\sharp$ , A

A $\sharp$ , B $\sharp$ , C $\sharp\sharp$ , D $\sharp$ , E $\sharp$ , F $\sharp\sharp$ , G $\sharp$ , G $\sharp\sharp$ , A $\sharp$

B $\flat$ , C, D, E $\flat$ , F, G, A $\flat$ , A, B $\flat$

B, C $\sharp$ , D $\sharp$ , E, F $\sharp$ , G $\sharp$ , A, A $\sharp$ , B

## Dominant Diminished

Step pattern: 1, 2, 1, 2, 1, 2, 1, 2

Ascending intervals: P1, Aug1, Aug2, M3, Aug4, P5, M6, m7, P8

C, C#, D#, E, F#, G, A, Bb, C

C#, C##, D##, E#, F##, G#, A#, B, C#

Db, D, E, F, G, Ab, Bb, Cb, Db

D, D#, E#, F#, G#, A, B, C, D

D#, D##, E##, F##, G##, A#, B#, C#, D#

Eb, E, F#, G, A, Bb, C, Db, Eb

E, E#, F##, G#, A#, B, C#, D, E

F, F#, G#, A, B, C, D, Eb, F

F#, F##, G##, A#, B#, C#, D#, E, F#

Gb, G, A, Bb, C, Db, Eb, Fb, Gb

G, G#, A#, B, C#, D, E, F, G

G#, G##, A##, B#, C##, D#, E#, F#, G#

Ab, A, B, C, D, Eb, F, Gb, Ab

A, A#, B#, C#, D#, E, F#, G, A

A#, A##, B##, C##, D##, E#, F##, G#, A#

Bb, B, C#, D, E, F, G, Ab, Bb

B, B#, C##, D#, E#, F#, G#, A, B

## Dorian and Aeolian Mixed

Step pattern: 2, 1, 2, 2, 1, 1, 1, 2

Ascending intervals: P1, M2, m3, P4, P5, m6, M6, m7, P8

C, D, Eb, F, G, Ab, A, Bb, C

C#, D#, E, F#, G#, A, A#, B, C#

Db, Eb, Fb, Gb, Ab, Bbb, Bb, Cb, Db

D, E, F, G, A, Bb, B, C, D

D#, E#, F#, G#, A#, B, B#, C#, D#

Eb, F, Gb, Ab, Bb, Cb, C, Db, Eb

E, F#, G, A, B, C, C#, D, E

F, G, Ab, Bb, C, Db, D, Eb, F

F#, G#, A, B, C#, D, D#, E, F#

Gb, Ab, Bbb, Cb, Db, Ebb, Eb, Fb, Gb

G, A, Bb, C, D, Eb, E, F, G

G#, A#, B, C#, D#, E, E#, F#, G#

Ab, Bb, Cb, Db, Eb, Fb, F, Gb, Ab

A, B, C, D, E, F, F#, G, A

A#, B#, C#, D#, E#, F#, F##, G#, A#

Bb, C, Db, Eb, F, Gb, G, Ab, Bb

B, C#, D, E, F#, G, G#, A, B

## Dorian Bebop

Step pattern: 2, 1, 1, 1, 2, 2, 1, 2

Ascending intervals: P1, M2, m3, M3, P4, P5, M6, m7, P8

C, D, Eb, E, F, G, A, Bb, C

C#, D#, E, E#, F#, G#, A#, B, C#

Db, Eb, Fb, F, Gb, Ab, Bb, Cb, Db

D, E, F, F#, G, A, B, C, D

D#, E#, F#, F##, G#, A#, B#, C#, D#

Eb, F, Gb, G, Ab, Bb, C, Db, Eb

E, F#, G, G#, A, B, C#, D, E

F, G, Ab, A, Bb, C, D, Eb, F

F#, G#, A, A#, B, C#, D#, E, F#

Gb, Ab, Bbb, Bb, Cb, Db, Eb, Fb, Gb

G, A, Bb, B, C, D, E, F, G

G#, A#, B, B#, C#, D#, E#, F#, G#

Ab, Bb, Cb, C, Db, Eb, F, Gb, Ab

A, B, C, C#, D, E, F#, G, A

A#, B#, C#, C##, D#, E#, F##, G#, A#

Bb, C, Db, D, Eb, F, G, Ab, Bb

B, C#, D, D#, E, F#, G#, A, B



## Half-diminished Bebop

Step pattern: 1, 2, 2, 1, 1, 1, 3, 1

Ascending intervals: P1, m2, m3, P4, dim5, P5, m6, M7, P8

C, Db, Eb, F, Gb, G, Ab, B, C

C#, D, E, F#, G, G#, A, B#, C#

Db, Ebb, Fb, Gb, Abb, Ab, Bbb, C, Db

D, Eb, F, G, Ab, A, Bb, C#, D

D#, E, F#, G#, A, A#, B, C##, D#

Eb, Fb, Gb, Ab, Bbb, Bb, Cb, D, Eb

E, F, G, A, Bb, B, C, D#, E

F, Gb, Ab, Bb, Cb, C, Db, E, F

F#, G, A, B, C, C#, D, E#, F#

Gb, Abb, Bbb, Cb, Dbb, Db, Ebb, F, Gb

G, Ab, Bb, C, Db, D, Eb, F#, G

G#, A, B, C#, D, D#, E, F##, G#

Ab, Bbb, Cb, Db, Ebb, Eb, Fb, G, Ab

A, Bb, C, D, Eb, E, F, G#, A

A#, B, C#, D#, E, E#, F#, G##, A#

Bb, Cb, Db, Eb, Fb, F, Gb, A, Bb

B, C, D, E, F, F#, G, A#, B

## Half-whole Tone Scale

Step pattern: 1, 2, 1, 2, 1, 2, 1, 2

Ascending intervals: P1, m2, m3, M3, Aug4, P5, M6, m7, P8

C, Db, Eb, E, F#, G, A, Bb, C

C#, D, E, E#, F##, G#, A#, B, C#

Db, Ebb, Fb, F, G, Ab, Bb, Cb, Db

D, Eb, F, F#, G#, A, B, C, D

D#, E, F#, F##, G##, A#, B#, C#, D#

Eb, Fb, Gb, G, A, Bb, C, Db, Eb

E, F, G, G#, A#, B, C#, D, E

F, Gb, Ab, A, B, C, D, Eb, F

F#, G, A, A#, B#, C#, D#, E, F#

Gb, Abb, Bbb, Bb, C, Db, Eb, Fb, Gb

G, Ab, Bb, B, C#, D, E, F, G

G#, A, B, B#, C##, D#, E#, F#, G#

Ab, Bbb, Cb, C, D, Eb, F, Gb, Ab

A, Bb, C, C#, D#, E, F#, G, A

A#, B, C#, C##, D##, E#, F##, G#, A#

Bb, Cb, Db, D, E, F, G, Ab, Bb

B, C, D, D#, E#, F#, G#, A, B

## Harmonic and Neapolitan Minor Mixed

Step pattern: 1, 1, 1, 2, 2, 1, 3, 1

Ascending intervals: P1, m2, M2, m3, P4, P5, m6, M7, P8

C, Db, D, Eb, F, G, Ab, B, C

C#, D, D#, E, F#, G#, A, B#, C#

Db, Ebb, Eb, Fb, Gb, Ab, Bbb, C, Db

D, Eb, E, F, G, A, Bb, C#, D

D#, E, E#, F#, G#, A#, B, C##, D#

Eb, Fb, F, Gb, Ab, Bb, Cb, D, Eb

E, F, F#, G, A, B, C, D#, E

F, Gb, G, Ab, Bb, C, Db, E, F

F#, G, G#, A, B, C#, D, E#, F#

Gb, Abb, Ab, Bbb, Cb, Db, Ebb, F, Gb

G, Ab, A, Bb, C, D, Eb, F#, G

G#, A, A#, B, C#, D#, E, F##, G#

Ab, Bbb, Bb, Cb, Db, Eb, Fb, G, Ab

A, Bb, B, C, D, E, F, G#, A

A#, B, B#, C#, D#, E#, F#, G##, A#

Bb, Cb, C, Db, Eb, F, Gb, A, Bb

B, C, C#, D, E, F#, G, A#, B

## Ichikotsucho

Step pattern: 2, 2, 1, 1, 1, 2, 2, 1

Ascending intervals: P1, M2, M3, P4, Aug4, P5, M6, M7, P8

C, D, E, F, F#, G, A, B, C

C#, D#, E#, F#, F##, G#, A#, B#, C#

Db, Eb, F, Gb, G, Ab, Bb, C, Db

D, E, F#, G, G#, A, B, C#, D

D#, E#, F##, G#, G##, A#, B#, C##, D#

Eb, F, G, Ab, A, Bb, C, D, Eb

E, F#, G#, A, A#, B, C#, D#, E

F, G, A, Bb, B, C, D, E, F

F#, G#, A#, B, B#, C#, D#, E#, F#

Gb, Ab, Bb, Cb, C, Db, Eb, F, Gb

G, A, B, C, C#, D, E, F#, G

G#, A#, B#, C#, C##, D#, E#, F##, G#

Ab, Bb, C, Db, D, Eb, F, G, Ab

A, B, C#, D, D#, E, F#, G#, A

A#, B#, C##, D#, D##, E#, F##, G##, A#

Bb, C, D, Eb, E, F, G, A, Bb

B, C#, D#, E, E#, F#, G#, A#, B

## Ionian and Lydian Mixed

Step pattern: 2, 2, 1, 1, 1, 2, 2, 1

Ascending intervals: P1, M2, M3, P4, dim5, P5, M6, M7, P8

C, D, E, F, Gb, G, A, B, C

C#, D#, E#, F#, G, G#, A#, B#, C#

Db, Eb, F, Gb, Abb, Ab, Bb, C, Db

D, E, F#, G, Ab, A, B, C#, D

D#, E#, F##, G#, A, A#, B#, C##, D#

Eb, F, G, Ab, Bbb, Bb, C, D, Eb

E, F#, G#, A, Bb, B, C#, D#, E

F, G, A, Bb, Cb, C, D, E, F

F#, G#, A#, B, C, C#, D#, E#, F#

Gb, Ab, Bb, Cb, Dbb, Db, Eb, F, Gb

G, A, B, C, Db, D, E, F#, G

G#, A#, B#, C#, D, D#, E#, F##, G#

Ab, Bb, C, Db, Ebb, Eb, F, G, Ab

A, B, C#, D, Eb, E, F#, G#, A

A#, B#, C##, D#, E, E#, F##, G##, A#

Bb, C, D, Eb, Fb, F, G, A, Bb

B, C#, D#, E, F, F#, G#, A#, B

## Ionian and Mixolydian Mixed

Step pattern: 2, 2, 1, 2, 2, 1, 1, 1

Ascending intervals: P1, M2, M3, P4, P5, M6, m7, M7, P8

C, D, E, F, G, A, Bb, B, C

C#, D#, E#, F#, G#, A#, B, B#, C#

Db, Eb, F, Gb, Ab, Bb, Cb, C, Db

D, E, F#, G, A, B, C, C#, D

D#, E#, F##, G#, A#, B#, C#, C##, D#

Eb, F, G, Ab, Bb, C, Db, D, Eb

E, F#, G#, A, B, C#, D, D#, E

F, G, A, Bb, C, D, Eb, E, F

F#, G#, A#, B, C#, D#, E, E#, F#

Gb, Ab, Bb, Cb, Db, Eb, Fb, F, Gb

G, A, B, C, D, E, F, F#, G

G#, A#, B#, C#, D#, E#, F#, F##, G#

Ab, Bb, C, Db, Eb, F, Gb, G, Ab

A, B, C#, D, E, F#, G, G#, A

A#, B#, C##, D#, E#, F##, G#, G##, A#

Bb, C, D, Eb, F, G, Ab, A, Bb

B, C#, D#, E, F#, G#, A, A#, B

## Magen Abot

Step pattern: 1, 2, 1, 2, 2, 2, 1, 1

Ascending intervals: P1, m2, Aug2, M3, dim5, m6, Aug6, M7, P8

C, Db, D#, E, Gb, Ab, A#, B, C

C#, D, D##, E#, G, A, A##, B#, C#

Db, Ebb, E, F, Abb, Bbb, B, C, Db

D, Eb, E#, F#, Ab, Bb, B#, C#, D

D#, E, E##, F##, A, B, B##, C##, D#

Eb, Fb, F#, G, Bbb, Cb, C#, D, Eb

E, F, F##, G#, Bb, C, C##, D#, E

F, Gb, G#, A, Cb, Db, D#, E, F

F#, G, G##, A#, C, D, D##, E#, F#

Gb, Abb, A, Bb, Dbb, Ebb, E, F, Gb

G, Ab, A#, B, Db, Eb, E#, F#, G

G#, A, A##, B#, D, E, E##, F##, G#

Ab, Bbb, B, C, Ebb, Fb, F#, G, Ab

A, Bb, B#, C#, Eb, F, F##, G#, A

A#, B, B##, C##, E, F#, F####, G##, A#

Bb, Cb, C#, D, Fb, Gb, G#, A, Bb

B, C, C##, D#, F, G, G##, A#, B

## Major Bebop

Step pattern: 2, 2, 1, 2, 1, 1, 2, 1

Ascending intervals: P1, M2, M3, P4, P5, Aug5, M6, M7, P8

C, D, E, F, G, G#, A, B, C

C#, D#, E#, F#, G#, G##, A#, B#, C#

Db, Eb, F, Gb, Ab, A, Bb, C, Db

D, E, F#, G, A, A#, B, C#, D

D#, E#, F##, G#, A#, A##, B#, C##, D#

Eb, F, G, Ab, Bb, B, C, D, Eb

E, F#, G#, A, B, B#, C#, D#, E

F, G, A, Bb, C, C#, D, E, F

F#, G#, A#, B, C#, C##, D#, E#, F#

Gb, Ab, Bb, Cb, Db, D, Eb, F, Gb

G, A, B, C, D, D#, E, F#, G

G#, A#, B#, C#, D#, D##, E#, F##, G#

Ab, Bb, C, Db, Eb, E, F, G, Ab

A, B, C#, D, E, E#, F#, G#, A

A#, B#, C##, D#, E#, E##, F##, G##, A#

Bb, C, D, Eb, F, F#, G, A, Bb

B, C#, D#, E, F#, F##, G#, A#, B



## Makam Hijaz

Step pattern: 1, 3, 1, 2, 1, 2, 1, 1

Ascending intervals: P1, m2, M3, P4, P5, m6, m7, M7, P8

C, Db, E, F, G, Ab, Bb, B, C

C#, D, E#, F#, G#, A, B, B#, C#

Db, Ebb, F, Gb, Ab, Bbb, Cb, C, Db

D, Eb, F#, G, A, Bb, C, C#, D

D#, E, F##, G#, A#, B, C#, C##, D#

Eb, Fb, G, Ab, Bb, Cb, Db, D, Eb

E, F, G#, A, B, C, D, D#, E

F, Gb, A, Bb, C, Db, Eb, E, F

F#, G, A#, B, C#, D, E, E#, F#

Gb, Abb, Bb, Cb, Db, Ebb, Fb, F, Gb

G, Ab, B, C, D, Eb, F, F#, G

G#, A, B#, C#, D#, E, F#, F##, G#

Ab, Bbb, C, Db, Eb, Fb, Gb, G, Ab

A, Bb, C#, D, E, F, G, G#, A

A#, B, C##, D#, E#, F#, G#, G##, A#

Bb, Cb, D, Eb, F, Gb, Ab, A, Bb

B, C, D#, E, F#, G, A, A#, B

## Maqam Shaddaraban

Step pattern: 1, 2, 1, 1, 1, 3, 1, 2

Ascending intervals: P1, m2, m3, M3, P4, dim5, M6, m7, P8

C, Db, Eb, E, F, Gb, A, Bb, C

C#, D, E, E#, F#, G, A#, B, C#

Db, Ebb, Fb, F, Gb, Abb, Bb, Cb, Db

D, Eb, F, F#, G, Ab, B, C, D

D#, E, F#, F##, G#, A, B#, C#, D#

Eb, Fb, Gb, G, Ab, Bbb, C, Db, Eb

E, F, G, G#, A, Bb, C#, D, E

F, Gb, Ab, A, Bb, Cb, D, Eb, F

F#, G, A, A#, B, C, D#, E, F#

Gb, Abb, Bbb, Bb, Cb, Dbb, Eb, Fb, Gb

G, Ab, Bb, B, C, Db, E, F, G

G#, A, B, B#, C#, D, E#, F#, G#

Ab, Bbb, Cb, C, Db, Ebb, F, Gb, Ab

A, Bb, C, C#, D, Eb, F#, G, A

A#, B, C#, C##, D#, E, F##, G#, A#

Bb, Cb, Db, D, Eb, Fb, G, Ab, Bb

B, C, D, D#, E, F, G#, A, B

### Messiaen Mode 2 (4 groups of 3 notes each)

Step pattern: 1, 2, 1, 2, 1, 2, 1, 2

Ascending intervals: P1, m2, m3, M3, Aug4, P5, M6, m7, P8

C, Db, Eb, E, F#, G, A, Bb, C

C#, D, E, E#, F##, G#, A#, B, C#

Db, Ebb, Fb, F, G, Ab, Bb, Cb, Db

D, Eb, F, F#, G#, A, B, C, D

D#, E, F#, F##, G##, A#, B#, C#, D#

Eb, Fb, Gb, G, A, Bb, C, Db, Eb

E, F, G, G#, A#, B, C#, D, E

F, Gb, Ab, A, B, C, D, Eb, F

F#, G, A, A#, B#, C#, D#, E, F#

Gb, Abb, Bbb, Bb, C, Db, Eb, Fb, Gb

G, Ab, Bb, B, C#, D, E, F, G

G#, A, B, B#, C##, D#, E#, F#, G#

Ab, Bbb, Cb, C, D, Eb, F, Gb, Ab

A, Bb, C, C#, D#, E, F#, G, A

A#, B, C#, C##, D##, E#, F##, G#, A#

Bb, Cb, Db, D, E, F, G, Ab, Bb

B, C, D, D#, E#, F#, G#, A, B

### Messiaen Mode 2 Inverse (4 groups of 3 notes each)

Step pattern: 2, 1, 2, 1, 2, 1, 2, 1

Ascending intervals: P1, M2, m3, P4, dim5, m6, M6, M7, P8

C, D, Eb, F, Gb, Ab, A, B, C

C#, D#, E, F#, G, A, A#, B#, C#

Db, Eb, Fb, Gb, Abb, Bbb, Bb, C, Db

D, E, F, G, Ab, Bb, B, C#, D

D#, E#, F#, G#, A, B, B#, C##, D#

Eb, F, Gb, Ab, Bbb, Cb, C, D, Eb

E, F#, G, A, Bb, C, C#, D#, E

F, G, Ab, Bb, Cb, Db, D, E, F

F#, G#, A, B, C, D, D#, E#, F#

Gb, Ab, Bbb, Cb, Dbb, Ebb, Eb, F, Gb

G, A, Bb, C, Db, Eb, E, F#, G

G#, A#, B, C#, D, E, E#, F##, G#

Ab, Bb, Cb, Db, Ebb, Fb, F, G, Ab

A, B, C, D, Eb, F, F#, G#, A

A#, B#, C#, D#, E, F#, F##, G##, A#

Bb, C, Db, Eb, Fb, Gb, G, A, Bb

B, C#, D, E, F, G, G#, A#, B

### Messiaen Mode 4 (2 groups of 5 notes each)

Step pattern: 1, 1, 3, 1, 1, 1, 3, 1

Ascending intervals: P1, m2, M2, P4, Aug4, P5, m6, M7, P8

C, Db, D, F, F#, G, Ab, B, C

C#, D, D#, F#, F##, G#, A, B#, C#

Db, Ebb, Eb, Gb, G, Ab, Bbb, C, Db

D, Eb, E, G, G#, A, Bb, C#, D

D#, E, E#, G#, G##, A#, B, C##, D#

Eb, Fb, F, Ab, A, Bb, Cb, D, Eb

E, F, F#, A, A#, B, C, D#, E

F, Gb, G, Bb, B, C, Db, E, F

F#, G, G#, B, B#, C#, D, E#, F#

Gb, Abb, Ab, Cb, C, Db, Ebb, F, Gb

G, Ab, A, C, C#, D, Eb, F#, G

G#, A, A#, C#, C##, D#, E, F##, G#

Ab, Bbb, Bb, Db, D, Eb, Fb, G, Ab

A, Bb, B, D, D#, E, F, G#, A

A#, B, B#, D#, D##, E#, F#, G##, A#

Bb, Cb, C, Eb, E, F, Gb, A, Bb

B, C, C#, E, E#, F#, G, A#, B

### Messiaen Mode 4 Inverse (2 groups of 5 notes each)

Step pattern: 3, 1, 1, 1, 3, 1, 1, 1

Ascending intervals: P1, m3, M3, P4, dim5, M6, m7, M7, P8

C, Eb, E, F, Gb, A, Bb, B, C

C#, E, E#, F#, G, A#, B, B#, C#

Db, Fb, F, Gb, Abb, Bb, Cb, C, Db

D, F, F#, G, Ab, B, C, C#, D

D#, F#, F##, G#, A, B#, C#, C##, D#

Eb, Gb, G, Ab, Bbb, C, Db, D, Eb

E, G, G#, A, Bb, C#, D, D#, E

F, Ab, A, Bb, Cb, D, Eb, E, F

F#, A, A#, B, C, D#, E, E#, F#

Gb, Bbb, Bb, Cb, Dbb, Eb, Fb, F, Gb

G, Bb, B, C, Db, E, F, F#, G

G#, B, B#, C#, D, E#, F#, F##, G#

Ab, Cb, C, Db, Ebb, F, Gb, G, Ab

A, C, C#, D, Eb, F#, G, G#, A

A#, C#, C##, D#, E, F##, G#, G##, A#

Bb, Db, D, Eb, Fb, G, Ab, A, Bb

B, D, D#, E, F, G#, A, A#, B

### Messiaen Mode 5 (2 groups of 5 notes each)

Step pattern: 1, 1, 3, 1, 1, 1, 3, 1

Ascending intervals: P1, m2, M2, P4, Aug4, P5, m6, M7, P8

C, Db, D, F, F#, G, Ab, B, C

C#, D, D#, F#, F##, G#, A, B#, C#

Db, Ebb, Eb, Gb, G, Ab, Bbb, C, Db

D, Eb, E, G, G#, A, Bb, C#, D

D#, E, E#, G#, G##, A#, B, C##, D#

Eb, Fb, F, Ab, A, Bb, Cb, D, Eb

E, F, F#, A, A#, B, C, D#, E

F, Gb, G, Bb, B, C, Db, E, F

F#, G, G#, B, B#, C#, D, E#, F#

Gb, Abb, Ab, Cb, C, Db, Ebb, F, Gb

G, Ab, A, C, C#, D, Eb, F#, G

G#, A, A#, C#, C##, D#, E, F##, G#

Ab, Bbb, Bb, Db, D, Eb, Fb, G, Ab

A, Bb, B, D, D#, E, F, G#, A

A#, B, B#, D#, D##, E#, F#, G##, A#

Bb, Cb, C, Eb, E, F, Gb, A, Bb

B, C, C#, E, E#, F#, G, A#, B

### Messiaen Mode 6 (2 groups of 5 notes each)

Step pattern: 2, 2, 1, 1, 2, 2, 1, 1

Ascending intervals: P1, M2, M3, P4, Aug4, Aug5, Aug6, M7, P8

C, D, E, F, F#, G#, A#, B, C

C#, D#, E#, F#, F##, G##, A##, B#, C#

Db, Eb, F, Gb, G, A, B, C, Db

D, E, F#, G, G#, A#, B#, C#, D

D#, E#, F##, G#, G##, A##, B##, C##, D#

Eb, F, G, Ab, A, B, C#, D, Eb

E, F#, G#, A, A#, B#, C##, D#, E

F, G, A, Bb, B, C#, D#, E, F

F#, G#, A#, B, B#, C##, D##, E#, F#

Gb, Ab, Bb, Cb, C, D, E, F, Gb

G, A, B, C, C#, D#, E#, F#, G

G#, A#, B#, C#, C##, D##, E##, F##, G#

Ab, Bb, C, Db, D, E, F#, G, Ab

A, B, C#, D, D#, E#, F##, G#, A

A#, B#, C##, D#, D##, E##, F###, G##, A#

Bb, C, D, Eb, E, F#, G#, A, Bb

B, C#, D#, E, E#, F##, G##, A#, B



# **Messiaen Mode 6 Inverse (2 groups of 5 notes each)**

Step pattern: 2, 2, 1, 1, 2, 2, 1, 1

Ascending intervals: P1, M2, M3, P4, dim5, m6, m7, M7, P8

C, D, E, F, Gb, Ab, Bb, B, C

C#, D#, E#, F#, G, A, B, B#, C#

Db, Eb, F, Gb, Abb, Bbb, Cb, C, Db

D, E, F#, G, Ab, Bb, C, C#, D

D#, E#, F##, G#, A, B, C#, C##, D#

Eb, F, G, Ab, Bbb, Cb, Db, D, Eb

E, F#, G#, A, Bb, C, D, D#, E

F, G, A, Bb, Cb, Db, Eb, E, F

F#, G#, A#, B, C, D, E, E#, F#

Gb, Ab, Bb, Cb, Dbb, Ebb, Fb, F, Gb

G, A, B, C, Db, Eb, F, F#, G

G#, A#, B#, C#, D, E, F#, F##, G#

Ab, Bb, C, Db, Ebb, Fb, Gb, G, Ab

A, B, C#, D, Eb, F, G, G#, A

A#, B#, C##, D#, E, F#, G#, G##, A#

Bb, C, D, Eb, Fb, Gb, Ab, A, Bb

B, C#, D#, E, F, G, A, A#, B

## Minor Bebop

Step pattern: 2, 1, 1, 1, 2, 2, 1, 2

Ascending intervals: P1, M2, m3, M3, P4, P5, M6, m7, P8

C, D, Eb, E, F, G, A, Bb, C

C#, D#, E, E#, F#, G#, A#, B, C#

Db, Eb, Fb, F, Gb, Ab, Bb, Cb, Db

D, E, F, F#, G, A, B, C, D

D#, E#, F#, F##, G#, A#, B#, C#, D#

Eb, F, Gb, G, Ab, Bb, C, Db, Eb

E, F#, G, G#, A, B, C#, D, E

F, G, Ab, A, Bb, C, D, Eb, F

F#, G#, A, A#, B, C#, D#, E, F#

Gb, Ab, Bbb, Bb, Cb, Db, Eb, Fb, Gb

G, A, Bb, B, C, D, E, F, G

G#, A#, B, B#, C#, D#, E#, F#, G#

Ab, Bb, Cb, C, Db, Eb, F, Gb, Ab

A, B, C, C#, D, E, F#, G, A

A#, B#, C#, C##, D#, E#, F##, G#, A#

Bb, C, Db, D, Eb, F, G, Ab, Bb

B, C#, D, D#, E, F#, G#, A, B

## Mixolydian and Dorian Mixed

Step pattern: 2, 1, 1, 1, 2, 2, 1, 2

Ascending intervals: P1, M2, m3, M3, P4, P5, M6, m7, P8

C, D, Eb, E, F, G, A, Bb, C

C#, D#, E, E#, F#, G#, A#, B, C#

Db, Eb, Fb, F, Gb, Ab, Bb, Cb, Db

D, E, F, F#, G, A, B, C, D

D#, E#, F#, F##, G#, A#, B#, C#, D#

Eb, F, Gb, G, Ab, Bb, C, Db, Eb

E, F#, G, G#, A, B, C#, D, E

F, G, Ab, A, Bb, C, D, Eb, F

F#, G#, A, A#, B, C#, D#, E, F#

Gb, Ab, Bbb, Bb, Cb, Db, Eb, Fb, Gb

G, A, Bb, B, C, D, E, F, G

G#, A#, B, B#, C#, D#, E#, F#, G#

Ab, Bb, Cb, C, Db, Eb, F, Gb, Ab

A, B, C, C#, D, E, F#, G, A

A#, B#, C#, C##, D#, E#, F##, G#, A#

Bb, C, Db, D, Eb, F, G, Ab, Bb

B, C#, D, D#, E, F#, G#, A, B

**Neveseri**

Step pattern: 1, 2, 3, 1, 1, 2, 1, 1

Ascending intervals: P1, m2, m3, dim5, P5, m6, m7, M7, P8

C, Db, Eb, Gb, G, Ab, Bb, B, C

C#, D, E, G, G#, A, B, B#, C#

Db, Ebb, Fb, Abb, Ab, Bbb, Cb, C, Db

D, Eb, F, Ab, A, Bb, C, C#, D

D#, E, F#, A, A#, B, C#, C##, D#

Eb, Fb, Gb, Bbb, Bb, Cb, Db, D, Eb

E, F, G, Bb, B, C, D, D#, E

F, Gb, Ab, Cb, C, Db, Eb, E, F

F#, G, A, C, C#, D, E, E#, F#

Gb, Abb, Bbb, Dbb, Db, Ebb, Fb, F, Gb

G, Ab, Bb, Db, D, Eb, F, F#, G

G#, A, B, D, D#, E, F#, F##, G#

Ab, Bbb, Cb, Ebb, Eb, Fb, Gb, G, Ab

A, Bb, C, Eb, E, F, G, G#, A

A#, B, C#, E, E#, F#, G#, G##, A#

Bb, Cb, Db, Fb, F, Gb, Ab, A, Bb

B, C, D, F, F#, G, A, A#, B

## Phrygian and Aeolian Mixed

Step pattern: 1, 1, 1, 2, 2, 1, 2, 2

Ascending intervals: P1, m2, M2, m3, P4, P5, m6, m7, P8

C, Db, D, Eb, F, G, Ab, Bb, C

C#, D, D#, E, F#, G#, A, B, C#

Db, Ebb, Eb, Fb, Gb, Ab, Bbb, Cb, Db

D, Eb, E, F, G, A, Bb, C, D

D#, E, E#, F#, G#, A#, B, C#, D#

Eb, Fb, F, Gb, Ab, Bb, Cb, Db, Eb

E, F, F#, G, A, B, C, D, E

F, Gb, G, Ab, Bb, C, Db, Eb, F

F#, G, G#, A, B, C#, D, E, F#

Gb, Abb, Ab, Bbb, Cb, Db, Ebb, Fb, Gb

G, Ab, A, Bb, C, D, Eb, F, G

G#, A, A#, B, C#, D#, E, F#, G#

Ab, Bbb, Bb, Cb, Db, Eb, Fb, Gb, Ab

A, Bb, B, C, D, E, F, G, A

A#, B, B#, C#, D#, E#, F#, G#, A#

Bb, Cb, C, Db, Eb, F, Gb, Ab, Bb

B, C, C#, D, E, F#, G, A, B

## Phrygian and Locrian Mixed

Step pattern: 1, 2, 2, 1, 1, 1, 2, 2

Ascending intervals: P1, m2, m3, P4, dim5, P5, m6, m7, P8

C, Db, Eb, F, Gb, G, Ab, Bb, C

C#, D, E, F#, G, G#, A, B, C#

Db, Ebb, Fb, Gb, Abb, Ab, Bbb, Cb, Db

D, Eb, F, G, Ab, A, Bb, C, D

D#, E, F#, G#, A, A#, B, C#, D#

Eb, Fb, Gb, Ab, Bbb, Bb, Cb, Db, Eb

E, F, G, A, Bb, B, C, D, E

F, Gb, Ab, Bb, Cb, C, Db, Eb, F

F#, G, A, B, C, C#, D, E, F#

Gb, Abb, Bbb, Cb, Dbb, Db, Ebb, Fb, Gb

G, Ab, Bb, C, Db, D, Eb, F, G

G#, A, B, C#, D, D#, E, F#, G#

Ab, Bbb, Cb, Db, Ebb, Eb, Fb, Gb, Ab

A, Bb, C, D, Eb, E, F, G, A

A#, B, C#, D#, E, E#, F#, G#, A#

Bb, Cb, Db, Eb, Fb, F, Gb, Ab, Bb

B, C, D, E, F, F#, G, A, B

## Phrygian and Neapolitan Minor Mixed

Step pattern: 1, 2, 2, 2, 1, 2, 1, 1

Ascending intervals: P1, m2, m3, P4, P5, m6, m7, M7, P8

C, Db, Eb, F, G, Ab, Bb, B, C

C#, D, E, F#, G#, A, B, B#, C#

Db, Ebb, Fb, Gb, Ab, Bbb, Cb, C, Db

D, Eb, F, G, A, Bb, C, C#, D

D#, E, F#, G#, A#, B, C#, C##, D#

Eb, Fb, Gb, Ab, Bb, Cb, Db, D, Eb

E, F, G, A, B, C, D, D#, E

F, Gb, Ab, Bb, C, Db, Eb, E, F

F#, G, A, B, C#, D, E, E#, F#

Gb, Abb, Bbb, Cb, Db, Ebb, Fb, F, Gb

G, Ab, Bb, C, D, Eb, F, F#, G

G#, A, B, C#, D#, E, F#, F##, G#

Ab, Bbb, Cb, Db, Eb, Fb, Gb, G, Ab

A, Bb, C, D, E, F, G, G#, A

A#, B, C#, D#, E#, F#, G#, G##, A#

Bb, Cb, Db, Eb, F, Gb, Ab, A, Bb

B, C, D, E, F#, G, A, A#, B

**Prokofiev**

Step pattern: 1, 2, 2, 1, 2, 2, 1, 1

Ascending intervals: P1, m2, m3, P4, dim5, m6, m7, M7, P8

C, Db, Eb, F, Gb, Ab, Bb, B, C

C#, D, E, F#, G, A, B, B#, C#

Db, Ebb, Fb, Gb, Abb, Bbb, Cb, C, Db

D, Eb, F, G, Ab, Bb, C, C#, D

D#, E, F#, G#, A, B, C#, C##, D#

Eb, Fb, Gb, Ab, Bbb, Cb, Db, D, Eb

E, F, G, A, Bb, C, D, D#, E

F, Gb, Ab, Bb, Cb, Db, Eb, E, F

F#, G, A, B, C, D, E, E#, F#

Gb, Abb, Bbb, Cb, Dbb, Ebb, Fb, F, Gb

G, Ab, Bb, C, Db, Eb, F, F#, G

G#, A, B, C#, D, E, F#, F##, G#

Ab, Bbb, Cb, Db, Ebb, Fb, Gb, G, Ab

A, Bb, C, D, Eb, F, G, G#, A

A#, B, C#, D#, E, F#, G#, G##, A#

Bb, Cb, Db, Eb, Fb, Gb, Ab, A, Bb

B, C, D, E, F, G, A, A#, B



## Ramkali

Step pattern: 1, 3, 1, 1, 1, 1, 3, 1

Ascending intervals: P1, m2, M3, P4, dim5, P5, m6, M7, P8

C, Db, E, F, Gb, G, Ab, B, C

C#, D, E#, F#, G, G#, A, B#, C#

Db, Ebb, F, Gb, Abb, Ab, Bbb, C, Db

D, Eb, F#, G, Ab, A, Bb, C#, D

D#, E, F##, G#, A, A#, B, C##, D#

Eb, Fb, G, Ab, Bbb, Bb, Cb, D, Eb

E, F, G#, A, Bb, B, C, D#, E

F, Gb, A, Bb, Cb, C, Db, E, F

F#, G, A#, B, C, C#, D, E#, F#

Gb, Abb, Bb, Cb, Dbb, Db, Ebb, F, Gb

G, Ab, B, C, Db, D, Eb, F#, G

G#, A, B#, C#, D, D#, E, F##, G#

Ab, Bbb, C, Db, Ebb, Eb, Fb, G, Ab

A, Bb, C#, D, Eb, E, F, G#, A

A#, B, C##, D#, E, E#, F#, G##, A#

Bb, Cb, D, Eb, Fb, F, Gb, A, Bb

B, C, D#, E, F, F#, G, A#, B

## Saurashtra

Step pattern: 1, 3, 1, 2, 1, 1, 2, 1

Ascending intervals: P1, m2, M3, P4, P5, m6, M6, M7, P8

C, Db, E, F, G, Ab, A, B, C

C#, D, E#, F#, G#, A, A#, B#, C#

Db, Ebb, F, Gb, Ab, Bbb, Bb, C, Db

D, Eb, F#, G, A, Bb, B, C#, D

D#, E, F##, G#, A#, B, B#, C##, D#

Eb, Fb, G, Ab, Bb, Cb, C, D, Eb

E, F, G#, A, B, C, C#, D#, E

F, Gb, A, Bb, C, Db, D, E, F

F#, G, A#, B, C#, D, D#, E#, F#

Gb, Abb, Bb, Cb, Db, Ebb, Eb, F, Gb

G, Ab, B, C, D, Eb, E, F#, G

G#, A, B#, C#, D#, E, E#, F##, G#

Ab, Bbb, C, Db, Eb, Fb, F, G, Ab

A, Bb, C#, D, E, F, F#, G#, A

A#, B, C##, D#, E#, F#, F##, G##, A#

Bb, Cb, D, Eb, F, Gb, G, A, Bb

B, C, D#, E, F#, G, G#, A#, B

## Shostakovich

Step pattern: 1, 2, 1, 2, 1, 2, 2, 1

Ascending intervals: P1, m2, m3, M3, dim5, P5, M6, M7, P8

C, Db, Eb, E, Gb, G, A, B, C

C#, D, E, E#, G, G#, A#, B#, C#

Db, Ebb, Fb, F, Abb, Ab, Bb, C, Db

D, Eb, F, F#, Ab, A, B, C#, D

D#, E, F#, F##, A, A#, B#, C##, D#

Eb, Fb, Gb, G, Bbb, Bb, C, D, Eb

E, F, G, G#, Bb, B, C#, D#, E

F, Gb, Ab, A, Cb, C, D, E, F

F#, G, A, A#, C, C#, D#, E#, F#

Gb, Abb, Bbb, Bb, Dbb, Db, Eb, F, Gb

G, Ab, Bb, B, Db, D, E, F#, G

G#, A, B, B#, D, D#, E#, F##, G#

Ab, Bbb, Cb, C, Ebb, Eb, F, G, Ab

A, Bb, C, C#, Eb, E, F#, G#, A

A#, B, C#, C##, E, E#, F##, G##, A#

Bb, Cb, Db, D, Fb, F, G, A, Bb

B, C, D, D#, F, F#, G#, A#, B

### Spanish Phrygian (add M3)

Step pattern: 1, 2, 1, 1, 2, 1, 2, 2

Ascending intervals: P1, m2, m3, M3, P4, P5, m6, m7, P8

C, Db, Eb, E, F, G, Ab, Bb, C

C#, D, E, E#, F#, G#, A, B, C#

Db, Ebb, Fb, F, Gb, Ab, Bbb, Cb, Db

D, Eb, F, F#, G, A, Bb, C, D

D#, E, F#, F##, G#, A#, B, C#, D#

Eb, Fb, Gb, G, Ab, Bb, Cb, Db, Eb

E, F, G, G#, A, B, C, D, E

F, Gb, Ab, A, Bb, C, Db, Eb, F

F#, G, A, A#, B, C#, D, E, F#

Gb, Abb, Bbb, Bb, Cb, Db, Ebb, Fb, Gb

G, Ab, Bb, B, C, D, Eb, F, G

G#, A, B, B#, C#, D#, E, F#, G#

Ab, Bbb, Cb, C, Db, Eb, Fb, Gb, Ab

A, Bb, C, C#, D, E, F, G, A

A#, B, C#, C##, D#, E#, F#, G#, A#

Bb, Cb, Db, D, Eb, F, Gb, Ab, Bb

B, C, D, D#, E, F#, G, A, B

## Symmetrical Diminished

Step pattern: 1, 2, 1, 2, 1, 2, 1, 2

Ascending intervals: P1, m2, m3, M3, Aug4, P5, M6, m7, P8

C, Db, Eb, E, F#, G, A, Bb, C

C#, D, E, E#, F##, G#, A#, B, C#

Db, Ebb, Fb, F, G, Ab, Bb, Cb, Db

D, Eb, F, F#, G#, A, B, C, D

D#, E, F#, F##, G##, A#, B#, C#, D#

Eb, Fb, Gb, G, A, Bb, C, Db, Eb

E, F, G, G#, A#, B, C#, D, E

F, Gb, Ab, A, B, C, D, Eb, F

F#, G, A, A#, B#, C#, D#, E, F#

Gb, Abb, Bbb, Bb, C, Db, Eb, Fb, Gb

G, Ab, Bb, B, C#, D, E, F, G

G#, A, B, B#, C##, D#, E#, F#, G#

Ab, Bbb, Cb, C, D, Eb, F, Gb, Ab

A, Bb, C, C#, D#, E, F#, G, A

A#, B, C#, C##, D##, E#, F##, G#, A#

Bb, Cb, Db, D, E, F, G, Ab, Bb

B, C, D, D#, E#, F#, G#, A, B

## Utility Minor

Step pattern: 2, 1, 2, 2, 1, 2, 1, 1

Ascending intervals: P1, M2, m3, P4, P5, m6, m7, M7, P8

C, D, Eb, F, G, Ab, Bb, B, C

C#, D#, E, F#, G#, A, B, B#, C#

Db, Eb, Fb, Gb, Ab, Bbb, Cb, C, Db

D, E, F, G, A, Bb, C, C#, D

D#, E#, F#, G#, A#, B, C#, C##, D#

Eb, F, Gb, Ab, Bb, Cb, Db, D, Eb

E, F#, G, A, B, C, D, D#, E

F, G, Ab, Bb, C, Db, Eb, E, F

F#, G#, A, B, C#, D, E, E#, F#

Gb, Ab, Bbb, Cb, Db, Ebb, Fb, F, Gb

G, A, Bb, C, D, Eb, F, F#, G

G#, A#, B, C#, D#, E, F#, F##, G#

Ab, Bb, Cb, Db, Eb, Fb, Gb, G, Ab

A, B, C, D, E, F, G, G#, A

A#, B#, C#, D#, E#, F#, G#, G##, A#

Bb, C, Db, Eb, F, Gb, Ab, A, Bb

B, C#, D, E, F#, G, A, A#, B

## Van der Horst Octatonic

Step pattern: 1, 2, 2, 1, 1, 2, 2, 1

Ascending intervals: P1, m2, m3, P4, dim5, P5, M6, M7, P8

C, Db, Eb, F, Gb, G, A, B, C

C#, D, E, F#, G, G#, A#, B#, C#

Db, Ebb, Fb, Gb, Abb, Ab, Bb, C, Db

D, Eb, F, G, Ab, A, B, C#, D

D#, E, F#, G#, A, A#, B#, C##, D#

Eb, Fb, Gb, Ab, Bbb, Bb, C, D, Eb

E, F, G, A, Bb, B, C#, D#, E

F, Gb, Ab, Bb, Cb, C, D, E, F

F#, G, A, B, C, C#, D#, E#, F#

Gb, Abb, Bbb, Cb, Dbb, Db, Eb, F, Gb

G, Ab, Bb, C, Db, D, E, F#, G

G#, A, B, C#, D, D#, E#, F##, G#

Ab, Bbb, Cb, Db, Ebb, Eb, F, G, Ab

A, Bb, C, D, Eb, E, F#, G#, A

A#, B, C#, D#, E, E#, F##, G##, A#

Bb, Cb, Db, Eb, Fb, F, G, A, Bb

B, C, D, E, F, F#, G#, A#, B

## Verdis Scala Enigmatica

Step pattern: 1, 3, 1, 1, 2, 2, 1, 1

Ascending intervals: P1, m2, M3, P4, dim5, m6, m7, M7, P8

C, Db, E, F, Gb, Ab, Bb, B, C

C#, D, E#, F#, G, A, B, B#, C#

Db, Ebb, F, Gb, Abb, Bbb, Cb, C, Db

D, Eb, F#, G, Ab, Bb, C, C#, D

D#, E, F##, G#, A, B, C#, C##, D#

Eb, Fb, G, Ab, Bbb, Cb, Db, D, Eb

E, F, G#, A, Bb, C, D, D#, E

F, Gb, A, Bb, Cb, Db, Eb, E, F

F#, G, A#, B, C, D, E, E#, F#

Gb, Abb, Bb, Cb, Dbb, Ebb, Fb, F, Gb

G, Ab, B, C, Db, Eb, F, F#, G

G#, A, B#, C#, D, E, F#, F##, G#

Ab, Bbb, C, Db, Ebb, Fb, Gb, G, Ab

A, Bb, C#, D, Eb, F, G, G#, A

A#, B, C##, D#, E, F#, G#, G##, A#

Bb, Cb, D, Eb, Fb, Gb, Ab, A, Bb

B, C, D#, E, F, G, A, A#, B



## Whole-half Tone Scale

Step pattern: 2, 1, 2, 1, 2, 1, 2, 1

Ascending intervals: P1, M2, m3, P4, dim5, m6, M6, M7, P8

C, D, Eb, F, Gb, Ab, A, B, C

C#, D#, E, F#, G, A, A#, B#, C#

Db, Eb, Fb, Gb, Abb, Bbb, Bb, C, Db

D, E, F, G, Ab, Bb, B, C#, D

D#, E#, F#, G#, A, B, B#, C##, D#

Eb, F, Gb, Ab, Bbb, Cb, C, D, Eb

E, F#, G, A, Bb, C, C#, D#, E

F, G, Ab, Bb, Cb, Db, D, E, F

F#, G#, A, B, C, D, D#, E#, F#

Gb, Ab, Bbb, Cb, Dbb, Ebb, Eb, F, Gb

G, A, Bb, C, Db, Eb, E, F#, G

G#, A#, B, C#, D, E, E#, F##, G#

Ab, Bb, Cb, Db, Ebb, Fb, F, G, Ab

A, B, C, D, Eb, F, F#, G#, A

A#, B#, C#, D#, E, F#, F##, G##, A#

Bb, C, Db, Eb, Fb, Gb, G, A, Bb

B, C#, D, E, F, G, G#, A#, B

## Zirafkend

Step pattern: 2, 1, 2, 2, 1, 1, 2, 1

Ascending intervals: P1, M2, m3, P4, P5, m6, M6, M7, P8

C, D, Eb, F, G, Ab, A, B, C

C#, D#, E, F#, G#, A, A#, B#, C#

Db, Eb, Fb, Gb, Ab, Bbb, Bb, C, Db

D, E, F, G, A, Bb, B, C#, D

D#, E#, F#, G#, A#, B, B#, C##, D#

Eb, F, Gb, Ab, Bb, Cb, C, D, Eb

E, F#, G, A, B, C, C#, D#, E

F, G, Ab, Bb, C, Db, D, E, F

F#, G#, A, B, C#, D, D#, E#, F#

Gb, Ab, Bbb, Cb, Db, Ebb, Eb, F, Gb

G, A, Bb, C, D, Eb, E, F#, G

G#, A#, B, C#, D#, E, E#, F##, G#

Ab, Bb, Cb, Db, Eb, Fb, F, G, Ab

A, B, C, D, E, F, F#, G#, A

A#, B#, C#, D#, E#, F#, F##, G##, A#

Bb, C, Db, Eb, F, Gb, G, A, Bb

B, C#, D, E, F#, G, G#, A#, B

## 9 – NOTE SCALES

## Blues Nonatonic (with M2)

Step pattern: 2, 1, 1, 1, 1, 1, 2, 1, 2

Ascending intervals: P1, M2, m3, M3, P4, dim5, P5, M6, m7, P8

C, D, Eb, E, F, Gb, G, A, Bb, C

C#, D#, E, E#, F#, G, G#, A#, B, C#

Db, Eb, Fb, F, Gb, Abb, Ab, Bb, Cb, Db

D, E, F, F#, G, Ab, A, B, C, D

D#, E#, F#, F##, G#, A, A#, B#, C#, D#

Eb, F, Gb, G, Ab, Bbb, Bb, C, Db, Eb

E, F#, G, G#, A, Bb, B, C#, D, E

F, G, Ab, A, Bb, Cb, C, D, Eb, F

F#, G#, A, A#, B, C, C#, D#, E, F#

Gb, Ab, Bbb, Bb, Cb, Dbb, Db, Eb, Fb, Gb

G, A, Bb, B, C, Db, D, E, F, G

G#, A#, B, B#, C#, D, D#, E#, F#, G#

Ab, Bb, Cb, C, Db, Ebb, Eb, F, Gb, Ab

A, B, C, C#, D, Eb, E, F#, G, A

A#, B#, C#, C##, D#, E, E#, F##, G#, A#

Bb, C, Db, D, Eb, Fb, F, G, Ab, Bb

B, C#, D, D#, E, F, F#, G#, A, B

## Blues Nonatonic (with M7)

Step pattern: 3, 1, 1, 1, 1, 2, 1, 1, 1

Ascending intervals: P1, m3, M3, P4, dim5, P5, M6, m7, M7, P8

C, Eb, E, F, Gb, G, A, Bb, B, C

C#, E, E#, F#, G, G#, A#, B, B#, C#

Db, Fb, F, Gb, Abb, Ab, Bb, Cb, C, Db

D, F, F#, G, Ab, A, B, C, C#, D

D#, F#, F##, G#, A, A#, B#, C#, C##, D#

Eb, Gb, G, Ab, Bbb, Bb, C, Db, D, Eb

E, G, G#, A, Bb, B, C#, D, D#, E

F, Ab, A, Bb, Cb, C, D, Eb, E, F

F#, A, A#, B, C, C#, D#, E, E#, F#

Gb, Bbb, Bb, Cb, Dbb, Db, Eb, Fb, F, Gb

G, Bb, B, C, Db, D, E, F, F#, G

G#, B, B#, C#, D, D#, E#, F#, F##, G#

Ab, Cb, C, Db, Ebb, Eb, F, Gb, G, Ab

A, C, C#, D, Eb, E, F#, G, G#, A

A#, C#, C##, D#, E, E#, F##, G#, G##, A#

Bb, Db, D, Eb, Fb, F, G, Ab, A, Bb

B, D, D#, E, F, F#, G#, A, A#, B

## Chromatic and Diatonic Dorian Mixed

Step pattern: 1, 1, 1, 2, 2, 1, 1, 1, 2

Ascending intervals: P1, m2, M2, m3, P4, P5, m6, M6, m7, P8

C, Db, D, Eb, F, G, Ab, A, Bb, C

C#, D, D#, E, F#, G#, A, A#, B, C#

Db, Ebb, Eb, Fb, Gb, Ab, Bbb, Bb, Cb, Db

D, Eb, E, F, G, A, Bb, B, C, D

D#, E, E#, F#, G#, A#, B, B#, C#, D#

Eb, Fb, F, Gb, Ab, Bb, Cb, C, Db, Eb

E, F, F#, G, A, B, C, C#, D, E

F, Gb, G, Ab, Bb, C, Db, D, Eb, F

F#, G, G#, A, B, C#, D, D#, E, F#

Gb, Abb, Ab, Bbb, Cb, Db, Ebb, Eb, Fb, Gb

G, Ab, A, Bb, C, D, Eb, E, F, G

G#, A, A#, B, C#, D#, E, E#, F#, G#

Ab, Bbb, Bb, Cb, Db, Eb, Fb, F, Gb, Ab

A, Bb, B, C, D, E, F, F#, G, A

A#, B, B#, C#, D#, E#, F#, F##, G#, A#

Bb, Cb, C, Db, Eb, F, Gb, G, Ab, Bb

B, C, C#, D, E, F#, G, G#, A, B

## Chromatic and permuted Diatonic Dorian Mixed

Step pattern: 1, 1, 2, 1, 2, 1, 1, 2, 1

Ascending intervals: P1, m2, M2, M3, P4, P5, m6, M6, M7, P8

C, Db, D, E, F, G, Ab, A, B, C

C#, D, D#, E#, F#, G#, A, A#, B#, C#

Db, Ebb, Eb, F, Gb, Ab, Bbb, Bb, C, Db

D, Eb, E, F#, G, A, Bb, B, C#, D

D#, E, E#, F##, G#, A#, B, B#, C##, D#

Eb, Fb, F, G, Ab, Bb, Cb, C, D, Eb

E, F, F#, G#, A, B, C, C#, D#, E

F, Gb, G, A, Bb, C, Db, D, E, F

F#, G, G#, A#, B, C#, D, D#, E#, F#

Gb, Abb, Ab, Bb, Cb, Db, Ebb, Eb, F, Gb

G, Ab, A, B, C, D, Eb, E, F#, G

G#, A, A#, B#, C#, D#, E, E#, F##, G#

Ab, Bbb, Bb, C, Db, Eb, Fb, F, G, Ab

A, Bb, B, C#, D, E, F, F#, G#, A

A#, B, B#, C##, D#, E#, F#, F##, G##, A#

Bb, Cb, C, D, Eb, F, Gb, G, A, Bb

B, C, C#, D#, E, F#, G, G#, A#, B

## Full Minor

Step pattern: 2, 1, 2, 2, 1, 1, 1, 1, 1

Ascending intervals: P1, M2, m3, P4, P5, m6, M6, m7, M7, P8

C, D, Eb, F, G, Ab, A, Bb, B, C

C#, D#, E, F#, G#, A, A#, B, B#, C#

Db, Eb, Fb, Gb, Ab, Bbb, Bb, Cb, C, Db

D, E, F, G, A, Bb, B, C, C#, D

D#, E#, F#, G#, A#, B, B#, C#, C##, D#

Eb, F, Gb, Ab, Bb, Cb, C, Db, D, Eb

E, F#, G, A, B, C, C#, D, D#, E

F, G, Ab, Bb, C, Db, D, Eb, E, F

F#, G#, A, B, C#, D, D#, E, E#, F#

Gb, Ab, Bbb, Cb, Db, Ebb, Eb, Fb, F, Gb

G, A, Bb, C, D, Eb, E, F, F#, G

G#, A#, B, C#, D#, E, E#, F#, F##, G#

Ab, Bb, Cb, Db, Eb, Fb, F, Gb, G, Ab

A, B, C, D, E, F, F#, G, G#, A

A#, B#, C#, D#, E#, F#, F##, G#, G##, A#

Bb, C, Db, Eb, F, Gb, G, Ab, A, Bb

B, C#, D, E, F#, G, G#, A, A#, B



## Ionian and Dorian Mixed

Step pattern: 2, 1, 1, 1, 2, 2, 1, 1, 1

Ascending intervals: P1, M2, m3, M3, P4, P5, M6, m7, M7, P8

C, D, Eb, E, F, G, A, Bb, B, C

C#, D#, E, E#, F#, G#, A#, B, B#, C#

Db, Eb, Fb, F, Gb, Ab, Bb, Cb, C, Db

D, E, F, F#, G, A, B, C, C#, D

D#, E#, F#, F##, G#, A#, B#, C#, C##, D#

Eb, F, Gb, G, Ab, Bb, C, Db, D, Eb

E, F#, G, G#, A, B, C#, D, D#, E

F, G, Ab, A, Bb, C, D, Eb, E, F

F#, G#, A, A#, B, C#, D#, E, E#, F#

Gb, Ab, Bbb, Bb, Cb, Db, Eb, Fb, F, Gb

G, A, Bb, B, C, D, E, F, F#, G

G#, A#, B, B#, C#, D#, E#, F#, F##, G#

Ab, Bb, Cb, C, Db, Eb, F, Gb, G, Ab

A, B, C, C#, D, E, F#, G, G#, A

A#, B#, C#, C##, D#, E#, F##, G#, G##, A#

Bb, C, Db, D, Eb, F, G, Ab, A, Bb

B, C#, D, D#, E, F#, G#, A, A#, B

**Kiourdi**

Step pattern: 2, 1, 2, 1, 1, 1, 1, 1, 2

Ascending intervals: P1, M2, m3, P4, dim5, P5, m6, M6, m7, P8

C, D, Eb, F, Gb, G, Ab, A, Bb, C

C#, D#, E, F#, G, G#, A, A#, B, C#

Db, Eb, Fb, Gb, Abb, Ab, Bbb, Bb, Cb, Db

D, E, F, G, Ab, A, Bb, B, C, D

D#, E#, F#, G#, A, A#, B, B#, C#, D#

Eb, F, Gb, Ab, Bbb, Bb, Cb, C, Db, Eb

E, F#, G, A, Bb, B, C, C#, D, E

F, G, Ab, Bb, Cb, C, Db, D, Eb, F

F#, G#, A, B, C, C#, D, D#, E, F#

Gb, Ab, Bbb, Cb, Dbb, Db, Ebb, Eb, Fb, Gb

G, A, Bb, C, Db, D, Eb, E, F, G

G#, A#, B, C#, D, D#, E, E#, F#, G#

Ab, Bb, Cb, Db, Ebb, Eb, Fb, F, Gb, Ab

A, B, C, D, Eb, E, F, F#, G, A

A#, B#, C#, D#, E, E#, F#, F##, G#, A#

Bb, C, Db, Eb, Fb, F, Gb, G, Ab, Bb

B, C#, D, E, F, F#, G, G#, A, B

## Lydian and Mixolydian Mixed

Step pattern: 2, 2, 1, 1, 1, 2, 1, 1, 1

Ascending intervals: P1, M2, M3, P4, dim5, P5, M6, m7, M7, P8

C, D, E, F, Gb, G, A, Bb, B, C

C#, D#, E#, F#, G, G#, A#, B, B#, C#

Db, Eb, F, Gb, Abb, Ab, Bb, Cb, C, Db

D, E, F#, G, Ab, A, B, C, C#, D

D#, E#, F##, G#, A, A#, B#, C#, C##, D#

Eb, F, G, Ab, Bbb, Bb, C, Db, D, Eb

E, F#, G#, A, Bb, B, C#, D, D#, E

F, G, A, Bb, Cb, C, D, Eb, E, F

F#, G#, A#, B, C, C#, D#, E, E#, F#

Gb, Ab, Bb, Cb, Dbb, Db, Eb, Fb, F, Gb

G, A, B, C, Db, D, E, F, F#, G

G#, A#, B#, C#, D, D#, E#, F#, F##, G#

Ab, Bb, C, Db, Ebb, Eb, F, Gb, G, Ab

A, B, C#, D, Eb, E, F#, G, G#, A

A#, B#, C##, D#, E, E#, F##, G#, G##, A#

Bb, C, D, Eb, Fb, F, G, Ab, A, Bb

B, C#, D#, E, F, F#, G#, A, A#, B

### Messiaen Mode 3 (3 groups of 4 notes each)

Step pattern: 2, 1, 1, 2, 1, 1, 2, 1, 1

Ascending intervals: P1, M2, m3, M3, Aug4, P5, m6, m7, M7, P8

C, D, Eb, E, F#, G, Ab, Bb, B, C

C#, D#, E, E#, F##, G#, A, B, B#, C#

Db, Eb, Fb, F, G, Ab, Bbb, Cb, C, Db

D, E, F, F#, G#, A, Bb, C, C#, D

D#, E#, F#, F##, G###, A#, B, C#, C##, D#

Eb, F, Gb, G, A, Bb, Cb, Db, D, Eb

E, F#, G, G#, A#, B, C, D, D#, E

F, G, Ab, A, B, C, Db, Eb, E, F

F#, G#, A, A#, B#, C#, D, E, E#, F#

Gb, Ab, Bbb, Bb, C, Db, Ebb, Fb, F, Gb

G, A, Bb, B, C#, D, Eb, F, F#, G

G#, A#, B, B#, C##, D#, E, F#, F##, G#

Ab, Bb, Cb, C, D, Eb, Fb, Gb, G, Ab

A, B, C, C#, D#, E, F, G, G#, A

A#, B#, C#, C##, D###, E#, F#, G#, G##, A#

Bb, C, Db, D, E, F, Gb, Ab, A, Bb

B, C#, D, D#, E#, F#, G, A, A#, B

## Modes of Major Pentatonic Mixed

Step pattern: 2, 1, 1, 1, 2, 1, 1, 1, 2

Ascending intervals: P1, M2, m3, M3, P4, P5, m6, M6, m7, P8

C, D, Eb, E, F, G, Ab, A, Bb, C

C#, D#, E, E#, F#, G#, A, A#, B, C#

Db, Eb, Fb, F, Gb, Ab, Bbb, Bb, Cb, Db

D, E, F, F#, G, A, Bb, B, C, D

D#, E#, F#, F##, G#, A#, B, B#, C#, D#

Eb, F, Gb, G, Ab, Bb, Cb, C, Db, Eb

E, F#, G, G#, A, B, C, C#, D, E

F, G, Ab, A, Bb, C, Db, D, Eb, F

F#, G#, A, A#, B, C#, D, D#, E, F#

Gb, Ab, Bbb, Bb, Cb, Db, Ebb, Eb, Fb, Gb

G, A, Bb, B, C, D, Eb, E, F, G

G#, A#, B, B#, C#, D#, E, E#, F#, G#

Ab, Bb, Cb, C, Db, Eb, Fb, F, Gb, Ab

A, B, C, C#, D, E, F, F#, G, A

A#, B#, C#, C##, D#, E#, F#, F##, G#, A#

Bb, C, Db, D, Eb, F, Gb, G, Ab, Bb

B, C#, D, D#, E, F#, G, G#, A, B

## Moorish Phrygian

Step pattern: 1, 2, 1, 1, 2, 1, 2, 1, 1

Ascending intervals: P1, m2, m3, M3, P4, P5, m6, m7, M7, P8

C, Db, Eb, E, F, G, Ab, Bb, B, C

C#, D, E, E#, F#, G#, A, B, B#, C#

Db, Ebb, Fb, F, Gb, Ab, Bbb, Cb, C, Db

D, Eb, F, F#, G, A, Bb, C, C#, D

D#, E, F#, F##, G#, A#, B, C#, C##, D#

Eb, Fb, Gb, G, Ab, Bb, Cb, Db, D, Eb

E, F, G, G#, A, B, C, D, D#, E

F, Gb, Ab, A, Bb, C, Db, Eb, E, F

F#, G, A, A#, B, C#, D, E, E#, F#

Gb, Abb, Bbb, Bb, Cb, Db, Ebb, Fb, F, Gb

G, Ab, Bb, B, C, D, Eb, F, F#, G

G#, A, B, B#, C#, D#, E, F#, F##, G#

Ab, Bbb, Cb, C, Db, Eb, Fb, Gb, G, Ab

A, Bb, C, C#, D, E, F, G, G#, A

A#, B, C#, C##, D#, E#, F#, G#, G##, A#

Bb, Cb, Db, D, Eb, F, Gb, Ab, A, Bb

B, C, D, D#, E, F#, G, A, A#, B

## Nine-note Scale

Step pattern: 2, 1, 1, 2, 1, 1, 1, 2, 1

Ascending intervals: P1, M2, m3, M3, dim5, P5, m6, M6, M7, P8

C, D, Eb, E, Gb, G, Ab, A, B, C

C#, D#, E, E#, G, G#, A, A#, B#, C#

Db, Eb, Fb, F, Abb, Ab, Bbb, Bb, C, Db

D, E, F, F#, Ab, A, Bb, B, C#, D

D#, E#, F#, F##, A, A#, B, B#, C##, D#

Eb, F, Gb, G, Bbb, Bb, Cb, C, D, Eb

E, F#, G, G#, Bb, B, C, C#, D#, E

F, G, Ab, A, Cb, C, Db, D, E, F

F#, G#, A, A#, C, C#, D, D#, E#, F#

Gb, Ab, Bbb, Bb, Dbb, Db, Ebb, Eb, F, Gb

G, A, Bb, B, Db, D, Eb, E, F#, G

G#, A#, B, B#, D, D#, E, E#, F##, G#

Ab, Bb, Cb, C, Ebb, Eb, Fb, F, G, Ab

A, B, C, C#, Eb, E, F, F#, G#, A

A#, B#, C#, C##, E, E#, F#, F##, G##, A#

Bb, C, Db, D, Fb, F, Gb, G, A, Bb

B, C#, D, D#, F, F#, G, G#, A#, B

**Pahadi**

Step pattern: 2, 2, 1, 2, 1, 1, 1, 1, 1

Ascending intervals: P1, M2, M3, P4, P5, m6, M6, m7, M7, P8

C, D, E, F, G, Ab, A, Bb, B, C

C#, D#, E#, F#, G#, A, A#, B, B#, C#

Db, Eb, F, Gb, Ab, Bbb, Bb, Cb, C, Db

D, E, F#, G, A, Bb, B, C, C#, D

D#, E#, F##, G#, A#, B, B#, C#, C##, D#

Eb, F, G, Ab, Bb, Cb, C, Db, D, Eb

E, F#, G#, A, B, C, C#, D, D#, E

F, G, A, Bb, C, Db, D, Eb, E, F

F#, G#, A#, B, C#, D, D#, E, E#, F#

Gb, Ab, Bb, Cb, Db, Ebb, Eb, Fb, F, Gb

G, A, B, C, D, Eb, E, F, F#, G

G#, A#, B#, C#, D#, E, E#, F#, F##, G#

Ab, Bb, C, Db, Eb, Fb, F, Gb, G, Ab

A, B, C#, D, E, F, F#, G, G#, A

A#, B#, C##, D#, E#, F#, F##, G#, G##, A#

Bb, C, D, Eb, F, Gb, G, Ab, A, Bb

B, C#, D#, E, F#, G, G#, A, A#, B



## Tcherepnin Nine-note Scale

Step pattern: 1, 2, 1, 1, 2, 1, 1, 2, 1

Ascending intervals: P1, m2, m3, M3, P4, P5, m6, M6, M7, P8

C, Db, Eb, E, F, G, Ab, A, B, C

C#, D, E, E#, F#, G#, A, A#, B#, C#

Db, Ebb, Fb, F, Gb, Ab, Bbb, Bb, C, Db

D, Eb, F, F#, G, A, Bb, B, C#, D

D#, E, F#, F##, G#, A#, B, B#, C##, D#

Eb, Fb, Gb, G, Ab, Bb, Cb, C, D, Eb

E, F, G, G#, A, B, C, C#, D#, E

F, Gb, Ab, A, Bb, C, Db, D, E, F

F#, G, A, A#, B, C#, D, D#, E#, F#

Gb, Abb, Bbb, Bb, Cb, Db, Ebb, Eb, F, Gb

G, Ab, Bb, B, C, D, Eb, E, F#, G

G#, A, B, B#, C#, D#, E, E#, F##, G#

Ab, Bbb, Cb, C, Db, Eb, Fb, F, G, Ab

A, Bb, C, C#, D, E, F, F#, G#, A

A#, B, C#, C##, D#, E#, F#, F##, G##, A#

Bb, Cb, Db, D, Eb, F, Gb, G, A, Bb

B, C, D, D#, E, F#, G, G#, A#, B

**Youlan**

Step pattern: 1, 1, 2, 1, 1, 1, 2, 1, 2

Ascending intervals: P1, m2, M2, M3, P4, dim5, P5, M6, m7, P8

C, Db, D, E, F, Gb, G, A, Bb, C

C#, D, D#, E#, F#, G, G#, A#, B, C#

Db, Ebb, Eb, F, Gb, Abb, Ab, Bb, Cb, Db

D, Eb, E, F#, G, Ab, A, B, C, D

D#, E, E#, F##, G#, A, A#, B#, C#, D#

Eb, Fb, F, G, Ab, Bbb, Bb, C, Db, Eb

E, F, F#, G#, A, Bb, B, C#, D, E

F, Gb, G, A, Bb, Cb, C, D, Eb, F

F#, G, G#, A#, B, C, C#, D#, E, F#

Gb, Abb, Ab, Bb, Cb, Dbb, Db, Eb, Fb, Gb

G, Ab, A, B, C, Db, D, E, F, G

G#, A, A#, B#, C#, D, D#, E#, F#, G#

Ab, Bbb, Bb, C, Db, Ebb, Eb, F, Gb, Ab

A, Bb, B, C#, D, Eb, E, F#, G, A

A#, B, B#, C##, D#, E, E#, F##, G#, A#

Bb, Cb, C, D, Eb, Fb, F, G, Ab, Bb

B, C, C#, D#, E, F, F#, G#, A, B

## 10 – NOTE SCALES

## Ionian and Aeolian Mixed

Step pattern: 2, 1, 1, 1, 2, 1, 1, 1, 1

Ascending intervals: P1, M2, m3, M3, P4, P5, m6, M6, m7, M7, P8

C, D, Eb, E, F, G, Ab, A, Bb, B, C

C#, D#, E, E#, F#, G#, A, A#, B, B#, C#

Db, Eb, Fb, F, Gb, Ab, Bbb, Bb, Cb, C, Db

D, E, F, F#, G, A, Bb, B, C, C#, D

D#, E#, F#, F##, G#, A#, B, B#, C#, C##, D#

Eb, F, Gb, G, Ab, Bb, Cb, C, Db, D, Eb

E, F#, G, G#, A, B, C, C#, D, D#, E

F, G, Ab, A, Bb, C, Db, D, Eb, E, F

F#, G#, A, A#, B, C#, D, D#, E, E#, F#

Gb, Ab, Bbb, Bb, Cb, Db, Ebb, Eb, Fb, F, Gb

G, A, Bb, B, C, D, Eb, E, F, F#, G

G#, A#, B, B#, C#, D#, E, E#, F#, F##, G#

Ab, Bb, Cb, C, Db, Eb, Fb, F, Gb, G, Ab

A, B, C, C#, D, E, F, F#, G, G#, A

A#, B#, C#, C##, D#, E#, F#, F##, G#, G##, A#

Bb, C, Db, D, Eb, F, Gb, G, Ab, A, Bb

B, C#, D, D#, E, F#, G, G#, A, A#, B

### Messiaen Mode 7 (2 groups of 5 notes each)

Step pattern: 1, 1, 1, 2, 1, 1, 1, 1, 2, 1

Ascending intervals: P1, m2, M2, m3, P4, Aug4, P5, m6, M6, M7, P8

C, Db, D, Eb, F, F#, G, Ab, A, B, C

C#, D, D#, E, F#, F##, G#, A, A#, B#, C#

Db, Ebb, Eb, Fb, Gb, G, Ab, Bbb, Bb, C, Db

D, Eb, E, F, G, G#, A, Bb, B, C#, D

D#, E, E#, F#, G#, G##, A#, B, B#, C##, D#

Eb, Fb, F, Gb, Ab, A, Bb, Cb, C, D, Eb

E, F, F#, G, A, A#, B, C, C#, D#, E

F, Gb, G, Ab, Bb, B, C, Db, D, E, F

F#, G, G#, A, B, B#, C#, D, D#, E#, F#

Gb, Abb, Ab, Bbb, Cb, C, Db, Ebb, Eb, F, Gb

G, Ab, A, Bb, C, C#, D, Eb, E, F#, G

G#, A, A#, B, C#, C##, D#, E, E#, F##, G#

Ab, Bbb, Bb, Cb, Db, D, Eb, Fb, F, G, Ab

A, Bb, B, C, D, D#, E, F, F#, G#, A

A#, B, B#, C#, D#, D##, E#, F#, F##, G##, A#

Bb, Cb, C, Db, Eb, E, F, Gb, G, A, Bb

B, C, C#, D, E, E#, F#, G, G#, A#, B

### Messiaen Mode 7 Inverse (2 groups of 5 notes each)

Step pattern: 2, 1, 1, 1, 1, 2, 1, 1, 1, 1

Ascending intervals: P1, M2, m3, M3, P4, dim5, m6, M6, m7, M7, P8

C, D, Eb, E, F, Gb, Ab, A, Bb, B, C

C#, D#, E, E#, F#, G, A, A#, B, B#, C#

Db, Eb, Fb, F, Gb, Abb, Bbb, Bb, Cb, C, Db

D, E, F, F#, G, Ab, Bb, B, C, C#, D

D#, E#, F#, F##, G#, A, B, B#, C#, C##, D#

Eb, F, Gb, G, Ab, Bbb, Cb, C, Db, D, Eb

E, F#, G, G#, A, Bb, C, C#, D, D#, E

F, G, Ab, A, Bb, Cb, Db, D, Eb, E, F

F#, G#, A, A#, B, C, D, D#, E, E#, F#

Gb, Ab, Bbb, Bb, Cb, Dbb, Ebb, Eb, Fb, F, Gb

G, A, Bb, B, C, Db, Eb, E, F, F#, G

G#, A#, B, B#, C#, D, E, E#, F#, F##, G#

Ab, Bb, Cb, C, Db, Ebb, Fb, F, Gb, G, Ab

A, B, C, C#, D, Eb, F, F#, G, G#, A

A#, B#, C#, C##, D#, E, F#, F##, G#, G##, A#

Bb, C, Db, D, Eb, Fb, Gb, G, Ab, A, Bb

B, C#, D, D#, E, F, G, G#, A, A#, B

## Minor Pentatonic with Ascending Chrom. Approach

Step pattern: 2, 1, 1, 1, 1, 1, 2, 1, 1, 1

Ascending intervals: P1, M2, m3, M3, P4, dim5, P5, M6, m7, M7, P8

C, D, Eb, E, F, Gb, G, A, Bb, B, C

C#, D#, E, E#, F#, G, G#, A#, B, B#, C#

Db, Eb, Fb, F, Gb, Abb, Ab, Bb, Cb, C, Db

D, E, F, F#, G, Ab, A, B, C, C#, D

D#, E#, F#, F##, G#, A, A#, B#, C#, C##, D#

Eb, F, Gb, G, Ab, Bbb, Bb, C, Db, D, Eb

E, F#, G, G#, A, Bb, B, C#, D, D#, E

F, G, Ab, A, Bb, Cb, C, D, Eb, E, F

F#, G#, A, A#, B, C, C#, D#, E, E#, F#

Gb, Ab, Bbb, Bb, Cb, Dbb, Db, Eb, Fb, F, Gb

G, A, Bb, B, C, Db, D, E, F, F#, G

G#, A#, B, B#, C#, D, D#, E#, F#, F##, G#

Ab, Bb, Cb, C, Db, Ebb, Eb, F, Gb, G, Ab

A, B, C, C#, D, Eb, E, F#, G, G#, A

A#, B#, C#, C##, D#, E, E#, F##, G#, G##, A#

Bb, C, Db, D, Eb, Fb, F, G, Ab, A, Bb

B, C#, D, D#, E, F, F#, G#, A, A#, B

## Sindhi-Bhairavi

Step pattern: 1, 1, 1, 1, 1, 2, 1, 2, 1, 1

Ascending intervals: P1, m2, M2, m3, M3, P4, P5, m6, m7, M7, P8

C, Db, D, Eb, E, F, G, Ab, Bb, B, C

C#, D, D#, E, E#, F#, G#, A, B, B#, C#

Db, Ebb, Eb, Fb, F, Gb, Ab, Bbb, Cb, C, Db

D, Eb, E, F, F#, G, A, Bb, C, C#, D

D#, E, E#, F#, F##, G#, A#, B, C#, C##, D#

Eb, Fb, F, Gb, G, Ab, Bb, Cb, Db, D, Eb

E, F, F#, G, G#, A, B, C, D, D#, E

F, Gb, G, Ab, A, Bb, C, Db, Eb, E, F

F#, G, G#, A, A#, B, C#, D, E, E#, F#

Gb, Abb, Ab, Bbb, Bb, Cb, Db, Ebb, Fb, F, Gb

G, Ab, A, Bb, B, C, D, Eb, F, F#, G

G#, A, A#, B, B#, C#, D#, E, F#, F##, G#

Ab, Bbb, Bb, Cb, C, Db, Eb, Fb, Gb, G, Ab

A, Bb, B, C, C#, D, E, F, G, G#, A

A#, B, B#, C#, C##, D#, E#, F#, G#, G##, A#

Bb, Cb, C, Db, D, Eb, F, Gb, Ab, A, Bb

B, C, C#, D, D#, E, F#, G, A, A#, B



## Symmetrical Decatonic

Step pattern: 1, 1, 2, 1, 1, 1, 1, 2, 1, 1

Ascending intervals: P1, m2, M2, M3, P4, dim5, P5, m6, m7, M7, P8

C, Db, D, E, F, Gb, G, Ab, Bb, B, C

C#, D, D#, E#, F#, G, G#, A, B, B#, C#

Db, Ebb, Eb, F, Gb, Abb, Ab, Bbb, Cb, C, Db

D, Eb, E, F#, G, Ab, A, Bb, C, C#, D

D#, E, E#, F##, G#, A, A#, B, C#, C##, D#

Eb, Fb, F, G, Ab, Bbb, Bb, Cb, Db, D, Eb

E, F, F#, G#, A, Bb, B, C, D, D#, E

F, Gb, G, A, Bb, Cb, C, Db, Eb, E, F

F#, G, G#, A#, B, C, C#, D, E, E#, F#

Gb, Abb, Ab, Bb, Cb, Dbb, Db, Ebb, Fb, F, Gb

G, Ab, A, B, C, Db, D, Eb, F, F#, G

G#, A, A#, B#, C#, D, D#, E, F#, F##, G#

Ab, Bbb, Bb, C, Db, Ebb, Eb, Fb, Gb, G, Ab

A, Bb, B, C#, D, Eb, E, F, G, G#, A

A#, B, B#, C##, D#, E, E#, F#, G#, G##, A#

Bb, Cb, C, D, Eb, Fb, F, Gb, Ab, A, Bb

B, C, C#, D#, E, F, F#, G, A, A#, B

## 12 – NOTE SCALES

## Chromatic

Step pattern: 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1

Ascending intervals: P1, Aug1, M2, Aug2, M3, P4, Aug4, P5, Aug5, M6, Aug6, M7, P8

C, C#, D, D#, E, F, F#, G, G#, A, A#, B, C

C#, C##, D#, D###, E#, F#, F##, G#, G###, A#, A##, B#, C#

Db, D, Eb, E, F, Gb, G, Ab, A, Bb, B, C, Db

D, D#, E, E#, F#, G, G#, A, A#, B, B#, C#, D

D#, D##, E#, E###, F##, G#, G###, A#, A##, B#, B##, C##, D#

Eb, E, F, F#, G, Ab, A, Bb, B, C, C#, D, Eb

E, E#, F#, F##, G#, A, A#, B, B#, C#, C##, D#, E

F, F#, G, G#, A, Bb, B, C, C#, D, D#, E, F

F#, F##, G#, G###, A#, B, B#, C#, C##, D#, D##, E#, F#

Gb, G, Ab, A, Bb, Cb, C, Db, D, Eb, E, F, Gb

G, G#, A, A#, B, C, C#, D, D#, E, E#, F#, G

G#, G###, A#, A##, B#, C#, C##, D#, D##, E#, E##, F##, G#

Ab, A, Bb, B, C, Db, D, Eb, E, F, F#, G, Ab

A, A#, B, B#, C#, D, D#, E, E#, F#, F##, G#, A

A#, A##, B#, B##, C##, D#, D##, E#, E##, F##, F###, G##, A#

Bb, B, C, C#, D, Eb, E, F, F#, G, G#, A, Bb

B, B#, C#, C##, D#, E, E#, F#, F##, G#, G###, A#, B

## Chromatic (Harmonic)

Step pattern: 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1

Ascending intervals: P1, m2, M2, m3, M3, P4, Aug4, P5, m6, M6, m7, M7, P8

C, Db, D, Eb, E, F, F#, G, Ab, A, Bb, B, C

C#, D, D#, E, E#, F#, F##, G#, A, A#, B, B#, C#

Db, Ebb, Eb, Fb, F, Gb, G, Ab, Bbb, Bb, Cb, C, Db

D, Eb, E, F, F#, G, G#, A, Bb, B, C, C#, D

D#, E, E#, F#, F##, G#, G##, A#, B, B#, C#, C##, D#

Eb, Fb, F, Gb, G, Ab, A, Bb, Cb, C, Db, D, Eb

E, F, F#, G, G#, A, A#, B, C, C#, D, D#, E

F, Gb, G, Ab, A, Bb, B, C, Db, D, Eb, E, F

F#, G, G#, A, A#, B, B#, C#, D, D#, E, E#, F#

Gb, Abb, Ab, Bbb, Bb, Cb, C, Db, Ebb, Eb, Fb, F, Gb

G, Ab, A, Bb, B, C, C#, D, Eb, E, F, F#, G

G#, A, A#, B, B#, C#, C##, D#, E, E#, F#, F##, G#

Ab, Bbb, Bb, Cb, C, Db, D, Eb, Fb, F, Gb, G, Ab

A, Bb, B, C, C#, D, D#, E, F, F#, G, G#, A

A#, B, B#, C#, C##, D#, D##, E#, F#, F##, G#, G##, A#

Bb, Cb, C, Db, D, Eb, E, F, Gb, G, Ab, A, Bb

B, C, C#, D, D#, E, E#, F#, G, G#, A, A#, B

## DUPLICATE REPORT

*Technical note: Scales that contain the same interval sequence, but have different names.*

**Table 16 – Duplicate Structures**

| <b>NUMBER<br/>OF NOTES</b> | <b>DUPLICATE<br/>GROUPS</b> | <b>STRUCTURES<br/>INVOLVED</b> |
|----------------------------|-----------------------------|--------------------------------|
| 5                          | 9                           | 19                             |
| 6                          | 2                           | 4                              |
| 7                          | 17                          | 42                             |
| 8                          | 6                           | 15                             |
| <b>Total</b>               | <b>34</b>                   | <b>80</b>                      |

## 5 – Note Scales

### Grup 1

Ascending intervals: P1, m3, P4, P5, m7, P8

- Aeolian Pentatonic (5th mode of major pentatonic)
- Minor Pentatonic (5th mode of major pentatonic)
- Tcherepnin Major Pentatonic Mode V

### Grup 2

Ascending intervals: P1, m2, P4, P5, m6, P8

- Akebono (suspended)
- Tcherepnin Minor Pentatonic Mode IV

### Grup 3

Ascending intervals: P1, m3, P4, dim5, m7, P8

- Blues Pentatonic
- Pentatonic Minor 7 b5

**Grup 4**

Ascending intervals: P1, M2, P4, P5, m7, P8

- Dorian Pentatonic (2nd mode of major pentatonic)
- Suspended Pentatonic (2nd mode of major pentatonic)

**Grup 5**

Ascending intervals: P1, M2, M3, P5, M7, P8

- Hamsadhvani
- Hansadhvani

**Grup 6**

Ascending intervals: P1, M2, m3, P5, m6, P8

- Japanese Mode
- Tcherepnin Minor Pentatonic Mode I

**Grup 7**

Ascending intervals: P1, M2, M3, P5, M6, P8

- Major Pentatonic (1st mode of major pentatonic)
- Tcherepnin Major Pentatonic Mode I

**Grup 8**

Ascending intervals: P1, M2, P4, P5, M6, P8

- Mixolydian Pentatonic (4th mode of major pentatonic)
- Tcherepnin Major Pentatonic Mode IV

**Grup 9**

Ascending intervals: P1, m3, P4, m6, m7, P8

- Phrygian Pentatonic (3rd mode of major pentatonic)
- Tcherepnin Major Pentatonic Mode III



## 6 – Note Scales

### Grup 1

Ascending intervals: P1, M2, M3, Aug4, Aug5, Aug6, P8

- Messiaen Mode 1 (6 groups of 2 notes each)
- Whole-tone

### Grup 2

Ascending intervals: P1, M2, P4, P5, M6, m7, P8

- Mixolydian Hexatonic (suspended)
- Yosen

## 7 – Note Scales

### Grup 1

Ascending intervals: P1, M2, M3, Aug4, P5, M6, m7, P8

- Acoustic Scale
- Bartok
- Lydian Dominant (4th mode of melodic minor)

### Grup 2

Ascending intervals: P1, M2, m3, P4, P5, m6, m7, P8

- Aeolian (6th mode of major)
- Minor

### Grup 3

Ascending intervals: P1, aug2, M3, Aug4, P5, M6, M7, P8

- Aeolian Harmonic (6th mode of harmonic minor)
- Lydian #2 (6th mode of harmonic minor)

**Grup 4**

Ascending intervals: P1, M2, M3, P4, P5, m6, m7, P8

- Aeolian Major (5th mode of melodic minor)
- Hindu scale (5th mode of melodic minor)
- Melodic Major
- Mixolydian b6 (5th mode of melodic minor)

**Grup 5**

Ascending intervals: P1, m2, m3, dim4, dim5, m6, dim7, P8

- Altered Diminished (7th mode of harmonic minor)
- Locrian Diminished (7th mode of harmonic minor)

**Grup 6**

Ascending intervals: P1, m2, m3, dim4, dim5, m6, m7, P8

- Altered Dominant (7th mode of melodic minor)
- Altered Scale (7th mode of melodic minor)
- Super Locrian (7th mode of melodic minor)

**Grup 7**

Ascending intervals: P1, m2, M3, P4, P5, m6, m7, P8

- Arabic (5th mode harmonic minor)
- Phrygian Dominant (5th mode of harmonic minor)

**Grup 8**

Ascending intervals: P1, m2, M3, P4, P5, m6, M7, P8

- Byzantine Scale
- Double Harmonic Major (5th mode harmonic minor)
- Major Gipsy

**Grup 9**

Ascending intervals: P1, m2, m3, P4, dim5, m6, dim7, P8

- Diminished b9 (7th mode of harmonic major)
- Locrian bb7

**Grup 10**

Ascending intervals: P1, m2, m3, P4, P5, M6, m7, P8

- Dorian b2 (2nd mode of melodic minor)
- Dorian b9 (2nd mode of melodic minor)
- Phrygian nat. 6 (2nd mode of melodic minor)

**Grup 11**

Ascending intervals: P1, m2, m3, P4, dim5, M6, m7, P8

- Dorian b2 b5 (2nd mode of harmonic minor)
- Locrian nat.6 (2nd mode of harmonic minor)

**Grup 12**

Ascending intervals: P1, M2, m3, Aug4, P5, m6, M7, P8

- Double Harmonic Minor
- Hungarian Minor
- Minor Gipsy

**Grup 13**

Ascending intervals: P1, M2, M3, P4, P5, m6, M7, P8

- Ethiopian
- Harmonic Major (1st mode harmonic major)
- Harmonic Major (3rd mode harmonic minor)

**Grup 14**

Ascending intervals: P1, M2, m3, P4, dim5, m6, m7, P8

- Half-diminished (6th mode of melodic minor)
- Locrian nat.2 (6th mode of melodic minor)

**Grup 15**

Ascending intervals: P1, M2, M3, P4, Aug5, M6, M7, P8

- Ionian #5
- Ionian Augmented (3rd mode of harmonic minor)

**Grup 16**

Ascending intervals: P1, M2, M3, P4, P5, M6, M7, P8

- Ionian (1st mode of major)
- Major

**Grup 17**

Ascending intervals: P1, m2, m3, P4, P5, m6, m7, P8

- Phrygian (3rd mode of major)
- Spanish Mode

## 8 – Note Scales

### Grup 1

Ascending intervals: P1, M2, M3, P4, P5, Aug5, M6, M7, P8

- Altered Mixolydian
- Major Bebop

### Grup 2

Ascending intervals: P1, M2, m3, P4, dim5, m6, M6, M7, P8

- Diminished
- Messiaen Mode 2 Inverse (4 groups of 3 notes each)
- Whole-half Tone Scale

### Grup 3

Ascending intervals: P1, M2, M3, P4, P5, M6, m7, M7, P8

- Dominant Bebop
- Ionian and Mixolydian Mixed

**Grup 4**

Ascending intervals: P1, M2, m3, M3, P4, P5, M6, m7, P8

- Dorian Bebop
- Minor Bebop
- Mixolydian and Dorian Mixed

**Grup 5**

Ascending intervals: P1, m2, m3, M3, Aug4, P5, M6, m7, P8

- Half-whole Tone Scale
- Messiaen Mode 2 (4 groups of 3 notes each)
- Symmetrical Diminished

**Grup 6**

Ascending intervals: P1, m2, M2, P4, Aug4, P5, m6, M7, P8

- Messiaen Mode 4 (2 groups of 5 notes each)
- Messiaen Mode 5 (2 groups of 5 notes each)



